

War and Peace in the Marketplace

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Preface

This book started, fifteen years ago, with a question I could not get the field's vocabulary to answer: why are the same states that compete fiercest in international markets also the same states that compete fiercest militarily, and why are the same states whose strategic situations are most constrained by both at once? The disciplines that study the wealth side and the power side of foreign policy have grown up apart, in different journals, with different vocabularies, asking different questions. But the states they study have not. Whatever else this book turns out to be, it is the book that tries to honor that observation as a structural fact about the international system rather than as a stylized fact about scholarly organization.

The substantive thesis is that the international political-economic system is one phenomenon, not two phenomena coupled, and that any analytical apparatus treating it as two phenomena coupled produces answers that diverge from the right answers structurally rather than as a matter of refinement. States make consumption decisions and deployment decisions jointly, because the prices that constrain consumption are themselves products of deployment, and the costs of deployment are denominated in goods whose prices the consumption decisions face. The international system is the object that emerges from those joint choices, and its analytical apparatus has to be built for that joint object specifically.

Gilpin [1975] called the substantive content of this thesis *the reciprocal and dynamic interaction in international relations of the pursuit of wealth and the pursuit of power*. The phrase, in roughly that form, has been with us for more than three centuries: from the seventeenth-century natural-law tradition through the eighteenth-century commerce-versus-conquest debates, the nineteenth-century political economy of war finance, the twentieth-century recovery of the *jealousy-of-trade* synthesis, and the contemporary international-political-economy program that the book engages in detail. What has been with us less continuously is the formal apparatus the joint pursuit requires. The book is that apparatus.

Who this book is for. The book is written for two audiences at once. The first is the formal-economics-and-formal-IR audience: general-equilibrium theorists, conflict economists, bargaining-theorists-of-war, mechanism-designers, and the readers whose shelves hold Mas-Colell, Schelling, Powell, and Skaperdas alongside Balasko and Hirsch. The second is the substantive-international-relations audience: scholars of international political economy, security studies, comparative political economy, and historical political economy, the readers who came to international relations through Gilpin, Krasner, Tilly, Hont, and Strange. The two audiences read different journals, attend different conferences, and have different burdens of proof. The book is written to be read by both, and structured so each can find what they need without reading every page of the other audience's material. A third constituency the book hopes to engage is the methodologists and philosophers of social science who think about how analytical traditions make their objects and what they leave out when they do.

How to read this book. Chapter 1, *The joint pursuit*, is the philosophical-and-methodological opening: it argues for the substantive thesis, traces its intellectual genealogy, engages the principal dissents, develops the framework’s joint-determination commitments, distinguishes the framework from the partialled traditions of formal IR and political economy, and reads the verbal-historical regime-change literature through the apparatus. It is also long, and intense by design. Readers who want the substantive argument can read it narrowly—Section 1.1, Section 1.2, and Section 1.3—and move directly to Chapter 2. Readers who want the structural-formal apparatus and the methodological positioning can either read the whole chapter or skip the genealogical and dissent material and pick up at Section 1.4. Section 1.7 maps the rest of the book chapter by chapter; readers who want a brief structural tour can start there.

Chapter 2 through Chapter 4 build the apparatus formally. Chapter 5 through Chapter 8 deploy it on welfare, donor aid and multilateral peacemaking, institutions and regime change, and sanctions. The deployment chapters each draw on Chapter 4 but are otherwise independent of one another. Sections marked with a trailing asterisk in any chapter are skippable on first reading; they cover the technical material—transversality, generic regularity from primitives, index theorems, catastrophe-theoretic stratification—that substantive readers can take on faith. Within each chapter, the dual-mandate structure lets smooth-general-equilibrium readers and formal-international-relations readers each find what they need without reading every paragraph of the other audience’s material.

Chapter 9, *Of wealth and power*, is the close. It certifies what the book has delivered, addresses the gap-naming-predecessor cluster from Section 1.1 directly, and flags what the framework holds fixed for the next generation of the program.

Semi-textbookiness. The book is not a textbook. It advances a substantive thesis and develops the formal apparatus the thesis requires, which is monograph work, not pedagogical work. But it is also not a traditional monograph: the dual mandate just described means that some readers will arrive with the formal apparatus already in their hands and others will need it built up from the inside. The compromise is something between the two—a *semi-textbook*. In that spirit, each chapter ends with a short list of exercises for readers who want to practice the apparatus on their own terms, and a brief mathematical appendix at the back of the book gathers the background material the technical sections rely on. Neither addition is intended to make the book a teaching object. Both are intended to make the apparatus more accessible to readers whose primary training is in one half of the dual mandate.

Chapter 1

The joint pursuit

The international political-economic system is one phenomenon, not two phenomena coupled. The standard analytical apparatus treats it as two phenomena coupled: exchange in one set of vocabularies, conflict in another, with their interaction taken as the object recovered when both vocabularies are deployed at once. The book's claim is that the two-phenomena treatment is not a worse approximation of the right theory; it is a different theory whose answers diverge structurally, not quantitatively. The book supplies the apparatus the one-phenomenon view requires, and demonstrates on three substantive applications that the apparatus generates results—counterfactuals, welfare gradients, institutional comparisons, regime-change classifications—that the two-phenomena apparatus cannot generate from any combination of refinements.

The thesis the apparatus formalizes is older than the apparatus. Gilpin called international political economy the joint and reciprocal pursuit of wealth and power. The same observation runs through the European political-philosophical tradition from Pufendorf's natural-law treatment of war, leagues, and commerce as one jurisprudential fabric, through Hont's synthesis of the eighteenth-century *jealousy of trade* debates. The international-relations and international-political-economy fields of the past half-century have repeatedly named the gap between the substantive thesis and the apparatus required to do work with it. The thesis is not new; the apparatus is.

The chapter develops the master claim by stages, in six substantive sections plus a closing. Section 1.1 states the sibling claim—that exchange and conflict are joint moments of one specific object—and locates the claim in the gap-naming-predecessor chain running from List through Mansfield. Section 1.2 traces the European political-philosophical genealogy of the joint phenomenon from the seventeenth century forward, as evidence that the joint object was visible long before the apparatus could be built. Section 1.3 engages the principal dissents within this tradition—Schumpeter's atavism thesis from one side, Rosenberg's historical-materialist critique from the other—and shows that the framework brackets between them. Section 1.4 develops the framework's joint-determination commitment at the agent level—states are economically embedded; the opportunity-cost of force takes prices as a primitive argument—and at the system level—the equilibrium object is one manifold, not two coupled. Section 1.5 develops the structural argument that distinguishes the framework from the partialled traditions: each tradition's apparatus is the joint comparative-static under a specific restriction; the framework lifts both restrictions, and the difference is structural rather than a tractability tradeoff. Section 1.6 closes with the substantive reading of regime structure as multiplicity in the joint manifold's component decomposition, and flags what the framework holds fixed for the conclusion's future-work scope.

The chapter’s master claim is one observation; the framework is one apparatus. The two together constitute what the rest of the book argues for, builds, and deploys.

1.1 The sibling claim, named

States that engage most actively in international markets tend to be the same states that engage most actively in international military affairs. The Dutch Republic in the seventeenth century was simultaneously the dominant maritime trading power and the dominant naval power of its era; eighteenth-century Britain extended both roles, and the British East India Company—one institutional artifact of this extension—is conventionally read as either an instrument of trade or an instrument of empire, with each reading suppressing the other. Twentieth-century Germany and Japan compressed an aggressive expansion into international markets and a comparable expansion into military projection within decades of each other. The contemporary United States combines the world’s largest military force with one of its most internationally exposed economies; contemporary China’s expansion into international markets is paired with a military modernization that the United States explicitly tracks as an indicator of geopolitical ambition. The pattern is too stable across too many centuries to be coincidence.

The standard analytical separation of these activities is a substantive theoretical commitment, not an innocent tractability move. A reader of the academic literature in international relations would encounter wealth-pursuits and power-pursuits in distinct journals, distinct course catalogs, and distinct subfields within distinct disciplines. The political-economic study of trade lives in international political economy and (more recently) economics; the study of military competition lives in security studies; the institutional study of cooperation lives in international organization scholarship. Each subfield has its own vocabulary, its own canonical texts, and its own benchmark questions, and a typical research career crosses one of these boundaries only as a deliberate methodological decision rather than as the default mode of work. Whatever else this division-of-labor reflects, it does not reflect a finding that the activities themselves are separable at the level of the international system.

The thesis the rest of the book defends is that the activities *are not* separable at the level of the international system—that exchange and conflict are joint moments of the same underlying competitive structure, and the analytical separation of wealth-pursuits from power-pursuits hides features of the system that exist independently of the analytical separation. The thesis has two readings, neither sufficient on its own. The weak reading is empirical: it observes that states which pursue wealth tend also to pursue power, and that the historical correlation between trade-engagement and military-engagement is robust enough to require explanation. The strong reading is structural: it claims that the joint pursuit is a property of the system’s underlying primitives, that any framework which treats wealth and power as separable will systematically fail to recover features of the system that operate through their joint determination, and that the correlation the weak reading observes is the empirical signature of a structural fact about how states optimize. The book defends the strong reading. The weak reading is the threshold the strong reading must clear; the strong reading is the substantive content the formal apparatus develops.

Gilpin’s articulation. The clearest modern statement of the joint-pursuit thesis in the international-relations vocabulary is Gilpin [1975, p. 43]: *“political economy in this study means the reciprocal and dynamic interaction in international relations of the pursuit of wealth and*

the pursuit of power.” Gilpin’s framing is sharper than the weak empirical reading: the two pursuits are not separately well-formed activities that happen to correlate; they are reciprocal moments of a single dynamic interaction, with the state’s strategic choices answerable to both at once. Keohane [1984, p. 21] receives Gilpin’s definition as a “working definition” for the field of international political economy, glossing it as the claim that “*causality is reciprocal rather than unidirectional: on the one hand, the distribution of power creates patterns of property rights within which wealth is produced and distributed; on the other hand, changes in productive efficiency and access to resources affect relations of power in the long term.*” The reciprocal-causality reading makes the structural content of the claim explicit: the distinction between economic and political analysis of international affairs is an artifact of academic specialization rather than a feature of the phenomenon being studied, and the field that takes seriously the joint pursuit will be the field that takes seriously the integrated analysis of states as political-economic actors.

What Keohane gets right is the further diagnostic move. “*In the neoclassical economic theory of value, wealth is not used as a primary category to explain demand or prices; on the contrary, value (hence wealth) is inferred from demand and supply, as reflected in price movements on markets. Thus the concepts of power and wealth have a common deficiency as the basis for explanations of behavior: to estimate the power of actors, or whether a given product, service, or raw material constitutes wealth, one has to observe behavior... To use the concepts of power and wealth to account for the behavior that allows us to identify their presence would be to engage in circular reasoning*” (Keohane, pp. 19–20). The diagnosis is precise and the framework inherits it: wealth and power as primary categories cannot do explanatory work because they have to be inferred from the very behavior they would explain. The implicit prescription is to focus on the *processes* that produce the behavior from which wealth and power are then identified. The framework’s joint-equilibrium structure is precisely such a process focus: prices and military deployments are simultaneously fixed by the strategic choices of states, and wealth and power are recovered from the equilibrium pair (p^*, m^*) rather than imposed on it.

Where Keohane departs from the framework is in the disjunction. The diagnosis names a “common deficiency”—wealth and power both inferred from behavior—and treats the inference as taking place in two parallel domains: “*in power relationships or in markets.*” The disjunction reinstates the separation the diagnosis had begun to question. The reciprocal-causality reading is then schematized in the next paragraph as Figure 2.1’s three-by-three matrix of high-medium-low entries on two axes, with international political economy at the cell where both wealth-orientation and power-orientation are “high.” The two-axis schematization is the careful institutionalist’s domestication of the joint-pursuit insight: it preserves the surface commitment that both axes matter, while reinstating the separability that allows them to be axes at all. The framework’s response is that it is not an *or*; it is not even an *and*; it is one thing. Prices and military deployments are not separate domains in which wealth and power can be inferred respectively; they are the joint coordinates of a single equilibrium object whose existence and structure are properties of the joint formulation. Section 1.4 returns to Figure 2.1 directly.

What Gilpin did not provide is the formal apparatus. The 1975 statement is an analytic commitment with empirical and historical content but no machinery for computing how wealth and power respond jointly to perturbations of the system’s parameters. The fields of international political economy and security studies absorbed the substantive thesis without acquiring the apparatus the thesis required, and the gap between thesis and apparatus has been named, repeatedly and openly, by the discipline’s senior figures across the two centuries before the present book.

The gap, two centuries deep. List [1841] pressed the distinction in his own vocabulary. There is, on his account, a *cosmopolitical* economy—the science of how the entire human race may attain prosperity, after Quesnay, Smith, and Say—and a *political* economy proper—the science of how a given nation pursues prosperity, civilization, and power. The cosmopolitical version cannot speak to the political dimension because it abstracts from the existence of distinct nations. The remedy List proposed was a separate science, with the same analytical clarity as its individualistic counterpart, organized around the productive powers and political-economic configurations that are bound to nations rather than exchangeable on the world market. List was clear about what was missing: not a different substantive thesis, but a parallel analytical apparatus.

Strange [1970], in the article Keohane [2009] identifies as inaugurating the field of international political economy, named the same gap as one of *mutual neglect*. The international economic system was outpacing the international political system, the two had to be studied jointly, and neither international-economics scholarship (politically inattentive) nor international-relations scholarship (economically uncurious) was equipped to do so. The field Strange founded was meant to be the joint study; its analytical apparatus was meant to refuse the disciplinary separation that constituted the gap. The field’s quantitative turn over the half-century that followed operationalized within the separation rather than against it.

Keohane and Nye [1987], looking back on the post-1977 program of *Power and Interdependence*, named the gap as a self-assessment. “*Less has been done with the liberal than the realist half of our attempted synthesis. We have only partially incorporated the liberal emphasis on institutions, interdependence, and regularized transnational contacts into a sophisticated, systematic analysis of process and structure in world politics . . . research at the systemic level alone may have reached a point of diminishing returns*” (pp. 752–753). The institutionalist self-diagnosis is precise: the synthesis was attempted, the liberal half was undersupplied, and the apparatus capable of integrating the two halves was not yet on hand.

Mansfield [1994], fifteen years after Keohane and Nye’s self-assessment, named the gap in the formal-empirical vocabulary the field had moved toward by then. “*The lack of a unified deductive framework that incorporates features of both the distribution of power and trade continues to hamper theories of this sort*” (Ch. 7). Mansfield’s diagnosis is the empirical-IR reformulation of what Strange and Keohane and Nye had said: the apparatus has not been supplied; the substantive thesis cannot be carried by the existing tools.

The gap-naming runs the length of the modern field. List in 1841, Strange in 1970, Gilpin in 1975, Keohane and Nye in 1987, Mansfield in 1994—five voices across a century and a half, in distinct vocabularies, naming the same absence. What each named, the framework supplies. Section 1.4 develops the apparatus at the level of definitions and structural commitments; Chapter 2 through Chapter 8 build it formally and deploy it on three substantive applications.

The sibling, not the alliance. Wealth-pursuits and power-pursuits are siblings, not allies. This distinction is substantive. An *alliance* between two analytical traditions reads each as separately well-formed and asks how they interact at the margin: trade theory describes the price system; conflict theory describes the contest; the political economy of international relations describes the interaction between them. The interaction-at-the-margin reading is what the standard field organization performs at its best. Each tradition retains its primitives; the interaction is the object that connects them. What the sibling claim refuses is the prior separability the alliance presupposes. Wealth-pursuits and power-pursuits are not separately

well-formed activities that interact at the margin; they are joint moments of a single competitive process, with the appearance of separable margin-interaction being an artifact of the analytical separation rather than a feature of the underlying system. The structural claim is that the primitives are joint, not that the interaction is large.

The contrast with three traditional positions on the relation between wealth and power makes the sibling claim's content clearer. *Realism* reads states as pursuing power, with wealth as an instrument of power-acquisition or a side-effect of power-projection. *Liberalism* reads states as pursuing wealth, with conflict as a friction on the gains-from-trade that institutional design can reduce. *Constructivism* reads states as pursuing identity, with wealth and power as objects whose meaning is constructed by the discourses through which states understand their own roles. Each tradition picks one primitive (power, wealth, identity) and reduces the others to it. The sibling claim refuses all three reductions: wealth and power are coequal primitives of states' strategic situations, neither reducible to the other, and the framework that takes them seriously as siblings produces structural results that the single-primitive readings cannot. The book's argument with each tradition is local, not total: the realist reading is correct about the primary role of military capability, but wrong about the reducibility of trade-pursuits to power-pursuits; the liberal reading is correct about the substantive role of trade-pursuits, but wrong about the reducibility of conflict to friction; the constructivist reading is correct about the discursive scaffolding that makes states' strategic situations interpretable, but wrong about the constructed-rather-than-primary status of wealth and power as objectives.

What the framework promises. The book defends the sibling claim by constructing a formal apparatus that takes the joint pursuit of wealth and power as its basic object, and by deploying that apparatus on three substantive questions in international relations. The basic object is a smooth equilibrium manifold $\tilde{E}$ on which prices and military deployments are simultaneously fixed by the strategic choices of states under standard assumptions on preferences, endowments, and conflict primitives. The apparatus's signature capability is the joint comparative-static $D_{\theta}(p^*, m^*) = -J^{-1}D_{\theta}e$, which computes the simultaneous response of wealth (prices) and power (deployments) to any parameter perturbation by one inversion of the joint Jacobian. The framework specializes to the existing partial frameworks in their respective parameter limits and produces new results in the union of what each leaves on the table.

The three substantive applications close the gap between the formal apparatus and the older tradition's empirical preoccupations. The first, Chapter 6's donor-intervention problem, shows what the framework says about peacemaking aid: transfers evaluated at fixed prices cannot reduce conflict when the contest operates through the price system, and the n -fold underprovision of fragmented bilateral aid delivers a resource-neutral rationale for multilateral peacemaking institutions. The second, the institutional theory of Chapter 7, shows what the framework says about the coordinative and constructive functions institutions perform on the equilibrium manifold: aggregation, gradient-field implementation, and regime change as bifurcation across the singular locus Σ . The third, the sanctions theory of Chapter 8, shows what the framework says about coercive instruments: sanctions as parameter perturbations on $\tilde{E}$, with the joint comparative-static apparatus delivering the structural counterfactual the empirical literature on sanctions effectiveness has been working without. Each application earns its place by showing that the framework produces results no partialled tradition can deliver—an empirically-relevant result, an institutionally-relevant result, a coercion-relevant result, in each case derived as specializations of the same apparatus and consistent with the older tradition's substantive claims about the

joint pursuit of wealth and power.

The next section traces the European political-philosophical genealogy of the joint phenomenon. The recurring observation across two centuries was that the joint object was visible to careful observers in non-formal vocabulary; what was missing was the apparatus the present chapter develops in Section 1.4.

1.2 Commerce and conquest: a genealogy

The European political-philosophical tradition recognized the joint pursuit of wealth and power as a single phenomenon long before the apparatus to compute its consequences was available. The chain of authors who recognized it spans more than three centuries: from Pufendorf in the late seventeenth century, through Hume, Smith, Montesquieu, and Kant in the eighteenth, Ricardo in the nineteenth and Pareto across the marginalist break, to Hirschman and Hont in the twentieth and twenty-first. The substantive claim is stable across the vocabularies in which the chain expresses it—seventeenth-century natural-law-of-nations, eighteenth-century political economy, mid-twentieth-century intellectual history, contemporary synthesis. What varies is the analytical apparatus brought to bear; the substance recurs in each idiom. The genealogy’s role in the present chapter is to document this stability. The joint phenomenon was visible to careful observers across two centuries of non-formal work, and the framework the rest of the book develops is the apparatus those observers’ substantive claims required but could not themselves produce.

The line is interrupted at the marginalist break of the 1870s, when nineteenth-century social science divided political economy into economics and political science, divided economics into general-equilibrium and trade-theoretic branches, and divided political science into security-studies and international-political-economy branches. What was a single object in the eighteenth-century political-philosophical tradition—the joint pursuit of wealth and power by competitive nation-states—became a set of mutually unintelligible partial objects in the disciplines that inherited it. The genealogy stops at the marginalist break and resumes in the twentieth century with the recovery work of Hirschman and Hont. The interruption is what makes the recovery substantive: the present book’s framework returns to a phenomenon the disciplines that succeeded the older tradition had lost the conceptual vocabulary to name.

Pufendorf and the natural-law substrate. Pufendorf [1672]’s *De Jure Naturae et Gentium* systematized the seventeenth-century natural-law tradition¹ into a treatise on the law of nations

¹The natural-law-of-nations tradition Pufendorf systematizes runs deeper than the seventeenth century. Its modern foundation is Grotius [1625]’s *De Jure Belli ac Pacis*, which gathers the medieval scholastic just-war tradition—Augustine, Aquinas, the sixteenth-century Salamanca school of Vitoria and Suárez—and recasts it as a secular natural-law treatise on war, peace, treaties, and the rights of nations. Behind Grotius lie classical sources: principally Polybius’s *Histories*, with the Polybian distinction between justifying-reasons and motives that the modern just-war tradition inherits, and Cicero’s *De Officiis* for the Roman just-war framework Augustine and Aquinas Christianize. The natural-law-of-nations apparatus Pufendorf inherits is already two millennia deep when he reaches it. What is distinctive to Pufendorf, for the present argument, is not the natural-law apparatus itself but its application: the extension of natural-law jurisprudence to specifically political-economic objects—commerce, leagues, the wealth and power of nations as joint juridical questions rather than separate ones. The chapter starts at Pufendorf because the joint-pursuit lineage starts there. Grotius is the natural-law trunk; the classical and patristic sources are its roots; the eighteenth-century commerce tradition of Hume, Smith, and Montesquieu inherits Pufendorf’s specifically political-economic application of the apparatus, not the natural-law apparatus directly.

that takes states' commercial, diplomatic, and military relations as a single jurisprudential object. The treatise's eight books cover the law of nature, the faculties of man, the duties of contract, sovereignty, the law of war, the law of leagues, and the natural duties of peace between nations as continuous chapters of a single investigation. What the treatise does not do is what the later disciplines do: it does not divide the law-of-nations into a commercial branch to be analyzed in one register and a war-and-peace branch to be analyzed in another. Pufendorf's text treats the rights of war, the rights of leagues, and the duties of commerce as expressions of the same underlying natural-law structure—a single fabric of obligation in which states' commercial and military activities are coordinated by the same juridical principles. For the present argument, the natural-law substrate matters because it is the seventeenth-century baseline from which the eighteenth-century commerce-tradition departs: the eighteenth-century authors who articulate the joint-pursuit thesis do so in conscious reference to the natural-law tradition, retaining its juridical scope while replacing its juridical idiom with a structural-political-economic one.

Hume on jealousy of trade. Hume [1752]'s *Of the Jealousy of Trade* is the canonical eighteenth-century statement of the joint-pursuit thesis in essay form. Hume observes that European states treat neighboring states' commercial successes as threats rather than as opportunities, and argues that this disposition—the “jealousy of trade” of the title—is a competitive logic under which the wealth-pursuits and power-pursuits of one state constrain those of its neighbors. The essay's argumentative structure is reconstructive: jealousy of trade is identified as a substantive empirical regularity in European foreign policy, traced to a specific competitive mechanism (the redistribution of commerce as a quasi-zero-sum contest among states), and then critiqued by a counterargument (the gains-from-trade are positive-sum at the global level, so the jealousy is locally rational but globally self-defeating). The companion essay, *Of the Balance of Trade*, develops the structural side: the price-specie-flow mechanism is a genuine equilibration process, and the mercantile policies that the jealousy of trade rationalizes are typically unable to produce sustained advantages because the equilibration corrects them. What Hume delivers, for the present argument, is the joint-pursuit phenomenon stated in eighteenth-century political-economic vocabulary as a structural feature of the international system, with explicit recognition that the commercial competition between states operates simultaneously as power competition.

Smith on the cusp. Smith [1776]'s *Wealth of Nations* contains the clearest pre-formalization statements of the wealth-and-power coupling. The Book IV critique of mercantilism is the canonical reading of Smith as the founding figure of liberal international political economy: the gains-from-trade argument, the impolicy of restraints upon importation, and the self-correcting character of the international division of labor. That reading is correct as far as it goes, but it is partial. Smith also says, with equal seriousness, that “defence is of much more importance than opulence” (WN IV.ii.30), that the wealth of a neighboring nation “is dangerous in war and politics, for it enables our enemies to maintain fleets and armies superior to our own” (WN II.v.31), and that “the great object of the political economy of every country is to increase the riches and power of that country” (WN II.v.31). The same author who states the gains-from-trade argument states the wealth-as-power-for-defense argument and the joint-objective characterization of political economy. Hont reads these passages as the cusp of the formalization the present book completes: Smith articulated the joint pursuit in mid-eighteenth-century political-economic vocabulary, with the conceptual apparatus to distinguish the gains-from-trade case from the defense-and-power case but without the formal machinery to compute their joint determination. The contemporary

reading that treats Smith as a proponent of trade-as-pacification reads the gains-from-trade chapters in isolation from the defense chapters; the reading this book takes is that all three statements belong to a single Smithian theory of the joint pursuit of wealth and power by competitive states.

Montesquieu and the doux-commerce thesis. Montesquieu [1748]’s *The Spirit of Laws* states the doux-commerce thesis in its canonical form: “it is almost a general rule that wherever the ways of man are gentle there is commerce; and wherever there is commerce, there the ways of men are gentle” (Bk XX Ch I). The thesis as commonly received is a claim that commerce pacifies international relations. The thesis as Montesquieu states it is more specific: commerce reorganizes the relationship between political structure and economic activity such that some combinations of polity and economy become unreachable. Book XXII Chapter XIV’s title makes this regime-theoretic claim explicit—“How the Exchange Is a Constraint on Despotic Power”—and the body of the chapter argues that the bill-of-exchange instrument is structurally inconsistent with despotic governance, so that a state which adopts the exchange-based commercial structure has narrowed the polity arrangements available to it. What Montesquieu contributes to the genealogy is a regime-theoretic reading of the joint pursuit: states’ commercial structures and political structures are mutually-constituting coordinates rather than independent margins, and the equilibria of one constrain the other. The framework’s reading of regime structure as multiplicity in the joint manifold’s component decomposition (Chapter 4 and Section 1.6) is the formal correlate of Montesquieu’s regime-incompatibility argument.

Kant on perpetual peace. Kant [1795]’s *Perpetual Peace* extends the doux-commerce thesis into a theory of the institutional conditions under which the joint pursuit of wealth and power can be stabilized cooperatively. The Preliminary and Definitive Articles of the treatise specify constitutional arrangements (republican government), federative arrangements (a league of free states), and cosmopolitan arrangements (universal hospitality) under which the competitive logic Hume described as jealousy of trade can be replaced by a coordinative logic. Kant’s argument is not that commerce alone pacifies, but that commerce together with appropriately designed political institutions can shift the international system from the competitive to the coordinative regime—a regime-change claim, in the present book’s vocabulary, with institutional parameters playing the role of bifurcation parameters. The institutional theory of Chapter 7 returns to Kant as the canonical statement of the institutional approach to regime change in the joint-pursuit-of-wealth-and-power tradition.

Ricardo on war finance. Ricardo [1817]’s *Principles of Political Economy and Taxation* and the later *Funding System* essay [Ricardo, 1820] extend the Smithian framework to a structural question about the wealth-and-power coupling: how a state finances its wars. The choice between current taxation and funded debt as instruments of war finance is not financially neutral, on Ricardo’s reading, but politically consequential. A state that pays for wars out of current taxation makes the cost visible to taxpayers in the period in which the war is fought; a state that issues debt defers the cost across generations and obscures it from the cohort that authorizes the war. Ricardo argued, against the prevailing British practice of funded-debt war finance, that the funding system systematically distorts the political economy of war. It creates a rentier class whose interests diverge from productive activity; it generates an unfunded liability that constrains future fiscal capacity; it gives governments an incentive structure under which they

under-internalize the true cost of war when authorizing it; and it generates the appearance of limitless fiscal capacity at the moment of mobilization that the underlying productive economy does not actually possess. The argument identifies a specific instrument—the choice of war-finance vehicle—that operates jointly on the wealth side (the productive structure of the economy, the distribution of income across rentier and productive classes) and on the power side (the political-economic feasibility of war and the fiscal constraints on its prosecution). Ricardo’s contribution to the genealogy is the demonstration that the eighteenth-century joint-pursuit thesis admits sharper, instrument-specific articulations in nineteenth-century political economy: it is not enough to claim that wealth and power are joint at the level of the system’s primitives; the joint pursuit operates through specific institutional instruments, and the political economy of those instruments is itself a structural question about how states pursue wealth and power.

Pareto on production and appropriation. Pareto [1906]’s *Manuale di economia politica* carries the joint-pursuit insight across the marginalist break that absorbed most of late-nineteenth-century political economy. Pareto, one of the founders of mathematical general-equilibrium theory, retained the structural distinction between the two modes of economic activity: “*the efforts of men are utilized in two different ways: they are directed to the production or transformation of economic goods, or else to the appropriation of goods produced by others*” (Ch. VII). The distinction is precise and structural rather than moral. A single agent allocates effort along the margin between productive activity and appropriative activity, with both modes fully economic in the sense of being subject to the same rational-self-interest analysis the marginalist tradition was in the process of formalizing. The rest of marginalist economics largely abstracted away from the appropriative mode, treating it either as a market failure or as outside the proper domain of economic analysis. Pareto did not. For the present argument, Pareto’s contribution is the demonstration that the joint-pursuit thesis survives the marginalist transition in the work of an author who was himself central to the marginalist program: the production-versus-appropriation margin is what general-equilibrium theory could have internalized, had it followed Pareto’s lead, and the formal conflict-economics tradition of the late twentieth century returns to Pareto’s distinction as its starting point.

Hirschman’s recovery. Hirschman [1977]’s *The Passions and the Interests* is the twentieth-century recovery of the eighteenth-century synthesis. Hirschman reconstructs the eighteenth-century argument that interests can be made to govern passions in the constitution of the modern commercial state, locating the synthesis as a specifically eighteenth-century achievement prompted by the Wars of Religion’s destabilization of the older just-war framework. For the present argument, Hirschman’s contribution is twofold. First, the book documents the existence of the eighteenth-century synthesis as a coherent intellectual project that nineteenth-century social science fragmented and lost. Second, it provides the methodological precedent for the reconstructive mode the present book operates in: the eighteenth-century synthesis is recovered not by intellectual historiography alone but by reconstructing in modern terms what the synthesis was attempting to deliver. The framework’s relationship to the eighteenth-century political-economic tradition is the same relationship Hirschman traced between his mid-twentieth-century reconstruction and the eighteenth-century originals: the older tradition’s substantive claims are recovered, the formal idiom is updated, and the structural content is preserved.

Hont’s synthesis. Hont [2005]’s *Jealousy of Trade* provides the synthesis-across-traditions

that the genealogy needs. Hont integrates four distinct intellectual streams into a single account of how the joint-pursuit thesis was articulated in early-modern Europe: the post-Machiavellian *reason of state* tradition, the natural-jurisprudence tradition descending from Pufendorf and Smith, the post-Hobbesian state-and-society tradition, and the post-Augustinian moralism that critiqued commercial sociability from within. The four streams converge in the eighteenth-century synthesis Hont names *jealousy of trade*: the recognition that the logic of trade and the logic of war are joint moments of a single competitive structure operating in the foreign policy of European nation-states from the late seventeenth century onward. Hont's gloss is that "*jealousy of trade was a mongrel idiom, describing a process of corruption. It implied that in modern politics the logic of trade was bent to the logic of war.*" The mongrel-idiom reading recovers the structural content of the joint-pursuit thesis as a single phenomenon that eighteenth-century authors articulated in a vocabulary that nineteenth-century social science decomposed into separate specialties. The vocabulary Hont reconstructs—reason of state, jealousy of trade, balance of wealth, national emulation, natural progress of opulence, retrograde and unnatural order, closed commercial state, commercial sociability, unsocial sociability, politics of necessity—is the technical apparatus of a tradition the present book's formal framework returns to. The book's title *War and Peace in the Marketplace* is from before the marginalist break of the 1870s; the tools are from after; the deployment is against the politics-economics separation the marginalist revolution instituted.

What the genealogy delivers. The genealogy supports three claims, each specific. First, the joint pursuit of wealth and power was articulated in non-formal terms by a chain of authors from Pufendorf through Hont, with the substantive claim stable across vocabularies as different as seventeenth-century natural law, eighteenth-century political economy, mid-twentieth-century intellectual history, and contemporary synthesis. Second, the analytical separation of wealth-pursuits from power-pursuits is recent and specific to the disciplinary fragmentation of nineteenth-century social science; it is not a feature of the underlying phenomenon, and the older tradition that preceded the fragmentation did not require it. Third, the formal apparatus this book develops is best understood as a structural-formal recovery of the older tradition's substantive object, with the apparatus playing the role of the structural-formal correlate of what the older tradition stated in non-formal terms. The framework, on this reading, is not a new claim; it is a new instrument for an old claim, and the older claim is the better-supported one.

The dissenting voices remain. The eighteenth-century synthesis was not monolithic, and the twentieth-century inheritors disagreed about what to keep from it. Schumpeter and Rosenberg are the clearest cases of voices in this tradition that argue against the joint-pursuit reading on grounds the present book takes seriously. Section 1.3 engages them.

1.3 The dissenting voices

A tradition defended without honoring its critics is a tradition defended poorly. The genealogy of Section 1.2 threads through authors who agreed that the joint pursuit of wealth and power is a structural feature of the international system, but the agreement is not unanimous. Two external dissents matter most. The *too-confident-monotonicity* dissent of Schumpeter argued that pure capitalism is intrinsically pacific, so that the joint-pursuit thesis is wrong about the system's underlying competitive structure. The *historical-materialist* dissent of Rosenberg argued that the joint-pursuit thesis as ordinarily stated naturalizes a politics-economics separation that is itself a historically-specific capitalist artifact. The two dissents come from opposite directions

and reach different conclusions about the framework’s validity. Both are anticipated, in part and in different vocabulary, by an internal-tradition critique that runs through Ferguson and Fenelon: the eighteenth-century synthesis was contested at its source, and the regime-theoretic complications the internal critics surfaced are antecedents of the later dissents the framework takes seriously. The present section engages the two external dissents at length and the internal antecedents briefly, and argues that the framework’s joint-equilibrium structure is what survives them.

Schumpeter and the atavism thesis. Schumpeter [1919]’s *The Sociology of Imperialisms* is the cleanest twentieth-century statement of the commerce-tames-war reading. Schumpeter argues that imperialism—defined narrowly as “objectless expansion,” expansion for its own sake beyond any concrete utilitarian aim—is an atavistic survival of pre-capitalist warrior aristocracies rather than a consequence of capitalist logic. Pure capitalism, on this account, is rational, calculating, and risk-averse in the warmaking dimension; territorial ambition and the warlike spirit are residual mentalities of the pre-modern social formations that capitalism is in the process of displacing. The argument is doubly polemical. On one side, it rejects the Marxist theories of imperialism (Hilferding, Luxemburg, Lenin) that tied imperialism to capitalist contradictions: in Schumpeter’s reading, capitalism does not require imperialism for its reproduction, and the Marxist theories misidentify the social sources of warlike behavior. On the other side, it sharpens the eighteenth-century commerce-pacifies-conflict thesis into a monotonic prediction: capitalism, more of it, weakens the atavistic warrior survivals that drive imperialism, and imperialism therefore declines as capitalism advances.

The framework rejects Schumpeter’s monotonicity without rejecting his analytic distinction. Hirschman’s reading of Schumpeter, which the present book endorses, is that Schumpeter captures the mid-eighteenth-century *doux-commerce* thesis in too clean a form—monotonic, unidirectional, and without the “knottiness” that authors within the eighteenth-century tradition itself already surfaced as inherent to the commerce-and-conflict relationship. Hume’s own essay on the jealousy of trade ended with a counterargument noting the persistence of jealous behavior even among states that ought rationally to favor liberalization; Kant treated the commercial disposition as one of several conditions for perpetual peace rather than a sufficient one. The internal-tradition critique of overconfident commercial-pacification is older than Schumpeter’s twentieth-century systematization, and the framework’s regime-theoretic reading inherits that internal critique. Where Schumpeter predicts a single trend (capitalism weakens imperialism), the framework predicts multiple regimes: a commercial-pacific regime is reachable on the joint manifold under specific parameter configurations, and so is a commercial-conflictual regime, and the regime the international system occupies at any given time is determined jointly by the system’s primitives rather than by a one-way trend in capitalist development. Schumpeter’s atavism thesis collapses into the framework as a boundary case: the parameter region in which the commercial-pacific regime is uniquely reachable.

The dissent within the tradition. The eighteenth-century synthesis was not monolithic, and the deeper internal dissents are older than Schumpeter’s twentieth-century version. Ferguson [1767]’s *Essay on the History of Civil Society* argued that the commercial regime’s capacity to produce internal tranquility was structurally ambiguous: tranquility could express the absence of external threat—the favorable case the *doux-commerce* thesis emphasizes—or the absence of the civic-republican virtue that makes the polity self-sustaining, and the same primitives

could support either reading. The distinction is regime-theoretic in the framework's vocabulary: two distinct equilibria, both pacific in the surface measure, generated by different parameter configurations on the joint manifold, with the same primitives capable of supporting either. What Schumpeter would later flatten into a monotonic prediction, Ferguson had already complicated structurally a century and a half earlier. [de Salignac de la Mothe-Fénelon \[1699\]](#)'s *Telemachus* pressed a deeper dissent against the commercial project itself, on moral and theological grounds the framework does not adjudicate. The framework inherits the existence of internal dissent, not its substantive content; what the dissent demonstrates is that the eighteenth-century tradition was sufficiently aware of its own costs to generate principled critique from within, and the regime-multiplicity reading the framework formalizes is the structural correlate of what Ferguson surfaced verbally.

Rosenberg and the historical-materialist dissent. [Rosenberg \[1994\]](#)'s *The Empire of Civil Society* mounts a different and more methodologically searching dissent. Rosenberg argues that the politics-economics separation taken for granted in realist and liberal international-relations theory is not a transhistorical feature of inter-state politics but a historically-specific artifact of capitalist modernity. Sovereignty and anarchy as ordinarily theorized are capitalism-specific social forms; the realist treatment of states-system anarchy as a universal baseline is a reification that loses the social-historical conditions that produce the politics-economics separation in the first place. The theoretical source for the substantive thesis is [Wood \[1981\]](#)'s argument that the differentiation of political functions under capitalism is the social-historical condition that produces the appearance of separable politics-and-economics spheres. On Rosenberg's reading, the canonical international-relations theories—realism, neoclassical international economics, and the institutionalist syntheses between them—all share an uncritical acceptance of the politics-economics separation, and their characteristic errors are downstream consequences of that shared inheritance. The diagnosis is not that the joint-pursuit thesis is wrong, but that it has been stated in a way that naturalizes the very separation it ought to call into question.

The framework converges with Rosenberg on the negative diagnosis and diverges on the positive reconstruction. The negative convergence is precise: the framework's insistence on joint equilibration in (p, m) space is the formal counterpart of Rosenberg's historical-materialist insistence that politics and economics are constitutive moments of a single social totality, not separable spheres that interact at the margin. Both projects refuse the politics-economics separation at the methodological root, and both locate international-relations-theoretic errors in its uncritical acceptance. The positive divergence is also precise. Rosenberg's reconstruction goes through Marx's *Capital* Volume I analysis of the commodity form and Wood's differentiation-of-political-functions thesis; the framework's reconstruction goes through the formal joint-equilibrium structure on a manifold of reachable regimes. The two reconstructions are not in competition: the framework treats anarchy as a specific reachable equilibrium type on the manifold rather than as a universal baseline, which is Rosenberg's structural-specificity claim restated in formal idiom. What the framework does not import from Rosenberg is the historical-specificity claim in its strong form—the claim that the joint-pursuit logic is specific to capitalism and absent in other social formations. The framework treats the joint-pursuit logic as operative wherever the system's primitives produce a non-trivial joint manifold, which is a more general condition than capitalist modernity, although the parametric instantiations of the framework most relevant to contemporary international relations will indeed correspond to capitalist-modern configurations.

What survives the dissents. The two external dissents bracket the framework from opposite sides. Schumpeter dissents from below: the joint-pursuit thesis is too pessimistic about capitalism’s pacifying tendency, and a more careful analysis of the economic logic shows that pure capitalism does not generate the warlike behavior the joint-pursuit reading attributes to it. Rosenberg dissents from above: the joint-pursuit thesis is too historically-naïve about the categories it uses, and a more careful analysis of the social-historical conditions shows that the politics-economics separation the thesis seems to take for granted is itself an explanandum. The framework is consistent with the part of each dissent that is correct and inconsistent with the part of each that is incorrect. From Schumpeter, the framework keeps the analytic distinction between rational-calculating commercial behavior and atavistic warlike behavior, and rejects the monotonic claim that the latter declines with the former; from the internal-tradition antecedents in Ferguson, the framework also keeps the regime-theoretic distinction between favorable and unfavorable tranquilities, which Schumpeter’s monotonic version flattens. From Rosenberg, the framework keeps the methodological refusal of the politics-economics separation, and rejects the historical-specificity claim in its strong form. The result is a position closer to Rosenberg than to Schumpeter on the methodology of the politics-economics relation, and closer to Schumpeter than to Rosenberg on the universality of the joint-pursuit logic across social formations. The framework is positioned in a region of theoretical space that neither external dissenter occupies, and that the internal-tradition antecedents anticipated without the apparatus to formalize.

The chapter now turns from the genealogy and its critics to the formal apparatus. Section 1.4 develops the framework’s joint-determination commitment at the agent level—states are economically embedded, the opportunity-cost of force takes prices as a primitive argument—and at the system level, where the equilibrium object is one manifold rather than two coupled. Section 1.5 develops the structural argument that distinguishes the framework from the partialled traditions: the joint apparatus and the partial methods are different theories, not different approximations to the same theory, and the specification fights that have run through the empirical-IR commercial-peace literature for half a century are signature of the structural difference rather than evidence of resolvable empirical disagreement.

1.4 Embeddedness and the joint manifold

The substantive name for the framework’s first commitment is older than the formal apparatus and well-known in twentieth-century political economy. Polanyi [1944] called it *embeddedness*: the claim that the human economy is not, and structurally cannot be, an autonomous system of self-regulating markets standing apart from politics and society. Polanyi’s diagnosis was that the nineteenth-century project of constructing a disembodied self-regulating market economy was *Utopian*—“*such an institution could not exist for any length of time without annihilating the human and natural substance of society*”—and that the historical record of the project, from the factory acts to the New Deal to fascism to the Soviet experiment, is the record of society’s protective response to the disembedding attempt. For the present argument, embeddedness is the substantive name for what the framework formalizes at two levels. At the *agent level*, embeddedness is the structural fact that no state’s optimization problem separates into a consumption decision at given prices and a deployment decision at given resources. At the *system level*, embeddedness is the structural fact that the equilibrium object is one manifold, not two coupled.

The two-level joint structure has been seen, in different vocabularies, by careful observers

across the twentieth century. Viner [1948]’s *Power versus Plenty as Objectives of Foreign Policy in the Seventeenth and Eighteenth Centuries* articulates the mid-twentieth-century version of the substantive joint-pursuit thesis as a four-proposition formulation of how power and plenty stand in relation to each other in foreign-policy analysis: power as an end, plenty as an end, power as a means to plenty, and plenty as a means to power. Viner’s argument was that the four propositions are jointly necessary in the foreign-policy reasoning of seventeenth- and eighteenth-century European states, and that any account treating wealth and power as independently pursued objectives misses the structural-historical record. Schelling [1960]’s articulation that strategy is concerned with “*the exploitation of potential force*” rather than “*the efficient application of force*” treats force as a coordinate of the strategic situation rather than an instrument applied to it—the conceptual predecessor of the framework’s *m*-as-coordinate move. Boulding [1962]’s *Conflict and Defense* draws (in Figure 1.6) the conflict line and the trading-set in the same Edgeworth diagram, treating them as joint structural features of the strategic situation. Strange [1988]’s four-primary-structures account of power—security, production, finance, knowledge—extends the joint-pursuit thesis into a multi-dimensional articulation that British IPE textbooks consolidated into a research program. Gilpin [1981]’s equilibrium-theoretic account of international systems change posits a structural balance between the costs and benefits of expansion that is itself a joint feature of the system’s wealth and power configurations. Tilly [1990]’s three-path model of state-formation reads European political development as a joint optimization over coercion-intensive and capital-intensive trajectories, with capitalized coercion as the convergent equilibrium under competitive pressure. Findlay and O’Rourke [2009]’s *Power and Plenty* recovers the macro-historical-economics evidence for the joint-pursuit thesis at book length, with the explicit framing that economic and military history have been structurally joint since the late medieval period.

What the framework does, against this verbal background, is supply the formal apparatus the predecessors gestured at but could not deliver. The apparatus is two-level by construction: it secures joint determination at the agent level via a single architectural choice about how a state’s optimization problem is set up, and at the system level via the geometric type of the equilibrium object. Both levels are necessary; the framework’s distinctive contribution is that it secures joint determination at both levels simultaneously, rather than at one level only.

The agent level: two coupled architectural choices. Each state i has, alongside preferences u_i , two structural primitives: an opportunity-cost function c_i that maps the available bundle of resources, prices, and strategic environment to the cost of allocating those resources between consumption uses and military-deployment uses, and a battle function s_i that maps the resulting deployment-and-consumption profile to the redistribution accruing to the state in the contest. The agent-level architectural commitment has two coupled parts. The opportunity-cost function c_i takes prices p as a *primitive argument*, not as a parameter to be solved separately and then plugged in afterward, so the cost of mobilization moves with the same prices the market will clear. The battle function s_i takes consumption-after-cost x as a *primitive argument*, not as a quantity that drops out once deployment levels are fixed, so the redistribution the contest produces depends on what is left to be redistributed once costs of mobilization have been paid. Together the two choices couple the consumption decision and the deployment decision through an entanglement matrix $R_i = I + D_{x_i}s_i$ that appears inside the state’s first-order conditions. The consumption FOC line cannot be solved without the deployment line, and the deployment line cannot be solved without the consumption line, because each carries the matrix R_i that the other contributes.

With $R_i = I$ the lines decouple and a textbook consumer-problem-then-deployment-subgame computation recovers the joint solution; with $R_i \neq I$ no such sequential computation does. The agent-level joint determination is the structural fact that, for a state operating in any non-trivial international market system, $R_i \neq I$ holds generically. Chapter 2 works the two architectural choices out in detail and shows that they are the single substantive commitment—two faces of one structural choice—from which the rest of the agent-level apparatus follows.

The agent-level commitment is what embeddedness becomes in framework vocabulary. Polanyi's claim that the disembodied economy is Utopian is the structural claim that the wedge $R_i - I = D_{x_i} s_i$ is non-zero generically rather than by stipulation: the disembodied special case $R_i = I$, in which consumption and deployment decouple and the partialled methods recover the joint solution, is the case in which the battle function ignores consumption—a knife-edge condition that no real international system satisfies. Polanyi was right about the substantive impossibility; the framework names where the impossibility lives, and Chapter 2 establishes the genericity of the wedge formally.

The system level: $\tilde{E}$ as one manifold. At the system level, the joint determination shows up as a property of the equilibrium object's geometric type. The framework's equilibrium object is the joint zero set $\tilde{E} := e^{-1}(0) \subset S_N \times M \times \Omega_0$, where $S_N \times M$ is the joint space of prices and military deployments and Ω_0 is the space of admissible endowment-and-parameter profiles. The map e is a single excess-demand-like function whose zero set encodes the joint conditions under which all states' optimization problems are simultaneously solved. What matters geometrically is that $\tilde{E}$ is a *single* smooth manifold, not the product of a price manifold and a deployment manifold with an interaction term modulating their relationship. The price-equilibrium and the deployment-equilibrium are coordinates on the same surface, and the surface's shape is determined by the system's primitives jointly rather than by two independent equilibration conditions adjusted by a coupling. Chapter 3 establishes the manifold structure of $\tilde{E}$ and proves the regularity properties on which the rest of the apparatus depends.

The two-axis foil: Keohane's diagram. The clearest pointable example of the formulation the framework rejects is Keohane's Figure 2.1 in *After Hegemony* (p. 21), which schematizes wealth and power as two orthogonal axes whose intersection is labeled *political economy*. Keohane introduces the figure immediately after quoting Gilpin's reciprocal-and-dynamic definition of political economy, glossed (as Section 1.1 reproduces) as the claim that causality between wealth and power is reciprocal rather than unidirectional. The diagram is a careful institutionalist's response to the joint-pursuit definition. Its analytical content is precisely that the joint-pursuit thesis is to be honored by treating wealth and power as separate analytic axes whose interaction is the new object of study, called *political economy* when both axes are considered at once. The diagrammatic move is not a misreading of Gilpin so much as a domestication of him. What Gilpin stated as a structural claim about the underlying primitives of state action, Keohane's figure rephrases as an analytic claim about the recommended dimensionality of the political-economic study object. The reformulation preserves the surface joint-pursuit commitment—the figure does treat both axes—while reinstating the separability the joint-pursuit commitment was meant to refuse. Most contemporary international political economy operates within Keohane's diagrammatic framing, and the framework this book develops is built to do work that the diagrammatic framing cannot.

Unification as generativity. The framework’s relationship to the two-axis formulation is not that the framework adds a new term to the two-axis interaction or specifies a richer interaction structure. The framework is what Carroll [2026c] calls a *unification* rather than a coproduct: the joint object $\tilde{E}$ is not the disjoint union of a price-equilibrium manifold and a deployment-equilibrium manifold glued at their shared rational-choice substrate, but a single manifold whose geometric type is irreducibly joint. The Carroll-2026 result is that unification, when it succeeds, is generative in a precise sense: a unified theory proves results that neither of its component theories proves and that the pushout of the components also fails to prove. The pushout corresponds to the careful institutionalist’s two-axis formulation—components glued at the shared rational substrate, with their interaction available for study. The unification corresponds to the framework—a single object whose properties are not deducible from the two components and their interaction at all. The book’s substantive results—the joint comparative-static formula, the regime structure with its singular locus Σ , the index theorem on the joint Jacobian, the bifurcation analysis, and the multiplicity-as-regimes reading of regime change—are generative in this technical sense. They are not entailed by any combination of pure trade theory and pure conflict theory, however refined the combination, and their absence from the partialled traditions is structural rather than incidental.

A reader-response question follows. Readers fluent in either of the partialled traditions will sometimes find that the framework’s claims look like elaborations of what the reader’s home tradition already provides. Carroll’s earlier result establishes a more general fact that explains this response without dismissing it: an observation family detects the value of unification only if it is not a left adjoint, and within-domain consequence operators are precisely left adjoints. The reader who occupies a single-domain observational stance will see the framework’s joint-equilibrium structure as “more theory in my domain” by a structural property of the observational stance, not by inattention. The framework asks for a non-adjoint observational stance. The questions the chapter and the rest of the book are written to address are of the joint type: *when does the joint formulation recover comparative-static results that the sequential formulations miss? which bifurcations are visible only at the joint level? which regime distinctions are coordinate-free properties of the joint manifold rather than artifacts of the factor-decomposition?*

One operator, four uses. The signature capability of the apparatus is the joint comparative-static

$$D_{\theta}(p^*, m^*) = -J^{-1}D_{\theta}e,$$

which computes the simultaneous response of military deployments and prices to any parameter perturbation by one inversion of the joint Jacobian. The same operator does work in four substantive contexts across the rest of the book—one apparatus chapter and three applications. Chapter 4 establishes the comparative-static apparatus itself.

In Chapter 5, J^{-1} delivers the welfare gradient $D_{\omega}\Delta^*$ —the cardinal welfare cost of conflict at any joint equilibrium, computed from the same joint inverse that delivers comparative statics. The gradient gives the framework’s normative content as a single matrix-vector operation rather than as a separate welfare-economic exercise.

In Chapter 6, J^{-1} delivers the donor’s gradient field—the direction in ω -space along which an external transfer reduces conflict—together with the multilateral coordination result that fragmented bilateral aid implements the same direction at $1/n$ magnitude. The application’s substantive headline—that partial-equilibrium thinking has zeroed out the only operative channel by construction—is the cleanest pedagogy of the master formula’s content.

In Chapter 7, J^{-1} delivers the institutional vector field $X^{\Delta^*} := -D_{\omega}\Delta^*$ —the welfare-improvement gradient that institutional design either implements or fails to—together with the homotopy structure of paths-on- $\tilde{E}$ that captures the regime-change content of institutions. What the chapter calls regimes are connected components of E^{reg} , and what it calls regime change is a path crossing the singular locus Σ .

In Chapter 8, J^{-1} delivers the joint sanctions response, with the substantive surprise that the bargaining-camp and exchange-side formulations recover as slices of the joint comparative-static. The difference between either slice and the joint formulation is a Schur-complement correction induced by the joint Jacobian’s off-diagonal blocks; on the canonical worked example, the correction is the entire joint response.

The same operator, applied against four parameter directions, delivers four substantive results that no combination of partialled methods produces. The chapter that introduces the operator is Chapter 4; the chapters that deploy it are Chapter 5 through Chapter 8.

What the two levels secure together. The agent-level and system-level commitments cannot be separated without losing the framework’s content. A formulation that secures joint determination at the agent level—each state’s optimization is genuinely joint over consumption and deployment—but treats the system-level equilibrium object as the product of two factor manifolds with an interaction term will reproduce the two-axis structure at the level it actually matters analytically: in comparative statics, in the regime classification, and in bifurcation analysis. A formulation that secures joint determination at the system level—the equilibrium object is a single manifold—but treats each state’s optimization as separable in the agent’s primitives will be unable to derive the system-level joint structure from agent foundations and will be vulnerable to the charge that the joint manifold is imposed by stipulation rather than by construction. Both levels are required because the joint-pursuit thesis the framework formalizes is a claim about the system’s underlying primitives, and primitives that operate at both levels of the analysis are the only kind that can support the formalization all the way through. The framework secures both, and the rest of the book is what the both-levels unification produces.

1.5 The structural argument: PE-vs-GE is structural

The previous section established that the framework secures joint determination at two levels. The present section establishes the structural argument that distinguishes the framework from the partialled traditions of formal IR and political economy. The argument is precise. The partialled methods are not less refined approximations of what the joint apparatus delivers; they are different theories asking different counterfactual questions, and the answers diverge structurally rather than quantitatively. Each tradition’s apparatus is the joint comparative-static under a specific restriction; the framework lifts the restrictions; the structural difference is what makes the framework necessary rather than ornamental.

The argument runs in three beats. The first is methodological: the partial-equilibrium-versus-general-equilibrium divide that the methodology of economics has spent a century debating is, on the framework’s reading, a structural difference rather than a tractability tradeoff. The second is empirical-IR: the multi-decade specification fight in the dyadic commercial-peace literature is signature of the structural difference, not noise to be resolved within the regression frame. The third is methodology-debate: the contemporary international-political-economy literature has its own version of the same procedure-pole problem, which the framework engages on its own terms.

The three beats converge on a master rhetorical move—each tradition’s apparatus is a slice of the joint comparative-static under a specific restriction—that runs through the substantive applications of the rest of the book.

The methodology beat: Walras, Friedman, and the Marshall–Walras divide. The structural-formal program of general equilibrium that the framework descends from has a primary source in Walras [1896]’s *Éléments d’économie politique pure*. Walras’s *Éléments* is the foundational mathematical statement of the equilibrium of exchange across many markets simultaneously, and it is the source-text of the entire general-equilibrium tradition the framework formalizes. The methodology-of-economics literature has spent a century arguing about the empirical content of Walras’s program, and the canonical statement of the partial-equilibrium-defense position is Friedman [1955]’s *American Economic Review* review of Jaffé’s 1954 translation of the *Éléments*. Friedman’s argument is precise. “Walras solved a different, though no less important, problem [than Cournot]. He emptied Cournot’s problem of its empirical content and produced a ‘complete and rigorous’ solution ‘in principle,’ making no pretense that it could be used directly in numerical calculations. His problem is the problem of form, not of content: of displaying an idealized picture of the economic system, not of constructing an engine for analyzing concrete problems” (Friedman 1955, p. 904). Friedman’s verdict was that “what needs now to be done is Marshallian rather than Walrasian in character” (p. 909).

Friedman’s methodological commitments were stated in concentrated form in the 1953 essay *The Methodology of Positive Economics* [Friedman, 1953]. Theories should be evaluated by predictive accuracy in observed contexts; the realism of assumptions is irrelevant; partial models that produce accurate predictions in domains where the relevant causal mechanisms are localized are licensed by predictive performance regardless of their structural simplicity. The methodology is instrumentalist on its surface and produces, across an enormous literature in twentieth-century economics, a research program that operates at the partial-equilibrium level and treats general-equilibrium analysis as a methodological backstop rather than as the natural object of empirical inquiry.

The contemporary scholarly engagement with Friedman’s methodology has complicated the easy reading. Mäki [2009]’s edited volume of fifty-years-on reassessments of *The Methodology of Positive Economics* contains chapters that argue Friedman is better read as a causal realist than as an instrumentalist. Mäki’s own Chapter 3 makes the case that Friedman’s “unrealistic assumptions” claim is properly read as an idealization-to-isolate-causal-structure move; Hoover’s Chapter 12 argues that Friedman’s methodology in practice was the methodology of causal realism applied to monetary economics; De Vroey’s Chapter 13 is exactly about the Marshall–Walras divide, with the verdict that Friedman was correct in conclusion—Marshall is the right methodologist for empirical economics—but wrong in justification, since the right reason has to do with whether the relevant causal mechanisms are localized rather than with Friedman’s instrumentalism.

The framework agrees with the contemporary readings against the easy Friedman-as-instrumentalist caricature, and it agrees with De Vroey on the localization-of-causal-mechanisms test. It disagrees with the conclusion that Marshall is the right methodologist for international political economy. The international political-economic system is precisely the case in which the relevant causal mechanisms are not localized. The joint manifold’s off-diagonal blocks J_{pm} and J_{mp} carry the cross-channel content that propagates parameter perturbations across the wealth-and-power coupling, and the partial-equilibrium frames that hold one side fixed while letting the other respond

zero out the cross-channel content by construction. The framework’s structural-counterfactual apparatus is what gives the general-equilibrium program empirical bite without subjecting it to Marshallian estimation: the comparative-static $-J^{-1}D_{\theta}e$ is the structural counterfactual, not a regression. Friedman’s diagnosis of the empirical-content problem in 1900-Walras is correct as a critique of the GE program as predictive-empirical exercise; the framework’s response is that the GE program is operative at the counterfactual level rather than the predictive level, and that the questions of interest in international political economy are counterfactual rather than predictive.

The argument also engages Polanyi from the methodology side. Polanyi’s claim that the disembodied self-regulating market is *Utopian* is a structural-impossibility claim about the partial-equilibrium frame. The disembodied economy that PE methodology takes as its baseline is the case in which the agent-level coupling $R_i = I$ holds, and Polanyi’s argument is that this case cannot be sustained: society’s protective response to disembedding is structural, not contingent. The framework formalizes Polanyi’s substantive impossibility as the formal claim that the wedge $R_i - I$ is non-zero generically. PE methodology assumes the disembodied baseline that the framework’s agent-level commitment shows is structurally inaccessible.

The empirical beat: the dyadic-logit specification fight. The international-political-economy field’s quantitative engagement with the joint-pursuit thesis runs through a specification fight that Strange would not have recognized as the form the field she founded would take. The specification was established by Polachek [1980]’s *Conflict and Trade*: for each directed dyad-year in a panel, regress an indicator of conflict onset on bilateral trade dependence, with controls for the standard dyadic configuration—capability ratio, alliance, contiguity, regime type. Polachek’s structural setup made the directional commitment explicit. Countries first choose production via comparative advantage; the trade pattern is determined; conflict appears as a demand response to the resulting price structure. Trade is theoretical input; conflict is theoretical output; estimation matches the structural ordering. The design has been the field’s workhorse for forty-five years.

Oneal and Russett [1997] extended Polachek’s design to the post-Cold-War dyadic dataset under the heading “the classical liberals were right.” Russett and Oneal [2001] consolidated the program in book-length form as the Kantian-tripod synthesis: democracy, economic interdependence, and international organizations entered the dyadic logit as separate covariates whose pacifying effects were estimated jointly within a single regression. The central finding—that bilateral trade dependence is associated with reduced dispute likelihood at the dyadic level—became the canonical empirical statement of the commercial-peace tradition.

The specification class is the problem. Barbieri [1996] had already run the same dyadic design with a different measure of trade interdependence, and found a *positive* relationship between bilateral trade and militarized-dispute onset—more trade, more conflict, directly opposite the OnealRussett finding. The OnealRussett response was to attribute the discrepancy to specification choice: trade concentration rather than trade-to-GDP, all dyads rather than politically relevant dyads. This is exactly the form the framework predicts the procedure-pole violation will take. The literature fights over which specification is correct because the underlying joint structure is invisible from the regression frame, and each specification picks up a different slice of the joint constraint and produces a different sign. Gartzke [2007] re-ran the design with capitalist variables added—financial openness, GDP per capita, UN-voting affinity—and found that the democracy effect dropped to statistical insignificance: same data, same dyad-year panel structure, opposite substantive conclusions about which leg of the Kantian tripod carries the empirical weight. Hegre et al. [2010] responded with a simultaneous-equations specification that incorporated traders’

rational expectations of war and recovered the trade-promotes-peace finding under stronger gravity-model controls.

The exchange demonstrates the structural problem rather than resolving it. The same panel admits opposite-sign findings depending on the specification's control set; rebuttals recover the desired findings under more disciplined controls; the specification fight does not converge. Within the dyadic-logit frame, this is unsatisfactory empirical practice—a research program that cannot pin down the sign of the relationship between its core variables. From outside the frame, the non-convergence is signature. A regression program designed to estimate causal effects under the assumption that trade is exogenous to conflict at the dyadic level cannot be expected to converge when the underlying phenomenon is jointly determined and the regime structure varies across the panel. The directional choice the dyadic logit imposes—trade on the right-hand side, conflict on the left—is the procedure-pole violation of the joint-pursuit thesis, and the multi-decade specification fight is what the framework predicts the violation will produce.

The political-science articulation of the same diagnosis runs through Gowa [1994]'s *Allies, Adversaries, and International Trade* and the article version [Gowa and Mansfield, 1993]. Gowa models bilateral trade as a prisoner's dilemma with a security externality: trade flows generate gains-from-trade but also generate the wealth that translates into adversaries' military capacity, so the equilibrium pattern of bilateral trade is jointly determined by alliance configurations rather than estimated in isolation. The book closes with the line that “*the pursuit of power and of plenty are two sides of the same coin*” (Gowa 1994, p. 121)—the political-science articulation of the joint-pursuit thesis stated in Viner-1948 vocabulary, with the methodological consequence that bilateral trade volumes themselves are endogenous to the security configuration the dyadic logit treats as a control. Gowa does not deliver structural-formal joint equilibration on a manifold; the apparatus is prisoner's-dilemma-with-coupling rather than smooth-manifold equilibrium. But the substantive diagnosis is the same: the right-hand-side variables in the dyadic logit are not exogenous to its left-hand-side variables, and any apparatus that takes the joint-pursuit thesis seriously will need to recover this fact rather than condition it away.

What the framework offers the dyadic-empirical literature is therefore not an additional specification. It is the structural counterfactual the dyadic logit cannot provide. Once the joint-equilibrium structure is in view, the dyadic correlations are recoverable as surface measurements of an underlying joint manifold whose comparative statics depend on the regime the system is in; the specification fight is recoverable as evidence of cross-regime variation that the within-regime regression treats as residual; and the gap Strange identified in 1970 is recoverable as the gap between the regression frame and the joint object the regression frame was built on top of rather than to close. The framework is positioned as the structural-formal answer to the founding IPE call Strange made and the field's quantitative turn did not yet deliver.

The methodology-debate beat: OEP and integrative pluralism. The contemporary methodological self-description of the international-political-economy field has its own version of the procedure-pole problem the framework engages. Lake [2013] contains the canonical OEP textual statement. “*Analysis within OEP proceeds from the most micro to the most macro level in a linear and orderly fashion, reflecting an implicit unidirectional conception of politics as flowing up from individuals to interstate bargaining*” (Lake 2013, p. 575). The procedure-pole violation is stated in its purest textual form, and Lake offers it as a virtue: OEP's strength is its three-stage linear structure—interests, then institutions, then international bargaining—and the linearity is what makes the program tractable.

The companion essay [Lake, 2011] establishes the three-I lexicon—interests, interactions, institutions—that became the OEP-rebranded version of the older three-stage structure. The “embrace partiality” prescription Lake offers as the methodological program of OEP is, on the framework’s reading, structurally incompatible with the joint-pursuit thesis. Partial methods produce results within their domains; the joint phenomenon is not within any partial method’s domain by construction.

Mearsheimer and Walt [2013]’s *Leaving Theory Behind*—published in the same *European Journal of International Relations* special issue as Lake’s—is the canonical 2013 scientific-realism manifesto. Theory’s primacy over hypothesis-testing-driven instrumentalism is defended on grounds the framework operates within. The framework’s structural-realism is in the same broad family as the scientific realism Mearsheimer and Walt defend; the disagreement is about which structures are real and what kinds of counterfactuals each apparatus computes, not about whether structural reasoning is the right register for international political theory.

Dunne et al. [2013], editing the same *EJIR* issue, propose *integrative pluralism* as a third route between Lake’s eclecticism and Mearsheimer-and-Walt’s scientific realism. The framework is, on its own self-positioning, the formal-and-rigorous instantiation of integrative pluralism: the partialled traditions are integrated by being recovered as slices of the joint apparatus, the pluralism is honored by treating each tradition’s domain as the parameter region in which its slice is exact, and the rigor comes from the structural-formal apparatus rather than from a methodological compromise.

Hont’s “two tunnels” passage names the anti-primacy commitment in scholarly form: the eighteenth-century tradition the genealogy of Section 1.2 traces refused the procedural primacy of either the political or the economic dimension, and the modern field’s split between political science and economics is the procedural primacy reinstated. Hont names; the framework operationalizes. The structural-formal apparatus of the framework is the formal vehicle for the anti-primacy commitment Hont surfaces in the eighteenth-century tradition.

The methodological-foundations precedent is Lucas [1976]. The Lucas critique observes that econometric models with parameters depending on policy regimes will produce inconsistent estimates under regime change; the structural alternative is to estimate the deep parameters that do not depend on the regime. The framework is partially Lucas-immune at the manifold level: the structural primitives (s_i , c_i , u_i) are deep in Lucas’s sense, and the comparative-static formula computes responses to parameter perturbations at these primitives rather than at policy-dependent reduced-form coefficients. The framework’s empirical content is therefore structural-counterfactual rather than reduced-form-predictive, and it inherits the methodological defensibility the Lucas critique identifies as the test for structural status.

Slice recovery: each tradition is a restriction. The three beats above converge on a master rhetorical move that runs through the substantive applications of the rest of the book: each tradition’s apparatus is the joint comparative-static under a specific restriction; the framework lifts the restrictions; the difference is structural. The slice-recovery argument is the framework’s signature engagement with the partialled traditions, and the most articulate version appears in Chapter 8.

The bargaining-camp slice fixes both prices and deployments across a parameter perturbation. The strategic environment is given; the bargaining game runs within it; the comparative-static is the response computed under the stipulation that prices and deployments are exogenous to the game. The exchange-side slice fixes deployments and lets prices respond from the price-clearing

block; the comparative-static is the response computed under the stipulation that the conflict-side parameters are exogenous to the price-clearing exercise. The joint formulation lifts both restrictions: the comparative-static $-J^{-1}D_{\theta}e$ computes the simultaneous response of military deployments and prices to the parameter perturbation, with the off-diagonal blocks J_{pm} and J_{mp} carrying the cross-channel content the slice computations zero out by construction.

On the canonical worked example, the slice-recovery argument has a striking quantitative form. Chapter 8 computes the joint response to a prize-share sanction parameter θ on the two-state Tullock contest of Chapter 3: each slice gives exactly zero on its respective response variable while the joint formulation gives non-zero on both. The cross-effect—the Schur-complement correction $J_{pm}J_{mm}^{-1} \cdot D_{\theta}m^*$ the partialled methods zero out—is the entire joint response, not a small correction to the slice analyses. PE doesn’t approximate GE poorly; PE has zeroed out the operative channel by construction.

The structural argument has a formal warrant. Carroll’s earlier result establishes that an observation family detects the value of unification only if it is not a left adjoint, and within-domain consequence operators are precisely left adjoints. The reader who occupies a single-domain observational stance—the bargaining-camp reader, the exchange-side reader, the OEP-trained reader—will detect the framework’s joint-equilibrium structure as “more theory in my domain” by a structural property of the observational stance, not by inattention. The slice-recovery argument is the operational form of the same theorem: each within-domain method recovers a slice of the joint apparatus, and the joint surplus is structurally invisible from the within-domain stance.

What the structural argument delivers. The argument is precise and not difficult. PE methodology and GE methodology are different theories asking different counterfactual questions, not different approximations to the same theory. The partialled methods compute responses under stipulations the joint apparatus refuses, and the answers diverge structurally rather than quantitatively. The decades of methodology-of-economics debate over Friedman’s instrumentalism, the multi-decade specification fight in the dyadic commercial-peace literature, and the contemporary methodology-debate cluster around Lake, around Mearsheimer and Walt, and around Dunne, Hansen, and Wight all engage versions of the same problem the framework is built to address.

The framework’s response to each is the same. Friedman is right that the GE program of pure form has no empirical content; the framework’s structural-counterfactual content is what gives GE empirical bite without subjecting it to Marshallian estimation. The dyadic-logit specification fight is signature, not noise; the joint apparatus is the structural counterfactual the regression frame cannot provide. The OEP linear-causal-chain is the procedure-pole violation in its purest textual form; the framework’s two-level joint determination is the structural-formal alternative.

The chapter now turns from the structural argument to the substantive reading of regime structure as multiplicity in the joint manifold’s component decomposition. Section 1.6 closes the chapter with the regime-change-is-geometric reading, and flags what the framework holds fixed for the conclusion’s future-work scope.

1.6 Regime change is geometric

What scholars of international relations describe as *regime change*—the qualitative shift from one configuration of the international system to another, distinct in kind rather than in degree—has

a formal correlate in the geometry of the joint manifold $\tilde{E}$. The previous sections have argued for the framework as the formal apparatus the joint-pursuit thesis required. The present section closes the chapter by reading the verbal-historical regime-change tradition through the apparatus, and by stating honestly what the framework holds fixed for the conclusion’s future-work scope. The substantive claim of the section is that regime change has geometric content the verbal tradition recognizes informally: the same primitives can support distinct equilibrium configurations, the system selects among them through history, and the transitions between configurations have structural character that the joint Jacobian’s geometry classifies. The framework supplies the geometry; the verbal tradition supplies the substantive instances.

The singular locus and the components. The framework’s regime-change apparatus is concentrated in one geometric object: the singular locus $\Sigma := \{(p, m, \omega) \in \tilde{E} : \det J = 0\}$ on which the joint Jacobian fails to be invertible. Under joint regularity, Σ is empty and $\tilde{E}$ is one global manifold. Under generic joint regularity—the framework’s working hypothesis from Chapter 3 forward— Σ is closed and Lebesgue-null, and the regular set $\tilde{E}^{\text{reg}} := \tilde{E} \setminus \Sigma$ partitions into connected components. Each connected component is a parameter-coverage region within which prices and military deployments respond smoothly to parameter perturbations and the master comparative-static formula applies. Crossing the singular locus is what the framework calls regime change. Chapter 4 establishes the apparatus formally: the index theorem on the joint Jacobian constrains how equilibria pair across folds, and the catastrophe-theoretic stratification classifies the crossing by codimension—folds at codimension one, cusps at codimension two, swallowtails and butterflies and umbilics at higher codimensions. Chapter 7 cashes the apparatus out substantively: regimes are connected components of $\tilde{E}^{\text{reg}}$; regime change is a path crossing Σ ; the index theorem identifies which equilibrium pairings can appear or disappear at a given crossing and which cannot.

Verbal traditions and their formal correlates. The verbal-historical IR literature on regime structure runs through several distinct traditions, each with its own vocabulary and substantive claims. The framework reads each as a verbal description whose formal correlate the joint manifold supplies.

Krasner [1982]’s *International Regimes* volume defined regimes as “sets of implicit or explicit principles, norms, rules, and decision-making procedures around which actors’ expectations converge in a given area of international relations,” and treated regime stability and regime change as the field’s central organizing problem. The framework’s reading is that Krasner’s regimes are connected components of $\tilde{E}^{\text{reg}}$. A regime persists where the system’s parameter trajectory stays within a single component; regime change occurs where the trajectory crosses Σ into a distinct component. Krasner’s verbal characterization—convergent expectations, shared principles, stable behavioral patterns—is what within-component motion looks like in the system’s surface behavior; what the verbal tradition calls regime change is what cross-component motion looks like.

Ikenberry [2001]’s *After Victory* treats the post-major-war institutional settlements—1815, 1919, 1945—as the canonical instances of qualitative shift in international order. Each settlement reorganized the system’s institutional architecture, redistributed power among states, and produced a new equilibrium pattern that persisted until the next major shock. The framework’s reading is that the post-war moments are Σ -crossings driven by parameter shifts of sufficient magnitude to take the system out of one regime component and into another. The institutional

architecture each settlement instantiates is the system's selection among the post-crossing equilibria available at the new parameter configuration; the persistence of the settlement is within-component stability under the new regime.

Tilly's three-path model of European state-formation, taken up already in Section 1.4, reads the development of the modern state as a joint optimization over coercion-intensive and capital-intensive trajectories, with capitalized coercion as the convergent equilibrium under competitive pressure. The framework's reading is that the three trajectories are distinct components of the system's joint manifold, with the selection between them driven by the parameter configuration each historical environment realizes. The convergence-under-competitive-pressure result is the framework's claim that some components are reachable from a wider parameter range than others; competitive pressure is the selection mechanism that drives parameter trajectories toward the parameter regions where the convergent component is uniquely reachable.

Acemoglu and Robinson [2019]'s *Narrow Corridor* reads political development as a precarious balance between state capacity and societal mobilization, with three reachable political regimes (*Despotic Leviathan*, *Absent Leviathan*, *Shackled Leviathan*) and a corridor between them within which liberty and prosperity are jointly sustainable. The framework's reading is that Acemoglu and Robinson's three regimes are components of $\tilde{E}$ projected onto a two-dimensional state-society axis, with the corridor being the parameter region within which the Shackled-Leviathan component is reachable. The Red Queen dynamics they describe—the continuous co-evolution of state capacity and societal mobilization that keeps the system within the corridor—is within-component motion under sustained parameter co-adjustment. Corridor exit is a Σ -crossing into the Despotic or Absent component, and the difficulty of corridor entry is the framework's claim that some components are reachable only from narrow parameter regions.

Gilpin's equilibrium-and-decline framework, taken up already in Section 1.4, reads international systems change as driven by the rising costs of status-quo maintenance against the slower-rising productive capacity that supports the maintenance. The framework's reading is that Gilpin's structural balance between costs and benefits of expansion is itself a joint feature of the system's wealth and power configurations on the manifold; when the balance tips, the system's parameter trajectory approaches Σ , and the qualitative shift Gilpin calls hegemonic decline is the Σ -crossing the framework classifies.

The 18c precedent for the regime-multiplicity-with-path-dependence reading runs through Smith's distinction between the *natural progress of opulence* and the *retrograde and unnatural order* (WN Bk III), reconstructed by Hont as part of the eighteenth-century vocabulary that the marginalist break abstracted away. Smith's argument was that the natural sequence of economic development runs from agriculture through manufactures to foreign commerce, but Europe's actual development followed the *unnatural* sequence with foreign commerce driving the rest—a multi-equilibrium claim about distinct reachable orders on shared primitives, with history selecting between them. The framework's reading is that Smith named the multi-equilibrium phenomenon at the level of structural-economic development in 1776 in language that anticipates the framework's component-decomposition reading: distinct components reachable on shared primitives, with the selection among them not given by the primitives alone. What Smith stated as a structural-historical observation in eighteenth-century vocabulary, the framework formalizes as component structure on $\tilde{E}$ under generic joint regularity.

What the framework supplies. The framework's contribution to the verbal-historical regime tradition is the formal apparatus the verbal tradition could not itself produce. The apparatus

has four pieces. First, regimes are connected components of $\tilde{E}^{\text{reg}}$, with structural definition rather than verbal stipulation. Second, the index theorem on the joint Jacobian constrains how equilibria can appear or disappear at Σ -crossings: the sum of equilibrium indices over a regular fiber is preserved, which means equilibria pair across folds in ways the index constraint identifies. Third, the catastrophe-theoretic stratification classifies the crossings by codimension, with the substantive consequence that codim-one crossings (folds) admit smooth path-following analysis up to the crossing while higher-codimension crossings do not. Fourth, the welfare-improvement gradient field $X^{\Delta^*} := -D_{\omega}\Delta^*$ defined on $\tilde{E}^{\text{reg}}$ supplies a substantive direction for institutional intervention: institutional design that implements the gradient is welfare-improving; institutional design that fails to implement the gradient or that crosses Σ in directions that worsen welfare is not. Chapter 4 and Chapter 7 develop the four pieces formally; the present section's purpose is to identify the verbal-tradition correlates each piece supplies.

What the framework holds fixed. The framework's regime-change apparatus is parameter-driven bifurcation on a fixed-primitive manifold. The structural primitives—the opportunity-cost function c_i , the battle function s_i , the preferences u_i —are held fixed; what moves are the parameters those primitives take, and the manifold's geometry over the parameter space is what the framework analyzes. This is a substantive scope decision rather than a hidden assumption, and it leaves four classes of regime-change-flavored phenomena outside the framework's reach. The chapter is honest about the bounds; the conclusion takes them up as future-work scope.

Technology change. The framework's c_i and s_i are smooth functions, but the *functional forms* are primitives. Technology change in the strong sense—the introduction of new contest media (nuclear, cyber, AI), the emergence of new commodity dimensions, the displacement of one social formation of mobilization by another—is shift in the form of c_i or s_i at the level of the primitive itself, rather than parametric movement within a fixed primitive. This is structurally different from the framework's Σ -crossings: the apparatus describes regime change driven by parameters within a fixed structural setup; strong-sense technology-driven regime change is regime change of the structural setup. The conclusion takes up the weaker version—technology parameterized as a joint-reciprocal shock to both c_i and s_i , in the manner the framework's apparatus already accommodates—as the natural application rather than the extension.

Financial-network topology. The framework's commodity space and contest structure capture trade and force as continuous-variable quantities; they do not capture the network topology of financial relations through which contemporary economic coercion increasingly operates. Section 8.6 flags dedollarization and weaponized-interdependence as candidate Σ -crossings the framework can identify structurally without analyzing in detail; the structural identification is the framework's contribution, and the network analysis is future work.

Explicit dynamics in time. The framework is comparative-statics with bifurcations, not a dynamical system over time. The path-following analysis on $\tilde{E}^{\text{reg}}$ treats institutional trajectories as one-parameter families, and the regime-change classification operates on the geometry of $\tilde{E}$ rather than on the system's trajectory through time. What this leaves outside the framework is the time-dynamic content of regime change: when a Σ -crossing occurs, how fast it propagates, whether the system overshoots, what oscillatory behavior follows. The framework gives the structural classification; the dynamics require additional apparatus the conclusion takes up.

Endogenous formation. The number of states I is exogenous; preferences u_i are primitive; the system's primitives do not endogenously change through interaction. State births and deaths, consolidation and fragmentation, preference-formation through cultural and discursive processes,

the constructivist program of identity-formation—all are held outside the framework’s apparatus by the choice to take the primitives as given. The framework’s claim is not that endogenous formation does not matter; it is that the apparatus the present book develops treats those questions as outside its scope so that the joint-determination apparatus can be developed in concentrated form.

The four classes of held-fixed phenomena—technology change, financial-network topology, explicit dynamics, endogenous formation—together constitute the framework’s future-work scope. The conclusion of the book returns to them as a research agenda that the joint-determination apparatus, once developed, can support but does not yet contain.

1.7 Looking ahead

The chapter has stated the master claim, traced its intellectual genealogy, engaged its principal dissents, and developed the framework’s two-level joint-determination commitment, the structural argument that distinguishes it from the partialled traditions, and the regime-change apparatus its geometry supplies. The rest of the book builds the formal apparatus and deploys it on three substantive applications. The map runs in two halves: four chapters that develop the apparatus formally, and three that deploy it.

Chapter 2 develops the agent level. The state’s problem under threat is set up as a single optimization over consumption and deployment. The two architectural choices identified in Section 1.4— c_i taking prices as a primitive argument and s_i taking consumption-after-cost as a primitive argument—produce the entanglement matrix R_i inside the first-order conditions, and the chapter shows that this is the single substantive commitment from which the rest of the agent-level apparatus follows.

Chapter 3 assembles the system level. The single excess-demand-like map $e : S_N \times M \times \Omega_0 \rightarrow \mathbb{R}^{L-1+I}$ has zero set $\tilde{E} := e^{-1}(0)$, and the chapter establishes the smoothness of the joint manifold under joint regularity, an existence theorem via homotopy from a Walrasian endpoint, and a global chart Φ exhibiting $\tilde{E}$ as diffeomorphic to $\mathbb{R}^{IL}$. Under generic joint regularity—the framework’s working hypothesis from this chapter forward— Σ is closed and Lebesgue-null, $\tilde{E}$ partitions into regime components, and the regime-change apparatus the present chapter previewed becomes formally available.

Chapter 4 develops the master comparative-statics formula $D_\theta(p^*, m^*) = -J^{-1}D_\theta e$ and the geometric machinery the formula deploys. Generic regularity is earned from primitives via parameterized Sard rather than imposed by axiom. The index theorem on the joint Jacobian constrains how equilibria can pair across folds in Σ . The chapter is the apparatus chapter the application chapters draw on.

Chapter 5 develops the welfare apparatus on the joint manifold. The master scalar Δ^* measures the cardinal welfare cost of conflict at any joint equilibrium, computed from the same J^{-1} that delivers comparative statics. The conditional First Welfare Theorem under joint determination relocates Fearon’s bargaining-model premise—that war is inefficient—from assumption to theorem under explicit conditions on the conflict technology. The chapter identifies an implementability gap between frontier-level Pareto improvements and joint-equilibrium implementations of those improvements, which the bargaining-theoretic apparatus has been silent on.

Chapter 6 deploys the apparatus on the donor-intervention problem. An existence theorem for optimal aid schedules without convexity assumptions inherits the manifold structure of Chapter 3; a transversality result establishes that aid evaluated at fixed prices cannot reduce conflict because

the partial-equilibrium direction is transversal to the peace component; and a coordination result establishes that fragmented bilateral aid implements the same welfare direction at $1/n$ magnitude, supplying a resource-neutral rationale for multilateral peacemaking institutions distinct from the transaction-cost and information rationales the existing literature provides.

Chapter 7 deploys the apparatus on institutional design. A Chichilnisky–Heal aggregation theorem operates globally on $\tilde{E}$ under joint regularity and componentwise under generic joint regularity, with cross-regime aggregation impossibility recovering Krasner’s regime concept as a topological boundary condition. The welfare-improvement gradient field $X^{\Delta^*} := -D_{\omega}\Delta^*$ supplies a substantive direction for institutional intervention. Bifurcations along Σ classify regime change formally, with the index theorem constraining the crossings. The (target, path) reading of an institution treats institutions as paths on $\tilde{E}$ with cohomology-class preferences.

Chapter 8 deploys the apparatus on coercive instruments. Sanctions are parameter perturbations on $\tilde{E}$, and the joint comparative-static formula delivers the structural counterfactual the empirical sanctions-effectiveness literature has been working without. The slice-recovery proposition establishes that the bargaining-camp and exchange-side formulations are restrictions of the joint comparative-static; the cross-effect proposition gives the gap between slice and joint as a Schur-complement correction; and on the canonical worked example, the slices give zero on their respective response variables while the joint formulation gives the entire response.

The chapter admits two readings. Read narrowly, it is the front matter the rest of the book consumes: it states the thesis, locates the framework against the partialled traditions, and previews the apparatus. Read broadly, the chapter is a self-contained essay on the structural conditions under which the international political-economic system became analytically tractable as a single phenomenon—older than the formal apparatus by two centuries, recovered repeatedly by careful observers across the modern field, and held back from formal expression by the disciplinary fragmentation the marginalist break of the 1870s set in motion. The framework the rest of the book develops is the apparatus the recovery requires; the chapter has been the case for why the apparatus is the right shape.

Exercises

Chapter 1 is a substantively philosophical chapter, and most of these exercises are essay prompts rather than computational problems. The expected response is a one-to-three-paragraph argument rather than a closed-form calculation. Where exercises invoke the formal apparatus of Chapters 2 to 4, the expectation is that the reader has read those chapters, or read the technical sections marked with a trailing asterisk in this chapter’s Section 1.7.

1. *A fourth historical case.* Section 1.1 marshals four historical cases as support for the sibling claim: the Dutch Republic in the seventeenth century, eighteenth-century Britain, twentieth-century Germany/Japan, and the contemporary United States and China. Identify a fifth case in which the pattern of joint trade-and-military engagement is conspicuous. Argue whether the case adds substantive support to the sibling claim or whether it is selection-on-the-dependent-variable. The distinction matters: a case selected because it exhibits the pattern is not evidence for the pattern’s structural status, whereas a case selected on independent grounds (say, by chronological completeness or by geographical spread) and exhibiting the pattern anyway is.
2. *Restate Gilpin in formal language.* The chapter quotes Gilpin [1975]’s formulation of the

joint pursuit as “the reciprocal and dynamic interaction in international relations of the pursuit of wealth and the pursuit of power.” Translate this informal formulation into the chapter’s formal apparatus. Specifically, write down a state’s optimization problem that takes wealth and power as joint arguments (the SPT of Chapter 2 is one such formulation), and identify the structural condition that distinguishes the joint formulation from a separable-with-margin-interaction one. The condition is what Section 1.4 names embeddedness; identify it formally in terms of the x -dependence of the battle function s_i and the price-dependence of the cost function c_i .

3. *Counterexample of separable wealth/power?* Locate a historical episode in which states’ wealth-pursuits and power-pursuits appear to be substantively separable: trade proceeded as if no military shadow loomed, or military competition unfolded without trade-policy entanglement. Argue two readings of the episode. First, treat it as a counterexample to the sibling claim: separability is the rule, and the chapter’s pattern is overstated. Second, treat it as a parameter region in which the joint Jacobian’s cross-channel blocks vanish (Section 4.1’s collapse condition): the joint formulation predicts separability locally as a special case, and the chapter’s claim is about which regime the system sits in. Which reading is more compelling, and what would distinguish the two empirically?
4. *Tracing the joint pursuit through one tradition.* Section 1.2 traces the joint-pursuit claim through seven centuries of intellectual history: the seventeenth-century natural-law tradition, the eighteenth-century commerce-versus-conquest debates, the nineteenth-century political economy of war finance, the twentieth-century recovery of the *jealousy-of-trade* synthesis, and the contemporary international-political-economy program. Pick one of these traditions and trace the claim through it in 2–3 paragraphs. What does the tradition take as the key formulation of the joint pursuit? What does it leave out that the framework’s formal apparatus captures? Where does the tradition’s vocabulary collapse into the chapter’s vocabulary, and where does the chapter’s vocabulary depart from the tradition’s?
5. *Jealousy of trade vs. commerce-and-conquest.* Two organizing rubrics in the genealogy are Hont [2005]’s “jealousy of trade” synthesis and the older “commerce versus conquest” framing of the seventeenth-century natural-law tradition. Compare the two rubrics. What does each emphasize about the joint pursuit, and what does each suppress? The jealousy-of-trade synthesis emphasizes that commerce and military competition are part of the same strategic logic; the commerce-versus-conquest framing emphasizes that they substitute. Which framing fits the framework’s formal apparatus better, and why?
6. *Inheritor of the joint-pursuit claim in twentieth-century IPE.* The contemporary international-political-economy program is the most direct intellectual ancestor of the chapter’s substantive claim. Identify which twentieth-century IPE figure most cleanly inherits the joint-pursuit claim from the earlier traditions. Candidates: Gilpin [1975, 1981], Krasner [1976], Strange [1988], Keohane and Nye [1987], or others. Defend your choice. The structural question is which figure’s formulation is closest to the chapter’s own substantive thesis, and whose framework needs the chapter’s formal apparatus to be fully articulated.
7. *Schumpeter’s atavism dissent in framework terms.* Section 1.3 engages Schumpeter [1919]’s atavism dissent: military competition is a residue of pre-modern aristocratic culture, not a structural feature of modern political economy. Reconstruct the dissent in the framework’s

formal terms. The Schumpeterian position predicts that as modernization proceeds, military competition decays toward a limit where deployment is purely vestigial, and the framework's contest functions s_i become trivially zero-sum. Argue for or against this prediction in 1–2 paragraphs. The empirical question is whether twentieth-century militarization (Germany, Japan, the US, China) is a transient atavism or a structural feature.

8. *Ferguson–Fenelon: when does commerce pacify?* The Ferguson–Fenelon dissent argues that commerce, properly organized, pacifies international relations: extensive trade ties create commercial peace, and the structural prediction is the absence of war among heavily-trading states. Reconstruct the dissent in framework terms. The Ferguson–Fenelon position corresponds to a parameter regime in which the cross-channel block $J_{p,m}$ acts as a stabilizer (more trade $\Rightarrow$ less arming) rather than as a destabilizer. Identify what structural condition on the contest primitives (s_i, c_i) would deliver this stabilization, and discuss whether the framework's apparatus permits a clean test.
9. *Rosenberg's market-civilization argument.* Section 1.3 engages Rosenberg [1994]'s market-civilization dissent: the modern state-system's market-civilization is a coherent historical formation that the sibling claim mis-decomposes by treating wealth and power as joint arguments. Assess the argument. Two responses are available: first, treat market-civilization as the substantive content of the framework's joint formulation (in which case the sibling claim and Rosenberg's reading converge); second, treat market-civilization as an aggregate claim that the framework's disaggregated formulation refuses to make. Which response is more compelling, and what is at stake?
10. *Embeddedness as a structural property.* Section 1.4 defines embeddedness (L1) as the structural condition that consumption and force decisions are made jointly, not sequentially. State this condition in two formulations: a verbal formulation in Polanyi [1944]'s tradition (embeddedness as the social character of economic relations), and a formal formulation in the framework's terms (the x -dependence of s_i in Chapter 2's SPT). Discuss what each formulation emphasizes that the other suppresses, and where the framework's apparatus operationalizes Polanyi's verbal claim.
11. *Joint determination as architectural condition.* The framework's joint structure (L2) is the architectural condition that the apparatus is built around the joint (p, m) -vector rather than around its components separately. Identify the load-bearing role this structure plays in Chapter 4's master comparative-static formula $D_\theta(p^*, m^*) = -J^{-1}D_\theta e$. Specifically: where in the formula does joint determination enter, and what would the formula look like if the apparatus were built around the components separately? The answer is in the cross-channel blocks J_{pm}, J_{mp} of the joint Jacobian, which the partialled formulations zero out by construction.
12. *When does the framework collapse to the partialled structure?* The chapter argues that the partialled formulations (bargaining-camp, GE-camp) are special cases of the joint framework where the cross-channel blocks vanish. Identify the structural conditions under which this collapse happens formally: the $J_{pm} = 0$ condition (deployment shifts do not affect aggregate excess demand) and the $J_{mp} = 0$ condition (price shifts do not affect Nash deployment). For each condition, identify a class of contest specifications that satisfies it. The collapse conditions are well-posed local questions; the substantive question is whether

they hold generically (in which case the partialled formulations are nearly right) or fail generically (in which case the joint formulation is needed).

13. *PE-vs-GE distinction in the framework's terms.* Section 1.5 distinguishes the PE-vs-GE distinction in two senses. The conventional sense: PE holds some equilibrium objects fixed and the GE generalizes by endogenizing them. The framework's sense: PE-vs-GE is *structural*, with the GE formulation operating on the joint (p, m) -equilibrium and the PE formulations being slices of it. Articulate the difference between the two senses in 1–2 paragraphs. What does the conventional sense collapse into the framework's sense? Where do the two senses diverge, and which divergence is substantively important?
14. *Master comparative-static and the partials.* The chapter introduces the master comparative-static formula $D_\theta(p^*, m^*) = -J^{-1}D_\theta e$ as the load-bearing identity of Chapter 4. Identify how the PE partials (bargaining-camp $D_\theta m^*$ at fixed prices, and GE-camp $D_\theta p^*$ at fixed deployments) arise as restrictions of this formula. Each partial is the master formula with one channel zeroed out, and the substantive content of the master formula is the cross-channel correction the partials miss.
15. *Slice recovery: when do the partials get the right answer?* The framework permits the partialled formulations to be exactly correct in special cases. Identify when each partial recovers the joint response. The bargaining-camp partial is exact when $J_{mp} = 0$ (the partial collapse from Exercise 12); the GE-camp partial is exact when $J_{pm} = 0$ (the symmetric collapse). Each condition picks out a specific structural configuration: in the first, deployment is invisible to prices; in the second, prices are invisible to deployment. Discuss which of the two configurations is closer to the historical record, and what the empirical answer would imply for which partialled formulation is closer to right in practice.
16. *Catastrophe theory and historical regime change.* Section 1.6 reads regime change as a geometric phenomenon: the singular locus Σ of the joint Jacobian is the locus of regime boundaries, and crossings of Σ are catastrophe-theoretic events (Remark 3.20 of Chapter 3). Identify the structural content of this reading. The geometric apparatus gives a precise mathematical correlate to substantive talk about regime change in IR theory, but it constrains the correlate in specific ways: the index theorem (Theorem 4.5 of Chapter 4) constrains which pre-/post-crossing equilibrium pairings are admissible, and the catastrophe classification constrains the local geometry of the crossing. Discuss whether this constraint is a feature or a bug.
17. *Candidate historical regime change as a Σ -crossing.* The chapter and Chapter 7 read several historical regime changes as candidate Σ -crossings: the 1815 Concert of Europe, the post-1945 Bretton Woods settlement, the post-1989 unipolar moment, the post-2018 Sino-American restructuring. Pick one of these. Argue whether the case fits the framework's formal apparatus as a Σ -crossing or whether it fits better as smooth motion within a regime component. The substantive question is whether the regime change involves a discrete topological reorganization (parameter crosses Σ , equilibria appear or disappear in pairs) or a continuous shift (parameter remains in $\Omega_0 \setminus \Sigma$, equilibrium varies smoothly).
18. *The geometric reading and rationalist explanations.* The bargaining-of-war literature (Fearon 1995) organizes rationalist explanations for war around three frictions: commitment problems, private information, and indivisibilities. The geometric reading of regime change

adds a fourth class of explanations: regime crossings, where the smooth machinery breaks at Σ . Discuss how the geometric reading constrains the rationalist explanations. Which of Fearon's three frictions correspond to features of the geometric apparatus, and which fall outside it? The framework's Σ -crossings cover the case where "rational" choice cannot continuously deform across regimes, which is structurally distinct from the three frictions Fearon identifies. The exercise is the chapter's deepest contact point with the IR canon, and a complete answer requires reading the apparatus and the bargaining tradition side by side.

Part I

Theory

Chapter 2

The state and its problem

How does a state, facing both market prices and rival military deployments, decide what to consume and how much force to project? Two formal traditions answer the question on partialled agents. Consumer theory asks how a price-taker chooses consumption to maximize utility given an income, with deployment absent and the budget exogenous. Contest theory asks how a force-projecting agent chooses deployment to maximize a payoff given a probability of victory and a cost of conflict, with consumption-and-allocation absent and the prices that determine deployment cost stipulated from outside. Each tradition holds half of the agent's problem fixed while solving the other half. This chapter takes a third picture.

The state's problem under threat is a single optimization that chooses consumption and deployment jointly, with the cost of force depending on prices from the outset. The chapter assembles the agent-level data the rest of the book consumes—preferences, endowments, deployments, opportunity costs, and battle—in a form whose joint character is visible already in one state's decision problem: the cost of deployment is price-dependent from the start, so the optimization over consumption and deployment is joint the moment a price exists.

What the chapter delivers is hard to find elsewhere. First, the state's problem under threat as a single optimization whose first-order conditions produce demand and deployment as a joint best response, rather than demand from a consumer problem and deployment from a separately posed contest. Second, an entanglement matrix R_i^{-1} inside the first-order conditions that couples the consumption decision and the deployment decision so that neither component solves without the other; the matrix is the agent-level seed of the joint Jacobian J that drives the equilibrium and comparative-statics machinery of the chapters that follow. Third, substantive readings of offense-defense balance, capability advantage, and bargaining power as endogenous consequences of the joint optimum rather than primitive parameters; the IR canon's treatment of these as exogenous structural conditions recovers as a partial-equilibrium special case of the joint treatment here.

The chapter is organized around the agent-level data the rest of the book consumes. Section 2.1 assembles the primitives—preferences, endowments, deployments, opportunity costs, and battle—with particular attention to the price-dependence of c_i as the agent-level seed of joint determination. Section 2.2 states the state's problem under threat and derives its first-order conditions, with the matrix R_i^{-1} entangling the consumption decision and the deployment decision so that neither solves without the other. Section 2.3 reads that entanglement in substantive terms, treating offense-defense balance, capability advantage, and bargaining power as endogenous consequences of the joint optimum rather than primitive parameters. Section 2.4 records the properties of the

resulting joint best-response pair (f_i, μ_i) that Chapter 3 will assemble across states. Section 2.5 hands the chapter's apparatus to the cross-state equilibrium concept that the rest of Part 1 builds on.

2.1 Primitives

Most of what follows is standard smooth general-equilibrium setup, adapted to a bounded-below consumption set. Preferences, endowments, deployments, and the battle function are recognizable to a reader of Mas-Colell or Balasko. The architectural commitment the rest of the book turns on is concentrated in one place—the opportunity-cost function c_i is a primitive of the model with prices as an argument. The price argument of c_i is the agent-level seed of joint determination. It is what makes the state's problem joint over consumption and force the moment the state faces a price.

2.1.1 States, goods, and prices

Take as primitive a finite list of *states*, indexed $i = 1, \dots, I$ with $I \geq 2$, and a finite list of *goods*, indexed $\ell = 1, \dots, L$ with $L \geq 2$. States are unitary actors. Goods may be differentiated by physical characteristics, time, location, or state of the world.

Each good carries a strictly positive price $p^\ell > 0$. Only relative prices are determined by the model, so we fix good L as numeraire and write

$$p \in S_N := \{p \in \mathbb{R}_{++}^L \mid p^L = 1\}.$$

The set S_N is the open numeraire-normalized simplex.

2.1.2 Consumption sets and endowments

State i 's consumption set is the open orthant

$$X_i := \prod_{\ell=1}^L (b_i^\ell, \infty),$$

where the bounds $b_i^\ell < 0$ are state-and-good-specific. The bounds encode the structural floor below which losses become catastrophic and the state ceases to operate as such; the strict negativity guarantees that the origin and the entire strictly positive orthant lie inside X_i . Bundles are vectors $x_i \in X_i$, and an allocation is a tuple $x = (x_1, \dots, x_I) \in X := \prod_i X_i$.

Each state enters with a strictly positive *endowment* $\omega_i \in \mathbb{R}_{++}^L$. The endowment profile is $\omega = (\omega_1, \dots, \omega_I)$, and the endowment space is

$$\Omega_0 := \mathbb{R}_{++}^{IL}.$$

The endowment $\omega \in \Omega_0$ is the model's principal parameter: the distribution of resources across the world.

2.1.3 Preferences

State i 's preferences over total consumption are represented by a utility function $u_i : X_i \rightarrow \mathbb{R}$.

Assumption 2.1 (Preferences). For each i , the function u_i is

1. smooth (C^∞) on X_i ;
2. smoothly strictly quasiconcave (the bordered Hessian of u_i is nonsingular with sign $(-1)^L$ everywhere on X_i);
3. smoothly monotone ($Du_i(x_i) \gg 0$ for all $x_i \in X_i$); and
4. boundary-blow-up: $u_i(x_i) \rightarrow -\infty$ as $x_i^\ell \rightarrow (b_i^\ell)^+$ for any ℓ .

Conditions (1)–(3) are the standard hypotheses of smooth general equilibrium theory, adapted to the bounded-below domain X_i . Condition (4) replaces, on X_i , the role played by the boundary of $\mathbb{R}_{++}^L$ in the classical setting: a catastrophic shortfall in any single commodity sends utility to $-\infty$, so optimal bundles remain in the interior of X_i regardless of how conflict redistributes resources at the margin. We write $\mathcal{U}_i$ for the class of utility functions satisfying Assumption 2.1 and treat u_i as the representation of the state's effective objective without committing here to any particular reading of where it comes from.

2.1.4 Deployments

State i 's level of military deployment is a scalar $m_i \in M_i$, where each $M_i \subset \mathbb{R}_+$ is a convex open set. The deployment profile is

$$m = (m_1, \dots, m_I) \in M := \prod_i M_i.$$

A deployment is a quantity of force; the technology that turns resources into force, and the conditions under which doing so is costly, are captured below in the opportunity-cost function.

Remark 2.2 (Why m_i is a scalar). Treating force as a scalar abstracts from the compositional structure of military capability—the army, the navy, the air force, and the doctrine that combines them into operational effect. Whether this multi-dimensional object can be reduced to a single scalar is a substantive question. Carroll [2026a] characterizes the conditions under which the reduction is justified and m_i is the appropriate unit of analysis. The present chapter takes the reduction as given.

2.1.5 Opportunity costs of deployment

Producing force consumes resources, and the resource bundle absorbed depends on the prices the state faces in the goods market. We collect this dependence into a single primitive object: the *opportunity-cost function*

$$c_i : M_i \times S_N \times \mathbb{R}_{++}^L \longrightarrow \mathbb{R}^L,$$

so that, when state i deploys m_i at prices p from endowment ω_i , the bundle remaining for market exchange is the *diminished endowment* $\omega_i - c_i(m_i, p, \omega_i)$.

Assumption 2.3 (Opportunity costs). For each i , the opportunity-cost function c_i is

1. smooth on $M_i \times S_N \times \mathbb{R}_{++}^L$;
2. zero at zero deployment: $c_i(0, p, \omega_i) = 0$ for all $(p, \omega_i) \in S_N \times \mathbb{R}_{++}^L$;
3. coercive in m_i : $\|c_i(m_i, p, \omega_i)\| \rightarrow \infty$ as $|m_i| \rightarrow \infty$, uniformly in (p, ω_i) on compacts; and
4. interior on the consumption set: $\omega_i^\ell - c_i^\ell(m_i, p, \omega_i) > b_i^\ell$ for every feasible (m_i, p, ω_i) and every ℓ .

The price argument of c_i is load-bearing for the architecture of the book. A state's mobilization plan optimizes against an internal cost of extraction. That cost is in part a market cost—wages paid, capital diverted, fuel consumed—and so the optimal plan, and therefore the resource bundle that producing m_i in fact costs, moves with prices. Recording c_i as a function of p at the level of primitives, rather than restoring price-dependence after the fact, is what makes the state's budget set price-dependent in a non-trivial way: not just through the price of each good, but through how prices reshape the cost of every level of force. This is the agent-level seed of joint determination. The same primitive that supplies the state's resource constraint is already coupled to the same price vector that the market will clear. The coupling is built in from the start, not introduced as a connection between two otherwise-separate problems.

Remark 2.4 (Why c_i is a primitive, not a derived object). The opportunity-cost function can be derived from a finer specification in which the state minimizes a price-dependent extraction cost subject to a smooth force-production technology; under standard regularity the resulting policy is smooth and inherits the convexity, coercivity, and price-dependence required by Assumption 2.3. We do not carry that derivation here. See Carroll [2026b]. The substantive content of the framework lives in how c_i enters the joint equilibrium problem, and that content is fixed by Assumption 2.3 alone.

2.1.6 Battle functions

Conflict redistributes goods across states. The *battle function* s_i records the redistribution accruing to state i as a function of the deployment profile, the allocation, and endowments.

Assumption 2.5 (Battle technology). For each i , the battle function

$$s_i : M \times X \times \Omega_0 \longrightarrow \mathbb{R}^L$$

is smooth, and total consumption respects the consumption set:

$$x_i^\ell + s_i^\ell(m, x, \omega) > b_i^\ell \quad \text{for all feasible } (m, x, \omega) \text{ and every } \ell.$$

The dependence of s_i on each of its arguments carries content. On m : how much force each side is projecting, the standard channel. On x : what has been produced, traded, or consumed, and so what is in play to be seized at the moment of conflict. On ω : what exists in the world to be contested in the first place. Writing s_i on (m, x, ω) jointly—rather than as a function of m alone—is what lets the function operate on *net flows*. Spoils proportional to $\omega_j - x_j$ for adversaries j , for instance, capture a battle that targets shipping rather than stockpiles, and the dependence on x is what makes “net flows” an equilibrium object rather than a parameter.

The downstream existence and structure theorems require one further condition on s_i , the *conflict-feedback bound*, which controls how strongly the spoils accruing to i respond to i 's own allocation. We state it in the form needed by the joint equilibrium function.

Assumption 2.6 (Conflict feedback). For each i and every (m, x, ω) in the relevant domain, the matrix

$$R_i(m, x, \omega) := I + D_{x_i} s_i(m, x, \omega) \in \mathbb{R}^{L \times L}$$

is invertible.

A sufficient condition for Assumption 2.6 is $\|D_{x_i} s_i\| < 1$ in the operator norm: spoils respond to the allocation, but less than one-for-one. Economically, this rules out the perverse configuration in which buying a unit of a good at market causes the state to lose more than a unit through conflict, which would unwind the logic of exchange itself. Assumption 2.6 reappears in the joint Jacobian of the equilibrium function as part of what makes the deployment block invertible (Chapter 3); we record it here so that the primitives already carry everything the existence argument will use.

Remark 2.7 (Why s_i is a primitive, not a derived object). The battle function can be derived from a finer specification in which a contest technology resolves competing claims on the allocation given the deployment profile; under standard regularity the resulting object is smooth and inherits the conflict-feedback bound and the interior condition required by Assumptions 2.5 and 2.6. We do not carry that derivation here, and we do not point to a companion paper for it: s_i is a primitive of this framework without a corresponding finer-specification treatment elsewhere in the author's work. The substantive content of the framework lives in how s_i enters the joint equilibrium problem, and that content is fixed by Assumptions 2.5 and 2.6 alone.

2.2 The state's problem under threat

The primitives of Section 2.1 let us state what each state actually does. A state i , observing prices p and the deployment profile of rivals m_{-i} , chooses jointly its market consumption $x_i \in X_i$ and its own deployment $m_i \in M_i$. Total consumption is what the state buys at market plus what it gains or loses through conflict; the budget set is the diminished endowment $\omega_i - c_i(m_i, p, \omega_i)$. We call this the *state's problem under threat* (SPT).

The section does two things. First, it states the SPT and derives the first-order conditions. These make the agent-level joint determination explicit, with a single matrix R_i^{-1} entangling the consumption decision and the deployment decision so that neither can be solved without the other. Second, it introduces the second leg of the agent-level small-complementarities setup. The added condition is a joint condition on c_i and s_i that pairs with Assumption 2.6 to deliver a unique smooth best-response pair (f_i, μ_i) .

We adopt strong forms of these conditions for tractable initial development. The framework's later chapters revisit which of them are strictly necessary for which structure theorems on the joint equilibrium manifold. The walking-back is what produces the regime-multiplicity story; the strong form is what makes the present-section picture clean.

2.2.1 Statement and first-order conditions

State i chooses $(x_i, m_i) \in X_i \times M_i$ to solve

$$\max_{(x_i, m_i)} u_i(x_i + s_i(m_i, m_{-i}, x, \omega)) \quad \text{s.t.} \quad p \cdot x_i \leq p \cdot (\omega_i - c_i(m_i, p, \omega_i)). \quad (2.1)$$

The constraint is Walrasian on the diminished endowment. The objective is the utility of *total* consumption, namely market purchases x_i plus spoils s_i .

Two features distinguish Equation (2.1) from a textbook consumer problem. The budget set depends on the choice variable m_i through c_i . The objective depends on m_i and on the full allocation x through s_i , which entangles state i 's consumption choice with its deployment choice via the x_i -dependence of s_i .

Let $x'_i := x_i + s_i(m_i, m_{-i}, x, \omega)$ denote total consumption. Differentiating the objective in x_i uses the chain rule through s_i :

$$\frac{\partial}{\partial x_i} u_i(x'_i) = Du_i(x'_i) \cdot R_i,$$

where $R_i = I + D_{x_i} s_i$ from Assumption 2.6. The matrix R_i is the *effective return* on a market purchase: a unit of good ℓ bought at market yields one unit of consumption plus the induced change in spoils through the allocation channel.

A necessary condition for an interior solution (x_i^*, m_i^*) to Equation (2.1) is the existence of a multiplier $\lambda_i^* > 0$ such that

$$\begin{aligned} Du_i(x'_i) \cdot R_i &= \lambda_i^* p, \\ Du_i(x'_i) \cdot D_{m_i} s_i &= \lambda_i^* p \cdot D_{m_i} c_i, \\ p \cdot x_i^* &= p \cdot (\omega_i - c_i(m_i^*, p, \omega_i)). \end{aligned} \tag{2.2}$$

The first line equates the marginal utility of total consumption, adjusted by the effective return, to the price ratio. The second equates the marginal utility of force to its marginal budget cost. The third is the budget identity, holding with equality by monotonicity (Assumption 2.1).

The two FOCs do not separate. Inverting R_i in the first line and substituting into the second yields

$$p \cdot R_i^{-1} \cdot D_{m_i} s_i = p \cdot D_{m_i} c_i. \tag{2.3}$$

The left side depends on x_i through R_i^{-1} , which is what makes the coupling visible. The optimal deployment cannot be solved without reference to the optimal consumption, and conversely. Equation (2.3) is the agent-level statement of joint determination.

What the partialled traditions miss. A textbook consumer problem holds m_i fixed and solves the first FOC line for the demand f_i . A textbook deployment subgame holds p and x_i fixed and solves the second FOC line for the best response μ_i . Each calculation uses one of the FOC lines and produces the corresponding response—but each ignores the matrix R_i that couples the two lines. With $R_i = I$ the lines uncouple and the textbook calculations recover the joint solution; with $R_i \neq I$ they do not. The wedge $R_i - I = D_{x_i} s_i$ measures how much the partialled answers diverge from the joint answer at the agent level.

R_i^{-1} as the algebraic locus of joint determination. Equation (2.3) concentrates the joint determination into a single matrix. R_i^{-1} stands between the marginal-spoils vector and the marginal-cost vector; without it, the equation reduces to a standard force-versus-resources tradeoff with no allocation channel. With it, marginal spoils are evaluated at *effective* prices that the state's own consumption choice has shifted from market value, and the resulting tradeoff is over the joint pair (x_i, m_i) rather than over m_i alone. The algebraic content is here. The substantive content of the coupling, what the wedge encodes about offense-defense balance, capability advantage, and bargaining power, is the work of Section 2.3. R_i^{-1} is the agent-level matrix that Chapter 3 will combine across states into the joint Jacobian J , and that Chapter 4 will invert to compute equilibrium responses from a single application of J^{-1} .

The tame primitives. Example 2.12 below makes the coupling computable. The conflict-feedback matrix in the tame setting is diagonal,

$$(R_i)_{\ell\ell} = 1 - \lambda^\ell \rho_i, \quad \rho_i(m) := \frac{\sum_j a_{ij} m_j}{1 + \sum_j m_j} \in (-1, 1),$$

where ρ_i is state i 's alliance-weighted deployment edge in the contest. Inverting,

$$(R_i^{-1})_{\ell\ell} = \frac{1}{1 - \lambda^\ell \rho_i},$$

and Equation (2.3) specializes to

$$\sum_\ell \frac{p^\ell}{1 - \lambda^\ell \rho_i} \cdot \frac{\partial s_i^\ell}{\partial m_i} = \sum_\ell p^\ell \cdot \frac{\partial c_i^\ell}{\partial m_i}. \quad (2.4)$$

The left side weights the marginal spoils in good ℓ by the per-good wedge factor $1/(1 - \lambda^\ell \rho_i)$ before contracting with the price vector. When $\rho_i > 0$ the wedge factor exceeds 1 on every good, raising the effective valuation of marginal spoils above market. When $\rho_i < 0$ the wedge factor is below 1, discounting marginal spoils relative to market. The state's own deployment edge is not a primitive in this framework; it is determined in equilibrium by the cross-state assembly of Chapter 3, and the partialled traditions cannot compute it without solving the joint problem.

Remark 2.8 (When the coupling vanishes). If $D_{x_i} s_i \equiv 0$ —the battle function does not depend on the allocation—then $R_i = I$ and Equation (2.3) reduces to $p \cdot D_{m_i} s_i = p \cdot D_{m_i} c_i$, determining m_i^* independently of x_i^* . The state's problem decomposes: choose deployment, then solve a standard consumer problem. This is the partialled architecture the framework refuses. The substantive force of the x -dependence in s_i is precisely that this collapse does not happen.

2.2.2 Diminishing returns to deployment

Assumption 2.6 controls how spoils respond to the allocation. The second leg of the agent-level small-complementarities setup controls how the *net return* to deployment behaves in the deployment variable itself.

Assumption 2.9 (Diminishing returns to deployment). For each i and every (m_{-i}, x, ω, p) in the relevant domain, the net return to deployment

$$m_i \mapsto p \cdot (s_i(m_i, m_{-i}, x, \omega) - c_i(m_i, p, \omega_i))$$

is strictly concave in m_i at any (x_i^*, m_i^*) solving Equation (2.1).

Assumption 2.9 is a joint condition on c_i and s_i . Neither alone delivers it. The curvature of c_i and s_i in m_i is unrestricted at the level of primitives, so the net concavity at the solution is a structural assumption about the joint technology of force and conflict. Read economically, the assumption says that after the cost of force has been netted out, additional deployment yields strictly diminishing marginal returns at the optimum. This rules out the increasing-returns-to-militarization configuration in which doubling deployment more than doubles spoils net of cost, under which the SPT would have a corner solution or no interior solution at all.

Assumption 2.9 is one of the load-bearing initial-tractability assumptions of the framework. It pairs with Assumption 2.6 to deliver smoothness and uniqueness of the joint best-response

below. Downstream, in Chapter 3, the same pair will reappear in the form of a Nash-contraction property, which is the agent-level mechanism through which the joint Jacobian becomes invertible. A weakened version of that downstream property—admitting Jacobian singularities over a thin set of endowments—is what allows the manifold to break into multiple components and produces the framework’s account of regime multiplicity. The strong form is adopted here so that the agent-level picture stays clean. The walking-back is recorded at the layer where the geometric consequence is visible.

2.2.3 Existence and smoothness of the joint best-response

The SPT has a unique interior solution that varies smoothly with the parameters it conditions on. The maps that record this dependence are the agent-level data the next chapter assembles.

Proposition 2.10 (Joint best-response). *Under Assumptions 2.1, 2.3, 2.5, 2.6 and 2.9, the SPT Equation (2.1) admits a unique interior solution*

$$(x_i^*, m_i^*) = (f_i(p, m_{-i}, \omega), \mu_i(p, m_{-i}, \omega))$$

on the open set of parameters where the FOC Jacobian is nonsingular. The maps f_i and μ_i are smooth on that set.

The proof is a standard application of the implicit function theorem to the FOC system Equation (2.2), which has $L + 2$ equations in the $L + 2$ unknowns (x_i, λ_i, m_i) . The Jacobian of Equation (2.2) decomposes into a consumption block, made invertible by the strict quasiconcavity of u_i (Assumption 2.1) together with R_i invertibility (Assumption 2.6), and a deployment block, made invertible by diminishing returns (Assumption 2.9). Implicit-function smoothness follows.

Remark 2.11 (f_i and μ_i as joint objects). f_i is the consumption state i demands *when it is also optimizing its own deployment*, and μ_i is the deployment it chooses *when it is also optimizing its own consumption*. These are not Walrasian demands corrected after the fact for conflict. They are joint best-responses derived from the SPT in one shot. The distinction matters in Chapter 3: the joint equilibrium function e is assembled from f_i and μ_i as joint objects, not from a Walrasian-demand layer composed with a separate Nash-deployment layer.

Example 2.12 (Tame primitives). A canonical illustration uses the tame primitives of Carroll [2026b], in which the inner force-production technology is log-linear and the inner extraction cost is linear in market prices. On the open set where all components are positive, c_i has the closed-form water-filling solution

$$c_i^\ell(m_i, p, \omega_i) = \mu_i \cdot \frac{\beta_i^\ell}{p^\ell} - 1,$$

where $\beta_i \in \mathbb{R}_+^L$ with $\sum_\ell \beta_i^\ell = 1$ is a state-specific share vector, $A_{\tau,i} > 0$ a state-specific scale parameter (both potentially ω_i -dependent), and the scalar $\mu_i = \mu_i(m_i, p, \omega_i)$ solves the implicit equation

$$m_i = A_{\tau,i} \log \mu_i + A_{\tau,i} \sum_\ell \beta_i^\ell \log \frac{\beta_i^\ell}{p^\ell}.$$

The price argument of c_i runs through μ_i and through the per-good ratio β_i^ℓ/p^ℓ .

A natural tame battle function takes the multi-state form

$$s_i^\ell(m, x, \omega) = \lambda^\ell \cdot \frac{\sum_j a_{ij} m_j}{1 + \sum_j m_j} \cdot \sum_j (\omega_j^\ell - x_j^\ell),$$

with $\lambda \in (0, 1)^L$ a vector of good-specific lootabilities and $a_i \in [-1, 1]^I$ a state-specific alliance vector encoding valence: positive a_{ij} for allies, whose deployment adds to i 's edge, and negative a_{ij} for rivals, whose deployment subtracts from it. The uniform-rival case $a_{ii} = 1$, $a_{ij} = -1/(I-1)$ for $j \neq i$ recovers the mean rival deployment $\bar{m}_{-i} = (I-1)^{-1} \sum_{j \neq i} m_j$ and reduces the numerator to $m_i - \bar{m}_{-i}$. The deployment ratio is bounded in $(-1, 1)$, smooth, and zero when $\sum_j a_{ij} m_j = 0$. The conflict-feedback matrix is diagonal,

$$(R_i)_{\ell\ell} = 1 - \lambda^\ell \cdot \frac{\sum_j a_{ij} m_j}{1 + \sum_j m_j} \in (0, 2),$$

and Assumption 2.6 holds with strict slack.

The agent-level conditions accumulated in Sections 2.1 and 2.2 are designed to deliver one substantive result: each state has a unique smooth best-response pair (f_i, μ_i) on the parameters (p, m_{-i}, ω) . We have not asserted anything about how f_i and μ_i across states fit together at equilibrium; that assembly is the work of Chapter 3.

The pair Assumption 2.6 and Assumption 2.9 is stronger than what the structure theorems strictly require. Their downstream consequence—that the deployment best-response is a contraction, and hence that the Nash component of the joint problem has a unique solution given (p, ω) —combines with regularity of the resulting reduced exchange economy to give a global form of joint regularity, under which the equilibrium manifold is a single smooth chart. A weaker form, admitting Jacobian singularities over a thin set of endowments, gives a componentwise picture in which each component is a regime. The framework's institutions chapter operates on a single component, and crossing between components is regime change. The agent-level data of this chapter is silent on which form to adopt. Both readings of the manifold consume (f_i, μ_i) identically. What differs is the global geometry assembled in Chapter 3, not the agent-level optimization assembled here.

2.3 Reading the coupling

Equation (2.3) states the agent-level joint determination as a matrix equation; this section unpacks what the matrix is doing. The conflict-theory tradition takes capabilities as exogenous and conflict as one-shot, and writes its equilibrium concept in terms of those primitives. Equation (2.3) refuses both partials at the agent level. The matrix R_i^{-1} is the formal apparatus that couples consumption and force decisions, and its substantive content is the agent-level form of what that tradition typically encodes in primitive parameters.

The effective price vector. The price vector p enters Equation (2.3) on both sides, but it enters differently. On the left, prices are pre-multiplied by R_i^{-1} before contracting with the marginal-spoils vector $D_{m_i} s_i$. On the right, prices contract with marginal extraction $D_{m_i} c_i$ at face value. Define the *effective price vector*

$$\tilde{p}_i := p \cdot R_i^{-1}.$$

Then Equation (2.3) reads

$$\tilde{p}_i \cdot D_{m_i} s_i = p \cdot D_{m_i} c_i.$$

The state values its marginal spoils at $\tilde{p}_i$, not at p . The prices governing the spoils side are effective prices that the state's own market choices have shifted from market value.

The wedge in the tame primitives. The wedge between $\tilde{p}_i$ and p has a direct reading in Example 2.12. With diagonal R_i having entries $(R_i)_{\ell\ell} = 1 - \lambda^\ell \rho_i$, the effective price ratio is

$$\frac{\tilde{p}_i^\ell}{p^\ell} = \frac{1}{1 - \lambda^\ell \rho_i}.$$

The sign of the wedge tracks the sign of ρ_i . Positive ρ_i —an alliance-weighted deployment edge in i 's favor—gives an effective price above the market. Negative ρ_i gives an effective price below the market. What the formal-IR literature variously names offense-defense balance, capability advantage, or relative power enters the agent-level FOC as the sign of ρ_i , and the framework treats it not as a primitive parameter but as a derived consequence of the joint deployment profile.

The conflict-theory translation. In the conflict-theory tradition, the probability of victory and the cost of conflict enter the model as primitives, and the equilibrium is written in terms of those primitives. The joint formulation here derives those quantities from (τ_i, κ_i, s_i) at the agent level. What the conflict-theory tradition treats as bargaining power, the joint formulation treats as the value of ρ_i at the joint optimum: a function of the equilibrium deployment profile, not an exogenous parameter. The matrix R_i^{-1} is the formal mechanism that delivers this endogeneity at the agent level; what Chapter 3 adds is the cross-state consistency that determines ρ_i in equilibrium.

Remark 2.13 (ρ_i as endogenous analog of probability of victory). In the conflict-theory tradition [cf. Fearon, 1995, Powell, 2006], the probability of victory π_i and the cost of conflict κ_i enter the model as exogenous primitives, fixed before any equilibrium concept is applied. In the tame-primitives setting of Example 2.12, the alliance-weighted deployment edge ρ_i defined in Equation (2.4) plays the analogous role: positive ρ_i signals a net advantage for state i , negative signals a disadvantage, but ρ_i is not a primitive. ρ_i is determined in equilibrium alongside prices and consumption bundles. The effective-price wedge $\tilde{p}_i^\ell/p^\ell = 1/(1 - \lambda^\ell \rho_i)$ translates this deployment edge into a per-good premium or discount on the spoils valuation; its sign and magnitude depend on both the equilibrium deployment profile m and the lootability λ^ℓ of each good. The formal-IR question “what is state i 's probability of victory?” is, in this framework, answered by computing ρ_i at the joint equilibrium of Chapter 3 rather than by supplying π_i as a model input.

Equation (2.3) is one equation. The language of offense and defense, of capabilities and consumption, of bargaining power and costs, is the substantive content the equation encodes. Reading the agent-level coupling is therefore not a separate exercise from the framework's main argument: it is the substantive content of the joint-determination claim, made concrete at the level where the SPT lives.

2.4 Demand and deployment

Proposition 2.10 delivers a unique smooth pair (f_i, μ_i) on the parameters (p, m_{-i}, ω) . This section records the properties of the pair that Chapter 3 will consume: the budget identity, homogeneity in prices, and the joint comparative-statics formula.

The budget identity is Equation (2.2) line three, evaluated at the solution. At the SPT optimum, the state spends its diminished endowment exactly. Define the *diminished wealth*

$$w_i(p, m_{-i}, \omega) := p \cdot (\omega_i - c_i(\mu_i(p, m_{-i}, \omega), p, \omega_i)),$$

the market value of state i 's endowment after the cost of force has been deducted. Walras' law for state i then reads

$$p \cdot f_i(p, m_{-i}, \omega) = w_i(p, m_{-i}, \omega).$$

The identity holds pointwise on the SPT solution and propagates, when summed across states, to the aggregate market-clearing condition that organizes Chapter 3.

The demand function under threat satisfies a boundary-behavior property that the existence argument of Chapter 3 will use to bound equilibrium allocations away from the boundary of the domain.

Lemma 2.14 (Desirability). *Let $(p^n, m_{-i}^n, \omega^n) \in S_N \times M_{-i} \times \Omega_0$ be a sequence converging to a boundary point of the domain—either $p^{0,\ell} = 0$ for some ℓ , or $\omega_i^{0,\ell} = b_i^\ell$ for some ℓ . Then $\|f_i(p^n, m_{-i}^n, \omega^n)\| \rightarrow \infty$.*

The argument runs through the two boundary cases. If $p^{n,\ell} \rightarrow 0$, the budget constraint allows market purchases of good ℓ up to $w_i/p^{n,\ell}$, an upper bound that diverges as $p^{n,\ell} \rightarrow 0$; strict monotonicity of u_i (Assumption 2.1) drives the state to exploit this slack, so the demand for good ℓ diverges. If $\omega_i^{n,\ell} \rightarrow (b_i^\ell)^+$, the boundary blow-up condition of Assumption 2.1 keeps total consumption bounded away from b_i^ℓ at the optimum while the budget shrinks in good ℓ . Maintaining both forces a component of f_i to diverge.

Homogeneity in prices is absorbed by the numeraire normalization $p^L = 1$. The pair (f_i, μ_i) is invariant to a uniform rescaling of the price vector by definition of the simplex S_N . What remains is how the pair responds to non-trivial perturbations of relative prices, of rival deployments, and of endowments—the joint comparative statics.

Differentiate the FOC system Equation (2.2) in any parameter $\theta \in \{p, m_{-i}, \omega\}$. The implicit function theorem yields

$$D_\theta(f_i, \lambda_i^*, \mu_i) = -J_i^{-1} \cdot D_\theta \text{FOC}_i,$$

with J_i the FOC Jacobian of Proposition 2.10 and $D_\theta \text{FOC}_i$ the parameter-derivative of the FOC system at the choice variables held fixed. The FOC Jacobian has the block structure named in Proposition 2.10: a consumption block invertible through u_i 's strict quasiconcavity and R_i 's invertibility, a deployment block invertible through diminishing returns. A single inverse therefore delivers the joint response of (f_i, μ_i) to every parameter shock. Neither component has a partial response computable without the other.

The headline of the joint formulation lives here. Three parameter shocks illustrate the apparatus, each computed from the same agent-level inverse J_i^{-1} against a different parameter direction.

Arms races ($\theta = m_{-i}$). A shift in rival deployment changes state i 's alliance-weighted edge ρ_i : a rival's arming pulls ρ_i down through $a_{ij} < 0$, an ally's arming pulls it up through $a_{ij} > 0$. The conflict-theory tradition treats arms-race responses as a Nash subgame in deployments, with consumption choices held outside the analysis. The exchange-theory tradition has no apparatus for the question, since rivals' deployments do not enter the consumption FOC at all. The joint formulation computes $D_{m_{-i}}(f_i, \mu_i) = -J_i^{-1} \cdot D_{m_{-i}}\text{FOC}_i$ and recovers both the deployment response and the consumption substitution that accompanies it. The cross-channel content runs through ρ_i . A rival's arming lowers ρ_i and therefore lowers the wedge factor $1/(1 - \lambda^\ell \rho_i)$ on every good, depressing the effective valuation of marginal spoils across the board—an adjustment the conflict-side partial misses because it computes the deployment response at market prices, not at the wedge-adjusted effective prices the joint formulation produces. The framework is silent on whether μ_i rises or falls in absolute terms. Whether i counter-arms or accommodates depends on the relative weight of the wedge adjustment against the direct $D_{m_{-i}}s_i$ term in the FOC. A state's arming response is its consumption response, and the two are computed by one inversion of J_i rather than two.

Resource booms ($\theta = \omega_i$). An increase in i 's endowment relaxes the budget and, through $A_{\tau,i}(\omega_i)$ and $\beta_i(\omega_i)$ in Example 2.12, can also shift the force-production technology. The exchange-theory tradition treats this as a pure income effect on demand, computable from a Slutsky decomposition with deployment held fixed. The conflict-theory tradition has nothing to say—endowments do not enter the contest primitives directly. The joint formulation computes $D_{\omega_i}(f_i, \mu_i) = -J_i^{-1} \cdot D_{\omega_i}\text{FOC}_i$, decomposing the response into a consumption flow and a military-buildup flow with their proportions determined by the sign structure of J_i at the equilibrium point. The cross-channel content has two pieces. First, the wealth effect on consumption shifts $D_{x_i}s_i$ through the chain rule, which propagates into R_i^{-1} and changes the effective price wedge. Second, when $A_{\tau,i}$ or β_i depends on ω_i , the technology shift alters $D_{m_i}c_i$ directly, changing the marginal cost of force at the same time the budget loosens. The exchange-side partial captures only the income effect. The conflict-side partial captures only the technology shift. The joint formulation captures both at once. The resource-curse debate in formal IR concerns the sign of the deployment response: whether resource windfalls flow to consumption or to military buildup. The framework supplies the apparatus to make the question precise, with the sign answered by computing $J_i^{-1}D_{\omega_i}\text{FOC}_i$ at the relevant equilibrium.

Market-price shocks ($\theta = p$). A change in relative prices alters both the cost of market consumption and, through the price argument of c_i , the cost of force production. The exchange-theory tradition treats this as a standard Slutsky decomposition on the consumption side, with deployment held fixed. The conflict-theory tradition has no apparatus for prices entering the contest at all, since force is priced in resource units that the cost-of-conflict parameter does not touch. The joint formulation computes $D_p(f_i, \mu_i) = -J_i^{-1} \cdot D_p\text{FOC}_i$ as a single object: substitution across goods on the consumption side, recalibration of the mobilization plan on the deployment side. The cross-channel content runs through $D_p c_i$. A price shift changes the resource bundle c_i absorbs at any given m_i , which shifts the diminished endowment $\omega_i - c_i(m_i, p, \omega_i)$, which feeds into f_i 's budget and thence into $D_{x_i}s_i$ and the wedge. The architectural commitment of Section 2.1.5— c_i as a primitive function of p rather than a derived object recovered after the fact—is what makes this propagation visible at the FOC level. A market shock and a security adjustment are not separate events in the framework but two components of a single shift.

Arms races, resource booms, and market shocks are not separate models in the framework but three parameter substitutions in the same J_i^{-1} calculation. Chapter 4 assembles these agent-level responses across states into the cross-state inverse J^{-1} , where the single-matrix discipline of Equation (2.3) is recovered at the system level.

2.5 Looking ahead

The chapter has delivered the agent-level data the rest of the book consumes. Each state's SPT (Equation (2.1)) admits a unique smooth solution (f_i, μ_i) that satisfies Walras' law in the form $p \cdot f_i = w_i$ and the desirability of Lemma 2.14, with comparative statics computed from a single matrix inverse. The architectural commitment is that this data is already joint— f_i and μ_i are entangled through R_i^{-1} at the agent level, before any cross-state consistency requirement is imposed.

Chapter 3 assembles this agent-level data into a system. The joint equilibrium function $e(p, m, \omega) = (\bar{d}, r)$ aggregates excess demand and Nash-residual blocks across states. Joint regularity—a single condition on the joint Jacobian of e —replaces the small-complementarities setup of Assumptions 2.6 and 2.9 as the operational hypothesis for the existence and structure theorems on the joint equilibrium manifold $\bar{E} := e^{-1}(0)$.

Two companion papers fill in specifications this chapter takes as given. Carroll [2026b] derives c_i from a finer state-production problem and characterizes the topology of the resulting space of states (cited in Remark 2.4). Carroll [2026a] characterizes when the compositional complexity of military capability reduces to a single scalar deployment (cited in Remark 2.2). The battle function s_i has no companion derivation; it is taken as a primitive.

Read as smooth-GE machinery, the chapter delivers the SPT and the joint best-response in a form that connects to standard demand-and-supply theory, with the price-dependence of c_i as load-bearing. Read as an engagement with the conflict-theory tradition, it names the agent-level coupling R_i^{-1} and translates it into the substantive content that tradition typically encodes in primitive parameters. Both readings are views of the same FOC system.

Exercises

1. *Operator-norm sufficient condition for feedback.* The conflict-feedback bound Assumption 2.6 requires the matrix $R_i = I + D_{x_i} s_i$ to be invertible at every (m, x, ω) in the relevant domain, but the assumption is stated as bare invertibility rather than as a contraction-style bound. Show that if $\|D_{x_i} s_i(m, x, \omega)\| < 1$ in the operator norm, then R_i is invertible and Assumption 2.6 holds. This is a sufficient (not necessary) form of the bound: spoils respond to the allocation, but less than one-for-one. (Hint: Neumann series.) Then identify a primitive specification for which Assumption 2.6 holds but the operator-norm bound fails, illustrating the gap between the necessary and sufficient conditions.
2. *Closed-form water-filling in the tame primitives.* Example 2.12 states the closed-form water-filling solution for the inner extraction-cost minimization in the tame primitives, $c_i^\ell(m_i, p, \omega_i) = \mu_i \beta_i^\ell / p^\ell - 1$, without deriving it. Set up the Lagrangian for the inner minimization problem and apply the FOCs in x_i^ℓ to recover this closed form. Identify the implicit equation that pins down $\mu_i(m_i, p, \omega_i)$ and verify it agrees with the form stated at

Example 2.12. The exercise is the standard Cobb-Douglas-style water-filling computation, presented as the tame-primitives instance of the general extraction-cost setup.

3. *Diagonal feedback in the tame setting.* For the tame primitives of Example 2.12, compute the diagonal entry $(R_i)_{\ell\ell} = 1 - \lambda^\ell \rho_i$, where $\rho_i := (\sum_j a_{ij} m_j) / (1 + \sum_j m_j)$ is state i 's alliance-weighted deployment edge. Verify that the off-diagonal entries vanish (each s_i^ℓ depends on x^ℓ only). Conclude that $\rho_i \in (-1, 1)$ together with $\lambda^\ell \in (0, 1)$ forces $(R_i)_{\ell\ell} \in (0, 2)$, so Assumption 2.6 holds with strict slack on every coordinate. The substantive content is that the tame-primitives setup automatically delivers the conflict-feedback bound by construction, with no need for auxiliary regularity assumptions.
4. *Decoupling when $D_{x_i} s_i \equiv 0$.* Remark 2.8 states that when the battle function does not depend on the allocation, the coupling matrix R_i reduces to the identity and the SPT decomposes into a Walrasian consumer problem and a standalone Nash deployment subgame. Verify this decomposition explicitly. With $D_{x_i} s_i \equiv 0$, $R_i \equiv I$, the joint FOC reduces to $p \cdot D_{m_i} s_i = p \cdot D_{m_i} c_i$, which determines m_i^* given (p, m_{-i}, ω) alone. The consumer problem at the resulting budget is the standard Walrasian one. Substantively: this is the partialled architecture the framework refuses by construction, and the exercise illustrates exactly which primitive condition the framework relaxes to recover joint determination.
5. *Sign and magnitude of the price wedge.* The effective-price wedge in the tame primitives is $\tilde{p}_i^\ell / p^\ell = 1 / (1 - \lambda^\ell \rho_i)$, where $\tilde{p}_i$ is the wedge-adjusted price the state faces in evaluating its spoils relative to the market price p . Analyze its sign and magnitude across the alliance-lootability regions. Limits to consider: $\rho_i \rightarrow 1$ (deployment edge in i 's favor); $\rho_i \rightarrow -1$ (against i); $\rho_i = 0$ (neutral, no edge); λ^ℓ near 0 (good ℓ is unlootable) versus near 1 (highly lootable). Interpret each substantively: positive ρ_i raises the effective price above market (state i values its spoils more than market does), and the wedge is amplified for highly lootable goods. This exercise is the smooth-version analogue of the kinks-in-the-budget analyses familiar from PE conflict theory, but recovered from the joint formulation.
6. *Endowment derivative of demand under log preferences.* Suppose $b_i^\ell = 0$ for all ℓ to keep the algebra clean (Assumption 2.1(4) is then satisfied by plain log utility on the positive orthant). For the tame primitives with log preferences $u_i(x_i) = \sum_\ell \alpha_i^\ell \log x_i^\ell$ (with $\sum_\ell \alpha_i^\ell = 1$), derive Walrasian demand $f_i^\ell = \alpha_i^\ell w_i / p^\ell$ at budget $w_i = p \cdot \omega_i + p \cdot s_i - p \cdot c_i$. Compute the endowment derivative $D_{\omega_i} f_i$ at the SPT solution and decompose it into three channels: the direct income channel through $p \cdot \omega_i$ (with $\partial(p \cdot \omega_i) / \partial \omega_i^k = p^k$); the spoils channel through $s_i(m, x, \omega)$ at the Nash deployment; and the cost channel through $c_i(m_i, p, \omega_i)$. The exercise is a worked income-effect calculation that anticipates the more elaborate three-channel decompositions of Chapter 5 and Chapter 6.
7. *From ρ_i to a Fearon-style π_i .* The bargaining-of-war literature uses a contest-success function $\pi_i \in [0, 1]$ to represent state i 's probability of winning a contest as a function of relative deployments. The framework's $\rho_i \in (-1, 1)$ plays a related but distinct role: it indexes the alliance-weighted deployment edge, not a probability. In a 2-state symmetric contest with uniform-rival alliance ($a_{ii} = 1$, $a_{ij} = -1$), express ρ_i in terms of the deployment difference $m_i - m_{-i}$. Construct an affine map from ρ_i to a candidate $\pi_i \in [0, 1]$ and identify a region of (m, λ) in which π_i tracks the deployment edge in the way the bargaining-of-war literature assumes; identify a region in which the mapping fails as a contest-success function

(sign, monotonicity, or bounds). The exercise is open-ended; the substantive question is where the framework's ρ_i is and is not isomorphic to the standard π_i .

8. *Architectural role of price-dependence in c_i .* A core architectural commitment of Assumption 2.3 is that the opportunity-cost function c_i depends on prices p , not just on deployment m_i and endowment ω_i . This exercise interrogates that commitment by removing it. Suppose, counter to the chapter setup, that c_i depends on (m_i, ω_i) but not on p . Building on Exercise 4 (so $D_{x_i s_i} \equiv 0$ as well), show that the SPT then fully decouples into a deployment subgame in m and a Walrasian consumer problem in x_i given the resulting spoils, recovering the partialled architecture of bargaining-of-war. Then, isolating the role of price-dependence alone, show that when c_i is price-independent but $D_{x_i s_i} \not\equiv 0$, prices still enter the deployment FOC only through the inner-product term $p \cdot D_{m_i s_i}$. Conclude that the price-dependence of c_i is one of two primitives that load the joint structure into the agent-level problem (the other being the x -dependence of s_i).
9. *Desirability via an explicit parameter sequence.* Lemma 2.14 establishes that as the parameter $(p^n, m_{-i}^n, \omega^n)$ approaches a boundary point of the domain $S_N \times M_{-i} \times \Omega_0$, the demand function diverges: $\|f_i(p^n, m_{-i}^n, \omega^n)\| \rightarrow \infty$. Verify the lemma for the tame primitives in two cases. First, fix (m_{-i}, ω) and let $p^{n,\ell} \rightarrow 0$ for some ℓ while the other components of p^n remain bounded. Show that w_i^n is bounded below away from zero, so the budget upper bound $w_i^n/p^{n,\ell}$ on demand for good ℓ diverges, and conclude by strict monotonicity of u_i (Assumption 2.1(3)) that $\|f_i(p^n, m_{-i}, \omega)\| \rightarrow \infty$. Second, fix (p, m_{-i}) and let $\omega_i^{n,\ell} \rightarrow (b_i^\ell)^+$ for some ℓ ; show that boundary blow-up of u_i (Assumption 2.1(4)) combined with the shrinking budget in good ℓ forces a component of f_i to diverge. The exercise instantiates the proof of Lemma 2.14 on a worked specification, making the boundary-behavior conditions of Assumption 2.1 concrete.

Chapter 3

Equilibrium

When does a price shock translate into an arms race, and when does a shift in rivals' military deployments translate into a market disruption? Two formal traditions prefer to keep these questions apart. Trade theory equilibrates the goods market with deployments absent; the conflict-theory tradition equilibrates the contest with prices held fixed. Each tradition holds half of a joint equilibrium fixed while solving the other half. The setup of Chapter 2 forecloses both factorings: when each state's optimization is joint over consumption and deployment, no cross-state equilibrium concept can hold one side of the joint determination fixed.

This chapter defines the joint equilibrium that neither factoring recognizes. The object is a single map $e(p, m, \omega)$ whose zeros simultaneously close the international goods market and the international military contest, inheriting its structure from the agent-level data of Chapter 2. The chapter establishes existence, smoothness, and a diffeomorphism with parameter space under a single regularity condition. A weakening of that condition admits regime multiplicity that subsequent chapters develop.

What the chapter delivers is hard to find elsewhere. The joint equilibrium concept names what the dual-competitive equilibrium ultimately is: not a Walrasian equilibrium with deployments added, not a Nash equilibrium with prices added, but a single object whose existence, uniqueness, and smoothness are properties of the joint formulation rather than of either parent. The chapter's most distinctive content is its componentwise structure. When the regularity hypothesis admits folds, the equilibrium manifold partitions into *regimes* separated by a singular locus, and regime change becomes geometric rather than metaphorical. The structure theorems that follow make a unified setting available to the positive-theory, welfare, and institutional machinery of subsequent chapters; the regime-multiplicity machinery places qualitatively distinct international orders within the same geometric frame.

The chapter is organized around the joint equilibrium and the structure it admits. Section 3.1 constructs the joint equilibrium function e from the agent-level data of Chapter 2 and records the dimension count that anticipates the structure theorems below. Section 3.2 computes the block structure of e 's Jacobian, identifies the exchange-side and conflict-side regularity conditions as Schur sub-pieces, and adopts the single joint condition as the framework's structural primitive. Section 3.3 cashes the dimension count out as smooth-manifold structure on $\tilde{E}$ under joint regularity. Section 3.4 establishes that every endowment profile admits a joint equilibrium, transporting Walrasian existence through a homotopy that turns conflict on continuously. Section 3.5 strengthens existence to a two-way correspondence between endowment profiles and equilibria, exhibiting a global chart on $\tilde{E}$. Section 3.6 weakens joint regularity to admit folds on a thin set

of endowments and partitions the regular manifold into regimes separated by the singular locus.

Sections marked with an asterisk are technical and may be skipped at first reading; each starred section opens with a summary paragraph stating the takeaway.

3.1 The joint equilibrium

In Chapter 2 each state solved its own SPT, producing a smooth joint best-response pair (f_i, μ_i) defined on the cross-state parameters (p, m_{-i}, ω) . The cross-state question begins where the agent-level analysis ends. A pair $(p, m) \in S_N \times M$ is a joint equilibrium for endowment profile ω when prices clear the international goods market and each state plays a best response to its rivals' deployments at those prices. The two conditions are simultaneous, not sequential. Trade theory equilibrates the goods market with deployments held fixed; the conflict-theory tradition equilibrates the contest with prices held fixed. Neither alone yields the equilibrium pair (p, m) . The joint formulation treats (p, m) as a single decision variable, and this section constructs the joint equilibrium as the zero set of a single map.

The market-clearing component. The market-clearing component aggregates net market purchases across states. For each good ℓ , set

$$d^\ell(p, m, \omega) := \sum_i [f_i^\ell(p, m_{-i}, \omega) - (\omega_i^\ell - c_i^\ell(m_i, p, \omega_i))].$$

Walras' law at the SPT optimum (Section 2.4) gives $p \cdot [f_i - (\omega_i - c_i(\mu_i, p, \omega_i))] = 0$ for each state, and aggregation implies $p \cdot d = 0$ at any configuration where every state plays a best response. The L -th component of d is therefore redundant at any joint equilibrium. We work with the truncation

$$\bar{d}(p, m, \omega) := (d^1(p, m, \omega), \dots, d^{L-1}(p, m, \omega)) \in \mathbb{R}^{L-1},$$

which loses no equilibrium information.

The Nash-consistency component. The Nash-consistency component records, for each state, the gap between its chosen deployment and its joint best-response:

$$r_i(p, m, \omega) := m_i - \mu_i(p, m_{-i}, \omega), \quad i = 1, \dots, I.$$

The collection $r = (r_i)_{i=1}^I \in \mathbb{R}^I$ vanishes precisely when every state plays a best response to its rivals at the prevailing prices. This is Nash equilibrium of the deployment subgame at (p, ω) .

The two components fit together as a single map.

Definition 3.1 (Joint equilibrium function). The *joint equilibrium function* is the smooth map

$$e : S_N \times M \times \Omega_0 \longrightarrow \mathbb{R}^{L-1+I}, \quad e(p, m, \omega) = (\bar{d}(p, m, \omega), r(p, m, \omega)).$$

A *joint dual-competitive equilibrium* for endowment profile ω is a pair $(p, m) \in S_N \times M$ with $e(p, m, \omega) = 0$. The *joint equilibrium manifold* is the zero set

$$\tilde{E} := e^{-1}(0) \subset S_N \times M \times \Omega_0.$$

The object Definition 3.1 introduces is wider in scope than the conventional equilibrium concept. The conventional question fixes a parameter profile ω and asks for the prices and deployments at which all markets and all contests settle. The joint formulation instead treats the entire zero set $\tilde{E}$ as a single object: a point of $\tilde{E}$ is a triple (p, m, ω) recording equilibrium prices, equilibrium deployments, and the parameter profile they equilibrate. The conventional concept is recovered as the fiber $\tilde{\pi}^{-1}(\omega) \subset \tilde{E}$ above ω under the projection

$$\tilde{\pi} : \tilde{E} \rightarrow \Omega_0, \quad \tilde{\pi}(p, m, \omega) := \omega.$$

Treating the zero set as a single object lets several questions about the equilibrium take a single geometric form. Existence becomes the question of whether $\tilde{\pi}$ is surjective; uniqueness becomes the question of whether each fiber $\tilde{\pi}^{-1}(\omega)$ is a singleton; regime structure becomes the question of whether $\tilde{E}$ has more than one connected component. The chapter's structure theorems answer the first two under Assumption 3.6 and the third under a weakening of it.

Remark 3.2 (Walras' law on the zero set). At every (p, m, ω) satisfying $r = 0$, we have $m_i = \mu_i$ for every i , and individual Walras' law (Section 2.4) gives $p \cdot f_i = p \cdot (\omega_i - c_i(m_i, p, \omega_i))$. Aggregation across states yields $p \cdot d = 0$. The truncation $\bar{d}$ is therefore lossless on $\tilde{E}$, but $p \cdot d = 0$ does not hold as an identity on the ambient space.

Dimension count. The dimension count motivates the structure theorems that follow. The domain $S_N \times M \times \Omega_0$ has dimension $(L - 1) + I + IL$. The codomain $\mathbb{R}^{L-1+I}$ has dimension $L - 1 + I$. A transversal zero set therefore has dimension

$$(L - 1) + I + IL - (L - 1 + I) = IL,$$

matching the dimension of the parameter space Ω_0 above which $\tilde{\pi}$ projects.

Example 3.3 (A two-state, two-good contest at equilibrium). The simplest economy on which e admits a closed-form zero has $I = 2$, $L = 2$, with prices written $p = (p, 1)$ and good 2 the numeraire. States have logarithmic preferences $u_i(y) = \log y^1 + \log y^2$ and reflected endowments $\omega_1 = (\omega_H, \omega_L)$ and $\omega_2 = (\omega_L, \omega_H)$ with $\omega_H > \omega_L > 0$. Deployment costs $k > 0$ units of good 1 per unit of force, $c_i(m_i, p, \omega_i) = (km_i, 0)$, so the budget cost in numeraire is kpm_i . The case $k = 1$ is the unit-cost normalization; arbitrary k controls how resource-intensive force production is. The contest is Tullock over a prize that combines a fixed component $V_0 > 0$ with a share $\theta \in [0, 1)$ of aggregate good-1 market consumption $T^1 := x_1^1 + x_2^1$,

$$s_i^1(m, x, \omega) = (V_0 + \theta T^1) \cdot \frac{m_i - m_{-i}}{2(m_1 + m_2)}, \quad s_i^2 \equiv 0.$$

The parameter θ governs how much of the contest is over the table: at $\theta = 0$, the spoils contest is over a fixed external prize and the economy reduces to the symmetric proportional Tullock setup of Beviá and Corchón [2010]; at $\theta > 0$, the size of the prize at stake is itself an equilibrium quantity that scales with what the two states bring to market.

The conflict-feedback matrix is diagonal with $(R_i)_{1,1} = 1 + \theta(m_i - m_{-i})/[2(m_1 + m_2)]$ and $(R_i)_{2,2} = 1$, satisfying Assumption 2.6 for any $\theta < 1$. The matrix equals I along the symmetric ray $m_1 = m_2$ but differs from I off it—the agent-level coupling of Equation (2.3) is live everywhere except the equilibrium itself. Solving the SPT and imposing market clearing in good 1 gives the closed-form symmetric joint equilibrium

$$m_1^* = m_2^* = \frac{V_0 + \theta W}{2k(2 + \theta)}, \quad p^* = \frac{(2 + \theta)W}{2W - V_0}, \quad W := \omega_H + \omega_L,$$

provided $V_0 < 2W$ and $V_0/(4k) < \omega_L$ so that prices are finite and diminished endowments lie in $\mathbb{R}_{++}^2$. At $\theta = 0$ this collapses to $m_i^* = V_0/(4k)$, recovering the equilibrium effort of Beviá and Corchón [2010, Case 3.1] for arbitrary cost parameter k , and to $p^* = 2W/(2W - V_0)$.

Three features of the equilibrium are worth recording. First, the agent-level fixed-point $m^* = (V_0 + \theta T^{1*})/(4k)$ holds at every θ : equilibrium deployment matches the BC 2010 formula (for cost parameter k) evaluated at the *equilibrium* prize, but the prize is itself endogenous through $T^{1*} = W - 2km^*$. Second, the contest is decided in good 1 alone, yet the equilibrium price p^* rises in V_0 and in θ : more force burns more good 1, good 1 becomes scarcer at market, and its relative price climbs. Third, an exchange-theoretic equilibration that held m fixed would price-clear at $p = 1$ by symmetry and miss the contest channel; a conflict-side equilibration that held p fixed would solve the Tullock contest correctly but treat the prize as exogenous data rather than as the consumption-determined quantity it is at $\theta > 0$. The joint equilibrium records the contest, the price-formation channel, and the consumption-prize coupling at once, in a single zero of e . Fourth, the equilibrium price p^* is independent of the cost parameter k : changes in k rescale m^* but hold the total resource burn $2km^* = (V_0 + \theta W)/(2 + \theta)$ fixed, so the scarcity of good 1 at market—and hence p^* —does not change. The two partial equilibria misstate the price in opposite directions: the exchange-side partial sets $p = 1$ (too low, ignoring that war burns good 1), while the conflict-side partial sets p at whatever value was assumed before conflict was introduced. The quantitative price gap at $\theta = 0$ is $p^* - 1 = V_0/(2W - V_0)$, rising in V_0 and falling in W : a more contested prize in a smaller economy creates a larger price distortion than either partial captures.

The next section supplies the regularity hypothesis under which the transversality the dimension count anticipates is realized.

3.2 Joint regularity

Whether $\tilde{E}$ inherits a smooth manifold structure from the ambient parameter space depends on the rank of the Jacobian of e at each of its zeros. The natural condition is invertibility of the projection of that Jacobian onto the price-deployment directions; the substantive content lies in what invertibility requires of the underlying objects. This section computes the relevant Jacobian's block structure, identifies the two pieces that the existing literature treats as separate regularity conditions, and names the single condition that the framework adopts as its structural primitive.

The map e has codomain blocks $\bar{d} \in \mathbb{R}^{L-1}$ and $r \in \mathbb{R}^I$, and domain blocks p, m, ω . At any $(p, m, \omega) \in \tilde{E}$ write the projection of De onto the (p, m) -directions as

$$J(p, m, \omega) := D_{(p, m)} e(p, m, \omega) = \begin{pmatrix} J_{pp} & J_{pm} \\ J_{mp} & J_{mm} \end{pmatrix} \in \mathbb{R}^{(L-1+I) \times (L-1+I)}, \quad (3.1)$$

with the four blocks

$$J_{pp} = D_p \bar{d}, \quad J_{pm} = D_m \bar{d}, \quad J_{mp} = -D_p \mu, \quad J_{mm} = I - D_m \mu,$$

all evaluated at the equilibrium point. The subscripts name the directions each block differentiates in: J_{pp} is the price-price block, J_{pm} is the price-deployment block, J_{mp} is the deployment-price block, and J_{mm} is the deployment-deployment block. The signs and the identity term in J_{mm}

follow from the construction of r in Section 3.1: the residual $r_i = m_i - \mu_i$ contributes $+I$ on the m -side and $-D\mu$ on the p - and m -sides.

Two of the four blocks carry standalone economic content already familiar from the partialling architectures the chapter introduction distinguished.

The deployment block. The block $J_{mm} = I - D_m\mu$ records how each state's joint best-response deployment μ_i shifts when the deployment profile changes. J_{mm} is invertible whenever the spectral radius of $D_m\mu$ is strictly less than one—the natural sufficient condition under which the deployment best-response is a contraction and the Nash component of the joint problem has a unique fixed point at given (p, ω) . This is the regularity condition the conflict-theory tradition imposes when it solves the deployment subgame in isolation.

The price block, after substitution. Treat $r = 0$ as defining m implicitly as a smooth function $m^*(p, \omega)$, and substitute into $\bar{d}$ to obtain the *reduced excess demand*

$$\hat{d}(p, \omega) := \bar{d}(p, m^*(p, \omega), \omega).$$

Differentiating $r(p, m^*(p, \omega), \omega) = 0$ in p gives $J_{mp} + J_{mm} \cdot D_p m^* = 0$, so $D_p m^* = -J_{mm}^{-1} J_{mp}$ when J_{mm} is invertible. The Jacobian of $\hat{d}$ in p is then

$$D_p \hat{d} = J_{pp} + J_{pm} \cdot D_p m^* = J_{pp} - J_{pm} J_{mm}^{-1} J_{mp} =: S,$$

the Schur complement of J_{mm} in J . Invertibility of S is the regular-economy condition on the reduced exchange economy obtained by substituting Nash deployments back into market clearing. This is the regularity condition the exchange-theory tradition imposes when it equilibrates the goods market with deployments held fixed at any prevailing profile.

The block-matrix determinant identity $\det J = \det J_{mm} \cdot \det S$ links the two sub-conditions to the single Jacobian.

Lemma 3.4 (Schur decomposition of joint regularity). *At every $(p, m, \omega) \in \tilde{E}$ where $J_{mm} = I - D_m\mu$ is invertible, the joint Jacobian J is invertible if and only if the Schur complement $S = J_{pp} - J_{pm} J_{mm}^{-1} J_{mp}$ is invertible. The Schur complement equals the Jacobian $D_p \hat{d}$ of the reduced excess demand in p , where $\hat{d}(p, \omega) := \bar{d}(p, m^*(p, \omega), \omega)$ and m^* is the Nash selection implicit in $r = 0$.*

Proof. The block-matrix determinant identity $\det J = \det J_{mm} \cdot \det(J_{pp} - J_{pm} J_{mm}^{-1} J_{mp})$ gives the equivalence. The identification $S = D_p \hat{d}$ is the calculation in the preceding paragraph. $\square$

Corollary 3.5 (Small complementarities imply joint regularity). *If $J_{mm} = I - D_m\mu$ is invertible and the Schur complement $S = J_{pp} - J_{pm} J_{mm}^{-1} J_{mp}$ is invertible, then J is invertible and Assumption 3.6 holds at (p, m, ω) .*

Proof. $\det J = \det J_{mm} \cdot \det S \neq 0$. $\square$

Lemma 3.4 exhibits the two-condition architecture as a sufficient route to a single Jacobian condition. J_{mm} invertible and S invertible together imply J invertible, and the converse holds whenever J_{mm} is invertible. The framework adopts the single condition as primitive.

Assumption 3.6 (Joint regularity). At every $(p, m, \omega) \in \tilde{E}$, the joint Jacobian $J(p, m, \omega) := D_{(p,m)} e(p, m, \omega)$ is invertible.

Remark 3.7 (JR is strictly weaker than the two-condition architecture). Imposing contraction on $D_m \mu$ and regularity on the reduced economy is sufficient for Assumption 3.6 but stronger than necessary. J can be invertible when one of the two diagonal sub-conditions fails locally, provided the off-diagonal blocks J_{pm} and J_{mp} supply the missing rank. The contraction-plus-regularity reading is what most of the prior literature actually verifies and is retained as the standard sufficient setup throughout the chapter; Assumption 3.6 is the strictly relevant condition for the structural arguments below.

Reading the blocks. The deployment block J_{mm} controls how rivals' deployments adjust to one another at fixed prices and determines whether the contest, taken in isolation, settles to a unique fixed point. The reduced-economy block S controls whether the price system sits at a regular point of the Walrasian map after deployments have been substituted in and determines whether the goods market, taken with the contest implicit, equilibrates at a non-degenerate point. The conflict-theory tradition imposes the first as its local regularity hypothesis; the exchange-theory tradition imposes the second.

The off-diagonal blocks carry the cross-channel content that distinguishes the joint formulation. The block $J_{pm} = D_m \bar{d}$ records how aggregate excess demand responds to a shift in deployments at given prices; the block $J_{mp} = -D_p \mu$ records how the Nash deployment profile responds to a shift in prices at given rivals' deployments. Each is invisible to the architecture that holds the other side fixed. A localized failure of contraction can be absorbed by these blocks whenever the price-response of deployment and the deployment-response of markets jointly carry the missing rank, and likewise on the exchange-side: a critical point of the reduced exchange economy can fail to be a critical point of the joint Jacobian once the cross-channel terms are restored. The parameter regions where each partialling tradition diagnoses equilibrium breakdown therefore strictly contain the region where Assumption 3.6 fails. The diagnostic value of the joint formulation lies as much in what it does *not* call breakdown as in what it does.

3.3 Smooth structure of the equilibrium manifold

The dimension count of Section 3.1 anticipated that $\tilde{E}$ has dimension IL , matching the parameter space Ω_0 . Under Assumption 3.6 the count is realized as actual smooth structure: $\tilde{E}$ is a smooth submanifold of the ambient space, and the natural projection onto endowments is a smooth map between manifolds of equal dimension.

Proposition 3.8 (Smooth structure of $\tilde{E}$). *Under Assumption 3.6, the joint equilibrium manifold $\tilde{E} \subset S_N \times M \times \Omega_0$ is a smooth submanifold of codimension $L - 1 + I$ and dimension IL .*

Proof. We verify that $0 \in \mathbb{R}^{L-1+I}$ is a regular value of e and apply the preimage theorem. At any $(p, m, \omega) \in \tilde{E}$, the projection $J = D_{(p,m)} e$ onto the price-deployment directions is invertible by Assumption 3.6, so J has rank $L - 1 + I$. The full Jacobian De contains J as a submatrix and therefore has rank at least $L - 1 + I$; the codomain has dimension $L - 1 + I$, capping the rank; hence De has rank exactly $L - 1 + I$, and 0 is a regular value. The preimage theorem yields $\tilde{E}$ as a smooth submanifold of codimension $L - 1 + I$ in an ambient manifold of dimension $(L - 1) + I + IL$, giving $\dim \tilde{E} = IL$. $\square$

The projection $\tilde{\pi}$ defined in Section 3.1 restricts the smooth ambient projection $S_N \times M \times \Omega_0 \rightarrow \Omega_0$ and is therefore smooth.

Corollary 3.9 ($\tilde{\pi}$ is a smooth map between equal-dimension manifolds). *Under Assumption 3.6, the natural projection $\tilde{\pi} : \tilde{E} \rightarrow \Omega_0$ is a smooth map between smooth manifolds of dimension IL .*

Proposition 3.8 and Corollary 3.9 promote the dimension count from a coincidence of integers to a structural fact. “Smooth submanifold” means $\tilde{E}$ comes equipped with tangent spaces, vector fields, smooth paths, and the entire apparatus of differential topology; the equilibrium set is no longer an unstructured collection of triples but an object with intrinsic geometry. The downstream chapters consume that geometry directly. Comparative statics in Chapter 4 are differentiation along smooth paths in Ω_0 , lifted through the chart machinery of Section 3.5 to smooth paths in $\tilde{E}$. Welfare arguments in Chapter 5 integrate along those paths. The institutional aggregation theorems of Chapter 7 consume the topology of $\tilde{E}$ as their input, with the contractibility recorded in Corollary 3.16 below the specific feature Chichilnisky–Heal aggregation relies on. The worked sanctions example of Chapter 8 uses the global chart Φ as the computational coordinate system for the joint comparative-statics calculation. What the rest of the book does on $\tilde{E}$, it does as a geometric object, not as a collection of isolated points.

The corollary also sets up the next two sections. Section 3.4 asks whether $\tilde{\pi}$ is surjective: whether every endowment profile $\omega \in \Omega_0$ admits at least one joint equilibrium. Section 3.5 asks whether $\tilde{\pi}$ is in fact a diffeomorphism, delivering $\tilde{E}$ as a global smooth chart on the parameter space.

3.4 Existence*

Section summary. The result is that every endowment profile $\omega \in \Omega_0$ admits at least one joint equilibrium: the natural projection $\tilde{\pi} : \tilde{E} \rightarrow \Omega_0$ is surjective with mod-2 degree one. The proof transports classical Walrasian existence to the joint formulation through a homotopy that turns conflict on continuously, with mod-2 degree theory carrying the surjectivity across. Substantive readers can take Theorem 3.12 as established and skip to Section 3.5.

The existence question is whether $\tilde{\pi}$ is surjective: whether every endowment profile admits at least one joint equilibrium. What is at stake is the well-definedness of the framework as a system. The joint equilibrium function e defines, for each ω , an equation system whose zeros are the joint equilibria at ω ; existence is the claim that this system has solutions for every $\omega \in \Omega_0$, that the definition has instances above every parameter profile. Without it, the comparative-statics, welfare, and institutional arguments of subsequent chapters would condition on a possibly-empty set of equilibria and lose their content.

In its parent traditions, existence is a known result from which applied work proceeds. Pure-exchange Walrasian general equilibrium has Arrow-Debreu existence [Arrow and Debreu, 1954]; finite-action non-cooperative games have Nash existence [Nash, 1950]; both are inherited downstream and used without the inheriting work re-proving them. The joint formulation is a new system—a continuum of deployment strategies coupled with a price-clearing condition, over a parameter space—and its existence theorem cannot be borrowed from either parent. The argument here proves existence in the joint system by tracking classical Walrasian existence through a homotopy that turns conflict on continuously, transporting the well-known result at the conflict-off endpoint to the joint formulation at the conflict-on endpoint.

The homotopy on the joint space scales the battle function from absent to full. At one endpoint, the deployment subgame collapses trivially and the joint problem reduces to a classical

Walrasian exchange economy; at the other, the joint problem is the framework's $e = 0$ system in full. The mod-2 degree of the natural projection is invariant under proper homotopy, so the value at the conflict-on endpoint inherits the value at the conflict-off endpoint, and a positive-degree proper map between equal-dimension manifolds is surjective.

For $t \in [0, 1]$, set $s_i^t := t \cdot s_i$ and let μ_i^t, f_i^t denote the joint best-response functions derived from the SPT (Equation (2.1)) with the scaled battle function s_i^t in place of s_i . The scaled equilibrium function and manifold are

$$e^t(p, m, \omega) := (\bar{d}^t(p, m, \omega), r^t(p, m, \omega)), \quad \tilde{E}^t := (e^t)^{-1}(0),$$

with natural projection $\tilde{\pi}^t : \tilde{E}^t \rightarrow \Omega_0$. At $t = 1$, $e^1 = e$ and $\tilde{\pi}^1 = \tilde{\pi}$; at $t = 0$, the battle function is identically zero.

Lemma 3.10 (Properness, uniform in t). *Suppose Assumption 3.6 holds for e^t at every $t \in [0, 1]$ and that the desirability and cost-coercivity assumptions of Chapter 2 hold uniformly in t . For every compact $K \subset \Omega_0$, the set*

$$Z_K := \{ (p, m, \omega, t) \in S_N \times M \times K \times [0, 1] \mid e^t(p, m, \omega) = 0 \}$$

is compact. In particular, each $\tilde{\pi}^t$ is proper.

Proof. Take a sequence $(p^n, m^n, \omega^n, t^n) \in Z_K$. Compactness of K and of $[0, 1]$ permits passing to a subsequence with $\omega^n \rightarrow \omega^0 \in K$ and $t^n \rightarrow t^0 \in [0, 1]$. Individual Walras' law and $\omega^n \in K$ bound $p^n \cdot f_i^{t^n}$ and hence each individual demand; cost-coercivity (uniform in t since c_i does not depend on t) bounds m^n from above; desirability (Lemma 2.14, uniform in t since s_i^t only rescales interior behavior) prevents a boundary escape of p^n . Every convergent subsequence has limit in $S_N \times M \times K \times [0, 1]$, and continuity of e^t in all arguments gives $e^{t^0}(p^0, m^0, \omega^0) = 0$. The set Z_K is sequentially compact and hence compact; restriction to a fixed t gives properness of $\tilde{\pi}^t$. $\square$

Lemma 3.11 (The $t = 0$ endpoint). *At $t = 0$:*

- (i) $\mu_i^0 \equiv 0$;
- (ii) $r^0(p, m, \omega) = m$, so $r^0 = 0$ if and only if $m = 0$;
- (iii) $\bar{d}^0(p, 0, \omega)$ coincides with the excess demand of the standard smooth exchange economy with demands $f_i^0(p, \omega) := f_i(p, 0, \omega)$;
- (iv) $\tilde{E}^0 = \{ (p, 0, \omega) \mid \bar{d}^0(p, 0, \omega) = 0 \}$, and $\tilde{\pi}^0 : \tilde{E}^0 \rightarrow \Omega_0$ is canonically identified with the natural projection of the classical Walrasian equilibrium manifold.

In particular, $\deg_2(\tilde{\pi}^0) = 1$ by Balasko [2009, Proposition 4.6.3].

Proof. At $s_i^0 \equiv 0$, the SPT objective reduces to $u_i(x_i)$, independent of m_i . Cost non-negativity with $c_i(0, p, \omega_i) = 0$ implies that any positive m_i strictly tightens the budget and strictly lowers u_i , so $\mu_i^0 = 0$ and (i) holds. Items (ii)–(iv) follow from (i) and Definition 3.1. The degree computation for the classical Walrasian equilibrium manifold is Balasko's and uses smoothness of f_i^0 , Walras' law, and desirability—all inherited uniformly from the SPT assumptions of Chapter 2 at $t = 0$. $\square$

Theorem 3.12 (Existence of joint dual-competitive equilibrium). *Under Assumption 3.6 (holding uniformly in $t \in [0, 1]$) and the SPT assumptions of Chapter 2, the natural projection $\tilde{\pi} : \tilde{E} \rightarrow \Omega_0$ has mod-2 degree one. For every endowment profile $\omega \in \Omega_0$, there exists a joint dual-competitive equilibrium $(p, m) \in S_N \times M$.*

Proof. The family $\{e^t\}_{t \in [0, 1]}$ is smooth in (p, m, ω, t) . Lemma 3.10 gives properness uniform in t , so $\{\tilde{\pi}^t\}_{t \in [0, 1]}$ is a proper homotopy between $\tilde{\pi}^0$ and $\tilde{\pi}^1 = \tilde{\pi}$. The mod-2 degree is invariant under proper homotopy [Milnor, 1965], so $\deg_2(\tilde{\pi}) = \deg_2(\tilde{\pi}^0) = 1$ by Lemma 3.11. A proper map of positive degree between smooth manifolds of equal dimension is surjective, so $\tilde{\pi}^{-1}(\omega)$ is non-empty for every $\omega \in \Omega_0$. $\square$

Remark 3.13 (What the homotopy scales). The homotopy parameter t scales the battle function continuously from absent ($t = 0$) to full ($t = 1$) and deforms the joint equilibrium system as a single object rather than reducing or factoring it. At $t = 0$, the deployment subgame collapses trivially and what remains is a classical Walrasian exchange economy on the diminished endowments—a setting whose equilibrium structure is well known and whose mod-2 degree is one. The joint formulation’s existence proof is therefore a continuation of the standard Walrasian existence argument, not a departure from it: the equilibrium structure that holds without conflict survives the introduction of conflict at every scale $t \in [0, 1]$ on which Assumption 3.6 holds. The contrast with partialled architectures is that the degree computation happens in the $(L - 1 + I)$ -dimensional joint codomain throughout, never collapsing to a reduced economy at any intermediate t .

Existence ensures that every endowment profile is reached by $\tilde{\pi}$. The next section asks whether $\tilde{\pi}$ is also a bijection, whether each fiber $\tilde{\pi}^{-1}(\omega)$ contains exactly one point, and whether the resulting inverse $\Omega_0 \rightarrow \tilde{E}$ is smooth—together giving $\tilde{E}$ a global coordinate chart on the parameter space.

3.5 The diffeomorphism

Existence in Section 3.4 gave a one-way result: every endowment profile ω admits at least one joint equilibrium. This section strengthens existence to a two-way correspondence. The claim is that under Assumption 3.6, every ω corresponds to exactly one equilibrium (p, m) , and the resulting map $\omega \mapsto (m, p)$ varies smoothly—a small change in ω produces a small, predictable change in equilibrium prices and deployments.

The bite is twofold. First, the framework defines an unambiguous comparative-statics object: there is a single equilibrium response to a parameter shift, not a multi-valued correspondence to pick from. Second, the entire wealth-and-power equilibrium structure across all parameter profiles forms a single smooth piece in 1-1 correspondence with Ω_0 , with no folds where the equilibrium suddenly multiplies, no isolated branches, no breaks. The technical statement that follows—that $\tilde{\pi}$ is a *diffeomorphism*—packages both pieces in one term.

We establish the diffeomorphism by exhibiting a global chart on $\tilde{E}$ and identifying its codomain with $\mathbb{R}^{IL}$. The chart uses *diminished wealth* as one of its coordinates. For $(p, m, \omega) \in S_N \times M \times \Omega_0$, set

$$w_i(p, m, \omega) := p \cdot (\omega_i - c_i(m_i, p, \omega_i)) \in \mathbb{R}. \quad (3.2)$$

At any joint equilibrium, $r = 0$ so $m_i = \mu_i$, and w_i coincides with the wealth quantity the SPT (Equation (2.1)) produces at the optimum. Off the manifold, w_i is a definitional convenience without economic content.

For each i , write $\bar{\omega}_i := (\omega_i^1, \dots, \omega_i^{L-1})$ for the projection of ω_i onto its first $L - 1$ components.

Definition 3.14 (Global chart on $\tilde{E}$). The *global chart* on the joint equilibrium manifold is

$$\Phi : \tilde{E} \longrightarrow S_N \times \mathbb{R}^I \times \mathbb{R}^{(L-1)(I-1)}, \quad \Phi(p, m, \omega) = (p, w(p, m, \omega), \bar{\omega}_1, \dots, \bar{\omega}_{I-1}),$$

where $w = (w_1, \dots, w_I)$ is the diminished-wealth profile.

Proposition 3.15 (Global diffeomorphism). *Under Assumption 3.6 and the SPT assumptions of Chapter 2, Φ is a smooth diffeomorphism, $\tilde{E}$ is diffeomorphic to $\mathbb{R}^{IL}$ as a smooth manifold, and the natural projection $\tilde{\pi} : \tilde{E} \rightarrow \Omega_0$ is itself a smooth diffeomorphism.*

Proof. The codomain of Φ has dimension $(L - 1) + I + (L - 1)(I - 1) = IL = \dim \tilde{E}$. Smoothness of Φ inherits from smoothness of w_i in (p, m, ω) , which inherits from smoothness of c_i .

We construct Φ^{-1} as a smooth map from any image tuple $(p, w, \bar{\omega}_1, \dots, \bar{\omega}_{I-1}) \in S_N \times \mathbb{R}^I \times \mathbb{R}^{(L-1)(I-1)}$ in three coupled steps.

- (1) For each $i = 1, \dots, I - 1$, the wealth equation $p \cdot (\omega_i - c_i(m_i, p, \omega_i)) = w_i$ determines ω_i^L given $\bar{\omega}_i$, m_i , and (p, w_i) .
- (2) The aggregate-resource identity $\sum_j \omega_j = \sum_j f_j(p, m_{-j}, \omega) + \sum_j c_j(m_j, p, \omega_j)$ holds on $\tilde{E}$ by the first $L - 1$ market-clearing equations and Walras, pinning down ω_I given $(p, m, \omega_1, \dots, \omega_{I-1})$.
- (3) The Nash condition $r(p, m, \omega) = 0$ is a smooth system in m at fixed (p, ω) . By Lemma 3.4 and Assumption 3.6, the m -Jacobian $J_{mm} = I - D_m \mu$ is invertible, and Theorem 3.12 delivers a solution, so m is uniquely determined.

Steps (1)–(3) form a coupled system whose joint solvability is the content of Assumption 3.6. The implicit function theorem applied to the coupled system gives Φ^{-1} as a smooth map; the reconstructed (p, m, ω) lies in $\tilde{E}$ by construction and has the prescribed image coordinates. The composite Φ is therefore a smooth diffeomorphism, and $\tilde{E}$ inherits from its codomain the structure of a smooth manifold of dimension IL . The standard simplex chart identifies S_N with $\mathbb{R}^{L-1}$, making the full codomain diffeomorphic to $\mathbb{R}^{IL}$, so $\tilde{E} \cong \mathbb{R}^{IL}$.

For $\tilde{\pi}$: at each $(p, m, \omega) \in \tilde{E}$, the differential $d\tilde{\pi}$ is invertible if and only if $J(p, m, \omega)$ is invertible, so under Assumption 3.6 the differential is invertible at every point and $\tilde{\pi}$ is a local diffeomorphism. Lemma 3.10 at $t = 1$ gives properness, and a proper local diffeomorphism is a covering map. $\tilde{E}$ is simply connected by the chart Φ , and Ω_0 is simply connected as a convex open subset of Euclidean space; the covering between simply-connected spaces has exactly one sheet. Hence $\tilde{\pi}$ is a smooth bijection with smooth inverse, that is, a diffeomorphism. $\square$

Corollary 3.16 (Topology of $\tilde{E}$). *Under Assumption 3.6, $\tilde{E}$ is connected, simply connected, and contractible.*

Proof. $\mathbb{R}^{IL}$ has all three properties, and a diffeomorphic image inherits them. $\square$

The diffeomorphism is the structural payoff of the chapter. Under Assumption 3.6, the joint equilibrium manifold is a single smooth piece in 1-1 correspondence with the parameter space: each endowment profile ω has exactly one wealth-and-power equilibrium, and that equilibrium varies smoothly with ω . Subsequent chapters operate on $\tilde{E}$ as a single chart $\Omega_0 \cong \mathbb{R}^{IL}$, and the contractibility recorded in Corollary 3.16 is the input the institutions chapter uses to discuss continuous deformations of equilibrium structure. The next section asks what survives when Assumption 3.6 is weakened to allow folds.

3.6 Weakening: from uniqueness to regimes

The diffeomorphism of Section 3.5 is a strong claim: under Assumption 3.6, the international system has exactly one wealth-and-power equilibrium for every parameter profile, and the equilibrium varies smoothly with parameters. Strong claims are easy to break. Assumption 3.6 is the global regularity hypothesis that buys the strength. It rules out folds in the equilibrium structure where multiple equilibria might coexist, branching points where the equilibrium might split, and singular endowment profiles where the manifold structure might break down. But IR theorists routinely describe the international system as having qualitatively distinct configurations—great-power peace, multipolar war, cold war, hegemony in decline—and discrete transitions between them. The smoothness machinery cannot describe transitions between qualitatively distinct configurations as long as the equilibrium structure is forced to be a single smooth chart.

This section weakens Assumption 3.6 to admit folds on a thin set of endowment profiles. The structure theorems do not collapse; they refactor. Existence, generic finiteness, and local smoothness survive without change. The global diffeomorphism is replaced by a componentwise statement: the regular part of $\tilde{E}$ is a smooth manifold whose connected components are the framework's *regimes*, and the diffeomorphism of Section 3.5 holds within each component when restricted to that component's parameter coverage.

The substantive content is a theory of regimes. A connected component of the regular equilibrium manifold is a region of the parameter-and-equilibrium space within which small parameter changes produce small smooth equilibrium changes—the comparative-statics machinery of the diffeomorphism applies inside the component. Discrete regime change is the crossing of the singular locus that separates components. The peaceful component, the connected component containing pure-Walrasian equilibria with no deployment, is the framework's distinguished regime; war regimes are other components, reachable from peace only through singular bifurcation, not through smooth paths in $\tilde{E}^{\text{reg}}$. The mathematical structure matches how IR scholars already talk about international order: gradual institutional evolution within a regime, punctuated by discrete regime shifts that conventional IR theory also treats as categorical ruptures [cf. Gilpin, 1981, Ikenberry, 2001].

Assumption 3.17 (Generic joint regularity). The set

$$\Sigma := \{ \omega \in \Omega_0 \mid \exists (m, p) \text{ with } e(p, m, \omega) = 0 \text{ and } J(p, m, \omega) \text{ singular} \}$$

is closed in Ω_0 and has Lebesgue measure zero.

Assumption 3.17 is strictly weaker than Assumption 3.6: it permits singularities of the joint Jacobian but requires them to sit over a thin set of endowment profiles. It is the standard regularity hypothesis of smooth equilibrium theory [Debreu, 1970, Balasko, 2009], and it follows

under the SPT assumptions of Chapter 2 from a transversality argument of the Debreu–Mas-Colell type applied to the joint map e .

The *regular part* of $\tilde{E}$ is $\tilde{E}^{\text{reg}} := \tilde{E} \cap \tilde{\pi}^{-1}(\Omega_0 \setminus \Sigma)$.

Proposition 3.18 (Regular part is a smooth IL -manifold). *Under Assumption 3.17, $\tilde{E}^{\text{reg}}$ is a smooth submanifold of $S_N \times M \times \Omega_0$ of dimension IL . The restricted projection $\tilde{\pi}|_{\tilde{E}^{\text{reg}}} : \tilde{E}^{\text{reg}} \rightarrow \Omega_0 \setminus \Sigma$ is a proper smooth covering map of mod-2 degree one. For every $\omega \in \Omega_0 \setminus \Sigma$, the preimage $\tilde{\pi}^{-1}(\omega) \cap \tilde{E}^{\text{reg}}$ is finite of odd cardinality.*

Proof. The preimage-theorem argument of Proposition 3.8 goes through at every $\omega \in \Omega_0 \setminus \Sigma$. Properness is Lemma 3.10. The covering property and degree are standard for a proper submersion of equidimensional manifolds; the degree is inherited from Theorem 3.12, where the homotopy made no use of Nash uniqueness. The cardinality count follows from the degree being odd. $\square$

Corollary 3.19 (Finite odd count of joint equilibria). *For every $\omega \in \Omega_0 \setminus \Sigma$, the number of joint dual-competitive equilibria is finite and odd.*

Proof. The equilibria at ω are the points of $\tilde{\pi}^{-1}(\omega) \cap \tilde{E}^{\text{reg}}$, which Proposition 3.18 establishes to be finite of odd cardinality. $\square$

Remark 3.20 (The geometry of the singular locus). Σ is precisely the set of critical values of the projection $\tilde{\pi}$, and Sard’s theorem applied to $\tilde{\pi}$ is what makes Assumption 3.17’s measure-zero requirement the generic case rather than a strong restriction. The internal structure of Σ —the local geometry along which regimes meet—is the territory of singularity theory and the Thom-classified elementary catastrophes (fold, cusp, swallowtail, $\dots$), generically organized as a stratified set with codimension-one fold strata and higher-codimension catastrophes along their intersections. The framework’s smooth machinery delivers Σ ’s existence as the locus along which regimes meet; the calibrated examples of the institutions chapter use catastrophe-theoretic vocabulary to classify what kind of crossing each particular regime boundary is.

The regular part can have several connected components. The peaceful equilibrium serves as a distinguished reference for the componentwise structure. At any joint equilibrium with $m = 0$, the Nash residual $r(p, 0, \omega) = -\mu(p, 0, \omega)$ vanishes if and only if $\mu_i(p, 0, \omega) = 0$ for every i ; combined with market clearing, this is the equilibrium of the classical Walrasian exchange economy at ω with no deployment.

Definition 3.21 (Peaceful component). Fix a reference endowment $\omega^\circ \in \Omega_0 \setminus \Sigma$ at which the peaceful equilibrium $(p^\circ, 0, \omega^\circ)$ is regular. The *peaceful component* $\tilde{E}_0 \subset \tilde{E}^{\text{reg}}$ is the connected component of $\tilde{E}^{\text{reg}}$ containing $(p^\circ, 0, \omega^\circ)$.

Proposition 3.22 (Structure of the peaceful component). *$\tilde{E}_0$ is a smooth IL -manifold, and the restricted projection $\tilde{\pi}|_{\tilde{E}_0} : \tilde{E}_0 \rightarrow \tilde{\pi}(\tilde{E}_0) \subset \Omega_0$ is a proper smooth covering map. If this covering is single-sheeted, then $\tilde{E}_0$ is diffeomorphic to $\tilde{\pi}(\tilde{E}_0)$, and to $\mathbb{R}^{IL}$ when $\tilde{\pi}(\tilde{E}_0) = \Omega_0$.*

Proof. Smoothness and the covering property follow from Proposition 3.18 restricted to $\tilde{E}_0$. A single-sheeted smooth covering of a simply-connected target is a diffeomorphism. $\square$

Three reachability scenarios. Whether $\tilde{\pi}|_{\tilde{E}_0}$ covers all of Ω_0 , and whether the covering is single-sheeted, decomposes the question of how the international system relates to its peaceful baseline.

- (i) *Global diffeomorphism on the peaceful component.* $\tilde{\pi}|_{\tilde{E}_0}$ covers every endowment uniquely, recovering the full statement of Proposition 3.15 on $\tilde{E}_0$. This is the strongest configuration; it obtains under Assumption 3.6.
- (ii) *Full coverage, multi-sheeted.* $\tilde{\pi}|_{\tilde{E}_0}$ covers every endowment, but some endowments carry multiple peaceful equilibria connected to the reference point by continuous paths in $\tilde{E}_0$. The international system exhibits hysteresis: continuous changes in ω can move it along any of several branches, and path-dependent history matters.
- (iii) *Partial coverage and war regimes.* $\tilde{\pi}|_{\tilde{E}_0}$ covers a strict subset of Ω_0 . Some endowments have no equilibrium reachable from peace by continuous deformation, and the equilibria they do carry live in other components of $\tilde{E}^{\text{reg}}$, the *war regimes*, separated from $\tilde{E}_0$ by the singular locus Σ .

Scenario (i) on the two-state contest. Example 3.3 sits in scenario (i). For every admissible W , the symmetric equilibrium (m^*, p^*) is unique, the map $W \mapsto (m^*, p^*)$ is smooth, and the equilibrium manifold of the example is a single smooth chart over the parameter range. The agent-level coupling is live everywhere off the symmetric ray (where $R_i \neq I$), but at the equilibrium itself $\rho_i = 0$ and the wedge factor reduces to one—the example illustrates the global-diffeomorphism scenario in its cleanest form.

Scenario (ii): asymmetric branches. Scenario (ii) requires asymmetry that the symmetric reflected-endowment construction rules out by assumption. Difference-form Tullock contests with sufficient valuation asymmetry are known to exhibit asymmetric Cournot equilibria alongside the symmetric one; Hirshleifer [1989] provides the canonical demonstration. The reflected-endowment specification collapses all such asymmetric branches to the symmetric ray, yielding a one-equilibrium-per-parameter picture rather than the multi-sheeted covering of scenario (ii). Recovering scenario (ii) within the Tullock framework requires either valuation asymmetry across goods or a non-symmetric production technology—both natural extensions deferred to the structural-empirical follow-up paper.

Scenario (iii): unreachable war regimes. Scenario (iii) requires a peaceful baseline equilibrium that is regular and distinct from the war regimes. Example 3.3 has no peaceful equilibrium at $V_0 > 0$, since the contest is always live. The natural construction treats V_0 as a parameter controlling whether peace is admissible: at $V_0 = 0$, the deployment FOC has the trivial solution $m^* = 0$ and the equilibrium reduces to a Walrasian exchange economy on the original endowments. As V_0 rises from zero, the symmetric arming equilibrium emerges; at some threshold the peaceful baseline ceases to be reachable from the arming regime by continuous deformation, and the equilibrium manifold partitions into peaceful and war components separated by Σ . The structural-empirical follow-up to Chapter 8 is the natural setting to identify which threshold values produce scenario (iii) under empirically plausible primitives.

Scenario (iii) is the substantively rich case for an applied theory of regimes. It corresponds to the claim that the international system exhibits qualitatively distinct configurations—great-power

peace, multipolar war, cold war—separated by transitions that the smooth machinery classifies as catastrophes or bifurcations. Whether the framework actually exhibits scenario (iii) is a model-checking question, depending on the specific battle functions, preferences, and costs in play; in calibrated form, it is what the institutions chapter calls *regime change*.

The framework’s institutional payoff lives within a component. Inside $\tilde{E}_0$, or any other contractible component of $\tilde{E}^{\text{reg}}$, the comparative-statics machinery of Section 3.5 applies and the componentwise structure underwrites the institutions chapter’s analysis of continuous deformations of equilibrium configuration. Cross-component movement is regime change in the substantive sense, and the smooth theory marks rather than describes it: the crossing happens at Σ , where the singular geometry that the smoothness machinery does not handle takes over. This division is, on our reading, a feature. It cleanly separates the two kinds of change that IR theory has always distinguished—incremental institutional evolution within an order, and discrete regime shifts between orders—into the framework’s smooth and singular geometry respectively.

3.7 Looking ahead

Two positions on the table. The chapter has constructed two formal positions on the structure of the joint equilibrium manifold, distinguished by which regularity hypothesis is in force.

Position	Hypothesis	Structure of $\tilde{E}$
Joint regularity	Assumption 3.6	$\tilde{E} \cong \mathbb{R}^{IL}$ globally; every endowment profile has a unique smooth equilibrium
Generic joint regularity	Assumption 3.17	$\tilde{E}^{\text{reg}} \rightarrow \Omega_0 \setminus \Sigma$ a covering map; $\tilde{E}$ partitions into regimes, each diffeomorphic to its parameter coverage

The two positions agree on existence (every endowment admits a joint equilibrium), generic finiteness (the equilibrium count is finite and odd), and local smoothness (each equilibrium varies smoothly with parameters within its regime). They differ on the global structure theorem. The first delivers $\tilde{E} \cong \mathbb{R}^{IL}$ as a universal claim and makes the institutional and welfare machinery downstream operate on a single chart. The second delivers it only on each regime separately, and accepts regime multiplicity as the geometric correlate of qualitative distinctions in international order. The choice between them is a modeling decision: subsequent chapters work within the strong position by default and return to the weaker one in specific settings where regime multiplicity is empirically forced.

What downstream chapters consume. The chapter’s machinery is the structural input for the rest of the book.

- *Positive theory* (Chapter 4). Comparative-statics arguments operate on $\tilde{E}$ as a smooth manifold, pushing the diffeomorphism $\tilde{\pi}$ along smooth paths in Ω_0 to read off equilibrium responses; the master formula $D_\theta(p^*, m^*) = -J^{-1}D_\theta e$ is the chapter’s load-bearing identity.

- *Normative theory* (Chapter 5). Welfare functionals are evaluated along smooth paths through $\tilde{E}$ via the chart Φ , and the contractibility of Corollary 3.16 supports comparison of equilibrium configurations across parameter profiles within a regime; the conditional First Welfare Theorem at r^* and the cardinal scalar Δ^* depend on this manifold structure.
- *Peacemaking* (Chapter 6). The donor's aid problem is an optimal-control problem on $\tilde{E}$: an absolutely continuous transfer schedule lifts through $\tilde{\pi}$ to a smooth path in $\tilde{E}^{\text{reg}}$, and the donor minimizes integrated welfare cost along that path. The chart Φ supplies the coordinate system the Pontryagin maximum principle consumes.
- *Institutions* (Chapter 7). The Chichilnisky–Heal aggregation theorem applies to a contractible component of $\tilde{E}^{\text{reg}}$; cross-component movement is regime change rather than institutional choice. The structure of the singular locus Σ , sketched at Remark 3.20, is the chapter's substantive object.
- *Sanctions* (Chapter 8). Sanctions are parameter shifts on $\tilde{E}$ that the smooth machinery handles within a regime via the master comparative-statics formula and that the singular machinery handles across regimes via Σ -crossings. The joint-vs-sliced response distinction is what sanctions-as-tariff-style PE analyses miss.

The two channels named in the chapter introduction map onto the apparatus as follows. The market-disruption channel—deployment shifts moving prices—is illustrated in closed form by Example 3.3, where contest-side parameter changes move equilibrium prices through the good-1 scarcity that force burning produces. The arms-race channel is the systematic comparative-statics question of how equilibrium deployment responds to parameter shifts that operate through the price system, and the next chapter takes it up using the manifold and chart machinery built here.

Exercises

1. *Comparative statics of the closed form in θ* . The endogeneity parameter θ in Example 3.3 controls how much of the contest prize tracks the size of the table: at $\theta = 0$ the prize V_0 is purely exogenous, and as $\theta \rightarrow 1$ the prize scales with aggregate good-1 consumption. Differentiate the closed-form expressions $m^*(\theta) = (V_0 + \theta W)/(2k(2 + \theta))$ and $p^*(\theta) = (2 + \theta)W/(2W - V_0)$ to confirm $dm^*/d\theta > 0$ and $dp^*/d\theta > 0$ on the admissible range $V_0 < 2W$, $\theta \in [0, 1)$. Interpret each sign substantively: a more endogenous prize raises both equilibrium arming and the equilibrium price, because the contest now feeds back into the size of what the contestants are fighting over.
2. *Price-invariance to k via the resource-burn identity*. The cost parameter k scales how resource-intensive force production is, and one might expect k to shift the equilibrium price by changing the resource cost of arming. The closed form shows otherwise. Verify that p^* in Example 3.3 does not depend on k by computing the total resource burn $2km^* = (V_0 + \theta W)/(2 + \theta)$, observing that this quantity contains no k , and noting that p^* depends only on (V_0, θ, W) . Conclude that k rescales deployment but holds total scarcity of good 1 at market fixed; the price adjusts to whatever resource burn the contest produces, not to the cost-per-unit-deployment of producing it.
3. *Quantitative price gap from the partialled benchmark*. The exchange-side partial of Example 3.3 holds deployments fixed at zero and price-clears the resulting exchange economy;

by symmetry of the reflected endowments $\omega_1 = (\omega_H, \omega_L)$, $\omega_2 = (\omega_L, \omega_H)$, this partial returns $p = 1$ regardless of V_0 or θ . The joint equilibrium disagrees. At $\theta = 0$, derive $p^* - 1 = V_0/(2W - V_0)$ from the closed form. Analyze the three limits $V_0 \rightarrow 0$, $V_0 \rightarrow 2W$, and $W \rightarrow \infty$ at fixed V_0 , and interpret each as a statement about how the price gap between the exchange-side partial and the joint equilibrium scales with the contested-prize-to-economy ratio. The exercise quantifies the substantive bias of partialling.

4. *Joint Jacobian at the symmetric equilibrium.* For Example 3.3, compute the four blocks of the joint Jacobian $J = D_{(p,m)}e$ defined in Equation (3.1) at the symmetric equilibrium $(p^*, m_1^*, m_2^*) = (p^*, m^*, m^*)$. Use the Schur factorization $\det J = \det J_{mm} \cdot \det S$ from Lemma 3.4 to split the verification: $J_{mm} = I - D_m \mu$ is invertible from the deployment-best-response contraction, and the Schur complement S is nonsingular at the symmetric point. Conclude that $\det J \neq 0$ on the admissible parameter range $V_0 < 2W$, $V_0/(4k) < \omega_L$, and that Assumption 3.6 holds at this equilibrium. The exercise instantiates joint regularity on the chapter's running example.
5. *Inverse design from primitives.* A natural inverse problem for the framework asks: given a target equilibrium (m^*, p^*) , what primitives produce it? Treating (V_0, θ, W) as design variables and the unit-cost normalization $k = 1$ as fixed, invert the closed-form equations $m^* = (V_0 + \theta W)/(2(2 + \theta))$ and $p^* = (2 + \theta)W/(2W - V_0)$ to recover (V_0, θ, W) from a target (m^*, p^*) together with one auxiliary identifying restriction (e.g., W is observable). Identify the parameter region in which the inversion is unique and admissible (i.e., yields $\theta \in [0, 1)$, $V_0 \in (0, 2W)$). Substantively: the closed form is identifying for the structural parameters, with one observable needed beyond (m^*, p^*) .
6. *Endowment derivative on Example 3.3.* A symmetric resource boom raises ω_H and ω_L by the same amount, so $W = \omega_H + \omega_L$ rises by dW while preserving the reflection symmetry. Compute $D_W(p^*, m^*)$ from the closed form for this case. The exchange-side partial predicts $dp^*/dW = 0$ (symmetric resource shifts cannot affect the symmetric Walrasian price $p = 1$). The joint formulation predicts a non-trivial sign: $dp^*/dW = -(2 + \theta)V_0/(2W - V_0)^2 < 0$. Interpret substantively: a richer economy lowers the contested-prize-to-economy ratio, dampening the joint-equilibrium price wedge that the contest produces.
7. *Three predictions, one example.* Fix the parameter point $(V_0, \theta, W, k) = (1, 1/2, 4, 1)$ in Example 3.3 and tabulate three predictions at this point: (a) the exchange-side partial that holds m exogenous at zero and price-clears; (b) the conflict-side partial that holds p exogenous at 1 (the Walrasian-baseline price) and runs the Tullock contest at exogenous prize $V_0 + \theta W$; (c) the joint equilibrium (p^*, m^*) . Compute closed-form numerical values for each, and comment on which features of (c) each partial reproduces and which it misses. The exercise concretizes the three-way comparison the chapter introduces in Section 3.1, on a numerical instance where the discrepancies are visible by inspection.
8. *The $t = 0$ endpoint as a Walrasian recovery.* Lemma 3.11 asserts that at the homotopy endpoint $t = 0$ the joint equilibrium function e^0 reduces to the excess-demand map of a smooth exchange economy on the original endowments, with $\mu_i^0 = 0$ and $r^0(p, m, \omega) = m$. Verify each clause for Example 3.3 at the corner $(V_0, \theta) = (0, 0)$, where the battle function s_i is identically zero. Confirm: (a) the SPT collapses to a Walrasian consumer problem with log preferences and reflected endowments ω_1, ω_2 ; (b) the deployment best-response is

identically zero, so $\mu_i^0 = 0$; (c) market clearing in good 1 gives $p^{\text{Walrasian}} = 1$ by symmetry of the reflected endowments; (d) the closed form $p^*(V_0, \theta) \rightarrow 1$ in the limit. The exercise instantiates the homotopy endpoint that the existence proof of Theorem 3.12 consumes.

9. *Joint regularity for a three-state extension.* Example 3.3 works in $I = L = 2$, but the framework's machinery extends to arbitrary I and L . Construct a symmetric three-state, three-good extension: take $I = L = 3$, log preferences, reflected endowments $\omega_1 = (\omega_H, \omega_L, \omega_L)$ and cyclic permutations, and a Tullock contest in good 1 satisfying Assumptions 2.5 and 2.6. Compute the symmetric equilibrium and verify Assumption 3.6 via Lemma 3.4: confirm J_{mm} is invertible at the symmetric point and that the Schur complement $S = J_{pp} - J_{pm}J_{mm}^{-1}J_{mp}$ is nonsingular. The exercise extends the chapter's two-state-two-good closed-form analysis into the higher-dimensional setting where the matrix algebra ceases to be trivial.
10. *Scenario (iii) construction with V_0 as bifurcation parameter.* Proposition 3.22 distinguishes three scenarios for the peaceful component $\tilde{E}_0$, the third of which (war regimes not reachable from peace by smooth path) is the substantively rich case for an applied theory of regimes. Construct an instance of scenario (iii) by modifying Example 3.3: treat V_0 as a parameter that controls whether peace is admissible. At $V_0 = 0$, verify that the deployment FOC has the trivial solution $m^* = 0$ and the equilibrium reduces to a Walrasian exchange economy on the original endowments. Show that as V_0 rises from zero, the symmetric arming equilibrium emerges as a second branch, and identify (numerically or analytically) the threshold value of V_0/W at which the peaceful and arming components are separated by the singular locus. The exercise is open-ended: a partial answer that locates the bifurcation regime without fully classifying the Sard-Thom stratification (Remark 3.20) is a complete solution.
11. *Schur identity in a worked instance.* Lemma 3.4 identifies the Schur complement S of the joint Jacobian with $D_p \hat{d}$, the p -derivative of the reduced excess demand $\hat{d}(p, \omega) = \bar{d}(p, m^*(p, \omega), \omega)$. Verify this identity on Example 3.3 by direct calculation. First, compute $\hat{d}(p, \omega)$ explicitly for the closed-form Nash deployment $m^*(p, \omega)$ at $\theta = 0$ (where the Tullock prize is exogenous and the deployment best-response has a clean closed form). Take the p -derivative. Then assemble the Schur complement $S = J_{pp} - J_{pm}J_{mm}^{-1}J_{mp}$ from the four blocks of Equation (3.1) computed in Exercise 4. The two expressions should coincide.
12. *Cardinality count under asymmetric perturbation.* Corollary 3.19 establishes that the number of joint equilibria over a regular value ω is finite and odd (1, 3, 5, ...). The chapter's running example sits in the single-equilibrium regime under Assumption 3.6. This exercise explores the multi-equilibrium regime via perturbation. First, perturb Example 3.3 by breaking endowment symmetry: let $\omega_1 = (\omega_H, \omega_L)$, $\omega_2 = (\omega_L + \delta, \omega_H)$ for small $\delta > 0$. For small δ , the perturbed equilibrium is locally unique by Proposition 3.15; the count is one, odd, consistent with Corollary 3.19. Then construct (or describe in qualitative terms) a perturbation regime in which the count rises to three: identify primitives that generate two stable branches and one unstable interior branch, and verify the odd-cardinality statement on this regime. Open-ended; a worked numerical instance suffices.

Chapter 4

The positive theory

How do prices and deployments respond when fundamentals shift? Two formal traditions answer the question on partialled equilibria. Trade-theory comparative statics asks how prices respond to endowment shifts, with deployments absent. Conflict-theory comparative statics asks how deployments respond to cost or contest-parameter shifts, with prices held fixed. Each tradition does its comparative statics on one half of the joint determination while holding the other half exogenous. Chapter 3 foreclosed both factorings by delivering the joint equilibrium manifold $\tilde{E} = e^{-1}(0)$ as a single object whose existence and smoothness are properties of the joint formulation rather than of either parent. This chapter takes the corresponding picture on the positive side.

The chapter’s master object is the joint comparative-statics formula

$$D_{\omega}(p^*, m^*) = -J^{-1} D_{\omega}e,$$

in which a single inversion of the joint Jacobian J from Chapter 3 delivers the response of (p^*, m^*) to any parameter perturbation the framework admits. The same matrix that played a regularity-diagnostic role upstream is here the comparative-statics workhorse; the price response, the deployment response, and the cross-channel coupling between them are read off the columns of $-J^{-1}D_{\omega}e$ together.

What the chapter delivers is hard to find elsewhere. First, the master formula itself: every comparative-statics question the framework admits—arms races, resource booms, market-price shocks, deployment-cost parameter shifts—is one matrix-vector arithmetic on $-J^{-1}D_{\omega}e$, with both the within-channel response and the cross-channel coupling delivered together. Second, generic regularity earned from primitives: a Sard–Debreu transversality argument upgrades the open set on which J is invertible to a topologically generic set in the parameter space, securing the formula’s domain of applicability without imposing it as an axiom. Third, worked comparative statics on three headline parameter shifts—arms races, resource booms, and market-price shocks—each tracked through the exchange-side partial, the conflict-side partial, and the cross-channel correction the joint formula adds; the partialled traditions recover as special cases where the cross-channel block of J is zero, and the chapter quantifies what they miss when it is not. Fourth, an index theorem on the regular set whose signed-sum identity collapses to uniqueness under Assumption 3.6 and constrains how equilibria pair across folds under Assumption 3.17; the pairing structure in turn fixes the formal correlate of regime change at the singular locus Σ and hands that reading to Chapter 7.

The chapter is organized around that one formula and what it makes possible. Section 4.1 develops the joint Jacobian J as the comparative-statics workhorse—the same matrix that played

a regularity diagnostic role in Chapter 3, lifted into its working use here. Section 4.2 names the open set on which J is invertible and records what fails on its complement. Section 4.3 establishes that this open set is generic in the parameter space, via the Sard-Debreu transversality argument that earns Assumption 3.17 from primitives. Section 4.4 runs the comparative-statics formula on three headline parameter shifts—arms races, resource booms, and market-price shocks—tracking each shift through both the exchange-side partial and the conflict-side partial and reading the cross-channel correction. Section 4.5 states the index theorem: the signed sum of equilibrium indices over any regular fiber equals one. Under Assumption 3.6 this collapses to uniqueness; under Assumption 3.17 it constrains how equilibria appear in pairs of opposite index at fold catastrophes. Section 4.6 reads the singular locus Σ as the formal correlate of regime change and the handoff to Chapter 7. Section 4.7 hands the chapter’s apparatus to the welfare and applications chapters that follow.

Sections marked with an asterisk are technical and may be skipped at first reading; each starred section opens with a summary paragraph stating the takeaway.

4.1 The joint Jacobian

The implicit function theorem is the engine of comparative statics, and the joint Jacobian J of Chapter 3 is the matrix on which it operates. At any point $(p^*, m^*, \omega^*) \in \tilde{E}$ where Assumption 3.6 holds, the IFT delivers a smooth local inverse to $\tilde{\pi}$ in a neighborhood of ω^* : equilibrium prices and deployments are smooth functions $p^*(\omega), m^*(\omega)$ of the endowment profile. Differentiating $e(p^*(\omega), m^*(\omega), \omega) \equiv 0$ in ω at ω^* and rearranging yields the master comparative-statics formula.

$$J(p^*, m^*, \omega^*) \cdot D_\omega(p^*, m^*) + D_\omega e(p^*, m^*, \omega^*) = 0, \quad (4.1)$$

or equivalently

$$D_\omega(p^*, m^*) = -J^{-1} \cdot D_\omega e. \quad (4.2)$$

Equation (4.2) is the chapter’s single load-bearing identity. Every parameter shift the framework admits—a perturbation of any state’s endowment, a change in any structural coefficient that conditions f_i or μ_i , a shift in any deployment-cost parameter—produces a response in (p^*, m^*) obtained by one application of J^{-1} to the direct-derivative vector $D_\omega e$. The matrix is computed once at the equilibrium point; the rest of the comparative statics is matrix-vector arithmetic.

The block structure of J records that arithmetic. Chapter 3 introduced

$$J = \begin{pmatrix} J_{pp} & J_{pm} \\ J_{mp} & J_{mm} \end{pmatrix}, \quad J_{pp} = D_p \bar{d}, \quad J_{pm} = D_m \bar{d}, \quad J_{mp} = -D_p \mu, \quad J_{mm} = I - D_m \mu,$$

with $J_{pp} \in \mathbb{R}^{(L-1) \times (L-1)}$, $J_{pm} \in \mathbb{R}^{(L-1) \times I}$, $J_{mp} \in \mathbb{R}^{I \times (L-1)}$, and $J_{mm} \in \mathbb{R}^{I \times I}$. The subscripts name the directions each block differentiates in. The four blocks split along the cross-channel/within-channel axis. J_{pp} records the response of aggregate excess demand to prices at fixed deployments: a within-exchange-channel quantity. J_{mm} records the response of the Nash residual to deployments at fixed prices: a within-contest-channel quantity. J_{pm} and J_{mp} are the cross-channel blocks. J_{pm} records how aggregate excess demand responds to a deployment shift at fixed prices, and J_{mp} records how the joint best-response deployment vector responds to a price shift at fixed rivals’ deployments. The cross-channel blocks are the formal locus of joint determination at the system level, mirroring the role of R_i^{-1} at the agent level (Equation (2.3)).

One inverse, not two. The methodological payoff of Equation (4.2) lies in what it does not require. A exchange-theoretic comparative statics, holding deployments fixed at their Nash values, would solve $D_p \hat{d} \cdot D_\omega p^* = -D_\omega \hat{d}$, where $\hat{d}(p, \omega) := \bar{d}(p, m^*(p, \omega), \omega)$ is the reduced excess demand of Lemma 3.4 and the derivative $D_p \hat{d} = S$ is the Schur complement. A conflict-side comparative statics, holding prices fixed at any prevailing profile, would solve $J_{mm} \cdot D_\omega m^* = -D_\omega r$ where $J_{mm} = I - D_m \mu$ is the deployment block. Each tradition owns one of these inverses and reports the corresponding partialled response as if it were the equilibrium response.

Equation (4.2) differs in two ways at once. First, the matrix being inverted is J , not its diagonal blocks J_{pp} or J_{mm} . J is built from all four blocks, two of which neither partialled tradition computes. Second, the response computed is the joint vector $(D_\omega p^*, D_\omega m^*) \in \mathbb{R}^{(L-1+I) \times IL}$, not either of its components separately. The formula does not reduce to either partial; it produces both together, and produces them from the same inverse.

The block inverse. The relationship between the joint inverse and the partialled inverses is transparent through the Schur block-inverse formula. Provided J_{mm} is invertible, the joint Jacobian factors as

$$J = \begin{pmatrix} I & J_{pm} J_{mm}^{-1} \\ 0 & I \end{pmatrix} \begin{pmatrix} S & 0 \\ 0 & J_{mm} \end{pmatrix} \begin{pmatrix} I & 0 \\ J_{mm}^{-1} J_{mp} & I \end{pmatrix},$$

with $S = J_{pp} - J_{pm} J_{mm}^{-1} J_{mp}$ the Schur complement. Inverting this factorization, and recalling Lemma 3.4's identification $S = D_p \hat{d}$,

$$J^{-1} = \begin{pmatrix} S^{-1} & -S^{-1} J_{pm} J_{mm}^{-1} \\ -J_{mm}^{-1} J_{mp} S^{-1} & J_{mm}^{-1} + J_{mm}^{-1} J_{mp} S^{-1} J_{pm} J_{mm}^{-1} \end{pmatrix}. \quad (4.3)$$

The four blocks of J^{-1} tell a clean story. The top-left block is S^{-1} , the inverse of the reduced exchange economy's price Jacobian: the exchange-theoretic partialled inverse. The bottom-right block has J_{mm}^{-1} as its leading term, recovering the conflict-side partialled inverse when the cross-channel corrections are switched off. The off-diagonal blocks of J^{-1} are pure cross-channel: they carry no within-channel content, only the propagation of a parameter shock from one channel into the other and back.

The collapse conditions are visible from Equation (4.3). When $J_{pm} = 0$, the joint best-response deployment is invisible to aggregate excess demand at fixed prices: deployments shift, but aggregate excess demand does not register the shift directly. When $J_{mp} = 0$, the deployment best-response is invisible to prices: prices move, but the Nash deployment profile does not respond. In either case the off-diagonal blocks of J^{-1} vanish and the joint inverse becomes block-diagonal, $\text{diag}(S^{-1}, J_{mm}^{-1})$. The exchange-theoretic and conflict-side comparative statics each return the correct equilibrium response in their own block, with no cross-channel correction needed. These are the configurations under which sequential comparative-statics analysis is licensed. Outside them, Equation (4.2) is the operational comparative-statics statement, and the partialled methods produce biased estimates whose magnitude is set by the cross-channel term $\|J_{pm} J_{mm}^{-1} J_{mp}\|$.

Reading the off-diagonal blocks. The two cross-channel blocks have substantive economic content that the partialled traditions cannot directly compute.

The block $J_{pm} = D_m \bar{d}$ records how aggregate excess demand responds to a shift in the deployment profile at given prices. Three channels carry the response. First, spoils accruing to

state i shift through $D_m s_i$, which alters i 's consumption-side budget through the chain rule on f_i . Second, force production absorbs a different bundle of resources through $D_m c_i$, shifting the diminished endowment that finances i 's market consumption. Third, rivals' deployments enter f_j through m_{-j} , propagating the deployment shock across states. J_{pm} is the cumulative consumption-side response. In the partialling traditions J_{pm} is unobservable because “consumption” is solved with m held fixed; in the joint formulation it is computable from the agent-level data of Chapter 2.

The block $J_{mp} = -D_p \mu$ records how the joint best-response deployment responds to a shift in prices. Chapter 2's analysis of the SPT computed this at the agent level: the FOC system Equation (2.2) delivers $D_p \mu_i$ as one component of the agent-level inverse $-J_i^{-1} \cdot D_p \text{FOC}_i$. J_{mp} is the cross-state assembly of those agent-level price responses. In the conflict-theory tradition $J_{mp} \equiv 0$ by construction: prices are held fixed at the Walrasian level and the deployment best-response is computed in their absence. In the joint formulation J_{mp} records that arming-cost shifts propagate through the price system into the optimal deployment plan, and that the propagation goes both ways through the R_i^{-1} coupling at the agent level.

The two blocks compose in J^{-1} through the operator $J_{pm} J_{mm}^{-1} J_{mp}$: the cross-channel correction to the exchange-theoretic inverse, $S^{-1} - J_{pp}^{-1}$, has $J_{pm} J_{mm}^{-1} J_{mp}$ as its kernel. Read as a sequence of arrows, the operator takes a parameter shock that hits the price system, propagates it back into deployments via J_{mp} , scales by the contest's contraction factor J_{mm}^{-1} , and outputs a market-side correction through J_{pm} . When $\|J_{pm} J_{mm}^{-1} J_{mp}\| \ll \|J_{pp}\|$, the partialled exchange-theoretic inverse J_{pp}^{-1} is a good approximation to S^{-1} and the sequential method is empirically defensible even though it is not formally licensed. When the cross-channel correction is large, the partialled estimates can disagree with the joint response by arbitrary amounts. The worked sanctions example of Chapter 8 exhibits both regimes: at the symmetric two-state Tullock equilibrium of Example 3.3, the slice approximations agree with the joint response on some parameter directions and disagree completely on others.

The cross-state assembly. Chapter 2 delivered an agent-level comparative-statics formula at the SPT optimum: the joint response of (f_i, μ_i) to any parameter θ is $-J_i^{-1} \cdot D_\theta \text{FOC}_i$, where J_i is the SPT's FOC Jacobian. That formula tells the analyst what state i does when its parameters shift, holding prices and rivals' deployments fixed.

Equation (4.2) is the cross-state assembly of those agent-level responses. Each J_i appears inside J as a building block: the columns of J_{pm} and J_{mp} that index state i are computed from J_i^{-1} applied to the corresponding parameter direction. The cross-state inverse J^{-1} then records the second-round adjustments through which the price system and rivals' deployment profile re-clear after i 's response is admitted. What an agent-level calculation answers as “what would state i do?” the joint calculation answers as “what does state i do given that everyone else is also re-optimizing and the price system is also re-clearing?”. The two calculations are computed by the same formula, an inverted Jacobian against a parameter derivative, applied at different levels of the system. The framework's claim is that the cross-state inverse is non-decomposable into agent-level inverses except in the collapse-condition limits described above.

A worked instance. Example 3.3 provided closed-form symmetric equilibrium values for the two-state Tullock economy with reflected endowments $\omega_1 = (\omega_H, \omega_L)$, $\omega_2 = (\omega_L, \omega_H)$:

$$m^* = \frac{V_0 + \theta W}{2k(2 + \theta)}, \quad p^* = \frac{(2 + \theta) W}{2W - V_0}, \quad W = \omega_H + \omega_L.$$

The equilibrium price and the symmetric deployment depend on ω only through the aggregate W , so the comparative statics in W records the entire joint response of the symmetric equilibrium to a uniform resource boom. Differentiating directly,

$$\frac{\partial m^*}{\partial W} = \frac{\theta}{2k(2 + \theta)}, \quad \frac{\partial p^*}{\partial W} = -\frac{(2 + \theta)V_0}{(2W - V_0)^2}.$$

Both signs are economically clean. A proportional increase in total wealth raises equilibrium deployment, with the size of the increase scaling in the prize-feedback parameter θ . And it lowers the equilibrium price of good 1: a larger pie dilutes the burn-share of conflict in good-1 markets.

The same two derivatives are produced by Equation (4.2): at the symmetric equilibrium J is computable in closed form from the SPT FOCs, and the master formula returns $\partial m^*/\partial W$ and $\partial p^*/\partial W$ as components of $-J^{-1} \cdot D_W e$. The interesting structure is in how the formula does so.

At $\theta = 0$, the prize $V_0 + \theta T^1$ reduces to the exogenous constant V_0 , and the deployment FOC at the symmetric point becomes $V_0 m_2 / (m_1 + m_2)^2 = k$ —an equation in deployments alone, with no p -dependence. Differentiating in p at the symmetric equilibrium therefore gives $D_p \mu_i = 0$ for each i , hence $J_{mp} = 0$ globally on the $\theta = 0$ slice. Equation (4.3) collapses to $J^{-1} = \text{diag}(S^{-1}, J_{mm}^{-1})$. The comparative statics in W separates: the price response $\partial p^*/\partial W$ is the exchange-theoretic partial computed in the reduced exchange economy where deployments are spectator, and the deployment response $\partial m^*/\partial W = 0$ is the conflict-side partial in the contest where prices are spectator. At $\theta = 0$ the exchange-side partial and the conflict-side partial each happen to be correct in their own channel, because the channel they ignore is structurally inert.

At $\theta > 0$, the prize $V_0 + \theta T^1$ depends on prices through aggregate good-1 consumption, the deployment FOC carries a p -dependent term, and $D_p \mu_i \neq 0$. J_{mp} is non-zero, and the off-diagonal blocks of J^{-1} activate. The exchange-side partialled inverse J_{pp}^{-1} now misses a θ -scaled correction through $J_{pm} J_{mm}^{-1} J_{mp}$, and the conflict-side partialled inverse J_{mm}^{-1} now misses a symmetric correction through $J_{mp} S^{-1} J_{pm}$. The directly-computed derivatives above absorb both corrections; the partialled methods absorb neither. The size of the correction scales smoothly in θ , and the $\theta = 0$ collapse is the formal articulation of the case in which the partialled methods coincide with the joint formulation. This is the structural form of the empirical observation Example 3.3 concluded with: the two partial equilibria misstate the price (and, for $\theta > 0$, the deployment) in opposite directions.

The example illustrates the chapter's general point at the level of a closed-form calculation. The joint Jacobian inverse J^{-1} is the operational object on which comparative statics live. Its block structure makes the partialled traditions visible as limiting cases (when the off-diagonal blocks of J vanish) and makes the joint formulation's corrections precise (when they do not). The same matrix machinery extends, without conceptual change, to the cross-state perturbations of arms races, asymmetric resource booms, and market shocks that Section 4.4 takes up.

4.2 Regular equilibria

Section 4.1 stated the master formula $D_\omega(p^*, m^*) = -J^{-1} D_\omega e$ as the apparatus by which every comparative-statics question gets answered. The formula has a precondition: it requires J invertible. This section names the open subset of $\tilde{E}$ on which the precondition holds and records what fails on its complement.

The dichotomy regular-vs-critical is the operational unit of positive-theory analysis. At a regular equilibrium, comparative statics is well-posed: the IFT delivers a smooth local inverse to

$\tilde{\pi}$, the master formula returns a finite response vector, and small perturbations in ω produce small smooth perturbations in (p^*, m^*) . At a critical equilibrium, none of that holds. The framework's positive predictions live on the regular open set; what happens on the singular complement is substantively informative but operates by different machinery.

Definitions. At any $(p^*, m^*, \omega^*) \in \tilde{E}$, the joint Jacobian $J = D_{(p,m)}e$ is either invertible or not. We name the two cases.

Definition 4.1 (Regular and critical equilibria). A joint equilibrium $(p^*, m^*) \in S_N \times M$ at parameter profile ω^* is *regular* if $J(p^*, m^*, \omega^*)$ is invertible, and *critical* if $J(p^*, m^*, \omega^*)$ is singular. The *regular set* is

$$\tilde{E}^{\text{reg}} := \{ (p, m, \omega) \in \tilde{E} \mid J(p, m, \omega) \text{ invertible} \}.$$

A parameter profile $\omega \in \Omega_0$ is a *regular value* of $\tilde{\pi}$ if every equilibrium at ω is regular, and a *critical value* otherwise. The *singular locus* $\Sigma \subset \Omega_0$ is the set of critical values of $\tilde{\pi}$.

This terminology specializes the standard differential-topology vocabulary to the joint equilibrium setting. Assumption 3.17 of Chapter 3 requires Σ to have measure zero in Ω_0 ; Section 4.3 below earns that requirement from the SPT primitives via Sard's theorem applied to $\tilde{\pi}$.

What fails at critical equilibria. At a regular equilibrium, the apparatus of Section 4.1 applies in full. At a critical equilibrium, five things break.

- (1) *The implicit function theorem fails locally.* At a regular point, the IFT applied to $e \equiv 0$ delivers a smooth local inverse to $\tilde{\pi}$: a neighborhood of ω^* on which the joint equilibrium varies as a smooth function of ω . At a critical point the invertibility hypothesis fails by definition, and no smooth local inverse is delivered. How (p^*, m^*) depends on ω near ω^* has no smooth answer; the dependence may be non-smooth, may be multi-valued, or may fail to exist at all.
- (2) *The master formula has no application.* Equation (4.2) writes the comparative-statics response as J^{-1} applied to a parameter-derivative vector, and with J singular, J^{-1} does not exist. The underlying linear equation $J \cdot D_\omega(p^*, m^*) = -D_\omega e$ exhibits one of two singular behaviors. If $D_\omega e$ has a component outside the range of J , no solution exists: the perturbation cannot be absorbed by any smooth response, and the equilibrium ceases to exist for ω near ω^* in that direction. If $D_\omega e$ lies in the range of J , solutions exist but are non-unique, parametrized by $\ker J$; multiple smooth branches of equilibria intersect at ω^* and the parameter shift selects between them. The two alternatives encode distinct geometric phenomena, equilibrium disappearance versus equilibrium branching.
- (3) *Local uniqueness fails.* Within $\tilde{E}^{\text{reg}}$, distinct equilibria are isolated from one another: a neighborhood of any regular point contains no other equilibria over the same ω . At a critical point this isolation breaks down. Multiple distinct regular equilibria can coalesce at ω^* as ω approaches a critical value, and re-emerge as ω moves off the singular locus in different directions. The smooth-manifold structure $\tilde{E}^{\text{reg}}$ inherits from Assumption 3.17 does not extend across the critical points; $\tilde{E}$ as a whole is a manifold-with-singularities, not a manifold simpliciter.

- (4) *The equilibrium count changes discontinuously.* Corollary 3.19 of Chapter 3 establishes that for every $\omega \in \Omega_0 \setminus \Sigma$ the number of joint equilibria is finite and odd, with odd cardinality the mod-2 degree of Theorem 3.12 read off the regular fiber. As ω crosses Σ , that count can change: equilibria appear, disappear, or merge in pairs as fold catastrophes resolve. The mod-2 degree is invariant under proper homotopy, so the count stays odd at every regular value on either side of the crossing, but the specific cardinality can jump from one odd number to another. These jumps are the formal correlate of regime-change moments at which the strategic landscape reorganizes.
- (5) *Singularity theory takes over.* The local geometry that controls these changes has been classified, but not by smooth-equilibrium theory. The elementary catastrophes—fold, cusp, swallowtail, butterfly, and the umbilics—describe the codimension-stratified normal forms a generic critical point exhibits, and Thom’s classification fixes their structure up to smooth change of coordinates. The smooth machinery of this chapter does not extend to the catastrophic geometry; the apparatus stops at the boundary of $\tilde{E}^{\text{reg}}$. Chapter 7 takes up the singular case separately, using catastrophe-theoretic vocabulary to classify what kind of crossing each particular regime boundary is.

The five failures share a single source—the singularity of J —and a single consequence: the regular-equilibrium apparatus of Section 4.1 stops at the boundary of $\tilde{E}^{\text{reg}}$, and what happens beyond is read by different machinery.

Reading the dichotomy. The substantive content of the regular-critical distinction is how the international system responds to perturbations of ω .

At a regular equilibrium, the response is smooth and proportionate. A small change in any state’s endowment produces a small smooth change in the joint (p^*, m^*) , computable from J^{-1} . “Small” here is the IR-substantive sense of incrementalism: the international order at $\omega + d\omega$ is recognizably the same as the order at ω , just with marginally different prices and arming.

At a critical equilibrium, the response is not proportionate. A small change in ω can produce no equilibrium change at all (if the shift is along a kernel direction the critical-point equilibrium does not feel), or a discontinuous jump (if the shift selects between branching equilibria), or a catastrophic re-organization (at higher-codimension catastrophes where multiple branches merge or split). The international order at $\omega + d\omega$ may be qualitatively distinct from the order at ω .

The framework’s positive theory is therefore graduated by regularity. Where J is invertible, the incremental-response story holds and Section 4.4 computes it. Where J is singular, the regime story holds and Chapter 7 takes it up. The boundary between the two—the singular locus—is the formal locus where the framework’s two readings of international change meet.

The open-set property. The set $\tilde{E}^{\text{reg}}$ is open in $\tilde{E}$. The determinant $\det J(p, m, \omega)$ is a continuous function of $(p, m, \omega) \in \tilde{E}$, and the regular set is exactly the preimage of $\mathbb{R} \setminus \{0\}$ under this continuous map, which is open. The singular set is correspondingly closed.

Openness has methodological consequences. A comparative-statics calculation performed at a regular equilibrium remains valid in a neighborhood; the analyst does not need to recompute the regular/critical status at every point along a smooth path through $\tilde{E}^{\text{reg}}$. And under Assumption 3.17 the regular set is dense in $\tilde{E}$, so a generic equilibrium is regular and the apparatus applies.

The two-state contest revisited. The two-state Tullock economy of Example 3.3 sits entirely in $\tilde{E}^{\text{reg}}$ on its admissible parameter range. At $\theta = 0$, the analysis of Section 4.1 showed that $J_{mp} = 0$, making J block-diagonal with J_{mm} invertible by the contest's contraction property and J_{pp} invertible for any $W > V_0/2$. At $\theta > 0$, the off-diagonal blocks activate but the determinant identity $\det J = \det J_{mm} \cdot \det S$ remains nonzero throughout the symmetric ray on which the closed-form equilibrium lives. The example illustrates the regular-equilibrium case in its cleanest form.

For an example exhibiting critical equilibria, the framework turns to richer settings. The worked sanctions example of Chapter 8 and the asymmetric-branching constructions of Chapter 7 both exhibit critical points where the apparatus of this chapter hands off to the regime-change machinery downstream.

4.3 Generic regularity*

Section summary. The result is that the singular locus Σ has Lebesgue measure zero in Ω_0 : critical equilibria are non-generic, and the apparatus of Section 4.1 applies for almost every parameter profile. The proof is a transversality argument of the Debreu–Mas-Colell type, applied to the joint equilibrium manifold via Sard's theorem. Substantive readers can take generic regularity as established and skip to Section 4.4.

Section 4.2 named the regular set $\tilde{E}^{\text{reg}}$ as the open subset of $\tilde{E}$ on which the master formula applies, and the singular locus $\Sigma \subset \Omega_0$ as the set of parameter profiles where it does not. The substantive question this section answers is how big the singular locus is. The answer, under the SPT assumptions of Chapter 2, is that Σ has Lebesgue measure zero in Ω_0 . The regular case is generic; the singular case is exceptional.

The result is the joint-equilibrium analogue of the *Sard-Debreu* transversality argument that founds the regular-economies tradition [Debreu, 1970, Mas-Colell, 1985, Balasko, 2009]. The classical argument applies Sard's theorem to the projection of the reduced equilibrium manifold onto the parameter space; the joint version applies Sard's theorem to the projection of the joint manifold $\tilde{E}$. The mathematical content is the same, but the geometric object is wider.

Sard's theorem. The technical input is a standard result from differential topology: for any smooth map $f : M \rightarrow N$ between smooth manifolds, the set of critical values has Lebesgue measure zero in N . A *critical value* of f is the image of a critical point, where the Jacobian fails to have maximal rank. Sard's theorem says critical values are rare even when critical points are not. Applied to the projection $\tilde{\pi} : \tilde{E} \rightarrow \Omega_0$, the theorem yields the genericity statement we want. Σ , the set of critical values of $\tilde{\pi}$, has measure zero.

The work that remains is showing that the application is warranted from primitives. Sard requires $\tilde{\pi}$ smooth, which Corollary 3.9 of Chapter 3 delivered; Σ closed, which proper-map theory delivers; and the joint equilibrium manifold itself well-defined as a smooth object, which the SPT primitives deliver via the regular-value criterion below. The regular-value criterion in turn requires that the joint best-response deployment respond nontrivially to endowment perturbations, which the next lemma extracts from the SPT primitives.

Lemma 4.2 (Generic non-vanishing of $D_\omega \mu$). *Under the SPT assumptions of Chapter 2 and the additional non-triviality condition $D_\omega s_i \neq 0$ on $M \times X \times \Omega$ for each i , the set*

$$\{(p, m_{-i}, \omega) \in S_N \times M_{-i} \times \Omega_0 \mid D_{\omega_i} \mu_i(p, m_{-i}, \omega) = 0\}$$

has Lebesgue measure zero in $S_N \times M_{-i} \times \Omega_0$ for each i .¹

Proof sketch. Differentiating the SPT first-order condition for state i at the optimum and applying the envelope theorem to the consumption response, the response $D_{\omega_i} \mu_i$ inherits direct ω -dependence from s_i through the term $D_{\omega_i} s_i$ entering the deployment FOC. Under the non-triviality condition this term is nonzero on a generic subset of the domain. The zero set of $D_{\omega_i} \mu_i$ is then a closed subvariety of strictly lower dimension and hence has Lebesgue measure zero. $\square$

Theorem 4.3 (Generic regularity from primitives). *Under the SPT assumptions of Chapter 2, the non-triviality condition of Lemma 4.2, and the properness hypothesis of Theorem 3.12, the singular locus*

$$\Sigma = \{ \omega \in \Omega_0 \mid \exists (m, p) \text{ with } e(p, m, \omega) = 0 \text{ and } J \text{ singular} \}$$

has Lebesgue measure zero in Ω_0 and is closed. Equivalently, Assumption 3.17 of Chapter 3 follows from the SPT primitives.

Proof sketch. Three steps.

Step 1: 0 is a regular value of e generically. The full Jacobian De has block structure $(D_p e \mid D_m e \mid D_\omega e)$ of size $(L - 1 + I) \times (L - 1 + I + IL)$. At any zero $(p, m, \omega) \in e^{-1}(0)$, the endowment block $D_\omega e$ contributes the rank we need: market-clearing rows of $\bar{d}$ respond to ω_j^ℓ perturbations through the direct $-\omega$ term in excess demand, and the Nash rows of r respond through $D_\omega \mu$, which is generically nonzero by Lemma 4.2. Outside a measure-zero set of $\omega \in \Omega_0$, the full Jacobian De therefore has rank $L - 1 + I$ at every zero of e , so 0 is a regular value of e on the corresponding regular part of the domain.

Step 2: Parameterized Sard. The parameterized Sard theorem says that if 0 is a regular value of a smooth family $e : S_N \times M \times \Omega_0 \rightarrow \mathbb{R}^{L-1+I}$, then for almost every $\omega \in \Omega_0$, 0 is a regular value of the restriction $e_\omega : S_N \times M \rightarrow \mathbb{R}^{L-1+I}$. Regularity of e_ω at a zero (p, m) is exactly the statement that $J = D_{(p,m)} e$ has maximal rank, that is, JR at ω . The parameter set on which JR fails is therefore measure zero in Ω_0 .

Step 3: Σ is closed. The set of critical points of $\tilde{\pi}$ in $\tilde{E}$ is the preimage of $\{0\}$ under the continuous map $\det J$, hence closed. Properness of $\tilde{\pi}$ (Lemma 3.10 of Chapter 3) ensures that the image of this closed set under $\tilde{\pi}$ is closed; that image is Σ . $\square$

The closedness step uses properness in a way the classical transversality argument also uses. Without properness, Σ could fail to be closed and the regular complement could fail to be open, breaking the smooth-covering structure of Proposition 3.18. Properness is delivered by the desirability and cost-coercivity conditions on the SPT primitives, transported through the homotopy of Theorem 3.12.

¹The non-triviality condition $D_\omega s_i \neq 0$ is the natural reading of Chapter 2's "net flows" formulation, in which spoils target the disputed pool $\omega_j^\ell - x_j^\ell$ remaining after consumption depletes endowment (Assumption 2.5 permits but does not require this dependence). If $D_\omega s_i \equiv 0$ and the extraction cost c_i likewise has no ω -dependence, then $D_\omega \mu_i = 0$ identically and Theorem 4.3 below requires $D_\omega \mu \neq 0$ generically as an explicit primitive condition rather than as a derived consequence.

What “generic” means. The theorem says that for almost every $\omega \in \Omega_0$, every joint equilibrium at ω is regular. “Almost every” here is the measure-zero sense: the parameter profiles for which the apparatus of Section 4.1 does not apply form a negligible set in the standard Lebesgue sense.

This is a statement about how easy it is to find a critical configuration by accident. If one perturbs ω randomly, one almost never lands at a critical value. The transversality argument is what licenses the modeling practice of working at a generic equilibrium and trusting that conclusions extend to nearby parameter profiles.

What the theorem does *not* say is that critical configurations are rare in observed international history. Random perturbation is one selection mechanism on the parameter space; strategic and historical selection mechanisms are not. Configurations that the smooth machinery diagnoses as critical may nevertheless be empirically common if the underlying dynamics push international systems toward regime boundaries rather than drawing ω from Lebesgue measure. The framework leaves room for both readings: “rare in the parameter space” and “empirically common” are compatible if the empirical sampling distribution is concentrated near Σ .

The Debreu-Mas-Colell adaptation. Theorem 4.3 adapts an argument in continuous use since Debreu [1970]. The classical version proves that the singular set of the projection from the reduced equilibrium manifold onto endowments has measure zero in Ω_0 . Our version proves the same for the projection from the joint equilibrium manifold $\tilde{E}$. The Sard step is unchanged; what differs is the manifold to which Sard is applied.

The adaptation has methodological content. The classical argument works on a manifold of dimension IL parameterized by endowments alone. The joint manifold is also IL -dimensional but lives in a higher-dimensional ambient space, and the codimension of the equilibrium condition counts differently— $L - 1 + I$ on the joint side, $L - 1$ on the reduced side—so the regularity condition that delivers a smooth manifold is correspondingly stronger. Assumption 3.17 is what the joint formulation requires; the classical condition is what the reduced formulation requires.

The chapter from here. The upshot of this section is that the apparatus of Section 4.1 applies generically. Section 4.4 runs the master formula on three headline parameter shifts—arms races, resource booms, market shocks—at the cross-state level, lifting the agent-level treatment of Section 2.4 to the joint formulation. Section 4.6 returns to the non-generic case and reads the singular locus as the formal handoff to Chapter 7.

4.4 Cross-state comparative statics

Section 2.4 ran the agent-level master formula

$$D_\theta(f_i, \mu_i) = -J_i^{-1} \cdot D_\theta \text{FOC}_i$$

on three parameter shocks—arms races, resource booms, and market-price shocks—holding prices and rivals’ deployments fixed in each case. The cross-state lifting of those agent-level calculations is the work of this section. Where Chapter 2 told a state how it would respond, Chapter 4 tells the international system what happens once the state’s response triggers second-round adjustments through the price system and rivals’ deployment choices.

The lifting is mechanical. The agent-level Jacobians J_i^{-1} appear inside the cross-state Jacobian J as building blocks of J_{pm} and J_{mp} , and J^{-1} records the second-round adjustments

through which the system re-clears. The substantive content is in the cross-channel correction, the difference between what the agent-level J_i^{-1} predicts in isolation and what the system-level J^{-1} delivers in equilibrium. That correction is what sequential analysis cannot recover.

Three shocks illustrate the apparatus. Each is a parameter direction θ , each produces a joint response $D_\theta(p^*, m^*) = -J^{-1} \cdot D_\theta e$, and each exhibits a different pattern of cross-channel propagation between the exchange and contest channels.

Arms races (θ a contest parameter). A shift in any contest parameter—the prize size V_0 , the lootability vector λ , the alliance vector a_{ij} , the arming-cost coefficient—can trigger a system-level arms race in which equilibrium deployments rise across multiple states. The joint response $D_\theta(p^*, m^*)$ tracks both the arming change and the price-level change the burn produces.

The conflict-side partial is the natural computation here and gets close to right. With prices fixed at the prevailing profile, the contest parameter's effect on deployments is $D_\theta m^*|_{\text{conf}} = -J_{mm}^{-1} D_\theta r$. What the partial misses is that the equilibrium prices are not the prevailing profile: as deployments rise, more resources are burned, and the equilibrium price of those resources rises. The cross-channel correction in J^{-1} adjusts the deployment response for this equilibrium price shift.

The exchange-side partial does not see an arms race at all. The contest parameter does not enter the consumption FOC directly, so the exchange-side prediction $D_\theta p^*$ is the empty calculation. The framework's joint formulation reveals what exchange-side analysis misses: an arms-race parameter shift produces a price-level signature visible at market, and the magnitude is determined by how much resource the equilibrium arming change consumes.

Example 3.3 gives closed forms for the arms-race comparative statics in the prize parameter V_0 . At $\theta = 0$,

$$\frac{\partial m^*}{\partial V_0} = \frac{1}{4k}, \quad \frac{\partial p^*}{\partial V_0} = \frac{2W}{(2W - V_0)^2}.$$

Both responses are positive: a larger contest prize draws more arming, which burns more good 1, which raises good 1's equilibrium price. The exchange-side partial does not see V_0 in the exchange economy's primitives and returns $\partial(p^*, m^*)/\partial V_0 = (0, 0)$. The conflict-side partial holds prices at the no-conflict baseline $p = 1$ and computes the contest's deployment response alone, giving $\partial m^*/\partial V_0 = 1/(4k)$ at $\theta = 0$, matching the joint value exactly while returning zero on $\partial p^*/\partial V_0$. The conflict-side partial's accidental correctness on $\partial m^*/\partial V_0$ at $\theta = 0$ is the same $J_{mp} = 0$ collapse Section 4.1 exhibited at the symmetric equilibrium; at $\theta > 0$ the cross-channel block reactivates and the partial diverges.

Resource booms ($\theta = \omega_i$). A unilateral increase in state i 's endowment relaxes its budget. The direct effect of the shock decomposes into a consumption-side piece (state i 's excess demand falls) and an SPT piece (state i 's arming may shift through budget relaxation and through any ω -dependence in the force-production technology). The joint response propagates each piece through J^{-1} .

The exchange-side partial computes only the consumption-side response, treating equilibrium arming as exogenous. Its prediction for $D_{\omega_i} p^*$ is the standard income-effect channel through demand at the no-conflict baseline.

The conflict-side partial computes only the arming response, treating equilibrium prices as exogenous. Its prediction for $D_{\omega_i} m^*$ is computed at fixed prevailing prices, missing the equilibrium price shift the boom triggers.

The joint formulation captures both at once, with the cross-channel correction running in both directions: equilibrium arming shifts feed back into prices through J_{pm} , and equilibrium price shifts feed back into arming through J_{mp} . The resource-curse debate in formal IR concerns the sign of $D_{\omega_i} m^*$ at the joint equilibrium. The exchange-side partial returns zero by construction. The conflict-side partial returns the unconditional contest response. The joint formulation returns the equilibrium response, which depends on whether the boom raises or lowers the price of the goods i extracts for force production.

A worked instance on Example 3.3 is available once the example is read at general endowments ω^1, ω^2 (totals across states) rather than only at the symmetric reflected baseline. At $\theta = 0$, the deployment FOC gives $m_1^* = m_2^* = V_0/(4k)$ regardless of endowment asymmetry, and market clearing in good 1 yields

$$p^* = \frac{2\omega^2}{2\omega^1 - V_0}.$$

The original symmetric formula is the case $\omega^1 = \omega^2 = W$. A unilateral resource boom in state 1's good-1 endowment then gives

$$\frac{\partial m^*}{\partial \omega_1^1} = 0, \quad \frac{\partial p^*}{\partial \omega_1^1} = -\frac{4\omega^2}{(2\omega^1 - V_0)^2} = -\frac{4W}{(2W - V_0)^2}$$

at the reflected baseline. The exchange-side partial, computing on the no-conflict economy where $p_{\text{exch}}^* = \omega^2/\omega^1$, returns $\partial p_{\text{exch}}^*/\partial \omega_1^1 = -1/W$ at the same baseline. The joint formulation differs from the exchange-side partial by a multiplicative factor:

$$\frac{\partial p^*/\partial \omega_1^1|_{\text{joint}}}{\partial p_{\text{exch}}^*/\partial \omega_1^1} = \frac{4W^2}{(2W - V_0)^2}.$$

At $V_0 = 0$ the ratio is one and the partial recovers the joint response. As V_0 rises, the ratio rises above one and conflict *amplifies* the price response: a resource boom under conflict moves prices more than the same boom in the no-conflict economy would, because the burn channel tightens supply and the response to a boom scales accordingly.

Market-price shocks (θ a exchange-side parameter). A shift in a exchange-side primitive—a productivity parameter in c_i , a demand-side preference, a market-access friction—changes equilibrium prices, which propagate into equilibrium arming through the cost-of-extraction channel.

The exchange-side partial computes the price response correctly on the equilibrium manifold (the Schur identification of Lemma 3.4), but it does not produce a deployment response. What it does not see is the price-induced arming response that propagates through the cost-of-extraction channel: a productivity change in c_i shifts equilibrium prices, which shifts the cost of force production, which shifts equilibrium arming.

The conflict-side partial does not see a market-price shock at all, since exchange-side primitives do not enter the contest. The framework's joint formulation reveals what conflict-side analysis misses: a productivity shock or a market-access change produces a contest-level signature visible in equilibrium deployments, mediated by the cost-of-extraction channel through which c_i absorbs the price change.

Example 3.3 as written has thin exchange-side primitives. A clean market-price-shock instance requires a small generalization: replace log preferences with the parameterized form

$u_i(y) = \alpha \log y^1 + (1 - \alpha) \log y^2$, $\alpha \in (0, 1)$, recovering the original Example at $\alpha = 1/2$. On this generalization at $\theta = 0$, the deployment FOC is unchanged ($m^* = V_0/(4k)$ symmetrically), and market clearing in good 1 yields

$$p^* = \frac{2\alpha \omega^2}{(1 - \alpha)(2\omega^1 - V_0)}.$$

A demand-side shift toward good 1 (an increase in α) then gives

$$\frac{\partial m^*}{\partial \alpha} = 0, \quad \frac{\partial p^*}{\partial \alpha} = \frac{2W}{(2W - V_0)(1 - \alpha)^2}$$

at the symmetric baseline. The exchange-side partial at the no-conflict baseline gives $p_{\text{exch}}^* = \alpha/(1 - \alpha)$ and $\partial p_{\text{exch}}^*/\partial \alpha = 1/(1 - \alpha)^2$. The joint-versus-partial ratio is

$$\frac{\partial p^*/\partial \alpha|_{\text{joint}}}{\partial p_{\text{exch}}^*/\partial \alpha} = \frac{2W}{2W - V_0},$$

the same conflict-amplification structure that appeared in the resource-boom worked instance, modulated by the same V_0 -driven distance from the existence boundary $V_0 \rightarrow 2W$. The joint formulation's $\partial m^*/\partial \alpha = 0$ at $\theta = 0$ recovers the same $J_{mp} = 0$ collapse the other two shocks exhibited; at $\theta > 0$, all three worked instances develop a non-zero deployment response and a correspondingly modified price response.

The pattern. The three worked instances exhibit the same structural pattern on Example 3.3 at $\theta = 0$. The deployment response is zero or partial-coincident (the collapse of Section 4.1), and the price response differs from the exchange-side partial by a multiplicative factor controlled by the system's distance from the existence boundary. The conflict-amplification factor $2W/(2W - V_0)$ recurs across the resource-boom and market-price worked instances and equals unity at $V_0 = 0$, recovering the no-conflict exchange-side prediction in the limit. At $\theta > 0$, all three develop richer cross-channel content and the partials diverge from joint in directions Section 4.1 characterized. Whatever the comparative-statics question—a resource discovery, a treaty change in the alliance vector, an oil-price shock, a productivity shift—the answer at the joint equilibrium is a single application of J^{-1} to the direct-derivative vector, with the partialled methods recovering fragments under collapse conditions and missing structural pieces in general. The chapters below consume this formula: Section 4.5 reads it through the index theorem, and Chapter 8 applies it to sanctions perturbations, recovering the bargaining-camp and GE-camp slice analyses as restricted special cases of the joint comparative-static.

4.5 The index theorem*

Section summary. The index of a regular equilibrium is a ± 1 sign attached to its contribution to the global count. The signed sum of indices over the equilibrium fiber at any regular value of $\tilde{\pi}$ equals one. Under Assumption 3.6 this delivers uniqueness as a corollary; under Assumption 3.17 it sharpens Corollary 3.19's mod-2 odd count into a signed-sum identity, which constrains how equilibria appear and disappear at fold catastrophes. Substantive readers can take this as established and skip to Section 4.6.

Corollary 3.19 of Chapter 3 established that, under generic regularity, every regular value of $\tilde{\pi}$ has finitely many preimages and that count is odd. The index theorem refines this. Each regular equilibrium carries a ± 1 index encoding the orientation $\tilde{\pi}$ has at the equilibrium, and the signed sum across the fiber equals one. Where the mod-2 result records “odd,” the index theorem records “+1 once orientation is accounted for.”

The refinement matters for two reasons. First, it sharpens uniqueness arguments. Under Assumption 3.6 the manifold has a single equilibrium per ω (Proposition 3.15), and the index theorem confirms its index is +1 globally. Second, under Assumption 3.17 the signed sum constrains how branches fit together: at a fold catastrophe across Σ , equilibria appear or disappear in pairs of opposite index, preserving the signed total.

Index of a regular equilibrium. At a regular equilibrium $(p^*, m^*, \omega^*) \in \tilde{E}^{\text{reg}}$, the joint Jacobian $J = D_{(p,m)} e(p^*, m^*, \omega^*)$ is invertible. The orientation $\tilde{\pi}$ inherits from the ambient manifold $S_N \times M \times \Omega_0$ assigns a sign to $\det(d\tilde{\pi})$ at the equilibrium, and that sign is the equilibrium’s index.

Definition 4.4 (Index of an equilibrium). The *index* of a regular equilibrium $(p^*, m^*, \omega^*) \in \tilde{E}^{\text{reg}}$ is

$$\text{ind}(p^*, m^*, \omega^*) := (-1)^{L-1+I} \cdot \text{sign}(\det J(p^*, m^*, \omega^*)) \in \{-1, +1\}.$$

The factor $(-1)^{L-1+I}$ aligns the sign convention with the orientation inherited from the ambient space and the standard smooth-GE convention [Mas-Colell, 1985, Balasko, 2009]. The substantive content is in the $\text{sign}(\det J)$ factor, which records whether the joint Jacobian preserves or reverses orientation at the equilibrium.

The theorem. With the index defined, the structural claim is that the signed sum is constant across regular fibers and equals one, regardless of how many regular equilibria the fiber contains.

Theorem 4.5 (Index theorem for joint dual-competitive equilibrium). *Under the SPT assumptions of Chapter 2 and Assumption 3.17, for every $\omega \in \Omega_0 \setminus \Sigma$,*

$$\sum_{(p^*, m^*) \in \tilde{\pi}^{-1}(\omega) \cap \tilde{E}^{\text{reg}}} \text{ind}(p^*, m^*, \omega) = +1.$$

Proof sketch. The signed degree of a smooth proper map between equal-dimension oriented manifolds is invariant under proper homotopy [Milnor, 1965]. The homotopy $\{e^t\}_{t \in [0,1]}$ of Theorem 3.12 is proper uniformly and connects $e^1 = e$ to e^0 , the conflict-off endpoint. At $t = 0$, $\tilde{\pi}^0$ is canonically identified with the natural projection of the classical Walrasian equilibrium manifold, and the standard signed-degree result for regular exchange economies [Balasko, 2009, Proposition 4.6.2] delivers signed degree +1. Properness of the homotopy transports this to $t = 1$.

The signed degree of $\tilde{\pi}$ at any regular value ω equals the sum of indices over $\tilde{\pi}^{-1}(\omega)$ [Milnor, 1965], which gives the conclusion. $\square$

Corollary: uniqueness under JR. The strongest case is Assumption 3.6, where the signed sum has a single term and uniqueness follows.

Corollary 4.6 (Uniqueness under JR). *Under Assumption 3.6 and the SPT assumptions of Chapter 2, for every $\omega \in \Omega_0$ there is exactly one joint dual-competitive equilibrium, and its index is +1.*

Proof. Assumption 3.6 implies $\tilde{E}^{\text{reg}} = \tilde{E}$ and $\Sigma = \emptyset$. Proposition 3.15 delivers $\tilde{\pi}$ as a global smooth diffeomorphism, so each fiber has cardinality one. Theorem 4.5 forces the unique equilibrium's index to be +1. $\square$

Reading the signed sum. The index-theorem identity carries substantive content under Assumption 3.17, where multiple equilibria coexist over generic ω . The signed sum constrains how those equilibria appear: in any odd-cardinality fiber, the count of +1-index equilibria exceeds the count of −1-index equilibria by exactly one, and a fold catastrophe at Σ creates or destroys equilibria in pairs of opposite index. The pairs arise because, at a fold, two regular equilibria of opposite index merge at a critical point and re-emerge (or annihilate) on the other side. The framework's regime-multiplicity story, which Chapter 7 takes up, has this signed-pairing structure as one of its formal features. The alternation of equilibrium indices across Σ is what makes regime crossings discrete: a system at a +1-index equilibrium and a system at a −1-index equilibrium cannot be connected by a smooth path in $\tilde{E}^{\text{reg}}$ without crossing the singular locus, and the signed-sum identity controls how the count shifts across that crossing.

4.6 The bifurcation locus and regime change

Section 4.2 named $\Sigma \subset \Omega_0$ as the singular locus where the master formula breaks. Section 4.3 established that Σ has measure zero. Section 4.4 ran the master formula on its complement. The remaining question is what happens at Σ itself, and what the geometry of crossings tells us.

The substantive payoff is that Σ is the formal correlate of regime change. Where the smooth machinery handles the regular set, the singular geometry of Σ carries the framework's account of what happens when international order tips. The mathematics is catastrophe-theoretic; the substance is regime change in the IR sense.

The locus. Σ is the set of ω at which J is singular for at least one equilibrium (p, m) . By Theorem 4.3, Σ has Lebesgue measure zero in Ω_0 . By the Sard-Thom analysis sketched in Remark 3.20 of Chapter 3, Σ is generically a stratified set: a codimension-one fold-stratum makes up most of Σ , with higher-codimension catastrophes (cusps, swallowtails, and the umbilics) along the intersections.

What the apparatus does and does not handle. The smooth machinery of this chapter applies on $\Omega_0 \setminus \Sigma$. On Σ itself, the master formula has no application: J^{-1} does not exist and the IFT does not deliver a smooth local inverse to $\tilde{\pi}$. Comparative-statics calculations on Σ are catastrophe-theoretic, classifying what kind of crossing each particular point on Σ is rather than producing smooth response vectors.

The catastrophe-theoretic classification provides normal forms. At a generic fold point of Σ , two regular equilibria of opposite index merge and re-emerge (or annihilate); at a cusp, three

equilibria meet; and so on, with Thom's classification controlling the local geometry at each codimension level. The framework's apparatus inherits this stratified structure on Σ but does not extend the smooth response calculations across it.

Regime change. The substantive content of Σ is regime change. A point on Σ is a parameter configuration at which the international order qualitatively reorganizes: equilibria appear or disappear in pairs, the system's component-of-residence shifts, and the comparative-statics intuitions developed inside a regime cease to apply. Where smooth motion in $\Omega_0 \setminus \Sigma$ corresponds to incremental change within a regime, motion across Σ corresponds to discrete regime transitions.

The two kinds of change line up with how IR theory has always talked about international order. Gradual institutional evolution within a regime is the smooth side; discrete regime shifts between orders are the singular side. The framework's smooth-versus-singular geometry gives the distinction a precise mathematical correlate.

The handoff to Chapter 7. Chapter 7 takes up the regime story in full. The institutions chapter is organized around the geometry of Σ as a substantive object: which kinds of catastrophes are realized in calibrated economies, what they correspond to historically (great-power transitions, hegemonic shifts, financial-revolution moments), and how institutional design either stabilizes a regime by keeping the system away from Σ or initiates regime change by pushing it across.

The positive theory developed in this chapter is what Chapter 7 consumes for its smooth-side analysis. Comparative statics on $\Omega_0 \setminus \Sigma$ tells the institutional designer how the system responds to a local intervention, and the index-theoretic count tells her how many equilibria she is choosing among. The singular-side analysis, where local geometry takes over from smooth response, is what Chapter 7 develops in full.

4.7 Looking ahead

The positive theory delivered by this chapter is, in summary form, a single matrix calculation. Given any parameter shift, $-J^{-1}D_{\theta}e$ produces the joint response of prices and deployments, valid on the open set $\tilde{E}^{\text{reg}}$ where J is invertible. Generic regularity (Theorem 4.3) ensures the apparatus applies almost everywhere. The signed-sum identity (Theorem 4.5) refines the equilibrium count. And the singular locus Σ marks the geometric transition to the regime story Chapter 7 takes up.

Downstream chapters consume this machinery in graduated ways.

Chapter 5 integrates welfare functionals along smooth paths in $\tilde{E}^{\text{reg}}$, using the chart machinery of Chapter 3 and the comparative-statics formula of Section 4.4 to track how equilibrium welfare responds to parameter shifts. Chapter 6 lifts the formula to an optimal-control problem on $\tilde{E}^{\text{reg}}$, where a donor's absolutely continuous transfer schedule lifts to a smooth path on the manifold and the donor minimizes integrated welfare cost along admissible paths. Chapter 7 consumes the index theorem and the singular-locus geometry to read regime change as a formal phenomenon. Chapter 8 applies the formula to sanctions perturbations, recovering the bargaining-camp and GE-camp slice analyses as restricted special cases of the joint comparative-static and identifying the cross-channel structure neither slice can deliver.

The chapter has done one thing in many directions. The joint Jacobian is the apparatus, the master formula is the operational tool, and the regular-set / singular-locus dichotomy is the geometric frame. Whatever question the next chapters take up—a welfare comparison, a

substantive application, a numerical calibration, an institutional design problem—the answer at the joint equilibrium is one application of J^{-1} , with the kind of question determining only what direction $D_{\theta}e$ points and what side of Σ the equilibrium sits on.

Exercises

1. *Schur block-inverse by direct multiplication.* The block-inverse formula Equation (4.3) expresses J^{-1} as a four-block expression in J_{pp} , J_{pm} , J_{mp} , J_{mm} and the Schur complement $S = J_{pp} - J_{pm}J_{mm}^{-1}J_{mp}$, and is the comparative-statics workhorse of the chapter. Verify the formula by direct multiplication: form the product $J \cdot J^{-1}$ from the four-block expressions and check that each block of the resulting product equals the corresponding block of the identity matrix. The verification is mechanical, but each cancellation has substantive content: the within-channel blocks (I, I) emerge from cancellations involving S , and the cross-channel blocks $(0, 0)$ emerge from cancellations between the within-channel and cross-channel terms.
2. *$J_{mp} = 0$ at the symmetric equilibrium.* At the symmetric equilibrium of Example 3.3 with $\theta = 0$, the cross-channel block $J_{mp} = -D_p\mu$ vanishes because the deployment best-response μ_i^* does not depend on prices when the Tullock prize V_0 is exogenous. Compute the joint Jacobian J at this equilibrium, verify $J_{mp} = 0$, and conclude that J^{-1} is block-diagonal there. Substantively, this is the structural reason the conflict-side partial returns the correct deployment response to a V_0 shock at $\theta = 0$: the cross-channel block vanishes, so the joint inverse and the partialled inverse coincide on the deployment direction. The exercise instantiates the block-diagonal collapse condition the chapter introduces in Section 4.1.
3. *Leading-order θ -correction at $\theta > 0$.* At $\theta > 0$ the Tullock prize $V_0 + \theta T^1$ becomes endogenous in equilibrium consumption, the cross-channel block $J_{mp} = -D_p\mu$ no longer vanishes, and the partialled inverse no longer equals the joint inverse. For Example 3.3 at small $\theta > 0$, compute the first-order θ -correction to the exchange-side partial's prediction for $\partial p^*/\partial V_0$. The exchange-side partial returns zero at every θ (since the contest parameter does not enter the consumption FOC); the leading correction in θ comes from the cross-channel block $J_{pm}J_{mm}^{-1}J_{mp}$ re-activating. The exercise quantifies how fast the partialled prediction's bias grows with θ .
4. *Approximation error of the partialled inverse.* The exchange-theoretic partial uses J_{pp}^{-1} to compute price responses; the joint formulation uses $S^{-1} = (J_{pp} - J_{pm}J_{mm}^{-1}J_{mp})^{-1}$. Show that the operator-norm ratio $\|J_{pm}J_{mm}^{-1}J_{mp}\|/\|J_{pp}\|$ controls the relative error of J_{pp}^{-1} as an approximation to S^{-1} . Use a first-order matrix expansion: when the cross-channel correction is small in operator norm relative to J_{pp} , $S^{-1} \approx J_{pp}^{-1}(I + J_{pm}J_{mm}^{-1}J_{mp}J_{pp}^{-1})$, and the correction term has the desired norm. The exercise gives a quantitative criterion for when partialling is approximately right and when it is not.
5. *$\tilde{E}^{\text{reg}}$ is open.* Definition 4.1 defines the regular set $\tilde{E}^{\text{reg}} = \{(p, m, \omega) \in \tilde{E} \mid J(p, m, \omega) \text{ invertible}\}$ as a subset of the equilibrium manifold $\tilde{E}$. Show that $\tilde{E}^{\text{reg}}$ is open in $\tilde{E}$. The argument is one line: exhibit $\tilde{E}^{\text{reg}}$ as the preimage of the open set $\mathbb{R} \setminus \{0\}$ under the continuous map $(p, m, \omega) \mapsto \det J(p, m, \omega)$, and openness in $\tilde{E}$ follows. Substantively: the regular set is an open submanifold, which lets the chapter's smooth-machinery results operate on it freely.

6. *det J in closed form on Example 3.3.* For Example 3.3 at $\theta = 0$, write $\det J$ at the symmetric equilibrium as a closed-form function of (V_0, k, W) , and verify $\det J \neq 0$ on the admissible parameter range $V_0 < 2W$, $V_0/(4k) < \omega_L$. Use the Schur factorization $\det J = \det J_{mm} \cdot \det S$ to split the verification: $J_{mm} = I - D_m \mu$ is invertible from the deployment-best-response contraction (Exercise 4 of Chapter 3), and $S = J_{pp} - J_{pm} J_{mm}^{-1} J_{mp}$ is invertible from the regularity of the reduced exchange economy. The exercise gives an explicit verification of joint regularity on the chapter's running example.
7. *When J_{mm} fails the contraction condition.* The deployment block $J_{mm} = I - D_m \mu$ is invertible whenever the spectral radius of $D_m \mu$ is strictly less than one—i.e., the deployment best-response is a contraction. This is the conflict-tradition's local regularity hypothesis. Construct a 2-state economy (modify Example 3.3) in which J_{mm} has a non-invertible point in the admissible parameter range. The natural construction varies a contest parameter past the point where the deployment best-response ceases to be a contraction (e.g., a high-stakes prize V_0 relative to cost k , where the marginal benefit of escalation overpowers the diminishing-returns curvature). Identify the parameter region where J_{mm} becomes singular and describe the local geometry of $\tilde{E}$ at the critical point. Open-ended; a qualitative description suffices.
8. *Odd parity preserved across fold catastrophes.* Corollary 3.19 of Chapter 3 establishes that the cardinality count of $\tilde{\pi}^{-1}(\omega) \cap \tilde{E}^{\text{reg}}$ is finite and odd at every regular value. Show that this parity is preserved across a fold catastrophe: when ω crosses Σ at a generic fold point, two regular equilibria merge and either annihilate or re-emerge, changing the count by ± 2 and preserving the parity. The argument uses Theorem 4.5 and the index-pairing property of folds: at a generic fold, two equilibria of opposite index merge, so the mod-2 count drops by 2 (parity preserved) and the signed sum drops by 0 (signed identity preserved). The exercise illustrates the geometric content of both the mod-2 cardinality and the signed-sum identities together.
9. *Sard for parameterized families.* The proof of Theorem 4.3 relies on the parameterized version of Sard's theorem rather than the single-map version. State the parameterized version in a form usable for the proof: if $f : X \times \Omega \rightarrow Y$ is smooth and 0 is a regular value of f on the full domain $X \times \Omega$, then for almost every $\omega \in \Omega$ (i.e., on the complement of a Lebesgue-measure-zero set), 0 is a regular value of the restricted map $f_\omega : X \rightarrow Y$. Compare to the unparameterized statement ("the set of critical values of a smooth map between manifolds has measure zero in the codomain"). The parameterized version is the input Theorem 4.3 needs; cf. Guillemin and Pollack [1974, Theorem 3.1] or Hirsch [1976, Section 17].
10. *Step 1 of Theorem 4.3 in detail.* The proof of Theorem 4.3 relies on Lemma 4.2, which establishes that $D_\omega \mu \neq 0$ generically under the SPT primitives plus the non-triviality condition $D_\omega s_i \neq 0$. Verify Step 1 of the genericity proof in detail. Identify the rows of $\bar{d}$ that respond to ω_j^ℓ perturbations through the direct $-\omega$ term in excess demand, and the rows of r that respond through $D_{\omega_i} \mu_i$ as delivered by Lemma 4.2. Confirm that the resulting endowment block $D_\omega e$ has rank $L - 1 + I$ at every zero of e outside a measure-zero set of ω . Identify the boundary conditions on the SPT primitives (desirability via Lemma 2.14 of Chapter 2, cost coercivity) that prevent the verification from breaking down at the boundary of Ω_0 .

11. *When 0 fails to be a regular value of e .* Construct an example primitive set in which 0 is not a regular value of e , so Theorem 4.3 fails to apply. Identify which SPT primitive condition breaks: the natural counterexample is exactly the degenerate case the footnote to Lemma 4.2 flags, where $D_\omega s_i \equiv 0$ and the extraction cost c_i has no ω -dependence. In that case, $D_\omega \mu = 0$ identically, the endowment block $D_\omega e$ has rank $L - 1$ (only the market-clearing rows respond to ω), and the parameterized-Sard step fails for that primitive set. Open-ended; identify the operational content of the violation in primitives terms.
12. *Empirical sampling and the singular locus.* Theorem 4.3 delivers measure-zero Σ in the Lebesgue sense; the body section Section 4.3 notes this is fully compatible with Σ being empirically common if the sampling distribution of international configurations is non-uniform. Discuss which historical or strategic mechanisms might concentrate observed ω near Σ . Examples to consider: arms-race dynamics that push systems toward bifurcation thresholds where deterrence equilibria become fragile; balance-of-power maintenance that targets indifference points where small shifts shift coalitions; coalition formation that rebalances after each shock and tends to settle near the boundaries of regime components. Open-ended; one paragraph per mechanism suffices.
13. *$D_{\omega_i}(p^*, m^*)$ under asymmetric perturbation.* At $\theta = 0$, the body of Section 4.4 delivers the closed form $p^* = 2\omega^2/(2\omega^1 - V_0)$ for general aggregate endowments $\omega^\ell = \omega_1^\ell + \omega_2^\ell$, which generalizes the symmetric case of Example 3.3. Differentiate to compute $D_{\omega_1^1} p^*$ and $D_{\omega_2^2} p^*$ separately, and verify that a unilateral boom in state 1's good-1 endowment (raising ω^1 , leaving ω^2 fixed) lowers p^* . Solve the asymmetric SPT for m_1^*, m_2^* at $\theta = 0$ and confirm both remain $V_0/(4k)$ regardless of the asymmetry: the deployment best-response is decoupled from endowments at the homotopy endpoint, and only the price responds to the perturbation.
14. *Conflict-side partial's accidental correctness.* At $\theta = 0$ in Example 3.3, the conflict-side partial returns the joint $D_{V_0} m^*$ exactly, even though the partialled and joint inverses generally differ. Show this is the $J_{mp} = 0$ collapse from Exercise 2 by computing both sides explicitly: the partialled response $-J_{mm}^{-1} D_{V_0} r$ at fixed p , and the joint response from $-J^{-1} D_{V_0} e$. Confirm the cross-channel correction $-J_{mm}^{-1} J_{mp} S^{-1} D_{V_0} e$ vanishes at $J_{mp} = 0$, forcing the joint and partialled deployment responses to coincide. The accident is structural, not numerical: it tracks the block-diagonal collapse condition.
15. *Cross-channel correction at $\theta > 0$.* For Example 3.3 at $\theta > 0$, the cross-channel block J_{mp} no longer vanishes (Exercise 3), and the conflict-side partial's deployment response acquires a correction. Compute the cross-channel correction $-(J_{mm}^{-1} J_{mp}) D_{V_0} p^*$, where $D_{V_0} p^*$ is the joint price response. Determine its sign: a V_0 shock raises both arming and prices at $\theta > 0$, so the cross-channel correction depends on whether higher prices feed back positively or negatively into the deployment FOC. The exercise quantifies one channel of the bias the conflict-side partial incurs once the contest prize becomes endogenous in consumption.
16. *Resource-curse sign reversal.* Construct an economy in which the exchange-side partial returns $\partial m^*/\partial \omega_i > 0$ (a resource boom raises arming) while the joint formulation returns $\partial m^*/\partial \omega_i < 0$ (a boom lowers arming), or vice versa. The cross-channel correction is what flips the sign: when the price-feedback channel and the direct-resource channel point in opposite directions, the joint equilibrium's sign can disagree with the partialled prediction.

Identify the parameter regime where this sign flip occurs. Open-ended; substantive interest is in connecting the sign flip to the empirical resource-curse debate.

17. *Orientation convention via a simple example.* Definition 4.4 fixes the orientation factor $(-1)^{L-1+I}$ in the index of a regular equilibrium, aligning the framework's convention with the standard smooth-GE convention of Mas-Colell [1985] and Balasko [2009]. Verify the factor is correct on a simple example: take a 2-state, 2-good Walrasian economy with no conflict, log preferences, and generic endowments (here $L - 1 + I = 1 + 2 = 3$, so the factor is $(-1)^3 = -1$). Compute $\det(d\tilde{\pi})$ at the unique Walrasian equilibrium and confirm $(-1)^3 \cdot \text{sign}(\det J) = +1$, agreeing with the standard index-+1 result for any unique regular Walrasian equilibrium [Balasko, 2009, Proposition 4.6.2].
18. *Index +1 for Example 3.3 at $\theta = 0$.* Under Assumption 3.6, Corollary 4.6 forces the unique joint equilibrium to have index +1. Verify this directly on Example 3.3 at $\theta = 0$: use the closed form for $\det J$ from Exercise 6 to compute $\text{sign}(\det J)$ at the symmetric equilibrium, and apply the orientation convention $\text{ind} = (-1)^{L-1+I} \cdot \text{sign}(\det J)$ with $L - 1 + I = 1 + 2 = 3$. The verification is routine instantiation; the substance is that the example sits in the unique-equilibrium scenario where the index theorem is trivial (one term, signed sum equals that term).
19. *Three-equilibrium index patterns.* Suppose an economy under Assumption 3.17 has exactly three regular equilibria over some $\omega \in \Omega_0 \setminus \Sigma$. Theorem 4.5 requires the signed sum of indices to equal +1. Use this to enumerate the possible distributions of indices across the three equilibria. (Hint: each index is ± 1 and the sum is +1, so exactly two indices are +1 and one is -1.) Discuss: which of the three is the “unstable” equilibrium under a natural local-stability reading? In standard smooth-GE settings, the -1-index equilibrium is the locally-unstable one between two stable branches—the characteristic three-equilibrium picture of, e.g., the Tullock contest with multiple symmetric solutions.
20. *Mod-2 versus signed cardinality.* Two cardinality identities live in the framework's apparatus: Corollary 3.19 of Chapter 3 (count is odd) and Theorem 4.5 (signed sum is +1). Discuss the relation: how does fold-catastrophe pairing of opposite-index equilibria realize the “signed” version of the mod-2 result? When two opposite-index equilibria collide at a fold and disappear, the cardinality drops by 2 (parity preserved) and the signed sum drops by 0 (signed identity preserved). Verify both identities across a fold and conclude that the signed identity is the finer of the two: it controls not just parity but the directed count. Open-ended; one paragraph suffices.
21. *Boundary catastrophe at $V_0 = 2W$.* The closed-form existence condition for Example 3.3 is $V_0 < 2W$; at $V_0 = 2W$ the equilibrium price diverges and the equilibrium ceases to exist. Identify this boundary crossing geometrically: is it a fold catastrophe in the Sard-Thom sense (Remark 3.20 of Chapter 3) within the framework's smooth machinery, or is it a degenerate boundary failure outside the catastrophe classification? Justify your answer by examining $\det J$ as $V_0 \rightarrow 2W$. Hint: a fold catastrophe involves $\det J \rightarrow 0$ at a critical point in the interior of the domain, and the equilibrium merges with another at the critical ω . The $V_0 = 2W$ case has $\det J \rightarrow 0$ but the equilibrium leaves the open simplex S_N rather than merging with another in the interior. Open-ended.

22. *Local fold-normal-form.* Fold catastrophes have the local normal form $f(x; \omega) = x^3 - \omega \cdot x$ in one local coordinate and one parameter, with the singular locus $\Sigma = \{0\}$ in ω -space. Remark 3.20 of Chapter 3 notes that body's regime structure is generically a fold stratum at codimension one. Construct a toy two-state, two-good primitive setting in which the joint equilibrium function e admits this normal form near a critical point: identify x (a local coordinate on $\tilde{E}$) and ω (a local parameter) such that the relevant component of e behaves cubically in (x, ω) . A sketch with the qualitative geometry suffices; full normal-form reduction is a substantial calculation (Arnold 1992 provides the systematic treatment). Open-ended.
23. *Well-posedness across Σ .* The institutional-design problem of Chapter 7 asks which ω a designer should target. When the designer's intervention shifts ω across Σ (the singular locus of Assumption 3.17), the joint equilibrium (p^*, m^*) is multi-valued or undefined at the crossing itself, so the comparative-statics objective ceases to be a smooth function. Discuss in what sense the optimization problem is well-posed at Σ . Possible answers: (a) restrict the designer's feasible set to a single regime, treating Σ as a hard constraint; (b) select one branch by an equilibrium-selection rule; (c) accept that optimization across Σ requires non-smooth methods (variational inequalities or generalized derivatives). Each formulation has substantive cost: (a) restricts the designer's options; (b) introduces an extra-equilibrium selection device; (c) expands the apparatus to non-smooth optimization. Open-ended; one paragraph per answer.
24. *Systemic conflict from trade openness.* A central commitment of the liberal-peace literature is that trade pacifies: states with substantial bilateral trade should be less likely to fight each other. The empirical evidence (Polachek 1980; Findlay and O'Rourke 2009) is correlative on bilateral flows. The framework gives a structural counterexample. Consider a peaceful joint equilibrium with I states and L goods, and add an "inert entrant" W with: a large endowment R of a new non-lootable good $L + 1$; strict preferences over goods $1, \dots, L$; and prohibitive cost of deployment (W never attacks). Show that for some original state i with profitably lootable good ℓ^* (in the sense that $\kappa_i^{\ell^*} < \lambda^{\ell^*}$ in the chapter's contest-cost vocabulary), there exists R sufficiently large such that the extended-economy equilibrium violates state i 's peace condition—and that this happens through the price channel alone, with no change in the bilateral flow between i and any potential adversary. Substantively: the liberal-peace literature's dyadic measurement misses systemic causation that the joint formulation makes precise. Open-ended construction; the substantive content is the price-pressure mechanism through the global market clearing condition.
25. *Pathology in the GDP-share statistic.* The Stockholm International Peace Research Institute compiles the GDP-share-of-military-expenditure statistic $\mu_i := (p^* \cdot c_i(m_i^*, p^*)) / (p^* \cdot \omega_i)$ as the standard scalar proxy for state i 's military effort (Tian et al. 2018). The framework's diffeomorphism $\tilde{E} \cong \mathbb{R}^{IL}$ guarantees that the full equilibrium is recoverable from endowments, but μ_i is a further projection onto a single real number, and it does not preserve the diffeomorphism's good behavior. Show that μ_i is generically non-monotone in m_i^* in any economy with price-dependent opportunity costs ($\partial c_i^\ell / \partial p^{-\ell} \neq 0$ for some i, ℓ): construct a smooth parameter path along which m_i^* strictly decreases while μ_i strictly increases. The mechanism is that as κ_i^ℓ rises, m_i^* falls but the equilibrium price p^* shifts, and the cost-weighted expenditure $p^* \cdot c_i$ can rise faster than m_i^* falls. Substantively: the

statistic registers price-feedback as if it were a change in the thing it is supposed to measure. Open-ended.

26. *Trade expectations and preventive conflict.* Copeland [2015] argues that the relevant variable for conflict onset is not the current level of bilateral trade but a state's *expectation* of the future trade environment: a state anticipating favorable future access remains at peace, while one anticipating cutoff fights—and fights sooner rather than later. Formalize Copeland's mechanism in the framework. For an anticipated parameter path $\omega(t) = \omega_0 + t\Delta\omega$ with $t \in [0, T]$, define the *trade expectation* $\tau_i := D_\omega \bar{V}_i(\omega_0) \cdot \Delta\omega$, where $\bar{V}_i$ is state i 's peaceful-equilibrium indirect utility. Show, using the master comparative-static formula of Section 4.4, that a state with $\tau_i < 0$ (declining peaceful utility along the anticipated path) has strictly higher current deviation gain $G_i(\omega_0) = V_i^* - \bar{V}_i$ than a state at the same allocation facing $\tau_i = 0$. The substantive content is that the framework recovers Copeland's preventive-conflict mechanism without requiring a dynamic model: trade expectations are a directional derivative on the peaceful-equilibrium value function, computable from the *static* comparative-statics formula. Open-ended.

Chapter 5

The normative theory

How much does conflict cost at a joint equilibrium, and when does it cost nothing? Standard welfare theory operates on a closed exchange economy at fixed resources. Standard conflict theory reads the welfare consequences of armed competition as the deadweight cost of deployment, evaluated at fixed prices. Each tradition computes a welfare object on a partialled equilibrium. Neither names the cost of conditioning each on the other, because neither sees both halves of the joint determination at once. This chapter names that cost.

The chapter's master scalar is $\Delta^*(\omega)$, the cardinal welfare gap between the joint equilibrium and the no-conflict counterfactual at the same endowment. Its geometric content is the displacement between two Pareto frontiers: the *conditional* frontier at the resources actually available after deployment costs are paid, and the *global* frontier at the full endowment. The chapter develops Δ^* , characterizes its sign and smoothness, identifies the class of equilibria for which it vanishes, and delivers its gradient as one application of the joint Jacobian inverse from Chapter 4.

What the chapter delivers is hard to find elsewhere. The First Welfare Theorem under joint determination becomes the conditional First Welfare Theorem, with Δ^* as the cardinal name of what conditioning costs. The bargaining model's standard premise—that war is inefficient—is recovered as a theorem under explicit conditions on the conflict technology and on equilibrium peace. The welfare gradient $D_\omega \Delta^*$ is the framework's joint-determination capability applied to welfare: a single matrix-vector operation produces the joint welfare response to any perturbation the framework admits. None of the three is available from either partial.

The chapter is organized around the externality, the cases where it vanishes, and the sensitivity formula that controls how it moves. Section 5.1 introduces Δ^* as the master comparison and develops the conditional and global frontiers as the geometric objects it measures between. Section 5.2 establishes the conditional First Welfare Theorem and identifies the structural source of the conflict externality as the gap between social-planner and Nash first-order conditions on deployment. Section 5.3 characterizes the case in which Δ^* vanishes globally: under zero- or negative-sum warfare, peaceful equilibria are globally Pareto optimal, and the bargaining model's premise that war is inefficient is recovered as a theorem under explicit conditions. Section 5.4 develops $D_\omega \Delta^*$ as the master sensitivity formula, decomposes it into three interpretable channels, and reads the welfare cost of conflict as a smooth scalar field on the regular set. Section 5.7 hands the chapter's apparatus to the institutions, applications, and scaling chapters that follow.

5.1 The conflict externality

The welfare cost of conflict at a joint equilibrium has a clean cardinal name. Fix an endowment profile $\omega \in \Omega_0$, a joint equilibrium (p^*, m^*, x^*) at ω , and the resulting total resource vector $r^* = \omega - \sum_i c_i(m_i^*)$ that remains for consumption after deployment costs are paid. The dual-competitive equilibrium operates on r^* , not on ω : deployment costs subtract from the resources available to the price system before allocation begins. The welfare cost of operating on r^* rather than on ω is the chapter's master scalar, and the rest of the chapter unpacks it.

The master comparison. For any total resource vector $r \in \Omega_0$ and any utility profile $u_{-I} \in \mathbb{R}^{I-1}$ for the first $I - 1$ states, define the value function

$$v(r, u_{-I}) := \max\{u_I(x_I) \mid (x_1, \dots, x_I) \in \Omega(r), u_i(x_i) \geq u_{-I,i} \text{ for } i < I\}, \quad (5.1)$$

where $\Omega(r) := \{x \in X \mid \sum_i x_i = r\}$ is the resource-feasible set. v is the maximal utility available to state I when the resources are r and the others are held at u_{-I} . Because preferences are smooth and strictly monotone (Chapter 2), v is smooth on the open subset of $X \times \mathbb{R}^{I-1}$ at which the constraint set is non-empty, and strictly increasing in r (Balasko 2009, Proposition A.5.1). Holding the other states' utilities fixed, more aggregate resources strictly increase what is available to state I .

Let $u_{-I}^* := (u_1(x_1^*), \dots, u_{I-1}(x_{I-1}^*))$ be the utility levels of the first $I - 1$ states at the joint equilibrium. The state- I utility achieved at the equilibrium is $u_I^* = v(r^*, u_{-I}^*)$, since the equilibrium allocation lies on the conditional Pareto frontier at r^* (Section 5.2 below). The state- I utility achievable at the same u_{-I}^* if the full endowment ω were available—the no-conflict counterfactual—is $v(\omega, u_{-I}^*)$. The difference is the chapter's master scalar.

Definition 5.1 (Conflict externality). The *conflict externality* at the joint equilibrium (p^*, m^*, x^*) at ω is the cardinal scalar

$$\Delta^*(\omega) := v(\omega, u_{-I}^*) - v(r^*, u_{-I}^*). \quad (5.2)$$

Equation (5.2) is the chapter's single load-bearing identity. Δ^* is a number in utility units of state I : the additional utility state I would attain if the resources consumed by deployment were instead available for allocation, holding the other states' utility levels fixed. The choice of state I as the residual claimant is an arithmetic convenience, not a substantive asymmetry; permuting which state is taken as residual produces a parallel scalar with the same sign and the same vanishing locus. The geometric content of Δ^* is the same in any permutation: a displacement between two Pareto frontiers parameterized by the same u_{-I} .

The two frontiers. The conditional and global Pareto frontiers are the geometric objects Δ^* measures between. For $r \in \Omega_0$, define

$$\begin{aligned} \mathcal{P}(r) &:= \{x \in \Omega(r) \mid x \text{ is Pareto optimal}\}, \\ \bar{\mathcal{U}}(r) &:= \text{cl}\{u(x) \in \mathbb{R}^I \mid x \in X, \sum_i x_i \leq r\}, \\ \partial\bar{\mathcal{U}}(r) &:= \bar{\mathcal{U}}(r) \setminus \mathcal{U}(r), \end{aligned}$$

where $\mathcal{U}(r) := \text{int}(\bar{\mathcal{U}}(r))$ is the open utility possibility set and $\partial\bar{\mathcal{U}}(r)$ is its boundary, the *utility possibility frontier* at r . Balasko [2009, Appendix A] establishes the structural facts the chapter

consumes (Balasko 2009, Appendix A): $\mathcal{P}(r)$ is a smooth $(I - 1)$ -manifold; the utility map $x \mapsto (u_1(x_1), \dots, u_I(x_I))$ restricts to a diffeomorphism $\mathcal{P}(r) \rightarrow \partial\bar{\mathcal{U}}(r)$; the frontier is the graph of $v(\cdot, r)$ over $\mathcal{U}_{I-1}(r)$, the projection of $\bar{\mathcal{U}}(r)$ to its first $I - 1$ coordinates.

The conditional frontier at r^* records what utility profiles are simultaneously achievable when the resources available for allocation are those that survive deployment costs. The global frontier at ω records the same when the full endowment is available. Strict monotonicity of v in r implies $\partial\bar{\mathcal{U}}(\omega) \succ \partial\bar{\mathcal{U}}(r^*)$ in the componentwise sense whenever $r^* < \omega$ in any coordinate: the global frontier strictly dominates the conditional one. Δ^* measures that dominance along the u_I -axis at the fixed u_{-I}^* .

When the gap is positive. Δ^* has the sign that intuition suggests, with the qualification that positivity depends on whether deployment costs are non-trivial.

Proposition 5.2 (Sign of the conflict externality). *At any joint equilibrium (p^*, m^*, x^*) at ω :*

- (1) $\Delta^*(\omega) \geq 0$.
- (2) $\Delta^*(\omega) > 0$ whenever $\sum_i c_i(m_i^*) > 0$ in some coordinate.
- (3) $\Delta^*(\omega) = 0$ if and only if $r^* = \omega$ coordinatewise.

Proof. v is non-decreasing in r at fixed u_{-I} : enlarging the resource constraint weakly enlarges the feasible set of the maximization in Equation (5.1). Hence $v(\omega, u_{-I}^*) \geq v(r^*, u_{-I}^*)$ whenever $r^* \leq \omega$ coordinatewise, which is Proposition 5.2(1). For (2), strict monotonicity of v in any coordinate of r at which u_I 's utility is locally non-satiated promotes the weak inequality to strict whenever $r^* < \omega$ in any coordinate, and the cost assumption (Chapter 2) gives $\sum_i c_i > 0$ coordinatewise whenever $m^* \neq 0$ along that coordinate. For (3), v is strictly increasing in r at any $u_{-I} \in \mathcal{U}_{I-1}(r)$ where the constraint set is non-empty; equality of $v(\omega, u_{-I}^*)$ and $v(r^*, u_{-I}^*)$ therefore forces $r^* = \omega$. $\square$

The proposition has a one-line gloss. Δ^* records the resource cost of deployment, evaluated in utility units along the conditional frontier. Whenever deployment is operative the cost is non-zero in resource units, and monotonicity of preferences carries that to a strictly positive gap in utility units. The vanishing locus $\Delta^* = 0$ coincides with $r^* = \omega$, which under standard conditions on the cost function coincides with $m^* = 0$.

Smoothness on the regular set. Δ^* inherits its smoothness from the smoothness of v and of the equilibrium selection.

Proposition 5.3 (Smoothness of Δ^*). *On the open set $\tilde{E}^{\text{reg}} \subset \tilde{E}$ of regular joint equilibria (Chapter 4), $\Delta^*(\omega)$ is a smooth function of ω .*

Proof. On $\tilde{E}^{\text{reg}}$ the joint Jacobian J is invertible, so the implicit function theorem makes $(p^*, m^*)(\omega)$ and hence $r^*(\omega) = \omega - \sum_i c_i(m_i^*(\omega))$ and $u_{-I}^*(\omega)$ smooth functions of ω . v is smooth on the open subset of $X \times \mathbb{R}^{I-1}$ where the maximization in Equation (5.1) has a non-empty constraint set [Balasko, 2009, Proposition A.5.1]. The composition $v(\omega, u_{-I}^*(\omega)) - v(r^*(\omega), u_{-I}^*(\omega))$ is therefore smooth in ω . $\square$

Comparative statics on the externality. The smoothness in Proposition 5.3 connects directly to the master formula of Chapter 4. Differentiating Equation (5.2) in ω yields

$$D_\omega \Delta^* = D_\omega v(\omega, u_{-I}^*) - D_\omega v(r^*(\omega), u_{-I}^*(\omega)), \quad (5.3)$$

in which the second term unpacks via the chain rule into a weighted combination of $D_\omega m^*$, $D_\omega p^*$, and $D_\omega x^*$, each of which is computed by one application of $-J^{-1}$ to the corresponding direct-derivative column of $D_\omega e$. The welfare cost of conflict is therefore a smooth scalar field on the regular set, and its gradient at any equilibrium is one matrix-vector arithmetic operation away once J^{-1} is in hand. The positive theory delivers the calculation; the normative content is reading the calculation as a welfare measurement.

Worked instance. The two-state contest of Example 3.3 makes the construction concrete. At the symmetric equilibrium with prize V_0 in good 1, quadratic deployment costs θm_i^2 in good 2, and endowments $\omega_A^2 = \omega_B^2 = W$, the equilibrium deployment is $m_A^* = m_B^* = m^*$ and the equilibrium prices satisfy $p^* - 1 = V_0/(2W - V_0)$. Total deployment cost is $2\theta(m^*)^2$ in good 2, so $r^* = \omega - (0, 2\theta(m^*)^2)$. At the symmetric Cobb-Douglas specification with equal weights, the value function v is closed-form and

$$\Delta^* = \frac{1}{2} [\log(2W) - \log(2W - 2\theta(m^*)^2)] = \frac{1}{2} \log \left(\frac{W}{W - \theta(m^*)^2} \right),$$

positive whenever $\theta(m^*)^2 > 0$ and approaching zero as the deployment cost vanishes. The expression is the cardinal welfare cost of the contest in state I 's utility units, computed entirely from primitives of the example. The gradient $D_{V_0} \Delta^*$ at this equilibrium is non-zero—a higher prize raises equilibrium deployment, raises the cost, and widens the gap—and the calculation runs through the chapter's master formula Equation (5.3) composed with the comparative-statics master formula $D_{V_0} m^* = -J^{-1} D_{V_0} e$ of Chapter 4.

What this section establishes. Δ^* is the chapter's master scalar. It is non-negative on $\tilde{E}$, strictly positive whenever conflict is operative, smooth on the regular set, and computable from the positive-theory comparative statics by composition. What it *does not* yet say is whether the price system at the joint equilibrium is doing its job—whether the residual allocation, conditional on the deployment profile m^* , is itself efficient. That is the conditional First Welfare Theorem of Section 5.2, which records that the exchange-side machinery is doing exactly what the partial tradition claims for it, but only conditionally. The conditioning is what Δ^* measures.

5.2 The conditional First Welfare Theorem

Section 5.1 named the welfare cost of conflict as the gap between two Pareto frontiers. This section establishes that the price system delivers the *closer* of the two: at any joint equilibrium, the allocation is Pareto efficient relative to the conditional frontier $\partial \bar{U}(r^*)$ at the resources actually available after deployment costs are paid. The result is the natural translation of the First Welfare Theorem into the joint setting, and it pins down the chapter's master scalar from the other side: Δ^* is the cost of *conditioning* on the equilibrium deployment profile, not a failure of the price system to do its job once that conditioning is in force.

What the First Welfare Theorem says. The First Welfare Theorem is the formal statement of a claim older than its proof: that competitive markets, left alone, allocate resources efficiently. The substantive content is that no individual or institution needs to coordinate the allocation; price signals are sufficient. Each participant maximizes own utility against own budget, and the prices that clear all markets simultaneously yield an allocation that no feasible reallocation can Pareto-improve. The result is the load-bearing positive claim of any version of laissez-faire. It is what Smith's invisible hand promises in formal coordinates, and what [Arrow \[1951\]](#) established with modern rigor in the wake of the Hicks-Samuelson revealed-preference apparatus. Without the First Welfare Theorem, the case for decentralized exchange has to be made on grounds other than efficiency, on liberty or custom or political sociology. With it, the case can be made on efficiency directly, as a theorem.

When the First Welfare Theorem fails. The result is also notoriously fragile. The welfare-economics literature is partly a catalogue of its failures: externalities (price systems that do not internalize spillovers do not clear), asymmetric information (Arrow–Debreu equilibria with private information are generically not constrained-Pareto-optimal, as [Greenwald and Stiglitz \[1986\]](#) make precise), missing markets (a state space too coarse for the relevant contingencies), and non-convexities at which supporting prices fail to exist. Each failure is taken seriously precisely because the First Welfare Theorem is taken seriously. A theorem that did not deliver much would not generate a large literature about its limits. The fragility is itself diagnostic. The effort to find the conditions under which the theorem holds, and the conditions under which it does not, is downstream of how much the theorem is doing on the days it does hold.

Smith's invisible hand under arms. The conflict externality is in this catalogue. The framework's primitives encode that each state's deployment m_i creates spillovers on every other state's total consumption through the battle function s , and the price system has no instrument to discipline the spillover. The clean way to register this is the same way [Greenwald and Stiglitz \[1986\]](#) register the information failure: keep the First Welfare Theorem, but qualify it. Proposition 5.4 below is exactly the qualified form. Conditional on the prevailing deployment profile m^* , the price system delivers the standard First Welfare Theorem result on the residual allocation problem. Smith's invisible hand operates as advertised, but only on what the other hand has left for it to act on. The conditioning has already happened by the time prices form, and Δ^* , introduced in Section 5.1, is the welfare cost of the conditioning itself.

Total consumption and the conditional economy. What enters utility at a joint equilibrium is not market purchases but post-conflict consumption. At a joint equilibrium (p^*, m^*, x^*) at ω , write

$$\bar{x}_i^* := x_i^* + s_i(m^*, x^*, \omega) \quad (5.4)$$

for state i 's *total consumption*, the bundle on which u_i is evaluated. The market clears in market-purchase coordinates, $\sum_i x_i^* = \sum_i (\omega_i - c_i(m_i^*, p^*, \omega_i))$, but welfare arguments are about $\bar{x}^* = (\bar{x}_1^*, \dots, \bar{x}_I^*)$. Summing Equation (5.4) over i and using market clearing,

$$\sum_i \bar{x}_i^* = \sum_i \omega_i - \sum_i c_i(m_i^*, p^*, \omega_i) + \sum_i s_i(m^*, x^*, \omega). \quad (5.5)$$

The right-hand side is the *effective resource vector* r^* , which already appeared as the conditioning argument of v in Equation (5.2). Under zero- or negative-sum warfare ($\sum_i s_i \leq 0$ componentwise),

Equation (5.5) simplifies to $r^* \leq \omega - \sum_i c_i(m_i^*, p^*, \omega_i)$; under arbitrary s the expression is the right one for the welfare argument because it tracks total consumption exactly.

The conditional economy at r^* is the closed exchange economy that consumes r^* in aggregate: feasible total-consumption profiles are $\{\bar{x} \mid \sum_i \bar{x}_i = r^*\} = \Omega(r^*)$. The conditional Pareto set $\mathcal{P}(r^*) \subset \Omega(r^*)$ records the efficient allocations of r^* across the I states. The conditional First Welfare Theorem is the claim that $\bar{x}^*$ lies in this set.

Conditional efficiency. The result is the standard FWT applied to the closed exchange economy at r^* , with one notational adjustment: each state i behaves as though it were maximizing $u_i(z + s_i^*)$ over a budget set in market-purchase coordinates, where $s_i^* := s_i(m_i^*, x^*, \omega)$ is the equilibrium spoils transfer held fixed.

Proposition 5.4 (Conditional First Welfare Theorem). *At any joint equilibrium (p^*, m^*, x^*) , $\bar{x}^* \in \mathcal{P}(r^*)$.*

Proof. Suppose for contradiction that $\bar{y} = (\bar{y}_1, \dots, \bar{y}_I) \in \Omega(r^*)$ Pareto-dominates $\bar{x}^*$: $u_i(\bar{y}_i) \geq u_i(\bar{x}_i^*)$ for all i and strict for some j . Define $y_i := \bar{y}_i - s_i^*$, the corresponding market-purchase profile that produces total consumption $\bar{y}_i = y_i + s_i^*$.

At the joint equilibrium, x_i^* solves state i 's problem with deployments m_i^* and spoils s_i^* fixed: it maximizes $u_i(z + s_i^*)$ subject to $p^* \cdot z \leq w_i^*$, where $w_i^* := p^* \cdot (\omega_i - c_i(m_i^*, p^*, \omega_i))$ is the equilibrium budget. By the standard revealed-preference argument, any alternative purchase y_i with $u_i(y_i + s_i^*) \geq u_i(x_i^* + s_i^*)$ satisfies $p^* \cdot y_i \geq w_i^*$, with strict inequality whenever the utility comparison is strict. Summing over i and using $u_j(\bar{y}_j) > u_j(\bar{x}_j^*)$ for at least one j ,

$$p^* \cdot \sum_i y_i > \sum_i w_i^* = p^* \cdot \sum_i (\omega_i - c_i(m_i^*, p^*, \omega_i)).$$

Adding $p^* \cdot \sum_i s_i^*$ to both sides,

$$p^* \cdot \sum_i \bar{y}_i > p^* \cdot \left[\sum_i \omega_i - \sum_i c_i(m_i^*, p^*, \omega_i) + \sum_i s_i^* \right] = p^* \cdot r^*,$$

using Equation (5.5). But $\bar{y} \in \Omega(r^*)$ implies $\sum_i \bar{y}_i = r^*$, hence $p^* \cdot \sum_i \bar{y}_i = p^* \cdot r^*$, contradicting the strict inequality. $\square$

The proof is the standard First Welfare Theorem argument with spoils s_i^* entering as a fixed lump-sum gift to state i in market-purchase coordinates. Once the conflict outcome is held fixed, the price system does exactly what it does in a Walrasian exchange economy at r^* : no reallocation of the available aggregate that respects market clearing improves on $\bar{x}^*$. The qualification “conditional” is substantive. The proposition does not say that the joint equilibrium is globally efficient—only that the residual allocation problem, once m^* is given, is solved by the price system without waste.

Welfare-weight representation. Proposition 5.4 places $\bar{x}^*$ on the conditional Pareto frontier; the dual representation of that frontier identifies the specific point. The Pareto set $\mathcal{P}(r^*)$ is parameterized smoothly by an $(I - 1)$ -simplex of welfare weights $\lambda \in \mathbb{R}_{++}^{I-1}$ via the social welfare problem

$$T(r, \lambda) := \arg \max_{\bar{x} \in \Omega(r)} \left[\sum_{i=1}^{I-1} \lambda_i u_i(\bar{x}_i) + u_I(\bar{x}_I) \right], \quad (5.6)$$

which delivers a smooth bijection $T(r^*, \cdot) : \mathbb{R}_{++}^{I-1} \rightarrow \mathcal{P}(r^*)$ under the strict-concavity assumption on u (Balasko 2009, Proposition A.6.1). At the joint equilibrium, the welfare weights are pinned down by the budget multipliers.

Corollary 5.5 (Welfare-weight representation). *At any joint equilibrium (p^*, m^*, x^*) , there exist unique welfare weights $\lambda^* \in \mathbb{R}_{++}^{I-1}$ such that $\bar{x}^* = T(r^*, \lambda^*)$. The weights satisfy $\lambda_i^* \propto 1/\mu_i^*$, where μ_i^* is the Lagrange multiplier on state i 's budget constraint at the equilibrium.*

Proof. $\bar{x}^* \in \mathcal{P}(r^*)$ by Proposition 5.4, and $T(r^*, \cdot)$ is a bijection onto $\mathcal{P}(r^*)$, so a unique λ^* realizes $\bar{x}^*$. The proportionality follows from comparing first-order conditions: the social-welfare problem Equation (5.6) has FOCs $\lambda_i^* Du_i(\bar{x}_i^*) = \nu^*$ for $i < I$ and $Du_I(\bar{x}_I^*) = \nu^*$, where ν^* is the multiplier on the resource constraint; the agent-level budget problem has FOCs $Du_i(\bar{x}_i^*) = \mu_i^* p^*$. Substituting yields $\lambda_i^* = \mu_I^* / \mu_i^*$, with $\nu^* = \mu_I^* p^*$. $\square$

The representation has a one-line gloss. The joint equilibrium produces a particular point on the conditional Pareto frontier, and the welfare weights at that point are the inverses of the agents' marginal utilities of income. States with low marginal utility of income at the equilibrium—states for whom an additional dollar would not help much—receive high welfare weights; the price system distributes resources to those whose marginal utility most warrants it. The welfare weights are not chosen by anyone; they emerge from the budget multipliers, which emerge from the joint equilibrium.

What the price system does not address. Proposition 5.4 establishes that prices clear the allocation problem at fixed deployments. What prices *do not* clear is the choice of deployments itself: each m_i^* is set by Nash logic, not by anything the price system disciplines. The gap between Nash logic and a social-optimal choice of m is the substantive content of Δ^* .

The comparison runs through first-order conditions. A social planner choosing $(\bar{y}, m')$ to maximize $\sum_i \alpha_i u_i(\bar{y}_i)$ subject to the global feasibility constraint $\sum_i \bar{y}_i \leq \omega - \sum_i c_i(m'_i, p, \omega_i) + \sum_i s_i(m', \bar{y}, \omega)$ would, at any interior optimum with the welfare weights of Corollary 5.5 as the α_i , require for each i

$$\sum_{j=1}^I \lambda_j^* Du_j(\bar{y}_j) \cdot D_{m_i} s_j(m', \bar{y}, \omega) = \lambda_i^* \mu'_i p \cdot D_{m_i} c_i(m'_i, p, \omega_i), \quad (5.7)$$

where μ'_i is the planner's shadow income for state i . The Nash condition at the joint equilibrium, derived in Chapter 2 from the state's problem under threat, requires only the single-state version

$$Du_i(\bar{x}_i^*) \cdot D_{m_i} s_i(m^*, x^*, \omega) = \mu_i^* p^* \cdot D_{m_i} c_i(m_i^*, p^*, \omega_i). \quad (5.8)$$

Subtracting Equation (5.8) from Equation (5.7) at a point where both could hold,

$$\underbrace{\sum_{j \neq i} \lambda_j^* Du_j(\bar{x}_j^*) \cdot D_{m_i} s_j(m^*, x^*, \omega)}_{\text{cross-state effect of } m_i} = 0. \quad (5.9)$$

Equation (5.9) is the *conflict externality* in first-order form. State i 's deployment changes other states' total consumption through the battle function; the Nash condition does not internalize that channel; the planner's condition does. Whenever the cross-state derivatives $D_{m_i} s_j$ for $j \neq i$ are non-zero at (m^*, x^*, ω) —which is the generic case—Equation (5.9) is not satisfied at any $m^* \neq 0$, and the joint equilibrium is not globally Pareto optimal.

Proposition 5.6 (Generic global inefficiency). *Generically over the space of battle functions satisfying Assumptions 2.5 and 2.6, no joint equilibrium with $m^* \neq 0$ is globally Pareto optimal.*

Proof. Global Pareto optimality requires Equation (5.9) for every i . This is a system of I scalar conditions on the cross-state derivatives $\{D_{m_i}s_j \mid j \neq i\}$ of the battle function at the equilibrium, evaluated at the welfare weights of Corollary 5.5. For generic smooth s satisfying Assumptions 2.5 and 2.6 the system holds on a measure-zero set in the space of battle function profiles, so generically no $m^* \neq 0$ is a global Pareto optimum. $\square$

Proposition 5.6 decomposes into two structural sources. The *resource externality* is the term in $D_{m_i}c_i$ entering through the right-hand side of Equation (5.7) alongside the Nash term: each state's deployment subtracts from the resources available to the exchange economy in coordinates $\sum_i c_i$, and the Nash condition internalizes only its own subtraction. The *security externality* is the cross-state term $\sum_{j \neq i} \lambda_j^* Du_j \cdot D_{m_i}s_j$: state i 's deployment shifts other states' total consumption through the battle function, and the Nash condition is silent on that channel. The two together pin down what Δ^* measures: the welfare equivalent, in state I 's utility units, of the cross-state effects that no price disciplines.

Reading the chapter so far. The chapter's two halves slot together at this point. Proposition 5.4 says that the price system is doing its job: at any joint equilibrium, the allocation $\bar{x}^*$ sits on the conditional Pareto frontier $\partial\bar{U}(r^*)$. Proposition 5.2 says that the conditional frontier generically lies strictly inside the global frontier $\partial\bar{U}(\omega)$, with the gap measured by Δ^* . Proposition 5.6 identifies what is generating the gap: cross-state effects of deployment that the Nash logic does not internalize. The price system is not failing; it is solving a residual allocation problem on resources that conflict has already shrunk. Section 5.3 characterizes the case where conflict is unable to shrink the resource base—zero- or negative-sum warfare—and shows that peaceful equilibria escape the externality entirely, recovering the bargaining model's standard premise that war is inefficient as a theorem rather than as an assumption.

5.3 Peaceful equilibria

Section 5.2 established that the price system clears the residual allocation problem at fixed deployments and identified the conflict externality as what no price disciplines. The natural next question is when the externality vanishes. The answer is geometric: under conditions that prevent the conflict technology from creating aggregate resources, peaceful equilibria sit on the global Pareto frontier rather than on a strictly interior conditional one, and $\Delta^* = 0$ at every such equilibrium. The result has a separate consequence in the formal-IR tradition. Bargaining models of war (Fearon 1995, Powell 2006, and the literature downstream of them) take as their starting premise that war is inefficient—that any allocation reachable by fighting is also reachable by some agreement. Under the framework's primitives the premise is a *theorem*, under explicit conditions on the conflict technology and on the existence of a peaceful equilibrium.

Zero- and negative-sum warfare. The relevant restriction is on the aggregate effect of conflict.

Definition 5.7 (Zero- and negative-sum warfare). The battle technology s is *zero-sum* if $\sum_i s_i^\ell(m, x, \omega) = 0$ for every good ℓ , deployment profile m , allocation x , and endowment profile ω . It is *negative-sum* if $\sum_i s_i^\ell(m, x, \omega) \leq 0$ for every ℓ, m, x, ω .

Zero-sum warfare says that any goods one state seizes through conflict are lost by other states; the aggregate is unchanged. Negative-sum warfare adds destruction: the resources consumed in fighting—weapons, lives, infrastructure—are gone from both sides. What the conditions rule out is that conflict be a productive technology at the social level. Definition 5.7 does not constrain the within-conflict distribution of spoils or losses, only the aggregate. A battle function in which A seizes from B 's net exports, or in which destruction is asymmetric, or in which the spoils schedule depends on prices and allocations through arbitrary smooth functions, is zero- or negative-sum exactly when its component sum has the right sign.

Peaceful equilibria. A joint equilibrium (p^*, m^*, x^*) is *peaceful* if $m^* = 0$. At a peaceful equilibrium, the cost assumption (Chapter 2) gives $c_i(0, p^*, \omega_i) = 0$, the natural normalization on s gives $s_i(0, x^*, \omega) = 0$, and the equilibrium relations collapse: total consumption equals market purchases, $\bar{x}_i^* = x_i^*$, and effective resources equal endowment, $r^* = \omega$. The conditional First Welfare Theorem (Proposition 5.4) then applies at ω , giving $x^* \in \mathcal{P}(\omega)$: at a peaceful equilibrium the allocation lies on the global Pareto frontier. What Proposition 5.4 delivered conditionally for a generic $r^* \leq \omega$, it delivers unconditionally at the peaceful case where $r^* = \omega$.

The substantive claim of this section is sharper. At a peaceful equilibrium the allocation is not merely on the frontier of the resources actually available; it is on the frontier of every alternative that the conflict technology can reach.

Global efficiency. The theorem states that peace is globally Pareto optimal whenever the conflict technology is zero- or negative-sum.

Theorem 5.8 (Global efficiency of peaceful equilibria). *Suppose the battle technology is zero- or negative-sum (Definition 5.7). Then any peaceful joint equilibrium $(p^*, 0, x^*)$ at ω is globally Pareto optimal: there is no globally feasible $(\bar{y}, m')$ with $u_i(\bar{y}_i) \geq u_i(x_i^*)$ for all i and strict for some j .*

Proof. Take any $(\bar{y}, m')$ globally feasible at ω , where $\bar{y}$ is a total-consumption profile and m' is some deployment profile. Global feasibility requires $\sum_i \bar{y}_i = \sum_i y_i + \sum_i s_i(m', y, \omega)$ where y is the corresponding market-purchase profile satisfying $\sum_i y_i \leq \sum_i \omega_i - \sum_i c_i(m'_i, p', \omega_i)$ at the supporting price p' .

Case 1: $m' = 0$. Cost and battle normalizations give $c_i(0, \cdot, \cdot) = 0$ and $s_i(0, \cdot, \cdot) = 0$, so $\bar{y} = y$ with $\sum_i y_i \leq \omega$. By Proposition 5.4 applied at $r^* = \omega$, the peaceful equilibrium allocation x^* lies in $\mathcal{P}(\omega)$; no $\bar{y} \in \Omega(\leq \omega)$ Pareto-dominates x^* .

Case 2: $m' \neq 0$. Under zero- or negative-sum warfare, $\sum_i s_i(m', y, \omega) \leq 0$ componentwise, so

$$\sum_i \bar{y}_i \leq \sum_i y_i \leq \omega - \sum_i c_i(m'_i, p', \omega_i).$$

The cost assumption (Chapter 2) implies $\sum_i c_i(m'_i, p', \omega_i) > 0$ in at least one coordinate whenever $m' \neq 0$, so $\sum_i \bar{y}_i < \omega$ in that coordinate. Hence $u(\bar{y}) \in \bar{\mathcal{U}}(r')$ for some $r' < \omega$ componentwise in at least one good, and by monotonicity of the utility-possibility set [Balasko, 2009, Proposition A.5.6], $\bar{\mathcal{U}}(r') \subset \mathcal{U}(\omega) = \text{int}(\bar{\mathcal{U}}(\omega))$. Therefore $u(\bar{y})$ lies in the interior of $\bar{\mathcal{U}}(\omega)$.

The peaceful equilibrium delivers $u(x^*) \in \partial \bar{\mathcal{U}}(\omega)$ by Case 1's reasoning together with the boundary-frontier diffeomorphism of Section 5.1. Interior and boundary are disjoint, so $u(\bar{y})$ cannot weakly Pareto-dominate $u(x^*)$: any componentwise weak inequality forced by Pareto domination would place $u(\bar{y})$ on the boundary, contradicting Case 2's interior result. $\square$

The proof has a one-line geometric reading. Peaceful equilibrium consumption sits on the boundary of the utility-possibility set at the full endowment, $\partial\bar{U}(\omega)$. Any non-peaceful alternative under zero- or negative-sum warfare consumes strictly fewer aggregate resources, placing its utility profile in the interior of $\bar{U}(\omega)$. Boundary points cannot be weakly Pareto-dominated by interior points, so peace dominates every conflictual alternative. The two structural ingredients are doing different work. The boundary-placement of peace comes from the conditional FWT applied at $r^* = \omega$. The interior-placement of conflict comes from the fact that, under the zero- or negative-sum condition, conflict cannot expand the resource base.

The bargaining model's premise. The result speaks to a substantive premise of the formal-IR bargaining tradition. Fearon [1995] organizes rationalist explanations for war around an inefficiency premise: because war destroys resources, any allocation reachable by fighting is also reachable by an agreement that splits the spared cost; rationalist explanations for war must therefore identify what prevents the parties from reaching that agreement (commitment problems, private information, indivisibilities). The premise is the engine of the puzzle. Subsequent work takes the premise as given and asks what blocks the agreement.

Theorem 5.8 relocates the premise. What Fearon assumes, the framework derives: under zero- or negative-sum warfare, peaceful equilibria are globally Pareto optimal, hence dominate every conflictual alternative, hence the bargaining surplus the agreement is supposed to split is real. The conditions under which the assumption is valid are explicit: the battle technology must satisfy Definition 5.7 (no aggregate gains from fighting), and a peaceful equilibrium must exist (the comparison is vacuous when no peaceful $(p^*, 0, x^*)$ solves the joint equilibrium equations). The applications and institutions chapters take up cases in which each condition can fail. A battle technology that creates positive externalities through deterrence, or one in which the cost of conflict is recouped through trade gains the conflict opens up, can violate the zero/negative-sum condition; an equilibrium configuration in which arming is dominant for every state regardless of others' choices can fail to admit any peaceful equilibrium. The chapter does not show that Fearon's premise always holds; it shows what must hold for it to hold, and exhibits the substantive geometry that makes the premise a theorem rather than an axiom.

The two-state contest in light of the theorem. Example 3.3 makes the content and the limits of Theorem 5.8 concrete. The contest's battle function is zero-sum: the prize V_0 in good 1 is redistributive, with $\sum_i s_i^1 = 0$ at every deployment profile, so the example satisfies Definition 5.7. The theorem therefore applies as a conditional statement: if the example admits a peaceful equilibrium, that equilibrium is globally Pareto optimal.

The example admits no peaceful equilibrium at $V_0 > 0$. At $m^* = 0$ each state's marginal benefit of deploying is the unilateral capture of the entire prize at marginal cost zero, and $m_i = 0$ is not a best response. The unique Nash equilibrium has interior $m_i^* = \sqrt{V_0/(4\theta)}$, and the theorem is therefore vacuous on this example—peace would be globally Pareto optimal if it could be sustained, but the contest structure of s rules out its sustainment.

The vacuity has substantive content. The structural feature blocking peace is that the marginal benefit of unilateral deployment at $m^* = 0$ is strictly positive: each state captures the entire prize by deviating to $m_i > 0$. This is a commitment obstruction in Fearon's taxonomy. The parties cannot reach the peaceful agreement that Theorem 5.8 certifies as efficient because no unilateral move toward peace is individually rational at any arming level. The cardinal welfare cost of that obstruction is exactly Δ^* at the contested equilibrium, computed in closed form

in Section 5.1. The institutions chapter takes up structural changes—side payments, repeated interaction, externally enforceable disarmament—under which the obstruction can be removed and a peaceful equilibrium reached as a different equilibrium of a different game.

Remark 5.9 (Fearon’s premise and the Coase intuition). The bargaining tradition’s intuition for the inefficiency premise is often stated in Coasean terms: rational parties bargain to the efficient allocation absent transactions costs; war is a transactions cost on the bargaining; rational parties therefore should not fight. The framework’s apparatus inverts the order. The Coasean intuition assumes the existence and uniqueness of an efficient bargaining allocation; Theorem 5.8 identifies the structural conditions under which one exists. The Coasean reading then becomes a corollary of the geometric result: when zero- or negative-sum warfare and equilibrium peace are jointly available, the agreement Coase invokes is the peaceful equilibrium itself.

5.4 Sensitivity of the conflict externality

Proposition 5.3 established that Δ^* is a smooth scalar field on the regular set $\tilde{E}^{\text{reg}}$. This section develops the gradient. The result is the chapter’s signature dual-mandate capability in explicit form: a closed-form expression for how the welfare cost of conflict responds to any perturbation the framework admits, computed by one application of J^{-1} once that matrix has been formed at the equilibrium of interest. Neither the exchange-side partial nor the conflict-side partial delivers this calculation; the joint formulation does, by direct inheritance from Chapter 4.

The sensitivity formula. The cardinal welfare cost $\Delta^*(\omega) = v(\omega, u_{-I}^*) - v(r^*(\omega), u_{-I}^*(\omega))$ depends on ω through three places where the joint equilibrium enters: the conditional resource vector $r^*(\omega)$, the utility levels $u_{-I}^*(\omega)$ of the residual states, and the direct effect of ω on the global frontier itself. Differentiating at a regular equilibrium and collecting,

$$\begin{aligned} D_\omega \Delta^* &= \underbrace{D_r v(\omega, u_{-I}^*)}_{\text{direct resource}} - \underbrace{D_r v(r^*, u_{-I}^*) D_\omega r^*}_{\text{deployment-cost feedback}} \\ &\quad + \underbrace{[D_u v(\omega, u_{-I}^*) - D_u v(r^*, u_{-I}^*)] D_\omega u_{-I}^*}_{\text{reallocation}}. \end{aligned} \tag{5.10}$$

Equation (5.10) is the chapter’s master sensitivity identity. Three of its four chain factors are computed by one application of J^{-1} in Chapter 4’s master formula: $D_\omega m^*$ and $D_\omega p^*$ enter $D_\omega r^*$ through the cost function $\sum_i c_i(m_i^*, p^*, \omega_i)$, and $D_\omega \bar{x}^*$ enters $D_\omega u_{-I}^*$ through the conditional-FWT chain (Corollary 5.5). The remaining factors are derivatives of the value function v , computed at the two arguments (ω, u_{-I}^*) and (r^*, u_{-I}^*) from the FOCs of the maximization in Equation (5.1).

Three channels. The decomposition in Equation (5.10) reads intuitively. The direct-resource channel captures how an extra unit of ω raises the global frontier $\partial \bar{U}(\omega)$ that Δ^* measures from. The deployment-cost feedback channel captures the indirect effect: an extra unit of ω changes the Nash-equilibrium deployment m^* , which changes the cost $\sum_i c_i$, which changes the conditional resources r^* that Δ^* measures to. The reallocation channel captures the residual: as the equilibrium reallocates, the base utility levels u_{-I}^* at which the comparison is evaluated shift, sliding the comparison along both frontiers. The first channel is exchange-economic; the second is the distinctive joint-determination content; the third is a within-frontier reweighting.

The signs of the three channels need not agree. The direct channel is non-negative: more endowment cannot reduce the global frontier. The deployment-cost feedback can have either sign: more endowment might raise the equilibrium intensity of conflict (when deployment is a normal input to contest payoffs) or lower it (when richer states arm less). The reallocation channel typically has the sign of the welfare-weight redistribution: shifts in ω that move welfare weight toward states with higher marginal contribution to u_I pull the comparison favorably for state I , and the opposite otherwise. The total derivative aggregates the three; whether Δ^* rises or falls with ω at any equilibrium is an empirical question the formula answers, not a sign result the framework imposes.

The two-state contest: how the prize moves the cost. Example 3.3 computes the formula in closed form, with the prize V_0 as the parameter of interest. Section 5.1 established

$$\Delta^*(V_0, W, \theta) = \frac{1}{2} \log \left(\frac{W}{W - \theta(m^*)^2} \right), \quad m^* = \sqrt{V_0/(4\theta)},$$

so $\theta(m^*)^2 = V_0/4$ and $\Delta^* = \frac{1}{2} \log(W/(W - V_0/4))$. Differentiating in V_0 ,

$$D_{V_0} \Delta^* = \frac{1}{8W - 2V_0},$$

positive on the admissible range $V_0 < 4W$ and rising as V_0 approaches the boundary at which the deployment cost exhausts the good-2 endowment. The sign is the master formula's prediction in three channels: the direct-resource channel at fixed V_0 and the reallocation channel are zero (the prize does not enter ω directly nor shift other states' utility weights at the symmetric equilibrium), so the sign of $D_{V_0} \Delta^*$ is entirely the deployment-cost feedback channel—more prize, more arming, more deadweight, larger conditional-versus-global gap. The closed-form derivative recovers the same answer the master formula Equation (5.10) delivers in general.

Welfare flows along smooth paths. The sensitivity formula promotes Δ^* to a smooth scalar field with a computable gradient on the regular set. Two consequences follow.

First, welfare changes between two parameter profiles $\omega^0, \omega^1 \in \Omega_0 \setminus \Sigma$ connected by a smooth path are recovered by integration:

$$\Delta^*(\omega^1) - \Delta^*(\omega^0) = \int_0^1 D_\omega \Delta^*(\omega(t)) \cdot \dot{\omega}(t) dt$$

along any smooth $\omega : [0, 1] \rightarrow \Omega_0 \setminus \Sigma$ with $\omega(0) = \omega^0$ and $\omega(1) = \omega^1$. The integrand is the master formula Equation (5.10) evaluated along the path; the integral is path-independent within each regular regime by smoothness. Welfare comparisons between distant parameter profiles are therefore well-posed within a regime and reduce to the same matrix arithmetic the comparative-statics formula uses pointwise.

Second, the gradient field $D_\omega \Delta^*$ is itself the input to several downstream constructions. Chapter 6 consumes the gradient as the donor's welfare-improvement direction in the optimal-control formulation of peacemaking, where an admissible aid path on the equilibrium manifold minimizes integrated welfare cost. Chapter 7 reads policy as motion along Ω_0 and uses the gradient to identify directions in which institutional intervention reduces the welfare cost of conflict. Chapter 8 evaluates the formula on the canonical sanctions perturbations, with the joint comparative-static $D_\theta(p^*, m^*) = -J^{-1} D_\theta e$ as the apparatus the bargaining-camp and GE-camp slice formulations have been operating without. The chapter's smooth scalar field is one input that the rest of the book reuses.

What no partial can compute. Equation (5.10) cannot be obtained from either of the two partialled traditions the framework triangulates against. An exchange-side partialling treats deployments as exogenous and delivers a $D_\omega p^*$ that holds m^* fixed; the deployment-cost feedback channel is then identically zero and Δ^* degenerates to the resource-gain term alone. A conflict-side partialling treats prices as exogenous and delivers a $D_\omega m^*$ that holds p^* fixed; the cost-of-deployment terms then misread the equilibrium response because c_i is price-dependent (Chapter 2) and the cost feedback runs through both m^* and p^* jointly. The joint formulation reads the cross-channel terms by direct inheritance from Chapter 4’s master formula, and the sensitivity identity is the welfare consequence. What Chapter 4 did for prices and deployments $D_\omega(p^*, m^*)$, the present section does for the welfare cost $D_\omega \Delta^*$: a single matrix-vector operation produces the joint response of the chapter’s master scalar to any perturbation the framework admits.

5.5 Existence of peaceful equilibria and the implementability gap

Theorem 5.8 delivers a strong conclusion under two conditions: that the conflict technology is zero- or negative-sum, and that a peaceful joint equilibrium exists. The first condition is a structural restriction on the battle function; Definition 5.7 states it cleanly and the chapter has so far operated under it. The second condition has been treated as a hypothesis rather than as a substantive question. This section asks when the second condition holds. The answer turns on a corner-FOC question that Chapter 3’s machinery already settles in passing, and its consequence for the chapter’s normative apparatus is the content of the *implementability gap*: a peaceful Pareto improvement over a war equilibrium can exist on the global frontier $\partial \bar{U}(\omega)$ without being implementable as a peaceful equilibrium of the joint system. The gap has nothing to do with bargaining frictions in the Fearon [1995] sense; it is a property of the joint equilibrium concept’s primitives.

The conditional-existence premise. A peaceful joint equilibrium at ω is a triple $(p^*, 0, x^*)$ in $\tilde{E}$. At any such point, $m^* = 0$ is the deployment best-response profile: each state’s optimization at $(p^*, m_{-i}^* = 0, \omega_i)$ returns $m_i^* = 0$. The cost normalization $c_i(0, p^*, \omega_i) = 0$ (Chapter 2) and the battle-function normalization $s_i(0, x^*, \omega) = 0$ then collapse the equilibrium relations: $r^* = \omega$, $\bar{x}_i^* = x_i^*$, and the joint zero of e reduces to a Walrasian zero on the original endowments. The conditional FWT (Proposition 5.4) places the allocation on $\mathcal{P}(\omega)$, and Theorem 5.8 promotes the placement to global Pareto optimality under the zero/negative-sum condition.

The chain has a single load-bearing premise: $m^* = 0$ must itself be the deployment best-response at the candidate price p^* . Whether it is depends on the structure of the contest at the corner $m = 0$, an object Chapter 2’s primitives leave open.

The corner-FOC question. At the corner $(m_1, \dots, m_I) = 0$, the deployment FOC of Equation (2.3) reads as a comparison between the marginal cost of deploying force and the marginal contest gain from doing so. For state i , the deployment best-response selects $m_i = 0$ when

$$\limsup_{m_i \rightarrow 0^+} \frac{\partial s_i(m, x^*, \omega)}{\partial m_i} \Big|_{m_{-i}=0} \leq \frac{\partial c_i(m_i, p^*, \omega_i)}{\partial m_i} \Big|_{m_i=0^+}. \quad (5.11)$$

The right-hand side is the marginal cost of deploying the first unit of force at the prevailing price. The left-hand side is the marginal contest gain at the corner; the limsup formulation accommodates contest functions that are discontinuous at $m = 0$, of which the Tullock family is the canonical example.

When Equation (5.11) holds for every state, the corner $m = 0$ is a fixed point of the joint deployment best-response, and the conditional-existence premise of Theorem 5.8 is satisfied at the supporting Walrasian price p^* . When Equation (5.11) fails for some state, no peaceful joint equilibrium exists at ω : that state's deployment best-response selects $m_i > 0$, and the symmetric collapse to a Walrasian exchange economy fails.

The condition is structural in primitives. It depends on the cost function's behavior at the origin, on the contest function's behavior at the corner, and on the prevailing price p^* , but not on any feature of the bargaining process, the information structure, or the agents' commitment ability. Whether peace is reachable through the joint equilibrium concept is decided by the framework's primitives alone.

The contest at the corner: difference-form Tullock as canonical case. Example 3.3's difference-form Tullock illustrates how Equation (5.11) fails generically. At symmetric $m_1 = m_2 = m$, the contest pays each state zero in good 1: the difference $m_i - m_{-i}$ in the numerator of s_i vanishes. But the marginal contest gain at the symmetric ray is $\partial s_i^1 / \partial m_i = V_0 / (4m)$, which diverges as $m \rightarrow 0^+$. Equivalently, at the corner $m_2 = 0$, any $m_1 = \epsilon > 0$ shifts s_1^1 from zero to $V_0/2$: a discrete jump in good-1 holdings for vanishing marginal cost. State 1's deployment best-response to $m_2 = 0$ is therefore $m_1 > 0$. Equation (5.11) fails for both states at any $V_0 > 0$, and no peaceful equilibrium exists.

The fact is already implicit in Example 3.3's discussion: at $V_0 > 0$, the contest is always live. What the present section adds is to make the implication load-bearing for the chapter's normative apparatus. Theorem 5.8 applies to the example in the trivial case $V_0 = 0$ (when the contest disappears and $m^* = 0$ is the deployment best-response by inspection); it does not apply at $V_0 > 0$. The chapter's comparison of war to a peaceful counterfactual is therefore comparing the war equilibrium to a Walrasian benchmark *that is not itself a joint equilibrium of the framework*.

Finiteness of marginal cost as the structural variable. Equation (5.11) is a comparison of two limits. Whether it holds is governed by which side blows up faster as $m_i \rightarrow 0$ from above. On the cost side, $\partial c_i / \partial m_i$ at $m_i = 0^+$ is finite when c_i is differentiable at the corner (linear or smooth-and-bounded cost), and infinite when c_i has a vertical tangent at the origin (e.g., $c_i \propto \sqrt{m_i}$ or any cost with $c'(0) = +\infty$). On the contest side, the limit is finite when the contest is smooth at the corner (some logit specifications) and infinite when the contest has a discrete jump (the Tullock family, and any difference- or ratio-form variant).

Three regimes are possible. First, both limits finite, with cost dominant: peaceful equilibrium exists. Second, both limits finite, with contest dominant: peaceful equilibrium fails. Third, contest limit infinite: peaceful equilibrium fails unless the cost limit is also infinite and large enough to dominate. Example 3.3 sits in the third regime with finite cost, hence peaceful equilibrium fails. The structural-empirical follow-up to Chapter 8 can identify which regimes prevail under empirically plausible primitives. The qualitative point is that the question is decided by primitives the framework's joint equilibrium concept already supplies, with no auxiliary apparatus required.

The implementability gap. When Equation (5.11) fails, the chapter’s machinery identifies a Pareto improvement over the war equilibrium without producing a peaceful joint equilibrium that implements it. The improvement lives on the global frontier $\partial\bar{U}(\omega)$; the war equilibrium lives on the strictly-interior conditional frontier $\partial\bar{U}(r^*)$ at $r^* = \omega - \sum_i c_i(m_i^*, p^*, \omega_i) < \omega$ (Proposition 5.2). The gap between the two frontiers is $\Delta^*(\omega)$, the master scalar of Definition 5.1. A bargaining-theoretic reading of Δ^* would assert that the gap should be closable: there exists a peaceful allocation on $\partial\bar{U}(\omega)$ that Pareto-dominates the war allocation on $\partial\bar{U}(r^*)$, and rational states should be able to reach it.

The reading is correct as a frontier comparison. It is silent on the implementation question. The peaceful counterfactual on $\partial\bar{U}(\omega)$ is supported by the Walrasian allocation at ω with $m = 0$; the joint equilibrium concept of Chapter 3 requires that this allocation be a fixed point of the deployment best-response, which Equation (5.11) answers. When the corner-FOC fails, the bargaining-Pareto improvement is a point on the global frontier with no representation in $\tilde{E}$: implementing it would require modifying the framework’s primitives, not selecting among existing equilibria.

We name the set

$$\widetilde{W}_p(\omega) := \{ u(x^*) \in \mathbb{R}^I \mid (p^*, 0, x^*) \in \tilde{E} \}$$

the *peaceful-implementable subset* of utility profiles at ω . The implementability gap is the difference between $\widetilde{W}_p(\omega)$ and the bargaining range $\partial\bar{U}(\omega) \cap \{u : u \succeq u(\text{war})\}$. On Example 3.3 at $V_0 > 0$, $\widetilde{W}_p(\omega) = \emptyset$ and the gap equals the entire bargaining range. On other primitives, the gap can be a strict subset of the bargaining range, with intermediate cases admitting some peaceful-implementable utility profiles but not the full Pareto-improvement set.

The gap is a scope claim, not an impossibility claim. It says: implementing a bargaining-theoretic Pareto improvement when Equation (5.11) fails requires *something more* than the framework’s joint equilibrium concept provides. What that something is—contingent transfers backed by enforcement, repeated-game cooperation, third-party intervention, institutional commitment—belongs to chapters and frameworks that augment the joint equilibrium with the missing primitives. Chapter 6 treats third-party intervention as a control problem on the manifold; Chapter 7 treats institutional construction as the apparatus that selects among augmented equilibria; Chapter 8 treats coercion as parameter perturbation under exactly the implementability question the present section names. The chapter’s contribution is to make the question visible, not to answer it.

Bargaining theory’s three replies. A reader trained in the bargaining-theoretic tradition has three natural replies to the implementability gap. The chapter’s response to each is structural: the (p, m) formulation makes the implicit assumption of each reply visible, and shows what the apparatus contributes when the assumption is made explicit.

The first reply is that bargaining itself can implement the peaceful improvement: states will negotiate to a peaceful allocation in the bargaining range, and the implementability gap disappears in the limit of frictionless bargaining. This reply is correct on its terms but underspecified. *Which* bargaining protocol? The literature on the bargaining foundations of competitive equilibrium—Yildiz [2003], Dávila and Eeckhout [2008], Penta [2011], Harstad [2023], Alon and Lehrer [2019]—establishes that the limiting outcome depends sharply on which protocol is in force. Some protocols converge to Walrasian payoffs; others select inefficient or pledge-determined outcomes. The bargaining-theoretic intuition assumes a single protocol and takes its limit; the

joint formulation, which equilibrates (m, p) jointly, asks across the protocol space. Section 5.6 maps the structure of that space and locates Fearon’s bargaining-theory-of-war within it.

The second reply is that contingent transfers backed by enforcement can sustain peace. At the corner $m = 0$, state i ’s gain from defection is bounded—in the difference-form Tullock specification of Example 3.3, defection at $m_{-i} = 0$ delivers state i a discrete jump of $V_0/2$ in good 1 for vanishing marginal cost. A finite transfer from the rest of the system can compensate state i for foregoing the defection. Beviá and Corchón [2010] establishes the partial-equilibrium version of the question: when does a peace-sustaining transfer schedule exist, given fixed prices? The answer is sometimes no—the bargaining range can be empty even with transfers when defection profits exceed the system’s budget capacity. The joint-formulation extension asks the same question with prices endogenous. The transfer that sustains peace shifts the equilibrium price vector through the deployment-cost feedback channel of Equation (5.10); the price shift in turn shifts the defection profits and the budget constraints; peace-sustainability becomes a fixed-point condition. The condition can fail in a richer set of circumstances than the partial-equilibrium analysis identifies, because price feedback adds structural dependences the partialled framework cannot see. What the joint formulation contributes here is exactly the master sensitivity formula’s apparatus, evaluated at the bargaining perturbation rather than at an exogenous endowment shift.

The third reply is that bargaining-theoretic concavity—each state’s primitive utility is a concave function of consumption—suffices to rule out demand for randomized peace-versus-war outcomes. The reply is technically correct but not exhaustive. Three concavity conditions are distinguishable, and they are not equivalent. Primitive concavity is the condition the bargaining-theory literature explicitly assumes: u_i concave on consumption bundles. Reduced-form concavity over single-state endowments is a *consequence* of primitive concavity in single-stage choice problems, but is a *property of the equilibrium structure* in the joint formulation: state i ’s payoff at the joint equilibrium of $\tilde{E}$ as a function of i ’s resources can fail to inherit concavity from primitive concavity, because the equilibrium correspondence has regime-boundary kinks where one contest-regime gives way to another. Path-value-function concavity is the condition that an agent’s value over trajectories through $\tilde{E}$ is concave; it is the condition the bargaining-theoretic argument extends to when the choice space is generalized from single allocations to trajectories on the manifold, and it is a separable additional assumption beyond primitive concavity. The bargaining-theoretic literature has been operating under path-value-function concavity implicitly while assuming only primitive concavity explicitly. The (p, m) framework reveals the gap: regime-boundary structure on $\tilde{E}$, of the kind Chapter 3’s scenario (iii) identifies and Chapter 7 elaborates, can break path-value-function concavity even when primitive concavity holds. The chapter does not develop a worked actualization mechanism for the resulting demand for randomized outcomes; it identifies the assumption and flags the question for downstream applied work. What the joint formulation contributes here is to make the hidden assumption explicit, and to relocate the bargaining-theoretic argument’s premises within a framework where each premise can be named and assessed.

The two-state contest in light of the section. Example 3.3 at $V_0 > 0$ exhibits the implementability gap in its cleanest form. The deployment FOC Equation (5.11) fails at the corner; the chapter has been silently assuming the second premise of Theorem 5.8; the assumption fails on the canonical example. The peaceful Walrasian counterfactual at ω is computable in closed form: at reflected endowments $\omega_1 = (\omega_H, \omega_L)$, $\omega_2 = (\omega_L, \omega_H)$, the symmetric Walrasian

price is $p^{\text{peace}} = 1$, the peaceful allocation is $x_i^{1,\text{peace}} = x_i^{2,\text{peace}} = W/2$, and $u_i^{\text{peace}} = 2 \log(W/2)$ at log preferences. The war equilibrium is the symmetric solution of Example 3.3 at $V_0 > 0$ with $p^* = 2W/(2W - V_0) > 1$ and asymmetric consumption (each state holds relatively more of its own-specialty good in equilibrium because the war price favors holders of good 1, the deployment-burned good).

A direct calculation establishes Pareto dominance. At any admissible $V_0 > 0$,

$$\sum_i u_i^{\text{peace}} - \sum_i u_i^{\text{war}} \geq 2 \log p^* > 0,$$

where the inequality uses the AM–GM bound on the equilibrium budgets ($B_1^{\text{war}} B_2^{\text{war}} \leq W^2$ with equality at the symmetric Walrasian) and the strict inequality uses $p^* > 1$. At numerical anchor $\omega_H = 3$, $\omega_L = 1$, $V_0 = 1$, $k = 1$, the war equilibrium has $p^* = 8/7$, $B_1^{\text{war}} = 29/7$, $B_2^{\text{war}} = 27/7$, with utilities $u_1^{\text{war}} \approx 1.323$ and $u_2^{\text{war}} \approx 1.179$; the peaceful counterfactual delivers $u_1^{\text{peace}} = u_2^{\text{peace}} = 2 \log 2 \approx 1.386$. Both states are strictly better off at the peaceful counterfactual, strict on both sides, with state 2 (the good-2-rich state) gaining more because the war price disproportionately raises the cost of its preferred good.

The Pareto improvement is real and computable. The peaceful counterfactual is not in $\tilde{E}$. The implementability gap is non-empty—in fact, on this example it is the entire bargaining range, since $\widehat{W}_p(\omega) = \emptyset$. The bargaining-theoretic question of how to close the gap is exactly the question of which augmenting framework to invoke: a transfer scheme as in Beviá and Corchón [2010], with the joint-formulation extension’s price-feedback fixed point; a bargaining protocol from the cluster of Section 5.6; or a dynamic / repeated-game enforcement primitive the chapter does not develop. Each is a substantive addition to the framework’s primitives. The chapter’s contribution is the diagnosis, not the cure.

What no partial can compute. The implementability question requires the joint equilibrium concept of Chapter 3. A bargaining-side analysis treats the bargaining range as a primitive object and asks why states fail to implement it; this is Fearon [1995]’s framing. A general-equilibrium analysis treats deployments as exogenous and asks for the Walrasian allocation conditional on those deployments; this is the trade-and-gravity tradition’s framing. Neither analysis can ask whether the bargaining range admits a representation in the joint equilibrium manifold, because neither analysis works with $\tilde{E}$ as an object. The joint formulation, by contrast, makes the implementability question precise: a peaceful Pareto improvement is implementable if and only if its supporting allocation is a fixed point of the joint equilibrium correspondence with $m = 0$. Equation (5.11) is the corner-FOC test that decides the question. What the (p, m) framework contributes is the test, the gap it makes visible, and the catalog of replies the bargaining-theoretic tradition has been making implicitly without naming.

5.6 Bargaining theory and the joint formulation

The previous section identified an implementability gap and sketched three bargaining-theoretic replies. This section develops the first reply at depth. The literature on bargaining foundations of competitive equilibrium maps a structural space, parameterized by the bargaining protocol and by the setup of the underlying choice problem, in which the bargaining outcome converges to Walrasian for some configurations and fails for others. The map locates Fearon [1995]’s bargaining-theory-of-war as one specific case in the space, and it identifies what additional content the joint formulation contributes.

Two axes of the bargaining-foundations literature. The bargaining-implements-Walrasian project has had two methodological starting points, each producing a different sense in which the bargaining outcome equals the competitive outcome. The strategic axis takes a bargaining game—typically an alternating-offers game in the Rubinstein [1982] tradition—and asks whether its subgame-perfect equilibria converge to Walrasian as bargaining frictions vanish. Yildiz [2003] establishes a positive answer for two-agent exchange under a particular protocol where proposers offer prices and responders demand quantities at the offered price. Dávila and Eeckhout [2008] extends the result by adding maximum trade constraints, showing that without the constraint Yildiz’s sufficient conditions are generically violated, and with the constraint the convergence is generic across exchange economies. Penta [2011] generalizes the construction to multilateral bargaining with arbitrary numbers of agents and a wide class of protocols, with the auctioneer-as-residual-claimant playing the role the Arrow–Debreu market participant plays in the original existence proof. Harstad [2023] works a different protocol—each party proposes only its own dimension of the outcome—and shows the resulting pledge-and-review equilibria are generically inefficient, producing systematic departures from Walrasian even with full information and no commitment problems. Slantchev [2003] pushes the failure-of-convergence direction further: even with full information, primitive concavity, and divisible issues, alternating-offer bargaining over a divisible pie can have inefficient subgame-perfect equilibria sustained by a Folk-theorem-style mechanism in which the bargaining game’s within-disagreement-stage conflict allows each side to threaten particularly disadvantageous continuation play. The mechanism Slantchev names *the power to hurt*—the bargainers’ joint ability to inflict and bear costs in the disagreement stage—supports a continuum of inefficient SPE outcomes alongside the efficient one. The strategic axis is therefore not a single-answer literature: some protocols deliver convergence to Walrasian, others deliver generic inefficiency, and a third class delivers efficient and inefficient SPE multiplicity simultaneously.

The axiomatic axis approaches the same question through Nash [1950]-style cooperative bargaining theory. Alon and Lehrer [2019] characterize a market bargaining solution that recovers competitive equilibrium from an axiomatic system operating on resources and ordinal preferences rather than on utility-image bargaining problems. The setup matters: the same payoff structure, posed as a utility-image bargaining problem, admits no Walrasian-recovering solution by the impossibility result of Sertel and Yildiz; posed as a resource-based ordinal problem in the Roemer [1988] tradition, the market bargaining solution exists and is uniquely characterized. The two axes are not redundant. The strategic axis answers *which protocols implement Walrasian*; the axiomatic axis answers *which setups admit Walrasian-recovering solutions*.

What the literature establishes, taken together, is that bargaining-implements-Walrasian holds along a specific cross-section of a two-axis space. Strategic protocols with prices and maximum-trade constraints converge to Walrasian (DE 2008, Penta 2011); strategic protocols with allocation-offers converge to Nash bargaining solutions (Rubinstein 1982); strategic protocols where each agent proposes own-dimension are generically inefficient (Harstad 2023). Axiomatic setups on resources admit market bargaining solutions that match Walrasian (Alon–Lehrer 2019); axiomatic setups on utility-images do not (Sertel–Yildiz, the impossibility companion to Yildiz 2003). The cross-section where bargaining implements Walrasian is nontrivial but specific.

Where Fearon and Powell sit. The bargaining-theory-of-war tradition—Fearon [1995], Powell [2006], the literature downstream of them—takes the bargaining-Pareto comparison as a primitive and asks what prevents states from negotiating a peaceful allocation in the bargaining

range. The three rationalist explanations Fearon names—private information about types, commitment problems over future behavior, and indivisibility of the issue—are wedges that prevent the bargaining game from delivering the Pareto-improving allocation it would deliver in their absence. The framing assumes a single bargaining protocol, in which states exchange offers over allocations and accept-or-reject in Rubinstein-flavored manner. Within that protocol, in the absence of frictions, the bargaining outcome is the symmetric Nash bargaining solution; with the frictions, war becomes a possible equilibrium of the bargaining game.

Located in the two-axis space, the bargaining-theory-of-war operates at a specific point: strategic protocol with allocation offers, axiomatic setup over utility-image bargaining problems, primitive concavity of preferences, no other structural features on the bargaining environment. The (p, m) framework's contribution at this point is twofold. First, the implementability question of Section 5.5 is logically prior to the frictions-prevent-Pareto-improvement question Fearon poses: if Equation (5.11) fails, no peaceful equilibrium of the joint system exists, and the bargaining-theoretic frictions are not the explanation for war (war is the unique joint equilibrium even without them). Second, the operating point is one of many. The cluster of bargaining-foundations results identifies configurations where bargaining converges to Walrasian without any of Fearon's frictions, and configurations where bargaining fails to converge to Walrasian even with full information, perfect commitment, and divisible issues—the Harstad pledge-and-review case being the cleanest example. The bargaining-theory-of-war's framing locates one specific configuration and reads its features as the structural answer to why peace fails; the (p, m) formulation reads the same configuration as one specimen in a structurally richer space.

A bargaining-on-DCE model in the spirit of Beviá and Corchón. Having located the bargaining-theory-of-war within the bargaining-foundations literature, we make the comparison concrete by writing down a bargaining game on the framework's primitives and asking the convergence question directly. The model is bilateral and transfer-based, in the spirit of Fearon [1995] and Beviá and Corchón [2010]: states A and B negotiate directly with each other over a peaceful agreement, with the war joint equilibrium of $\tilde{E}$ as the disagreement payoff and the joint equilibrium of $\tilde{E}$ replacing the partial-equilibrium exchange economy as the underlying environment. Chapter 6 extends the model to the richer case in which a third-party donor sponsors the transfer; the bilateral case here is what that multi-actor extension specializes to when the donor is folded into one of the belligerents, and the present section's analysis of feasibility and enforceability carries over directly.

Setup. States A and B have concave preferences u_A, u_B , endowments ω_A, ω_B , and conflict primitives (s_i, c_i) as in Chapter 2. The war joint equilibrium $(p^*, m^*, x^*) \in \tilde{E}$ is the default outcome at $\omega = \omega_A + \omega_B$. The war payoffs are $u_i^{\text{war}} := u_i(x_i^*)$.

Bargaining proceeds in a single stage. A proposes a transfer $\tau \geq 0$ in numeraire to B . B accepts or rejects. On acceptance, both states commit to $m_A = m_B = 0$, the post-transfer endowments $(\omega_A - \tau \mathbf{e}_2, \omega_B + \tau \mathbf{e}_2)$ clear at the supporting Walrasian price $p^{\text{Walras}}(\tau)$, and the resulting peace allocation is implemented. On rejection, the war joint equilibrium plays out and each state receives u_i^{war} .

The model is one-shot for clarity; the alternating-offer extension in the Rubinstein [1982] tradition with discount $\delta \rightarrow 1$ delivers the same structural answer with the bargaining surplus allocated according to the relative-impatience rule, and the substantive question we address is independent of that allocation.

The model has the deliberate property that the bargaining *outcome* is a Walrasian allocation

at the post-transfer endowments and the bargaining *disagreement payoff* is a joint equilibrium of $\tilde{E}$. The joint formulation's contribution sits exactly at this asymmetry.

Failure I: feasibility under endogenous prices. The first question for any transfer-based bargain is whether an admissible transfer exists. B 's individual rationality constraint is

$$u_B^{\text{peace}}(\tau) := u_B(x_B^{\text{Walras}}(\tau)) \geq u_B^{\text{war}},$$

where $x_B^{\text{Walras}}(\tau)$ is B 's Walrasian allocation at post-transfer endowments and the supporting Walrasian price $p^{\text{Walras}}(\tau)$. Under primitive concavity, u_B^{peace} is continuous in τ on the admissible domain. A 's individual rationality is symmetric: $u_A^{\text{peace}}(\tau) \geq u_A^{\text{war}}$. The bargaining range is $\mathcal{T}(\omega) := \{\tau \geq 0 : u_i^{\text{peace}}(\tau) \geq u_i^{\text{war}} \text{ for } i \in \{A, B\}\}$. Beviá and Corchón [2010] establish that $\mathcal{T}(\omega)$ can be empty in partial-equilibrium settings where the Walrasian price is held fixed and only B 's post-transfer wealth changes with τ . The joint-formulation extension asks the same question with prices endogenous: $p^{\text{Walras}}(\tau)$ shifts as τ shifts, and both u_A^{peace} and u_B^{peace} respond through both the direct wealth channel and the indirect price channel.

Proposition 5.10 (Feasibility under endogenous prices). *The bargaining range $\mathcal{T}(\omega)$ is the solution set of a fixed-point condition on τ in which the Walrasian price $p^{\text{Walras}}(\tau)$ enters both individual rationality constraints. There exist primitives for which $\mathcal{T}(\omega) \neq \emptyset$ at fixed prices but $\mathcal{T}(\omega) = \emptyset$ under endogenous prices. The structural distinction is governed by the master sensitivity formula Equation (5.10) evaluated at the bargaining perturbation: when the deployment-cost feedback channel and the reallocation channel together push the price response in a direction that tightens both individual rationality constraints faster than the direct-resource channel relaxes them, the fixed-point condition has no solution.*

Proof sketch. The fixed-point structure is immediate: τ enters u_B^{peace} both directly (via B 's post-transfer wealth) and indirectly (via $p^{\text{Walras}}(\tau)$, which clears the post-transfer market). Differentiating the individual rationality constraint in τ yields the three-channel decomposition of Equation (5.10) applied at the bargaining perturbation, with the bargaining range's boundary being the locus where one of the constraints binds. The non-emptiness question reduces to the existence of an admissible τ satisfying the joint fixed-point. For the existence claim, take a primitive specification in which A is rich in good 1 (the deployment-burned good) at war and the post-transfer Walrasian shifts the price unfavorably for A : the price-channel cost to A then dominates the wealth-channel benefit to B for any positive τ , and no admissible transfer exists. The construction is parametric and is the joint-equilibrium analog of Beviá and Corchón's empty-bargaining-range parameter region. $\square$

Proposition 5.10's practical content is that the Beviá and Corchón result generalizes to GE under endogenous prices, and the generalization can fail in a richer parameter region than the partial-equilibrium analysis identifies. The price feedback adds structural dependences the partialled framework cannot see. What the joint formulation contributes is the master-sensitivity apparatus to characterize when feasibility holds, with Equation (5.10)'s three channels supplying the linearization.

Failure II: enforceability via the corner-FOC. Suppose the bargaining range is non-empty and a transfer $\tau^* \in \mathcal{T}(\omega)$ is selected by the bargaining protocol. The agreement specifies that both states deploy $m_i = 0$ and trade at the supporting Walrasian price. The framework's joint

equilibrium concept admits the agreed allocation as a peaceful joint equilibrium iff Equation (5.11) holds at the supporting Walrasian price. When the corner-FOC condition fails, the agreement requires each state to commit to $m_i = 0$ at a configuration where $m_i > 0$ is its deployment best-response. The agreement is not self-enforcing within the framework's primitives.

Proposition 5.11 (Enforceability via the corner-FOC). *For any $\tau^* \in \mathcal{T}(\omega)$, the agreed peace allocation $(p^{\text{Walras}}(\tau^*), 0, x^{\text{Walras}}(\tau^*))$ is an element of $\tilde{E}$ if and only if Equation (5.11) holds at $p^{\text{Walras}}(\tau^*)$. On primitives where Equation (5.11) fails, the bargaining game's outcome (when it exists) is a Walrasian allocation outside $\tilde{E}$: implementing it requires augmenting the framework with enforcement primitives the joint equilibrium concept does not include.*

Proof sketch. The if-and-only-if is direct from the definition of a peaceful joint equilibrium in Section 5.5: the corner $m = 0$ is a fixed point of the deployment best-response at $p^{\text{Walras}}(\tau^*)$ iff Equation (5.11) holds, and a peaceful joint equilibrium is exactly a fixed point of the joint best-response correspondence at the supporting Walrasian price. On Example 3.3 at $V_0 > 0$, the corner-FOC fails for both states, so no τ^* delivers a DCE. The bargaining outcome (assuming feasibility) is a peaceful Walrasian allocation that lives outside $\tilde{E}$. $\square$

The two propositions together say that bargaining-converges-to-Walrasian in the Yildiz [2003] / Dávila and Eeckhout [2008] / Penta [2011] sense delivers an outcome the joint formulation does not represent as a joint equilibrium. The bargaining-foundations literature is operating in a framework that takes the bargaining problem as primitive and asks for an implementation; the joint formulation is operating in a framework that takes the joint equilibrium as primitive and asks whether the bargaining-Pareto improvement has a representation. Proposition 5.10 answers when the bargaining problem itself is well-posed under endogenous prices; Proposition 5.11 answers when the resulting agreement, if reached, is in $\tilde{E}$.

Where Fearon's frictions sit on this model. A reader trained in the bargaining-theory-of-war tradition would locate Fearon's three frictions on this model as follows. The *commitment problem* appears as the corner-FOC failure of Proposition 5.11: the Fearonian observation that states cannot credibly commit to $m_i = 0$ is, in the joint formulation, the structural observation that Equation (5.11) is not satisfied at the supporting Walrasian price. The friction is recoded into framework-level primitives: deployments, contest functions, and cost structures jointly determine whether the corner is a fixed point of the deployment best-response. The *private information* friction appears as uncertainty about the disagreement payoffs u_i^{war} , which in the joint formulation are functions of $\tilde{E}$'s component structure: when $\tilde{E}$ has multiple regime components separated by Σ (Chapter 3 scenario (iii)), private information about which component is operative becomes structural uncertainty about the manifold's regime structure. The *indivisibility* friction appears as the topology of the bargaining range $\mathcal{T}(\omega)$: when the conditional Pareto frontier $\partial\bar{U}(r^*)$ has a discrete jump at a contested-good threshold, $\mathcal{T}(\omega)$ can fail to be connected even when non-empty, and the bargaining game has no single-allocation outcome that both sides prefer to war. A fourth friction, named in the formal literature after Fearon's trio and not on Fearon's list, is the *power-to-hurt* mechanism of Slantchev [2003]: even with all three of Fearon's frictions absent, the bargaining game can have inefficient SPE supported by within-stage conflict play that threatens particularly disadvantageous continuation values to deviators. On the joint formulation, the power to hurt is the contest-and-cost primitives (s_i, c_i) themselves: state i 's ability to inflict costs on j via deployment, and i 's ability to bear j 's deployments, are exactly the primitives the deployment best-response and the conflict-feedback matrix R_i of Chapter 2 encode. Slantchev

[2003]’s mechanism is recoded as a structural property of (s_i, c_i) supporting punishment threats within the bargaining game’s internal SPE-multiplicity. The recoding sits at a third structural layer of the model, distinct from the corner-FOC failure of Proposition 5.11 (joint-equilibrium fixed-point level) and from the feasibility failure of Proposition 5.10 (transfer-feasibility level). None of the four frictions disappears on the joint formulation; each is recoded as a specific structural condition the framework names directly, and the three layers of the model accommodate the recodings without collapsing them into a single mechanism.

The two-state contest in light of the model. Example 3.3 at $V_0 > 0$ provides a worked instance of both propositions. On feasibility (Proposition 5.10): with log preferences and reflected endowments, the post-transfer Walrasian price retains symmetry and is independent of τ ; the fixed-point condition for $\mathcal{T}(\omega)$ collapses to the linear bound of partial equilibrium, and the bargaining range is non-empty for sufficient V_0 . The example does not exhibit feasibility failure—the log-Cobb-Douglas symmetry suppresses the price-feedback channel Proposition 5.10 relies on. On enforceability (Proposition 5.11): the corner-FOC fails at any $V_0 > 0$, as Section 5.5 establishes; therefore no τ^* delivers a DCE, and the bargaining game’s outcome (if reached) is a peaceful Walrasian allocation outside $\tilde{E}$.

The example exhibits the second failure cleanly while showing the first only in its limiting form. A primitive specification with non-log preferences and asymmetric endowments—the natural extension—would activate the price-feedback channel of Proposition 5.10 and deliver an example where both failures bind simultaneously. The structural-empirical follow-up to Chapter 8 is the natural setting for such an example; the present chapter’s purpose is to identify the questions, not to exhaust the parameter space.

What the joint formulation adds. The joint formulation contributes, beyond what the bargaining-foundations literature establishes on its own, the prior question of whether the bargaining-Pareto allocation has a representation in $\tilde{E}$. The cluster of bargaining results operates on a framework that takes the bargaining problem as primitive and asks for an implementation; the (p, m) formulation operates on a framework that takes the joint equilibrium as primitive and asks whether the bargaining-Pareto improvement has a representation. The two readings are not in conflict; their primitives differ, and the relationship between them is the substantive question Section 5.5 and the present section have named at framework level (Propositions 5.10 and 5.11) and at literature level (the cluster, Fearon’s three frictions, and Slantchev’s fourth).

Path-value-function concavity beyond the bargaining game. The path-value-function distinction introduced in Section 5.5 matters precisely where the bargaining-foundations cluster operates. A bargaining game’s outcome is a single allocation; primitive concavity of preferences delivers the standard risk-aversion-prefers-certainty argument that supports the single-allocation Pareto comparison. A trajectory through $\tilde{E}$ is not a single allocation but a path across regime components; the value an agent assigns to the path depends on which component the trajectory terminates in, and the component-boundary structure can introduce non-concavities in the path-value function even with primitive concavity. The bargaining-theory-of-war’s argument extends from single-allocation choice to bargaining-over-paths only if path-value-function concavity holds, an assumption Fearon’s apparatus does not name and the joint formulation reveals can fail. What this opens is a class of questions about bargaining over trajectories that the literature has not formulated; the chapter does not develop the worked apparatus, but flags the class as a

natural extension of the framework's normative machinery.

The bargaining-theory bridge, recovered. Theorem 5.8 relocated Fearon [1995]'s inefficiency-of-war assumption from premise to theorem, under explicit conditions on the conflict technology and on the existence of a peaceful equilibrium. Section 5.5 has examined the second condition and made the corner-FOC test that decides it explicit. The present section has located the bargaining-theory-of-war within the bargaining-foundations literature and identified the substantive content the joint formulation contributes: the prior implementability question, the path-value-function distinction, and the structural map of when bargaining converges to a peaceful joint equilibrium.

Two readings of the bridge are coherent. Read narrowly, the chapter establishes that bargaining theory's inefficiency-of-war claim is correct as a frontier comparison and that its implementation question has substantive structural content the bargaining literature has been operating without. Read broadly, the chapter relocates the question of why states fail to bargain to peace from the level of frictions on the bargaining game to the level of primitives on the joint equilibrium concept. Whichever reading the reader prefers, the chapter's purpose at this point is achieved: the apparatus the rest of the book develops—third-party intervention as control on the manifold (Chapter 6), institutions as augmentation primitives (Chapter 7), sanctions as parameter perturbation under the implementability question (Chapter 8)—inherits a normative theory that names its own scope, identifies its own implicit assumptions, and stakes its substantive contribution in the gap between primitive concavity and the bargaining-theory-of-war's full argument.

5.7 Looking ahead

The normative theory delivered by this chapter is, in summary form, a single cardinal scalar. $\Delta^*(\omega)$ records the welfare cost of conflict at the joint equilibrium, the gap between two Pareto frontiers: the global frontier at the full endowment, and the conditional frontier at resources actually available after deployment costs are paid. Proposition 5.4 places the equilibrium on the conditional frontier. Proposition 5.6 names what conditioning costs as the gap between the social planner's and the Nash first-order conditions on deployment. Theorem 5.8 identifies the class of cases where the cost vanishes. Equation (5.10) delivers the gradient as one application of J^{-1} . The chapter's content is one scalar in many positions.

Subsequent chapters consume the apparatus in graduated ways. Chapter 6 runs Δ^* as the donor's welfare-improvement target in the optimal-control formulation of peacemaking, with the gradient field $D_\omega \Delta^*$ supplying the natural direction of welfare-improving aid. Chapter 7 reads $D_\omega \Delta^*$ as a gradient field on the regular set and uses its geometry to identify directions in which institutional intervention reduces the welfare cost of conflict. The same chapter consumes the peaceful-efficiency theorem as the geometric content of regime change: a transition between a peace regime where $\Delta^* = 0$ and a conflict regime where $\Delta^* > 0$ is a bifurcation across the singular locus of the joint manifold. Chapter 8 evaluates Δ^* and its gradient on the canonical sanctions perturbations, deriving the sender's welfare cost $\partial \Delta_S^* / \partial \theta$ and target's $\partial \Delta_T^* / \partial \theta$ as joint comparative-statics rather than as exogenous parameters.

The chapter admits two readings. Read as welfare economics, it is a translation of the First Welfare Theorem into the joint setting. The price system clears the residual allocation problem at fixed deployments, the welfare cost of conditioning takes a Greenwald and Stiglitz form, and the externality vanishes on a structurally-identifiable class. Read as international relations, the

chapter relocates the bargaining model's standard premise from assumption to theorem and makes its implementation question structurally precise. Theorem 5.8 establishes Fearon [1995]'s inefficiency-of-war claim as a result of the joint formulation rather than a postulate. Section 5.5 characterizes when the theorem's existence-of-peaceful-equilibrium premise holds and identifies the implementability gap when it fails. Section 5.6 locates the bargaining-theory-of-war within the bargaining-foundations literature, develops a bilateral bargaining-on-DCE model whose two propositions name feasibility and enforceability as distinct structural failures, recodes Fearon's three rationalist frictions plus Slantchev [2003]'s fourth as conditions on framework primitives, and names the prior implementability question and path-value-function distinction the joint formulation contributes. Both readings are views of the same smooth scalar field Δ^* , augmented now with explicit scope conditions on its substantive claims.

Exercises

1. Δ^* in closed form on Example 3.3. Proposition 5.2 establishes that the conflict externality $\Delta^*(\omega)$ is non-negative at every joint equilibrium, and strictly positive whenever the joint equilibrium has positive deployment costs. Example 3.3 provides closed forms for the joint equilibrium (p^*, m^*) , so we can compute Δ^* directly. Use the closed-form deployment $m^* = (V_0 + \theta W)/(2k(2 + \theta))$ and equilibrium price $p^* = (2 + \theta)W/(2W - V_0)$ to compute the post-conflict resource profile $r^* = \omega - \sum_i c_i(m_i^*)$, and from this compute $v(\omega, u_{-I}^*)$ and $v(r^*, u_{-I}^*)$ via the log-preference Walrasian solution. Verify $\Delta^*(\omega) > 0$ on the admissible parameter range, in agreement with Proposition 5.2.
2. *Monotonicity of the value function in resources.* The value function $v(r, u_{-I})$ defined in Equation (5.1) is the maximal utility available to state I when the resource profile is r and the other states' utilities are held at u_{-I} . Body cites Balasko [2009, Proposition A.5.1] for monotonicity of v in r . Verify $D_r v(r, u_{-I}) > 0$ at any interior (r, u_{-I}) where the constraint set defining v is non-empty. The argument runs through strict monotonicity of u_i from Assumption 2.1 of Chapter 2: at the optimum of Equation (5.1), relaxing r in any coordinate ℓ slackens the resource constraint, and strict monotonicity of u_I makes the relaxation strictly utility-improving. Standard envelope theorem plus strict monotonicity.
3. *Singleton on the conditional Pareto frontier.* The conditional Pareto frontier $\mathcal{P}(r)$ at resource profile r is the set of allocations that are Pareto optimal subject to $\sum_i x_i = r$. Show that $\mathcal{P}(r) \cap \{x : u_i(x_i) = u_{-I,i}^*\}_{i < I}$ is a singleton under strict concavity of every u_i (which Assumption 2.1(2) of Chapter 2 delivers via smooth strict quasiconcavity). The argument is a standard convex-analysis one: the constraint set $\{u_i(x_i) = u_{-I,i}^*\}$ is a convex level set under strict concavity, $\mathcal{P}(r)$ at fixed r is convex and compact, and the intersection has a unique maximizer of u_I by strict concavity. The singleton property is what makes Δ^* well-defined as a scalar.
4. *Welfare weights λ^* on the symmetric equilibrium.* Corollary 5.5 identifies the welfare weights λ^* as the supporting hyperplane normal to the conditional Pareto frontier $\mathcal{P}(r^*)$ at the joint equilibrium allocation x^* . Compute λ^* explicitly at the symmetric equilibrium of Example 3.3. By the reflection symmetry $\omega_1 = (\omega_H, \omega_L)$, $\omega_2 = (\omega_L, \omega_H)$, the equilibrium allocation x^* is also symmetric, and the supporting hyperplane has equal weights: $\lambda_1^* = \lambda_2^* = 1/2$ after normalization. Verify this directly. The exercise illustrates how reflection

symmetry forces the welfare weights to symmetric values, eliminating the reallocation channel of the sensitivity decomposition (Exercise 11).

5. *Conditional FWT in a 2-state, 2-good Cobb-Douglas case.* Proposition 5.4 establishes that the Walrasian equilibrium at the post-conflict resource profile r^* lies on the conditional Pareto frontier $\mathcal{P}(r^*)$, the Smith-style first-welfare-theorem statement applied to the diminished-resource economy. Verify the proposition by direct calculation in a $I = 2$, $L = 2$ Cobb-Douglas specialization: preferences $u_i(x_i^1, x_i^2) = (x_i^1)^{\alpha_i} (x_i^2)^{1-\alpha_i}$, generic endowments. Solve the Walrasian equilibrium at the post-conflict resource profile r^* explicitly: prices, demands, and allocation. Verify the resulting allocation lies on $\mathcal{P}(r^*)$ by checking the FOC $\nabla u_i(x_i^*) \parallel p^*$ across all states. Routine; the exercise instantiates the conditional FWT on a worked specification.
6. *Cross-state externality FOC nonzero.* Proposition 5.6 establishes that the joint equilibrium is generically Pareto-inefficient, with the inefficiency tracked by the cross-state externality FOC Equation (5.9). The substance of the inefficiency is that each state's deployment choice imposes externalities on others' utility, and the equilibrium does not internalize these. For Example 3.3, compute the cross-state derivatives $D_{m_i} s_j$ for $j \neq i$ at the symmetric equilibrium directly from the Tullock contest formula. Substitute into Equation (5.9) and verify the FOC is non-zero; in particular, $D_{m_i} s_j = -D_{m_i} s_i$ at the symmetric two-state contest by zero-sum, so the cross-state externality is the negative of the own-arming benefit. Conclude that the conflict externality of Proposition 5.6 is substantive.
7. *Beyond zero/negative-sum: when can war "create"?* Theorem 5.8 requires the battle technology to be zero- or negative-sum (Definition 5.7) for peace to be globally Pareto optimal. The hypothesis is restrictive: it rules out the possibility that warfare itself creates aggregate resources. Explore this hypothesis. Construct a battle function s violating zero- and negative-sum (so $\sum_i s_i(m, x, \omega) > 0$ in some component for some $m \neq 0$), and exhibit a non-peaceful joint equilibrium that is globally Pareto optimal. Substantively: the construction asks when warfare can make the aggregate strictly better off, and the answer requires a non-trivial coordination benefit from coercive deployment that exceeds its resource cost (e.g., a positive-sum public-good production through joint defense capacity). Open-ended.
8. *Strict inequality version of Theorem 5.8.* The statement of Theorem 5.8 delivers *weak* global Pareto optimality of peace: no globally feasible $(\bar{y}, m')$ weakly Pareto-dominates the peaceful allocation x^* . A natural strengthening is strict Pareto optimality. Extend Theorem 5.8 to strict inequality under *strict* negative-sum warfare: $\sum_i s_i(m, x, \omega) < 0$ strictly for $m \neq 0$. Show that peace strictly Pareto-dominates every non-peaceful alternative. Identify where Case 2 of the body proof needs the strict-inequality adjustment: the inclusion $\bar{\mathcal{U}}(r') \subset \mathcal{U}(\omega)$ becomes a strict containment $\bar{\mathcal{U}}(r') \subsetneq \mathcal{U}(\omega)$, so $u(\bar{y})$ is strictly inside, and any weak Pareto domination by an interior point contradicts the boundary placement of $u(x^*)$.
9. *Zero-sum verification on Example 3.3.* Theorem 5.8's conclusion—peace is globally Pareto optimal—requires the battle function to be zero- or negative-sum (Definition 5.7). Verify directly that the Tullock contest of Example 3.3 is zero-sum in good 1: compute $s_1^1(m, x, \omega) + s_2^1(m, x, \omega)$ and confirm it equals zero (modulo the prize component proportional to V_0 , which is a transfer not a resource). Conclude using Theorem 5.8 that any peaceful

- equilibrium would be globally Pareto optimal at ω . Then show that no peaceful equilibrium exists at $V_0 > 0$ in this example (the corner-FOC fails—see Exercise 14), and conclude that the example sits in the interesting regime where peace is optimal but not implementable.
10. *$D_W \Delta^*$ and the three channels.* Equation (5.10) decomposes the endowment sensitivity of Δ^* into three channels: direct (resource-base shift), cost-feedback (price-induced shift in extraction cost), and reallocation (welfare-weight shift). Compute $D_W \Delta^*$ for Example 3.3 in closed form using the closed-form $r^* = \omega - 2km^*$ from Chapter 3. Decompose the result into the three channels and identify which contribute non-trivially. The reallocation channel vanishes by the symmetric weights of Exercise 4; the direct and cost-feedback channels are non-trivial and have determinate signs. The exercise quantifies how the conflict externality responds to a uniform resource boom on the chapter's running example.
 11. *Activating the reallocation channel.* The reallocation channel of $D_\omega \Delta^*$ vanishes under reflection symmetry of Example 3.3 because the symmetric welfare weights λ^* are unchanged by a uniform resource shift. Construct an example in which the reallocation channel has nontrivial sign. Three sources of asymmetry to consider: (a) preference asymmetry across states (unequal Cobb-Douglas weights α_i); (b) endowment asymmetry breaking the reflection $\omega_1 = (\omega_H, \omega_L)$, $\omega_2 = (\omega_L, \omega_H)$; (c) contest-parameter asymmetry (asymmetric Tullock alliance vector a_{ij}). Identify the minimal asymmetry that activates the channel, and compute its sign on the resulting example. Open-ended.
 12. *Bergson-Samuelson representation of Δ^* .* Corollary 5.5 identifies λ^* as the supporting hyperplane normal to $\mathcal{P}(\omega)$ at the peaceful counterfactual. Show that under a Bergson-Samuelson social welfare functional $\mathcal{W}(u_1, \dots, u_I) = \sum_i \lambda_i^* u_i$ with these weights, the welfare gap a planner would compute between peace and the joint equilibrium equals Δ^* . The argument follows from linearity of $\mathcal{W}$: at the peaceful counterfactual, $\mathcal{W}^{\text{peace}} = \sum \lambda_i^* u_i^{\text{peace}}$; at the joint equilibrium, $\mathcal{W}^{\text{war}} = \sum \lambda_i^* u_i^{\text{war}}$; the difference $\mathcal{W}^{\text{peace}} - \mathcal{W}^{\text{war}}$ equals Δ^* by the supporting-hyperplane geometry of Corollary 5.5. The exercise certifies that the chapter's cardinal scalar has a Bergson-Samuelson interpretation.
 13. *Path-independence of Δ^* within a regime.* Proposition 5.3 establishes that Δ^* is a smooth single-valued function on each regular component of $\tilde{E}^{\text{reg}}$, so $D_\omega \Delta^*$ is well-defined and the integrated change in Δ^* along any path internal to a regime equals the difference of endpoint values. Verify this directly. Pick two endowment profiles $\omega^0, \omega^1 \in \Omega_0$ in the same regular component of $\tilde{E}^{\text{reg}}$. Integrate $D_\omega \Delta^*$ along a smooth path $\omega(t)$ from ω^0 to ω^1 and verify $\int_0^1 D_\omega \Delta^*(\omega(t)) \cdot \dot{\omega}(t) dt = \Delta^*(\omega^1) - \Delta^*(\omega^0)$ independently of the path chosen. The result is the fundamental theorem of calculus on a smooth manifold; the substantive content is that crossings of the singular locus Σ are the only way to break path-independence.
 14. *Corner-FOC verification on Example 3.3.* Equation (5.11) is the corner-FOC condition for peace to be a joint equilibrium: at $m = 0$, the marginal benefit of starting to deploy must not exceed the marginal cost. Verify the condition for Example 3.3 at $V_0 = 0$ (peace exists, the corner-FOC must hold) and at $V_0 > 0$ (peace fails, the corner-FOC must fail). The contest's difference-form denominator $1/(m_1 + m_2)$ produces an unbounded marginal spoil at the corner $m = 0$, which is what fails the corner-FOC at $V_0 > 0$. The exercise identifies the specific structural feature of the Tullock contest that defeats the implementability of peace.

15. *Vertical-tangent cost defeats the corner.* The $1/(m_1 + m_2)$ singularity of the Tullock contest at $m = 0$ creates an unbounded marginal spoil that defeats the corner-FOC. A natural remedy is a cost function with vertical tangent at $m = 0$, so that the marginal cost of deploying matches the unbounded marginal spoil. Construct a contest function s together with a cost c_i satisfying $c'_i(0) = +\infty$ (e.g., $c_i(m_i) = m_i^{1/2}$ near $m = 0$) such that the corner-FOC condition holds for all V_0 in an open interval. Verify a peaceful joint equilibrium exists on that interval. The exercise illustrates how a fixed-mobilization cost in c_i (Assumption 2.3 of Chapter 2) can defeat the marginal-spoil incentive at the peaceful corner. Substantively: the structural variable that controls implementability is the relative behavior of marginal benefit and marginal cost at $m = 0$.
16. *Walrasian counterfactual and the $2 \log p^*$ bound.* On Example 3.3 at $\theta = 0$, peace is globally Pareto optimal (Exercise 9) but not implementable (Exercise 14), so the Walrasian equilibrium at the original endowments is the natural counterfactual. Compute it: solve the symmetric Walrasian on $\omega_1 = (\omega_H, \omega_L)$, $\omega_2 = (\omega_L, \omega_H)$ to get $p^{\text{Walras}} = 1$ and $x_i^{\text{Walras}} = (W/2, W/2)$ for both states. Verify this Pareto-dominates the joint war equilibrium directly by comparing utilities. Compare the welfare gap to the AM-GM bound $2 \log p^*$ (where $p^* = 2W/(2W - V_0)$ is the war equilibrium price), and identify the slack between the substantive welfare gap and the AM-GM bound. The exercise quantifies the welfare cost of war on the chapter's running example.
17. *Implementability gap as strict subset of bargaining range.* Example 3.3 sits in the regime where the implementability gap (Section 5.5) coincides with the entire bargaining range: every feasible bargain is unenforceable as a joint equilibrium. This is a feature of the $1/(m_1 + m_2)$ singularity. Identify a primitive specification where the gap is a strict subset of the bargaining range, so that some feasible bargains are enforceable as joint equilibria. Hint: a smooth contest function at the corner (s_i smooth at $m = 0$ rather than the difference-form singularity) opens an intermediate regime where the corner-FOC holds for some but not all $\tau \in T(\omega)$. Open-ended.
18. *Contingent transfer schedule and the enforcement primitive.* The implementability gap creates a substantive role for contracts: state 1 can offer to transfer τ to state 2 conditional on $m_2 = 0$, providing a non-equilibrium mechanism that closes the gap. Sketch this contingent transfer schedule on Example 3.3 with linear cost. Compute the minimum τ that makes state 2 indifferent between disarming under the contract and defecting (best-responding to the transfer-conditional schedule). Then discuss what enforcement primitive the transfer requires that the joint equilibrium concept does not include: third-party verification of m_2 (the framework has no audit technology), binding-commitment technology on the transfer (the framework has no contract enforcement), or both. The substantive content is that closing the implementability gap requires extra-equilibrium machinery the (p, m) framework deliberately does not formalize. Open-ended.
19. *Yildiz vs. Rubinstein convergence.* The bargaining-foundations literature contains two canonical strategic-protocol models that yield different convergence results in the patient limit. Yildiz [2003]'s alternating-offer-with-prices SPE converges to the Walrasian allocation as $\delta \rightarrow 1$; Rubinstein [1982]'s allocation-offer SPE does not. On a two-good Cobb-Douglas exchange economy, verify both convergence results directly. Identify which feature of the protocol (offer space includes prices, or is restricted to allocations) delivers Walrasian

convergence in the first case but not the second. The exercise locates the framework's natural connection point with strategic-foundations bargaining theory: the protocols that converge to Walrasian are the ones that price the trade explicitly.

20. *Davila-Eeckhout convergence region.* Dávila and Eeckhout [2008]'s max-trade-constraint protocol provides an alternative route to Walrasian convergence that applies under conditions complementary to Yildiz's. Compute a parameter region in which the Davila-Eeckhout protocol converges to the Walrasian allocation on a two-good economy where Yildiz's conditions are violated. Hint: pick endowments off the anti-diagonal of the Edgeworth box, where the protocols' restrictions on offer space work differently. The exercise illustrates the substitutability of restrictive features in the strategic-protocol axis: trading offer-space size for trade-size restrictions yields the same Walrasian limit on a different parameter region.
21. *Locating Fearon's three frictions.* Fearon [1995] organizes rationalist explanations for war around three frictions: commitment problems, private information, and indivisibilities. Place these on the two-axis space of Section 5.6 (strategic-protocol axis vs. axiomatic-setup axis). Identify which combinations rule out which frictions: do Davila-Eeckhout/Penta protocols rule out commitment problems by the structure of the disagreement continuation game? Do Alon-Lehrer axiomatic setups rule out indivisibility by the divisibility postulate of the bargaining domain? Open-ended; one paragraph per friction. The exercise locates each friction in the bargaining-tradition literature and connects to the chapter's two-axis architecture.
22. *Non-concavity of path value on a multi-component $\tilde{E}$.* The path value of a smooth path through $\tilde{E}^{\text{reg}}$ is the integral of a smooth function on the path, hence smooth and generically well-behaved on each regular component. Across the singular locus Σ (under Assumption 3.17), the path-value function may inherit non-concavity even when primitive utilities are concave. Construct an example in which the path value on a multi-component $\tilde{E}$ is non-concave. The construction is open: see Carroll's working paper on financial-equilibrium sunspots for a 1-D specialization with added financial-market machinery that exhibits this phenomenon. The general construction is an open question of the framework. Open-ended.
23. *Bargaining range $T(\omega)$ for Example 3.3.* Proposition 5.10 establishes that the bargaining range $T(\omega)$ of the bargaining-on-DCE model is non-empty in the log-Cobb-Douglas case. Verify the claim on Example 3.3 by direct computation. Derive closed forms for $u_B^{\text{peace}}(\tau)$ and $u_A^{\text{peace}}(\tau)$ under log preferences, where τ is the transfer parameter; identify the binding individual-rationality constraint (whichever of A, B has the tighter IR at the war equilibrium); compute the maximum admissible τ . Verify $T(\omega)$ is non-empty by computing the interval of admissible τ explicitly. The exercise instantiates Proposition 5.10 on the chapter's running example.
24. *Activating the price-feedback channel in $T(\omega)$.* Under log preferences and symmetric endowments, the supporting Walrasian price is invariant in τ , and the price-feedback channel of Proposition 5.10 is dormant. Construct a primitive specification with non-log preferences and asymmetric endowments that activates the channel. Compute $T(\omega)$ under fixed prices (the partialled bargaining range) and under endogenous prices (the joint range); exhibit a parameter region where $T(\omega)$ is non-empty under the first and empty under the second. Substantive content: ignoring price-feedback can make the bargaining range

look feasible when it is not, and the joint formulation diagnoses the discrepancy explicitly. Open-ended.

25. *Enforceability fails on Example 3.3.* Proposition 5.11 establishes that even when the bargaining range $T(\omega)$ is non-empty, no $\tau \in T(\omega)$ delivers a peaceful joint equilibrium because the corner-FOC fails at the supporting Walrasian price. Verify the proposition on Example 3.3 by direct calculation. Pick any $\tau^* \in T(\omega)$ from Exercise 23, compute the supporting Walrasian price $p^{\text{Walras}}(\tau^*)$, and check that the corner-FOC condition fails at $p^{\text{Walras}}(\tau^*)$ for both states. Conclude that no τ^* delivers a peaceful joint equilibrium even though every τ^* delivers a feasible bargain. The exercise instantiates the chapter's implementability-gap proposition on a worked example.
26. *Fearon commitment problem in the bargaining-on-DCE model.* Fearon's commitment problem is the friction that bargained agreements may not be self-enforcing once parties have committed the bargained transfers. Trace the friction through the bargaining-on-DCE model. Identify the specific corner-FOC condition (Equation (5.11)) that makes the agreed $m = 0$ self-enforcing on a primitive specification with vertical-tangent costs $c'_i(0) = +\infty$ (Exercise 15). Show that the commitment problem disappears on this specification: the corner-FOC holds, $m = 0$ is a Nash response to itself, and the bargaining outcome is enforceable. The exercise locates Fearon's commitment problem in the framework's own corner-FOC vocabulary, and identifies the structural primitive that resolves it.
27. *Fearon private-information friction with regime uncertainty.* Fearon's private-information friction is that parties may have private types governing payoffs, and a bargained transfer schedule that screens types may also produce inefficient outcomes. Adapt the friction to the framework. In a multi-component $\tilde{E}^{\text{reg}}$ with regimes separated by Σ (under Assumption 3.17), formalize "uncertainty about which regime is operative" as a probability distribution over components, and derive the bargaining-on-DCE outcome under ex-ante uncertainty about u_i^{war} . Specifically: parameterize the prior, compute the ex-ante expected war payoff $\mathbb{E}[u_i^{\text{war}}]$ as the integral of regime-conditional payoffs against the prior, and show the bargaining outcome shifts with the prior. Open-ended; the formalization itself is the substance.
28. *Slantchev (2003) adapted to the (p, m) framework.* Slantchev [2003]'s two-state alternating-offer-with-disagreement -game model embeds a Nash deployment subgame as the disagreement continuation, generating inefficient SPEs under appropriate punishment-threat conditions. Adapt the model to the joint (p, m) framework. Replace Slantchev's disagreement game Γ with the deployment best-response game from Chapter 2, and replace the payoff structure $r_i(a_1, a_2)$ with $u_i(x_i^*(m, p))$ at the corresponding joint equilibrium. Verify that Slantchev's inefficient-SPE construction goes through with the modified primitives, provided the modified disagreement game has bounded minimax payoffs (Exercise 29) and the punishment-threat machinery still works. Open-ended; the adaptation is substantive but mechanical.
29. *Slantchev minimax on Example 3.3.* The Slantchev mechanism requires bounded minimax payoffs in the disagreement continuation game. Compute the minimax payoffs v_i for Example 3.3's contest: what is the worst payoff state j can impose on state i via its choice of deployment? Compute $v_i := \max_{m_j} \min_{m_i} u_i(x_i^*(p^*, m))$. For the Tullock contest at

fixed p^* , the worst case for i is $m_j \rightarrow \infty$, which sends $\rho_i \rightarrow -1$ and minimizes i 's spoils. Compare v_i to $u_i(x_i^*)$ at the symmetric war equilibrium and to $v_i = 0$ (autarky), and identify the parameter region in which the punishment threat is large enough to sustain inefficient SPEs through the Slantchev mechanism.

30. *Power to hurt as a property of R_i .* The Slantchev mechanism requires the range $\bar{v}_i - v_i$ (max minus min payoff state j can impose on state i) to be large enough to support SPE-multiplicity. Identify the structural condition on (s_i, c_i) under which this holds, and state it as a property of the contest-feedback matrix R_i from Assumption 2.6 of Chapter 2. Specifically: how does the operator norm $\|R_i^{-1} - I\|_{\text{op}}$ (which controls the wedge between effective and market prices) bound the power to hurt? A natural formulation is $\bar{v}_i - v_i \leq \kappa \cdot \|R_i^{-1} - I\|_{\text{op}}$ for some constant κ depending on preferences, which states the power-to-hurt bound as a property of the agent-level coupling matrix the chapter introduced in Chapter 2. Open-ended.
31. *Welfare cost of alliance shifts.* Resource booms (Exercise 10) and contest-parameter shifts in V_0 are endowment- and prize-side perturbations of the welfare gradient. Alliance shifts are a different parameter class: a perturbation of the alliance vector a_{ij} from Example 2.12 of Chapter 2, capturing changes in who is counted as ally vs. rival in the contest-feedback structure. For Example 3.3 extended to three states with a non-degenerate alliance vector, compute $D_{a_{ij}} \Delta^*$ at the symmetric equilibrium using the sensitivity formula Equation (5.10). Decompose into the three channels (direct-resource via the contest-feedback shift, cost-feedback via the equilibrium price response, reallocation via the welfare-weight shift), and identify which channels contribute non-trivially. Substantive interpretation: an alliance shift toward a former rival lowers ρ_i for the shifted dyad, dampening the contest's mutual-loss intensity and reducing Δ^* . The exercise instantiates the master sensitivity formula on the alliance-vector parameter direction; the substantive content is that the framework's apparatus handles alliance shifts directly without requiring a separate formal model.

Part II

Applications

Chapter 6

Peacemaking on the cheap

When the international community ships grain to a war economy, local grain prices fall, real wages for subsistence laborers rise, and the opportunity cost of soldiering rises with them. When it ships fuel, fuel prices fall, military logistics become cheaper, and fighting may intensify. The effect of a commodity transfer on conflict runs through equilibrium prices, and its sign depends on which commodity is transferred, in what quantity, and in what sequence. The empirical evidence confirms the mechanism: [de Ree and Nillesen \[2009\]](#) find that a ten-percent increase in foreign aid reduces the continuation probability of an ongoing conflict by roughly eight percentage points; [Doyle and Sambanis \[2000\]](#) document that net current transfers per capita are among the strongest predictors of peacebuilding success. The sign depends on deployment, however: [Nunn and Qian \[2014\]](#) show that undirected food transfers can prolong existing conflicts by providing resources to armed factions. The substantive donor question is therefore not how much aid to deliver but how to design the transfer—which commodities, on what schedule, in pursuit of which equilibrium response.

Two pictures of that design problem dominate the policy and academic literatures on peacemaking aid. The first treats the recipient as a fixed-price economy and aid as a relaxation of the resource constraint ([Collier and Hoeffler 2002](#), [Savun and Tirone 2012](#)): the donor selects which transfers best fund peacekeeping, postwar reconstruction, or relief, with prices held at their pre-aid level for the analytic exercise. The second treats the donor as a third player in a bargaining game and aid as a side payment that shifts the belligerents' outside options ([Beviá and Corchón 2010](#), [Chassang and Padró i Miquel 2010](#)): the donor selects which payments alter the bargaining incentives, with the recipient economy abstracted to a payoff matrix. Each picture computes the donor's intervention on a partialled recipient. The first ignores the price response that aid induces—the very mechanism the empirical evidence locates—and the second ignores the equilibrium-manifold structure of the recipient economy. A growing body of work establishes that commodity prices affect conflict through general-equilibrium channels: [Bó and Bó \[2011\]](#) embed conflict in a Heckscher–Ohlin economy and identify the Stolper–Samuelson mechanism; [Dube and Vargas \[2013\]](#) document in Colombia that international price increases for oil intensify conflict while price increases for coffee reduce it; and [Garfinkel et al. \[2008\]](#) characterize how trade regimes shape conflict intensity in a static Walrasian model. This chapter takes a third picture.

The donor's problem is navigation on the equilibrium manifold. An aid schedule is a path $\gamma : [0, 1] \rightarrow \tilde{E}$ on the joint equilibrium manifold of Chapter 3. The donor controls a transfer schedule that drives the parameter profile ω along an associated path in Ω_0 , and the inverse

function theorem of Chapter 4 lifts that path to a smooth path on $\tilde{E}^{\text{reg}}$. The target submanifold is the *peace component* $\tilde{E}_p \subset \tilde{E}$ where conflict deployment vanishes; under the zero- or negative-sum conditions of Chapter 5, peaceful equilibria are globally Pareto optimal. The donor’s task is to choose γ subject to budget, speed, and reachability constraints so that γ reaches $\tilde{E}_p$ at minimum cost.

What the chapter delivers is hard to find elsewhere. First, an existence theorem for an optimal aid schedule that requires no convexity assumption, inheriting the manifold structure of Chapter 3 and the submanifold structure of Chapter 5 §5.3. Second, a transversality result: aid evaluated at fixed prices cannot reduce conflict, because the partial-equilibrium direction in ω -space is not tangent to $\tilde{E}_p$ at any equilibrium where the conflict primitive operates through prices. The result is the welfare-side counterpart of the sensitivity-formula three-channel decomposition of Chapter 5 §5.4: the deployment-cost-feedback channel is exactly what partial-equilibrium thinking zeros out, and exactly what delivers the welfare effect of aid. Third, a coordination result: when multiple donors intervene non-cooperatively, each free-rides on the peace trajectory the others provide; fragmented bilateral aid delivers peace exactly n times more slowly than a cooperative arrangement with the same total budget. The result is a resource-neutral rationale for multilateral peacemaking institutions, distinct from the transaction-cost and information rationales that ground the existing literature.

The chapter is organized around the donor’s problem and the three results, with a structural characterization of optimal schedules running through the middle. Section 6.1 specializes the framework primitives to a recipient economy with a third-party donor, defines the aid schedule as a path on $\tilde{E}$, identifies the peace component $\tilde{E}_p$ as the target submanifold, and records the feasibility conditions on schedules. Section 6.2 establishes existence of an optimal aid schedule via a manifold-Arzelà–Ascoli argument, with no convexity assumption. Section 6.3 establishes the partial-equilibrium-zero transversality result and reads it through the three-channel decomposition of Chapter 5 Section 5.4. Section 6.4 characterizes optimal schedules via a manifold version of Pontryagin’s principle, with bang-bang behavior on absolute objectives and gradualism on convex efficiency-cost objectives. Section 6.5 extends the apparatus to multiple donors as a Nash equilibrium problem on the joint strategy space, with the n -times-slower coordination cost as the chapter’s institutional headline. Section 6.6 hands the chapter’s apparatus to the institutions chapter and the third Part 2 chapter that follow.

6.1 The donor problem

The donor problem specializes the framework’s primitives to a recipient economy with a third party that wants peace, and formalizes the donor’s instrument as a path on the equilibrium manifold of Chapter 3. Three architectural moves in this section set up the rest of the chapter. Aid is recast from “transferring resources” to “navigating $\tilde{E}$ ”, with the equilibrium response of the recipient contributing the navigation rather than blocking it. The peace component $\tilde{E}_p \subset \tilde{E}$ is identified as the target submanifold the donor steers toward. The donor’s preferences are recorded as a 1-form on $\tilde{E}$, with Δ^* from Chapter 5 a natural concrete choice.

States, donor, and the recipient economy. Specialize the framework’s primitives. Designate two states $A, B \in \{1, \dots, I\}$ as *belligerents* and a third state C as the *donor*; the remaining states W comprise the rest of the world. Belligerents and donor differ only in role—all four sets of states are subject to the framework’s primitives of Chapter 2—but the donor’s role is to

influence the recipient economy through transfers rather than to deploy force, so the donor's deployment is fixed at $m_C = 0$ throughout. The endowment profile remains $\omega \in \Omega_0$, and the joint equilibrium $(p^*, m^*, x^*) \in \tilde{E}$ is the joint equilibrium of Chapter 3.

The conflict primitive operates on the belligerent pair (A, B) : m_A^*, m_B^* may be non-zero at equilibrium and the battle function s redistributes goods between them. The donor's deployment $m_C^* = 0$ ensures C is not party to the contest; what C contributes is the transfer instrument.

Transfer schedules. A *transfer schedule* is an absolutely continuous function

$$\tau : [0, 1] \rightarrow \mathbb{R}_{\geq 0}^L \times \mathbb{R}_{\geq 0}^L,$$

$\tau(t) = (\tau_A(t), \tau_B(t))$, recording the cumulative goods transferred from C to each belligerent through time t , with $\tau(0) = 0$. Non-negativity reflects the donor relationship: C gives resources to belligerents, not the reverse. A constant $\eta \geq 1$ records aid efficiency: each unit delivered to a belligerent costs C a quantity η , with the residual $\eta - 1$ lost in transit. At $\eta = 1$ the transfer is pure redistribution; at $\eta > 1$ aggregate resources shrink as aid flows.

The schedule induces an endowment path $\omega(\cdot) : [0, 1] \rightarrow \Omega_0$ via

$$\omega_A(t) = \omega_A^0 + \tau_A(t), \quad \omega_B(t) = \omega_B^0 + \tau_B(t), \quad \omega_C(t) = \omega_C^0 - \eta(\tau_A(t) + \tau_B(t)). \quad (6.1)$$

Endowments of states in W remain fixed at ω_W^0 . The rate of transfer $\tau'(t)$ records the speed at which C moves resources; $\|\tau'(t)\|$ is its magnitude.

From transfers to equilibria. The endowment path $\omega(\cdot)$ traces a smooth curve in Ω_0 , but the substantive object is the corresponding equilibrium response. The natural projection $\tilde{\pi} : \tilde{E} \rightarrow \Omega_0$ of Chapter 3 is generically locally invertible: at any regular equilibrium $e^0 = (p^0, m^0, \omega^0) \in \tilde{E}^{\text{reg}}$, the inverse function theorem produces a smooth local section of $\tilde{\pi}$ on a neighborhood of ω^0 . Following that local section along the endowment path $\omega(t)$ and reapplying the theorem at each step yields a unique smooth path $\gamma : [0, 1] \rightarrow \tilde{E}^{\text{reg}}$ with $\gamma(0) = e^0$ and $\tilde{\pi}(\gamma(t)) = \omega(t)$ for all t , provided the path stays clear of the singular locus $\Sigma \subset \Omega_0$. By Sard's theorem, Σ has measure zero in Ω_0 (Chapter 4, Theorem 4.3), so a generic transfer schedule lifts uniquely to a smooth path on $\tilde{E}^{\text{reg}}$.

When the path passes through Σ —when it crosses a bifurcation locus where equilibria fold, merge, or appear in new pairs—the local section breaks down and the lift is non-unique. Chapter 7 reads such crossings as regime change. The present chapter works within the regular set; transfer schedules that approach Σ are admissible only when the analysis explicitly handles the bifurcation.

Definition 6.1 (Aid schedule). An *aid schedule* is a Lipschitz path $\gamma : [0, 1] \rightarrow \tilde{E}$ with $\gamma(0) = e^0$, the initial equilibrium. γ is *induced by* the transfer schedule τ if $\tilde{\pi}(\gamma(t)) = \omega(t)$ for all t , with $\omega(\cdot)$ given by Equation (6.1).

The donor's instrument is the transfer schedule τ ; the aid schedule γ is the equilibrium trajectory the transfer induces. The chapter works with γ as the primary object because the equilibrium manifold is the natural space for welfare analysis, and because the geometric structure that makes the existence, optimality, and coordination results clean is structure on $\tilde{E}$, not on Ω_0 .

The war region. A donor's operational task is to leave a *war region*. Identify conflict intensity with a smooth function $\phi : \tilde{E} \rightarrow \mathbb{R}$ and define

$$\mathcal{W} := \{e \in \tilde{E} \mid \phi(e) > 0\}. \quad (6.2)$$

$\mathcal{W}$ is open with smooth boundary $\partial\mathcal{W} = \phi^{-1}(0)$ when 0 is a regular value of ϕ . The leading example is a militarization threshold: $\phi(e) = \|m^*(e)\| - \bar{m}$ for some $\bar{m} > 0$, so that $\mathcal{W}$ collects equilibria at which aggregate deployment exceeds $\bar{m}$. Other natural specifications take ϕ proportional to $\Delta^*(\tilde{\pi}(e))$ from Chapter 5 (the welfare cost of conflict), or to $\sum_i c_i(m_i^*, p^*, \omega_i)$ (aggregate deployment cost in resources). The chapter takes ϕ as a primitive of the donor's problem; its specification reflects the donor's substantive objective, not the framework.

The relation between $\mathcal{W}$ and the peace component $\tilde{E}_p$ of Chapter 5 Section 5.3 is direct. At any peaceful equilibrium $m^* = 0$, so $\|m^*\| = 0$ and the threshold-specification ϕ takes the value $-\bar{m} < 0$; peaceful equilibria are not in $\mathcal{W}$. For threshold-flavored ϕ the inclusion $\tilde{E}_p \subset \mathcal{W}^c$ is strict; $\mathcal{W}^c$ may contain low-conflict equilibria with $m^* \neq 0$ but $\|m^*\| < \bar{m}$ that the donor accepts as “peace enough.” For the $\phi = \Delta^*$ specification, $\mathcal{W}^c$ contains exactly the equilibria where the welfare cost of conflict is at most a tolerated level, and under zero- or negative-sum warfare the boundary $\partial\mathcal{W}$ approaches $\tilde{E}_p$ as the tolerance shrinks to zero. The donor's problem is to leave $\mathcal{W}$, not specifically to reach $\tilde{E}_p$; what the relation buys is that Chapter 5's peaceful-efficiency theorem applies on the $\tilde{E}_p$ portion of $\mathcal{W}^c$ as a special case.

Assumption 6.2 (Initial war). $e^0 \in \mathcal{W}$: the initial equilibrium lies in the war region.

The donor's task is to find an aid schedule that moves the equilibrium out of $\mathcal{W}$ at minimum cost—making peace on the cheap.

Feasibility. An aid schedule γ is *feasible* if three conditions hold.

- (i) *Peace.* $\gamma(1) \notin \mathcal{W}$: the terminal equilibrium has exited the war region.
- (ii) *Budget.* Writing τ for the transfer schedule that induces γ , $\eta(\tau_A^\ell(t) + \tau_B^\ell(t)) \leq \bar{\tau}^\ell$ for each good ℓ and all t , where $\bar{\tau} \in \mathbb{R}_{++}^L$ is an exogenous transfer ceiling bounded above by the donor's good- ℓ endowment $\omega_C^{0,\ell}$.
- (iii) *Speed.* $\|\gamma'(t)\| \leq M$ for almost every t , where $\|\cdot\|$ is the Euclidean norm on $\tilde{E}$ under the identification of Chapter 3, and $M > 0$ is given.

The peace condition fixes the terminal: at $t = 1$ the equilibrium must lie outside the war region. The budget condition records that C cannot transfer resources it does not have, with a per-good ceiling that is at most the donor's endowment in that good. The speed condition records the absorptive capacity of the recipient economy: aid delivered too rapidly cannot be productively integrated into the recipient market structure, so the donor's control over the trajectory is limited by a hard physical ceiling on how fast equilibrium prices and deployments can adjust. A separate, softer cost of rapid delivery enters the donor's preference functional below.

Write $\mathcal{F}$ for the set of all feasible aid schedules.

Assumption 6.3 (Reachability). $\mathcal{F} \neq \emptyset$: the donor's budget and speed constraints jointly suffice to reach peace.

Assumption 6.3 is a scope condition. It holds whenever the budget ceiling and the speed bound are large enough that some path from e^0 to $\partial\mathcal{W}$ exists at admissible disbursement rate. Substantively, it says the donor is large enough to matter and the conflict is reachable at the available speed of intervention—neither premise is automatic, and a donor whose budget ceiling is below the threshold required to leave $\mathcal{W}$ falls outside the chapter’s scope.

The preference functional. The donor cares about the entire trajectory, not just the terminal equilibrium. A flow cost is paid at each t as resources move; intermediate equilibria may carry independent welfare weight when the donor weighs civilian welfare during transition; aid efficiency may depend on current equilibrium prices. The natural framework for path-dependent preferences over trajectories is line integration of a 1-form.

For any smooth 1-form α on $\tilde{E}$ and any aid schedule γ , the donor’s evaluation is

$$\mathcal{V}(\gamma) := \int_{\gamma} \alpha = \int_0^1 \alpha_{\gamma(t)}(\gamma'(t)) dt. \quad (6.3)$$

α is taken as a primitive of the donor’s problem; different choices accommodate different objectives. A specification $\alpha = -d(p \cdot \omega_C)$ captures the flow cost to C . A specification $\alpha = d(u_A + u_B)$ tracks the welfare gain to the belligerents. A specification $\alpha = -d\Delta^* \circ \tilde{\pi}$, using the welfare cost of conflict from Chapter 5, makes the donor a welfare maximizer in the cardinal scalar of Definition 5.1; gradient descent on Δ^* along γ is the natural welfare-economic donor problem. The chapter does not commit to a specific α .

The Euclidean structure of $\tilde{E}$ under Assumption 3.6 clarifies when the path itself matters as opposed to only the destination. If $\alpha = dV$ for some smooth $V : \tilde{E} \rightarrow \mathbb{R}$ —if α is a gradient—then

$$\mathcal{V}(\gamma) = V(\gamma(1)) - V(\gamma(0)),$$

and the evaluation depends only on the terminal equilibrium. This is natural for a donor whose sole objective is to reduce militarization below a threshold or to drive Δ^* to zero, regardless of the transition. When α is not closed—when its exterior derivative $d\alpha$ is non-zero—intermediate equilibria carry independent welfare weight and the path matters. On the simply connected manifold $\tilde{E} \cong \mathbb{R}^{IL}$ under Assumption 3.6, every closed 1-form is automatically the gradient of a function, so closed and path-independent are synonymous. Preferences are genuinely path-dependent precisely when α is not closed.

The donor’s problem in this chapter is

$$\min_{\gamma \in \mathcal{F}} \mathcal{V}(\gamma). \quad (6.4)$$

Grants and loans. Two cases of the donor’s problem deserve mention. An aid schedule γ is a *grant* if $\tilde{\pi}(\gamma(1)) \neq \omega^0$: the terminal endowment profile differs from the initial one and the transfer is not returned. It is a *loan* if $\tilde{\pi}(\gamma(1)) = \omega^0$: the transfer schedule satisfies $\tau(1) = 0$, the belligerents’ endowments are fully restored at $t = 1$, and C recoups its resources.

A loan traces a loop in endowment space, but the corresponding aid schedule γ need not be a loop in $\tilde{E}$. Under Assumption 3.6, every loop in Ω_0 lifts to a loop in $\tilde{E}$ because the natural projection is a diffeomorphism; under Assumption 3.17 (Chapter 3 weakening), a loop in Ω_0 may lift to a path that exits the regime component containing e^0 and terminates at a different equilibrium above ω^0 . The continuous path traced through $\tilde{E}$ determines which equilibrium the

economy reaches at the end of the loan—a form of path-dependent equilibrium selection invisible at the level of Ω_0 . Chapter 7 develops the topological characterization of this phenomenon as part of its regime-change apparatus.

Compactness of the feasible set. The existence theorem of Section 6.2 requires the feasible set $\mathcal{F}$ to be compact in a suitable topology. The structure of $\tilde{E}$ delivers compactness directly.

Lemma 6.4 (Compactness of $\mathcal{F}$). *The set $\mathcal{F}$ of feasible aid schedules is compact in $C^0([0, 1], \tilde{E})$ equipped with the uniform topology.*

Proof sketch. The budget condition confines the endowment path $\omega(\cdot)$ to a compact subset of Ω_0 . Properness of $\tilde{\pi}$ (Chapter 3) carries this to a compact preimage in $\tilde{E}^{\text{reg}}$. The speed condition makes every feasible γ an M -Lipschitz path in this compact subset. The Arzelà–Ascoli theorem gives pre-compactness in the uniform topology. Closure follows from continuity of ϕ at the terminal condition: a uniform limit of paths each ending in $\mathcal{W}^c$ ends in $\mathcal{W}^c$. Full details follow the standard manifold-Arzelà–Ascoli argument. $\square$

Lemma 6.4 reduces the donor’s problem Equation (6.4) to minimization of a continuous functional on a compact set, the standard input to existence arguments in the calculus of variations. The next section runs that argument.

The framework in perspective. Three features of the setup shape the rest of the chapter.

Aid as navigation. The donor steers the equilibrium through a sequence of joint equilibria, with prices and deployments adjusting at every step through the joint determination of Chapter 3. The same transfer can reach different equilibria depending on the path through $\tilde{E}$, and the equilibrium response of the recipient can amplify or reverse the direct effect of endowment movements—which is the substantive content of Section 6.3.

Process and outcome. When α is closed, the donor evaluates only the terminal equilibrium and any path to the same point on $\partial\mathcal{W}$ is equivalent. When α is not closed, intermediate equilibria carry welfare weight and the optimal schedule adjusts continuously along the trajectory—which is the substantive content of Section 6.4.

Multilateral coordination. A single donor with the resources to leave $\mathcal{W}$ is one case of the donor problem. The substantively important case is multiple donors with limited budgets each, intervening non-cooperatively—which is the substantive content of Section 6.5 and the chapter’s institutional headline.

6.2 Existence on the manifold

The donor’s problem has an answer. For any donor preferences representable by a smooth 1-form on $\tilde{E}$, an optimal aid schedule exists in $\mathcal{F}$. The result is unsurprising in shape—existence theorems are the floor of optimization theory—but the route to it is what makes the apparatus visible. Standard variational existence arguments lean on convexity of the objective or the constraint set; the donor’s problem has neither. What does the work instead is the manifold structure delivered by Chapter 3, the properness of the natural projection, and the compactness of the feasible set established in Lemma 6.4. The section states the theorem, runs the proof, and reads the apparatus working.

What variational existence normally requires. The classical setting is Tonelli’s direct method. A functional $\mathcal{V} : \mathcal{F} \rightarrow \mathbb{R}$ on a set $\mathcal{F}$ of admissible paths attains its infimum when $\mathcal{V}$ is lower-semicontinuous on $\mathcal{F}$ and $\mathcal{F}$ is compact in a topology fine enough to make $\mathcal{V}$ lower-semicontinuous. In flat $\mathbb{R}^n$ with paths in a Sobolev class, the standard recipe to deliver both is convexity. Lower-semicontinuity of $\mathcal{V}(\gamma) = \int L(\gamma, \dot{\gamma}) dt$ in the weak topology requires the Lagrangian $L(\cdot, v)$ to be convex in v , the velocity argument; growth conditions on L deliver weak compactness of the sub-level sets. The textbook treatment (Dacorogna 2008, Buttazzo et al. 1998) reads: convexity in the velocity, plus growth at infinity, plus a coercive boundary term, and minima are guaranteed.

The donor’s problem fails the convexity premise on every dimension that matters for the substantive content of peacemaking aid. Donor preferences over trajectories are typically non-convex: the welfare value of a trajectory that hovers just inside the war region for the entire horizon and then drops below the threshold at $t = 1$ is not the convex average of the welfare values of two trajectories that drop below the threshold at $t = 1/2$ and stay there. The conflict primitive does not produce convex Lagrangians in general: the non-monotonicities of bang-bang versus gradualist optimal schedules (Section 6.4) are exactly non-convexities the framework expects to encounter. The aid efficiency parameter η and the speed bound M together produce a feasible set $\mathcal{F}$ that is compact in the uniform topology (Lemma 6.4) but is not convex in path-space under any natural Sobolev structure. The classical recipe is not available.

What the manifold structure delivers instead. The compactness of $\mathcal{F}$ in the uniform topology, established on entirely different grounds in Lemma 6.4, suffices. Compactness in the uniform topology comes from three ingredients, each contributed by an upstream chapter. Chapter 3 delivers the manifold structure that makes $\tilde{E}^{\text{reg}}$ a smooth IL -dimensional submanifold of $S_N \times M \times \Omega_0$, with a global chart under Assumption 3.6 or a covering of mod-2 degree one under Assumption 3.17. Lemma 3.10 of Chapter 3 delivers properness of the natural projection $\tilde{\pi} : \tilde{E} \rightarrow \Omega_0$, which converts a budget-bounded endowment path into a manifold-bounded equilibrium path. The speed bound M delivers M -Lipschitz regularity on every $\gamma \in \mathcal{F}$, and Arzelà–Ascoli on the manifold delivers pre-compactness in $C^0([0, 1], \tilde{E})$. Closure of $\mathcal{F}$ follows from continuity of the conflict-intensity function ϕ at the terminal endpoint.

The key contrast with the classical setting is what is doing the compactness work. In the classical Tonelli argument, convexity of the Lagrangian is what makes the weak-topology limit of a minimizing sequence admissible. In the present argument, the equilibrium manifold’s geometric structure is what makes the uniform-topology limit of a minimizing sequence admissible: the limit of paths in $\tilde{E}^{\text{reg}}$ stays in $\tilde{E}$ because $\tilde{E}$ is closed in its ambient space, the limit is M -Lipschitz because a uniform limit of M -Lipschitz paths is M -Lipschitz, and the limit respects the budget condition because budgets are continuous in the path’s endowment trace. The manifold structure does the work convexity does. This is a standing pattern in the framework: the global manifold-structure replaces convexity arguments throughout Chapters 3 to 5, and the donor’s problem is one further substantive instance.

The theorem. With compactness of $\mathcal{F}$ in hand and continuity of $\mathcal{V}$ on $\mathcal{F}$ delivered by the line-integral representation Equation (6.3), the existence of a minimum is the extreme-value theorem.

Theorem 6.5 (Existence of an optimal aid schedule). *Suppose Assumptions 6.2 and 6.3 hold and the donor’s preferences are representable by a smooth 1-form α on $\tilde{E}$. Then $\mathcal{V}(\gamma) = \int_\gamma \alpha$*

attains its infimum on $\mathcal{F}$.

Proof. $\mathcal{F}$ is compact in $C^0([0, 1], \tilde{E})$ by Lemma 6.4. The functional $\mathcal{V} : \mathcal{F} \rightarrow \mathbb{R}$ is continuous in the uniform topology because α is smooth and integration is uniformly stable for Lipschitz paths in a fixed compact manifold neighborhood. The extreme-value theorem applied to a continuous function on a compact set delivers the infimum as a minimum. $\square$

The proof itself is one paragraph. The chapter’s substantive content lives one level above the proof, in what the apparatus is doing.

Reading the apparatus. Five ingredients enter Theorem 6.5 in different ways.

Chapter 3’s manifold theorem makes $\tilde{E}^{\text{reg}}$ the natural ambient space for aid schedules. Without Proposition 3.8, $\tilde{E}$ would be a generic zero set of an equation system, and “Lipschitz path on $\tilde{E}$ ” would not have a clear meaning. Under Assumption 3.6, $\tilde{E} \cong \mathbb{R}^{IL}$ globally, so the ambient space is as well-behaved as flat space; under Assumption 3.17, $\tilde{E}^{\text{reg}}$ is a covering and the ambient space remains well-behaved on regime components. Either condition makes “the donor steers along $\tilde{E}^{\text{reg}}$ ” a precise statement.

Lemma 3.10 of Chapter 3 converts the donor’s budget condition into compactness on the manifold. Without properness, a transfer schedule satisfying the budget condition could induce an equilibrium path that escapes to infinity in price coordinates while remaining bounded in endowment coordinates. Properness rules this out: bounded ω -traces correspond to bounded $\tilde{E}$ -traces, and the budget condition’s compactness on Ω_0 pulls back to compactness on $\tilde{E}^{\text{reg}}$. The economic content is direct: a finite donor endowment produces a finite range of equilibrium responses. The mathematical content is what Lemma 6.4 needed.

Chapter 4’s generic-regularity theorem (Theorem 4.3) makes the IFT-on-manifold lift of Definition 6.1 a generic operation. The singular locus Σ has measure zero in Ω_0 , so a generic transfer schedule never crosses it, and the lift is unique along the entire schedule. At a non-generic schedule that crosses Σ , the lift fails to be unique at the crossing instant, and the chapter’s analysis breaks into pieces: each side of the crossing handled separately, with regime change at the boundary. This is exactly the bifurcation reading of Chapter 7.

Chapter 5’s peaceful-efficiency theorem (Theorem 5.8) is what makes the target $\mathcal{W}^c$ substantively meaningful. Without it, exiting the war region would be a definitional operation on the conflict-intensity function ϕ . With it, exiting $\mathcal{W}$ via the peace component $\tilde{E}_p \subset \mathcal{W}^c$ delivers a globally Pareto-optimal outcome under zero- or negative-sum warfare; the bargaining-model premise that war is inefficient is not assumed but recovered from the apparatus. The donor’s problem is therefore not just “leave $\mathcal{W}$ ” but “leave $\mathcal{W}$ to a configuration the apparatus certifies as efficient.”

Chapter 5’s sensitivity formula (Equation (5.10)) makes the donor’s natural welfare-economic objective concrete. The specification $\alpha = -d\Delta^* \circ \tilde{\pi}$ takes the welfare cost of conflict from Chapter 5 as the donor’s running loss, with $\mathcal{V}(\gamma)$ delivering the integrated welfare cost along the trajectory. Under this specification Theorem 6.5 delivers a schedule that minimizes total welfare cost of conflict subject to budget and speed constraints. A donor who wants to minimize welfare-cost-of-conflict need not do anything more than apply the apparatus; the smooth-scalar-field machinery of Chapter 5 Section 5.4 does the rest.

Why no convexity matters. The result is not just a technical convenience. Substantive donor preferences over peacemaking trajectories are generally non-convex. A donor whose primary

objective is to drive conflict intensity ϕ below a threshold at the terminal time, regardless of the intermediate trajectory, has step-function preferences over $\phi(\gamma(1))$: a unit improvement at the threshold delivers the entire welfare gain, while improvements above the threshold deliver nothing. A donor whose primary objective is to minimize integrated welfare cost has Δ^* as the running loss, which is positive everywhere on $\mathcal{W}$ and zero on $\tilde{E}_p$ (Theorem 5.8); the resulting Lagrangian Δ^* along the trajectory is smooth on $\tilde{E}^{\text{reg}}$ but typically not convex along arbitrary paths. A donor whose objective combines a flow cost on its own resources and a flow benefit from the belligerents' welfare faces a Lagrangian that may be convex in some regions and concave in others, with the regime depending on equilibrium prices.

Theorem 6.5 accommodates all of these. The argument runs through the manifold structure rather than through convexity of the Lagrangian, so non-convex preferences over trajectories cause no difficulty. The substantive consequence is that the framework can analyze realistic donor problems—where preferences are step-function or composite or sign-changing—without requiring the donor to adopt artificial convexified preferences for analytical tractability.

What no partial can compute. A partialled treatment of the existence question delivers a different theorem with a different content. An exchange-side partialling holds equilibrium prices fixed at their pre-aid value p^0 and treats the donor's transfer as a shift in belligerents' budget constraints alone. The induced “aid schedule” lives in Ω_0 rather than on $\tilde{E}$, and the existence theorem is a standard finite-dimensional optimization in $\mathbb{R}^{IL}$ subject to budget and speed constraints. The theorem is true but answers a different question: it computes the optimal endowment trajectory holding prices fixed, which by the transversality result of Section 6.3 below is disconnected from the actual welfare effect of the transfer.

A conflict-side partialling holds the price vector at its equilibrium value pointwise but solves the militarization Nash equilibrium afresh at each point along the path. The induced trajectory is a curve in $\mathbb{R}^{IL} \times M$ rather than on $\tilde{E}$, with the constraint that prices satisfy market-clearing not enforced along the path. The existence theorem in this partialling requires the militarization game to admit a unique smooth equilibrium at every parameter profile and the Nash-residual map to be Lipschitz in the parameters, both of which are stronger conditions than Theorem 6.5 requires under Assumption 3.17.

Theorem 6.5 works on $\tilde{E}$ jointly because the joint-equilibrium apparatus of Chapter 3 does the work that the partialled approaches push onto auxiliary conditions. The existence theorem is one form-of-statement of what Chapter 3's manifold-structure architecture buys.

6.3 Why general equilibrium

The chapter's central substantive claim runs through this section. Aid intervention works through prices. The framework makes the claim precise: when the conflict primitive operates through the price system, transfers evaluated at fixed prices have zero effect on militarization, and the welfare effect of aid on the conflict externality of Chapter 5 is exactly the channel that partial-equilibrium thinking zeros out. This section states the result, recasts it as a transversality statement on the equilibrium manifold, and reads it through the three-channel sensitivity decomposition of Section 5.4. A two-good Tullock illustration verifies the qualitative pattern in closed form.

The corollary. The donor's transfer schedule shifts the recipient's endowment profile $\omega(t)$. Equilibrium deployment m^* depends on ω both directly (through the parameters of the conflict

primitive that take ω as an argument) and indirectly (through the equilibrium prices $p^*(\omega)$ that the conflict primitive also depends on). For a wide class of conflict specifications the indirect channel is the only operative channel: the equilibrium deployment is a function of ω only through prices, $m^*(\omega) = \hat{m}(p^*(\omega))$ for some smooth $\hat{m} : S_N \rightarrow M$.

Corollary 6.6 (PE transfers cannot reduce conflict). *Suppose equilibrium deployment is a function of ω only through equilibrium prices: $m^*(\omega) = \hat{m}(p^*(\omega))$ for a smooth $\hat{m} : S_N \rightarrow M$. Then for any pair of endowment profiles $\omega, \omega' \in \Omega_0$ with $p^*(\omega) = p^*(\omega')$, $m^*(\omega) = m^*(\omega')$.*

Proof. $m^*(\omega) - m^*(\omega') = \hat{m}(p^*(\omega)) - \hat{m}(p^*(\omega')) = 0$ when $p^*(\omega) = p^*(\omega')$. $\square$

The proof is one line. The substantive content is in the hypothesis and where it holds. The hypothesis is satisfied in the standard battlefield-model specification of Chapter 2: when the marginal benefit of deployment $\partial s_i^j / \partial m_i \cdot p^j$ depends on p but not on ω at fixed p , and the marginal cost $p \cdot \partial c_i / \partial m_i$ depends similarly, the Nash equilibrium m^* is determined by p alone. The price-dependence-of-cost architecture of Chapter 2 delivers this for a wide class of contest specifications by construction, including the canonical contest-success-function families: the ratio-form Tullock contest of Tullock [1980] and the axiomatic characterization of Skaperdas [1996], the difference-form contests of Hirshleifer [1989], and the all-pay specifications of Hodler and Yektaş [2012]. The hypothesis fails when the contest's payoff depends directly on endowments—a wealth-dependent contest in which the prize size or the cost technology scales with belligerents' holdings of a particular good. The substantive content of Corollary 6.6 is that *conditional on the hypothesis*, transfers held at fixed prices have no effect on the contest at all.

The geometric content. The corollary has a clean geometric reading on $\tilde{E}$. At any joint equilibrium $e \in \tilde{E}$, the tangent space $T_e \tilde{E}$ records the directions in which e moves under perturbations. The natural projection $\tilde{\pi} : \tilde{E} \rightarrow \Omega_0$ has differential $D\tilde{\pi}$ that is an isomorphism at any regular point under Assumption 3.6 (Chapter 3). Each direction $\dot{\omega} \in T_\omega \Omega_0$ maps to a unique tangent direction $(\dot{p}, \dot{m}, \dot{\omega}) \in T_e \tilde{E}$ with the joint-determination relations $\dot{p} = D_\omega p^* \cdot \dot{\omega}$ and $\dot{m} = D_\omega m^* \cdot \dot{\omega}$.

The *PE direction* at e is the subspace of $T_e \tilde{E}$ on which $\dot{p} = 0$. Under the conflict-through-prices hypothesis of Corollary 6.6, the chain rule $\dot{m} = D\hat{m} \cdot \dot{p}$ forces $\dot{m} = 0$ on the PE direction. The PE direction is therefore the subspace of $T_e \tilde{E}$ tangent to a single (p^*, m^*) pair: variations in ω that hold both prices and deployments fixed.

The peace component $\tilde{E}_p \subset \tilde{E}$ is the level set $\{e \in \tilde{E} \mid m^*(e) = 0\}$. At any non-peaceful equilibrium e with $m^*(e) \neq 0$, the donor's task is to move e in a direction that decreases $\|m^*\|$, and any such direction must have $\dot{m} \neq 0$. The PE direction has $\dot{m} = 0$ identically. The PE direction is therefore *transversal* to $\tilde{E}_p$ in the sharper sense that PE motion does not just fail to reach the peace component; it cannot move toward it at all.

The reading sharpens Corollary 6.6. For any smooth conflict-intensity function $\phi : \tilde{E} \rightarrow \mathbb{R}$ that depends on e only through m^* —the threshold-flavored $\phi = \|m^*\| - \bar{m}$ of Equation (6.2), the welfare-flavored $\phi = \Delta^* \circ \tilde{\pi}$ from Chapter 5, the resource-cost flavored ϕ proportional to $\sum_i c_i(m^*, p^*, \omega)$ —the PE direction is in the kernel of $d\phi$. PE transfers slide e along level sets of every sensible conflict-intensity specification. This is not a property of any one specification; it is a property of what PE-thinking is.

Reading through the three-channel decomposition. Section 5.4 decomposed the welfare effect of an endowment perturbation into three channels. Equation (5.10) restated:

$$D_\omega \Delta^* = \underbrace{D_r v(\omega, u_{-I}^*)}_{\text{direct resource}} - \underbrace{D_r v(r^*, u_{-I}^*) D_\omega r^*}_{\text{deployment-cost feedback}} + \underbrace{[\dots] D_\omega u_{-I}^*}_{\text{reallocation}}.$$

The deployment-cost-feedback channel operates through $D_\omega r^* = I - D_\omega [\sum_i c_i(m^*, p^*, \omega)]$, which involves $D_\omega m^*$ and $D_\omega p^*$ via the chain rule on c_i .

PE-thinking holds p fixed by assumption: $D_\omega p^* = 0$ along the PE direction. Under the conflict-through-prices hypothesis of Corollary 6.6, this forces $D_\omega m^* = 0$ as well. The deployment-cost-feedback channel zeros out completely: under PE evaluation, $D_\omega r^* = I$, every dollar transferred translates into a dollar of resource shift, and the cost-feedback term contributes nothing to $D_\omega \Delta^*$.

What remains under PE-thinking is the direct-resource channel plus the reallocation channel. The direct channel records how an extra unit of total ω raises the global frontier $\partial \bar{U}(\omega)$; for a redistribution between recipients with $\sum_i \omega_i$ held fixed, this channel is zero. The reallocation channel records how the welfare-weight representation of Corollary 5.5 shifts as the equilibrium reallocates; for transfers between belligerents, this channel can be non-zero.

The PE-thinking reading of an aid intervention is therefore the reallocation channel alone. This is what PE analysis computes when it reports that aid “affects the welfare distribution among recipients.” What it does not compute, because the deployment-cost-feedback channel is zeroed out by construction, is the welfare effect of aid *on the conflict itself*.

The three-channel reading connects directly to Corollary 6.6. The corollary says PE transfers leave m^* invariant. The decomposition says this means the welfare-effect-on-conflict portion of aid is exactly the deployment-cost-feedback channel, and that channel is exactly what PE-thinking zeros out. The equivalence is two readings of the same statement: aid that does not move prices cannot move conflict, and the welfare contribution of moving conflict is the channel that PE zeroes out.

Example 6.7 (A two-good Tullock contest under aid). A symmetric Tullock contest in two goods makes the picture concrete in closed form. Take two belligerents A, B symmetric with endowments (a, W) each in (good 1, good 2), a donor C with endowment (c, W_C) , and rest-of-world states with whatever ω_W clears the markets. The contest is over a prize $V_0 < 2a + c$ in good 1, with Tullock contest-success $\sigma_i(m) = m_i/(m_A + m_B)$ at $m \neq 0$ and $\sigma_i = 1/2$ at $m = 0$. State i 's share of the prize is $\sigma_i V_0$. Deployment cost is θm_i^2 in good 2 (numéraire). Belligerent i 's wealth at deployment m is $w_i = p^1 a + W + p^1 \sigma_i V_0 - \theta m_i^2$. All states have Cobb–Douglas preferences with weight α_i^1 on good 1.

At symmetric Nash deployment $m_A^* = m_B^* = m^*$, the first-order condition $\partial w_i / \partial m_i = 0$ gives

$$m^* = \sqrt{\frac{p^1 V_0}{8\theta}}. \quad (6.5)$$

The deployment depends on ω *only through* p^1 : this is the conflict-through-prices hypothesis of Corollary 6.6, satisfied here in closed form. The price elasticity of the deployment is one-half: $d \log m^* / d \log p^1 = 1/2$. Any transfer that moves p^1 moves m^* proportionally; any transfer that does not move p^1 leaves m^* exactly where it was.

NUMERICAL ANCHOR. Take $a = 1$, $W = W_C = 2$, $c = 1$, $V_0 = 1$, $\theta = 1$, with all states symmetric Cobb–Douglas $\alpha_i^1 = 1/2$ and rest-of-world endowments balancing the aggregate.

The pre-aid Walrasian price clears at $p^1 = 1$, and the Nash deployment from Equation (6.5) is $m^* = 1/\sqrt{8} \approx 0.354$. Consider three counterfactual aid schedules of equal numeraire value $\Delta = 1/2$, each with delivery efficiency $\eta = 1$:

- (a) *Distribution-neutral redistribution.* C transfers good 1 to A and B in equal shares. Aggregate good-1 supply is unchanged; aggregate good-1 demand is proportional to total wealth, which is also unchanged (the transfer is between Cobb–Douglas agents with identical weights). p^1 remains at 1, and by Equation (6.5), m^* remains at $1/\sqrt{8}$. PE and GE agree: nothing changes.
- (b) *Asymmetric-weight redistribution.* Replace the donor’s preferences with $\alpha_C^1 = 1/4 < 1/2$ (less interested in good 1 than the belligerents). C transfers good 1 to A and B . Aggregate good-1 demand rises (recipients value good 1 more than the donor did), pulling p^1 upward. Solving the post-transfer Walrasian gives $p^{1,\text{post}} \approx 1.143$, and Equation (6.5) gives $m^{*,\text{post}} = \sqrt{1.143/8} \approx 0.378$. Aid *intensifies* the contest by 7% in this Cobb–Douglas case. PE-thinking, holding $p^1 = 1$, would predict no change.
- (c) *Numeraire transfer.* C transfers good 2 to A and B in equal shares. Aggregate good-1 supply unchanged; aggregate good-1 demand rises by $\alpha^1 \cdot \Delta = 1/4$ (recipients spend a fraction of the new wealth on good 1), pulling p^1 upward. Solving the post-transfer Walrasian gives $p^{1,\text{post}} \approx 1.063$, and Equation (6.5) gives $m^{*,\text{post}} \approx 0.364$, a 3% increase. PE-thinking, again holding $p^1 = 1$, predicts zero change.

The framework’s GE evaluation in (b) and (c) is bounded above by what the supply-side aid efficiency $\eta > 1$ would deliver if the transferred good is shrunk in aggregate before distribution, which would push p^1 further upward and intensify the contest more sharply.

Reading the example. The donor who relies on PE thinking would conclude from Example 6.7 that aid is ineffective for peacemaking and at best a distributional intervention; the donor who reads the GE channel would identify which interventions move p^1 favorably and target those. The structural failure of PE-thinking in this domain is that it zeroes out the only channel through which aid affects conflict, not that it computes that channel imprecisely. Example 6.7’s case (b) and case (c) make the gap between PE and GE quantitative: PE gives a 0% change, GE gives 3% to 7% depending on the asymmetry the transfer crosses, and the empirical evidence from Dube and Vargas [2013] and Bazzi and Blattman [2014] reads as the PE-thinking counterpart of these structural predictions: the numbers depend on which commodity, in what direction, and against what preference asymmetry.

What this means structurally. Corollary 6.6, the geometric reading, the three-channel decomposition, and the Tullock illustration all say the same thing. PE-thinking is, by construction, motion along the equilibrium manifold along level sets of conflict intensity. The structural failure of PE-thinking in donor-intervention analysis is in its definition; no PE specification, however sophisticated about the recipient economy’s heterogeneity or the donor’s instruments, escapes it.

The contrast with the joint formulation is sharp. The framework’s apparatus, applied to the donor’s problem, reads the deployment response to a transfer as one application of J^{-1} to the appropriate direct-derivative column of $D_{\omega e}$ from Chapter 4. The welfare response is one further application via the sensitivity formula Equation (5.10). Each is a closed-form calculation that

PE-thinking cannot perform because PE-thinking has zeroed out the cross-channel terms before the calculation begins.

The methodological claim is that this is what “general equilibrium matters” means in the donor-intervention context. It is not that GE adds quantitative precision to a qualitatively correct PE picture; it is that GE supplies the only operative channel. Chapter 5’s positioning of the conflict externality as a Greenwald-Stiglitz-style FWT failure was the welfare-economic form of this point. Corollary 6.6 is the policy-relevant form: aid that ignores prices does not reach conflict. The empirical evidence is consistent with this reading. Dube and Vargas [2013] document opposite-signed effects of oil and coffee price shocks on Colombian conflict intensity, exactly the differentiation Corollary 6.6’s contrapositive predicts: when prices move, conflict moves with them, and the direction depends on which commodity moves. Bazzi and Blattman [2014] review the broader pattern across country-commodity panels. Acemoglu et al. [2012] show the analogous claim dynamically: resource price paths shape war incentives along trajectories. What unifies the empirics is the price channel that PE analysis is structurally blind to.

6.4 The structure of optimal schedules

Theorem 6.5 guarantees an optimal aid schedule exists; this section characterizes its shape. The structure runs through Pontryagin’s maximum principle on the smooth manifold $\tilde{E}$. Two ideal-typical patterns emerge. When the donor’s preferences treat aid speed as costless up to the absorptive-capacity ceiling, the optimal schedule is *bang-bang*: the donor delivers at the maximum speed M almost everywhere, and the interesting content is direction, not pace. When the donor’s preferences include a strictly convex cost of speed, *gradualism* replaces bang-bang: the optimal speed responds smoothly to the donor’s shadow value of the equilibrium position, with three regimes (pause, interior, maximum) sorted by the magnitude of that shadow value. The two patterns together cover the substantive donor’s problem.

Pontryagin’s principle on a smooth manifold. The maximum principle is the continuous-time analogue of the Lagrange multiplier conditions in static optimization. A static optimization problem characterizes a vector x^* that maximizes an objective subject to constraints on x ; the Lagrangian carries the constraint-multiplier λ , and optimality is the first-order condition that the Lagrangian’s gradient vanish. A continuous-time optimal-control problem characterizes a trajectory $\gamma^* : [0, 1] \rightarrow M$ that maximizes an objective subject to constraints on γ and on the control $\gamma'(t)$; the Hamiltonian carries the costate $\psi(t)$, and optimality is a system of conditions involving the Hamiltonian, the costate evolution, and a transversality condition at the terminal time. The structural analogy is direct: the costate ψ is a continuous-time multiplier; the Hamiltonian is a Lagrangian for the trajectory.

The setting is on a smooth manifold M rather than on flat $\mathbb{R}^n$. The state $\gamma(t) \in M$, the control $\gamma'(t) \in T_{\gamma(t)}M$ (a tangent vector at the current state), the costate $\psi(t) \in T_{\gamma(t)}^*M$ (a covector, dual to the tangent at the current state). The Hamiltonian $H : T^*M \times \text{control-space} \rightarrow \mathbb{R}$ acts on the cotangent bundle. On a flat manifold $M = \mathbb{R}^n$, the cotangent bundle is $\mathbb{R}^n \times \mathbb{R}^n$ and the Hamiltonian formulation reduces to the standard textbook setup. On a non-flat manifold, the formulation is intrinsic: the costate is a 1-form, and Hamilton’s equations are formulated on the cotangent bundle.

The framework’s apparatus delivers the manifold structure at the level of detail the donor’s problem requires. Under Assumption 3.6 (Chapter 3), $\tilde{E}$ is globally diffeomorphic to $\mathbb{R}^{IL}$ via

the chart $\Phi : \tilde{E} \rightarrow \mathbb{R}^{IL}$. The cotangent bundle is therefore globally trivial, costates are $\mathbb{R}^{IL}$ -valued, and the manifold-PMP reduces in coordinates to the flat-space textbook formulation [Liberzon, 2011b]. Under Assumption 3.17, $\tilde{E}^{\text{reg}}$ is a covering of $\Omega_0 \setminus \Sigma$, and the chart Φ exists componentwise: the manifold-PMP operates on each regime component separately, with the costate fields well-defined on each component but not on $\tilde{E}$ as a whole. The restriction matters when an optimal schedule crosses a bifurcation in the singular locus Σ ; Chapter 7 takes up the regime-change content of such crossings, and the PMP analysis here is component-internal.

The optimal control problem. Under the chart $\Phi : \tilde{E} \rightarrow \mathbb{R}^{IL}$ of Chapter 3, write the donor's problem in flat coordinates. The state is $x(t) := \Phi(\gamma(t)) \in \mathbb{R}^{IL}$, the control is $u(t) := \dot{x}(t) \in \mathbb{R}^{IL}$, the dynamics are trivial $\dot{x} = u$, and the objective is the line-integral form of Equation (6.3) pulled through Φ :

$$\mathcal{V}(x, u) = \int_0^1 \langle a(x(t)), u(t) \rangle dt, \quad (6.6)$$

where $a : \mathbb{R}^{IL} \rightarrow \mathbb{R}^{IL}$ collects the components of the 1-form α in the Φ -coordinates. Constraints: initial $x(0) = \Phi(e^0)$; terminal peace $\phi(x(1)) \leq 0$; speed $\|u(t)\| \leq M$ almost everywhere. The budget constraint is treated as interior in the present analysis; a binding budget admits a parallel treatment with an additional costate multiplier on the budget constraint, omitted here.

The Hamiltonian and the switching function. The Pontryagin Hamiltonian for the donor's problem is

$$H(x, \psi, u) := \langle a(x) + \psi, u \rangle, \quad (6.7)$$

where $\psi(t) \in \mathbb{R}^{IL}$ is the costate. Substantively, $\psi(t)$ is the donor's shadow value of the equilibrium position $x(t)$: the marginal welfare gain to the donor of the economy being at $x(t)$ rather than at a slightly different equilibrium. The Hamiltonian combines the direct preference gain $\langle a(x), u \rangle$ from moving in direction u with the shadow gain $\langle \psi, u \rangle$ of having moved.

The Hamiltonian is linear in the control u . Maximizing H over $\|u\| \leq M$ therefore drives u to the boundary along the gradient direction. Define the *switching function*

$$S(t) := a(x^*(t)) + \psi(t). \quad (6.8)$$

$S(t)$ is the donor's instantaneous urgency: the combined direct and shadow marginal value of moving the equilibrium forward at time t . Maximization of H over $\|u\| \leq M$ at time t yields $u^*(t) = M S(t)/\|S(t)\|$ when $S(t) \neq 0$: the direction is the unit vector in the urgency direction, and the speed is M . When $S(t) = 0$, the maximization is degenerate at first order and a singular-arc analysis is required; the bang-bang characterization rules this out generically.

The Pontryagin theorem. The maximum principle gives necessary conditions every optimal schedule satisfies.

Theorem 6.8 (Pontryagin characterization). *Let γ^* be an optimal aid schedule and $x^* = \Phi \circ \gamma^*$ its trajectory in flat coordinates. There exist an absolutely continuous costate $\psi : [0, 1] \rightarrow \mathbb{R}^{IL}$ and a multiplier $\lambda \geq 0$ such that for almost every t :*

- (i) Costate evolution. $\dot{\psi}(t) = -Da(x^*(t))^\top u^*(t)$, where Da is the Jacobian of a in Φ -coordinates.

- (ii) Maximality. $u^*(t) \in \arg \max_{\|u\| \leq M} \langle a(x^*(t)) + \psi(t), u \rangle$.
- (iii) Transversality. $\psi(1) = \lambda \nabla \phi(x^*(1))$, with $\lambda \phi(x^*(1)) = 0$.

The switching function evolves according to

$$\dot{S}(t) = \iota_{u^*(t)} d\alpha \big|_{x^*(t)}, \quad (6.9)$$

the interior product of the optimal control with the exterior derivative of α .

Proof sketch. Conditions (i)–(iii) are the standard Pontryagin maximum principle for an optimal control problem with a terminal inequality constraint [Liberzon, 2011b]. The switching-function evolution follows from differentiating $S = a(x^*) + \psi$ along the trajectory and identifying the resulting antisymmetric Jacobian piece $(Da - Da^\top)u^*$ with the interior product $\iota_{u^*} d\alpha$ in coordinates. $\square$

The substance of Theorem 6.8 lives in the switching-function evolution Equation (6.9). When the donor's preference 1-form α is closed ($d\alpha = 0$), the switching function is constant along the trajectory: the donor's urgency does not change as the economy moves. When α is not closed, $d\alpha$ is a non-zero 2-form on $\tilde{E}$, and the switching function evolves with the donor's direction of travel. The two regimes correspond to closed and non-closed preferences—path-independent and path-dependent donor welfare functionals, respectively. This is the structural meaning of the closed-vs-non-closed distinction introduced in Section 6.1: closed preferences make the optimal schedule's commodity mix constant along the trajectory; non-closed preferences make it shift.

Bang-bang under absolute objectives. The first ideal-typical pattern is bang-bang: the donor delivers at maximum speed almost everywhere.

Corollary 6.9 (Bang-bang). *At any time t with $S(t) \neq 0$, the optimal control satisfies $\|u^*(t)\| = M$. Singular arcs (intervals on which $S \equiv 0$) require $u^*(t)$ to lie in the characteristic distribution of the 2-form $d\alpha$ at $x^*(t)$. For generic donor preferences, the characteristic distribution is trivial on $\tilde{E}$ of dimension $IL \geq 2$ even, and no singular arcs at positive speed exist.*

Proof. Maximality of Theorem 6.8 gives $u^* = MS/\|S\|$ when $S \neq 0$, hence $\|u^*\| = M$. When $S = 0$, the switching evolution Equation (6.9) forces $\iota_{u^*} d\alpha = 0$, placing u^* in the kernel of the contraction map $v \mapsto \iota_v d\alpha$, i.e., the characteristic distribution of $d\alpha$. A 2-form on a vector space of even dimension is generically non-degenerate (symplectic), with trivial characteristic distribution; the donor's preference 1-form α generic with respect to this non-degeneracy condition has no singular arcs at positive speed. $\square$

The policy reading is direct. Under generic donor preferences and a binding speed constraint (absorptive capacity), optimal aid is never gradual by choice. The donor pushes the equilibrium at the maximum admissible rate until the peace component is reached; pacing the aid at less than maximum speed is suboptimal unless the donor happens to be exactly indifferent at the margin across all commodity dimensions, which is non-generic. The substantive choice variable is direction, not pace.

The non-degeneracy condition has a clean reading. A non-degenerate 2-form on an even-dimensional space is a symplectic structure; symplectic structures are the generic case on $\mathbb{R}^n$ for even n . For odd-dimensional state spaces, the corollary's non-degeneracy condition fails

identically, and singular arcs become non-degenerate phenomena. The donor's problem on $\tilde{E}$ has dimension IL , and IL is even unless one of I or L is unity. For $I \geq 2$ states and $L \geq 2$ goods (the substantive range), $IL \geq 4$ is even, the non-degeneracy condition is generic, and bang-bang is the typical regime.

Gradualism under convex efficiency cost. The bang-bang result depends on the Hamiltonian being linear in the control. When the donor pays a convex cost for speed—faster aid is proportionally more expensive—the Hamiltonian becomes strictly concave in u , and the maximum is interior.

Suppose the donor's objective includes a speed-dependent efficiency cost:

$$\mathcal{V}_C(\gamma) = \int_0^1 [\langle a(\gamma(t)), \gamma'(t) \rangle - C(\|\gamma'(t)\|)] dt, \quad (6.10)$$

with $C : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ smooth, strictly convex, increasing, and $C(0) = 0$. The cost $C(\|u\|)$ records the additional inefficiency of rapid disbursement [Rajan and Subramanian, 2008]: a slower schedule has lower per-unit overhead. The Hamiltonian for the modified problem is $H_C(x, \psi, u) = \langle a(x) + \psi, u \rangle - C(\|u\|)$, strictly concave in u , with the maximum at an interior point when $\|S\|$ falls below the marginal cost at the speed ceiling.

Proposition 6.10 (Gradualism under convex cost). *Under the modified objective Equation (6.10) with C strictly convex and increasing, the optimal direction remains $u^*/\|u^*\| = S/\|S\|$ (whenever $S \neq 0$), and the optimal speed is*

$$\|u^*(t)\| = \begin{cases} 0 & \|S(t)\| \leq C'(0), \\ (C')^{-1}(\|S(t)\|) & C'(0) < \|S(t)\| < C'(M), \\ M & \|S(t)\| \geq C'(M). \end{cases}$$

The optimal speed is strictly increasing in $\|S\|$ on the interior regime.

Proof. Fixing direction $v = u/\|u\|$, the reduced problem maximizes $\|S\|s - C(s)$ over $s \in [0, M]$. The objective is strictly concave in s by strict convexity of C ; the unconstrained maximum is at $C'(s) = \|S\|$, i.e., $s = (C')^{-1}(\|S\|)$, feasible when $C'(0) \leq \|S\| \leq C'(M)$. Outside this interval, the optimum is at the boundary $s = 0$ or $s = M$. The optimal direction maximizes $\langle S, v \rangle$ over unit vectors, giving $v = S/\|S\|$. $\square$

The proposition produces a three-regime characterization. *Pause:* when the donor's urgency is below the marginal cost at zero speed ($\|S\| \leq C'(0)$), the optimal control is zero—no aid flows because the marginal welfare gain is insufficient to cover the marginal efficiency loss. *Interior:* when the urgency is intermediate, the optimal speed is the smooth function $(C')^{-1}(\|S\|)$ of urgency, and the donor paces the schedule. *Maximum:* when the urgency exceeds the marginal cost at the speed ceiling, the optimal control is M , and the absorptive-capacity constraint binds.

The three regimes describe a donor whose pace responds to the state of the conflict. When the equilibrium is far from the peace component and the preference gradient is steep, the donor pushes hard. As the equilibrium approaches the boundary of $\mathcal{W}$, the gradient shrinks and the donor slows; eventually the optimal schedule pauses entirely, having reached a local minimum of the objective without quite leaving $\mathcal{W}$ at minimum cost. The bang-bang regime is the special case in which the urgency remains uniformly above $C'(M)$ throughout the trajectory.

Reading the apparatus. Three ingredients of the framework's apparatus do specific work in the PMP analysis.

The manifold structure of Chapter 3 delivers the chart $\Phi : \tilde{E} \rightarrow \mathbb{R}^{IL}$ that reduces manifold-PMP to flat-space PMP under Assumption 3.6. Without that diffeomorphism, costates would have to be developed intrinsically as 1-forms on $\tilde{E}$, and the standard textbook formulation [Liberzon, 2011b] would not apply directly. Under Assumption 3.17 the chart exists componentwise; the PMP analysis operates on each regime component separately, and the donor's problem becomes a manifold-PMP problem with a non-trivial topology between components. Chapter 7 reads cross-component motion as regime change.

The properness of the natural projection Lemma 3.10 of Chapter 3 ensures the costate trajectories are bounded. A divergent costate would correspond to an unbounded shadow value of the equilibrium position, which would in turn correspond to an unbounded sensitivity of the donor's welfare to the equilibrium position. Properness rules this out by bounding equilibrium responses to bounded parameter perturbations.

The peaceful-efficiency theorem Theorem 5.8 of Chapter 5 makes the transversality condition concrete. At the terminal time, $\psi(1) = \lambda \nabla \phi(x^*(1))$ records that the costate at the moment of reaching peace is proportional to the gradient of the conflict-intensity function. When ϕ is the welfare-flavored $\Delta^* \circ \tilde{\pi}$, the gradient is exactly the master sensitivity formula Equation (5.10), and the transversality condition records that the donor's terminal shadow value is the welfare gradient of the conflict externality. The PMP transversality is therefore a statement about how the donor's marginal value of peace at the moment of reaching it relates to the welfare structure of Chapter 5.

6.5 Donor coordination

The preceding sections treat the donor as a single actor. In practice peacemaking aid flows from many sources at once: bilateral donors, multilateral institutions, NGOs, private foundations. Olson and Zeckhauser [1966] established that alliance public goods are systematically undersupplied when members contribute independently. This section extends that result from the static defense-spending setting to the dynamic peacemaking setting of the chapter. The substantive output is sharp. When n donors share the preference structure of the donor's problem and contribute non-cooperatively, the aggregate peace trajectory is delivered at exactly $1/n$ the cooperative speed. Fragmented bilateral aid reaches peace n times more slowly than a cooperative arrangement with the identical aggregate budget. The result has institutional content. It is a resource-neutral rationale for multilateral peacemaking institutions, distinct from the transaction-cost rationale of Keohane [1984] and the centralization-and-information rationale of Abbott and Snidal [1998].

Setup. Each donor $j \in \{1, \dots, n\}$ independently chooses a transfer schedule $\tau_j : [0, 1] \rightarrow \mathbb{R}_{\geq 0}^L \times \mathbb{R}_{\geq 0}^L$ with $\tau_j(0) = 0$, budget ceiling $\bar{\tau}_j \in \mathbb{R}_{++}^L$, efficiency parameter $\eta_j \geq 1$, and speed limit $M_j > 0$. Write $\mathcal{T}_j$ for donor j 's feasible-transfer set, defined exactly as in Section 6.1 with j -specific parameters; $\mathcal{T} := \mathcal{T}_1 \times \dots \times \mathcal{T}_n$ is the joint strategy space.

The aggregate transfer is $\tau := \sum_j \tau_j$, with the joint endowment path

$$\omega_A(t) = \omega_A^0 + \sum_j \tau_{j,A}(t), \quad \omega_B(t) = \omega_B^0 + \sum_j \tau_{j,B}(t),$$

and donor endowments shrink by $\sum_j \eta_j(\tau_{j,A} + \tau_{j,B})$ in aggregate. The induced aid schedule $\gamma : [0, 1] \rightarrow \tilde{E}$ is the unique lift of the aggregate endowment path through Definition 6.1; it is a function of τ alone, not of the individual τ_j separately. This is the structural content of the public-good observation: each donor's payoff

$$V_j(\tau_1, \dots, \tau_n) := \int_{\gamma} \alpha_j - \int_0^1 C_j(\|\tau'_j(t)\|) dt$$

depends on the aggregate trajectory γ through the shared preference functional $\int_{\gamma} \alpha_j$, while the private cost C_j depends on the donor's own contribution alone. A donor that increases τ_j pays the marginal cost privately and contributes the marginal benefit to all n donors.

Nash equilibrium among donors. The strategic interaction is captured by a standard Nash equilibrium concept on the joint strategy space.

Definition 6.11 (Nash equilibrium among donors). A profile $(\tau_1^*, \dots, \tau_n^*) \in \mathcal{T}$ is a *Nash equilibrium* if for each j ,

$$V_j(\tau_j, \tau_{-j}^*) \leq V_j(\tau_j^*, \tau_{-j}^*) \quad \text{for all } \tau_j \in \mathcal{T}_j.$$

Existence of a Nash equilibrium follows from a standard fixed-point argument under a quasi-concavity assumption on each donor's payoff in their own strategy.

Assumption 6.12 (Quasi-concave payoffs). For each donor j and each fixed profile τ_{-j} , the payoff $V_j(\tau_j, \tau_{-j})$ is quasi-concave in τ_j .

Quasi-concavity is the standard sufficient condition for well-behaved best-response correspondences in continuous-strategy games. Two sufficient conditions for Assumption 6.12 are worth recording. First, when all goods are gross substitutes for every actor in the recipient economy, the equilibrium is globally unique at each endowment profile (Arrow–Block–Hurwicz), the map $\tau_j \mapsto \gamma$ is a smooth function, and V_j inherits quasi-concavity from the strict convexity of $-C_j$. Second, for small transfers—when $\|\sum_j \tau_j\|$ is small relative to initial endowments—the equilibrium path remains in a neighborhood of e^0 where the inverse function theorem ensures $\tau_j \mapsto \gamma$ is approximately affine and quasi-concavity holds locally. The chapter takes Assumption 6.12 as a maintained assumption.

Theorem 6.13 (Existence of Nash equilibrium). *Under Assumption 6.12, the multi-donor game has a Nash equilibrium.*

Proof sketch. Each $\mathcal{T}_j$ is convex and compact: budget and speed constraints are linear in τ_j , and Lemma 6.4 applies componentwise. Continuity of V_j in the product topology follows from Theorem 6.5 applied to the aggregate path. Combined with quasi-concavity, this gives best-response correspondences with non-empty convex compact values that are upper hemicontinuous via Berge's maximum theorem. The Fan–Glicksberg theorem delivers a fixed point, which is a Nash equilibrium. $\square$

The framework's apparatus contributes the existence argument's ingredients in three places. Chapter 3's manifold structure makes the aid-schedule space well-behaved. Lemma 6.4 delivers compactness of the single-donor feasible set; the multi-donor compactness is the product. Theorem 6.5, applied to the aggregate, delivers continuity of V_j in each donor's strategy. The fixed-point theorem then runs as in any standard Nash-existence argument.

Nash–Pontryagin conditions. Each donor’s problem, taking τ_{-j}^* as given, is an optimal control problem of the form analyzed in Section 6.4. Applying Theorem 6.8 to each donor produces a coupled system of necessary conditions.

Proposition 6.14 (Nash–Pontryagin conditions). *At a Nash equilibrium $(\tau_1^*, \dots, \tau_n^*)$ there exist costates $\psi_j : [0, 1] \rightarrow \mathbb{R}^{IL}$ and multipliers $\lambda_j \geq 0$ such that for almost every t :*

- (i) Costates. $\dot{\psi}_j(t) = -Da_j(x^*(t))^\top u^*(t)$, where $u^*(t) = \sum_k (\tau_k^*)'(t)$ is the aggregate velocity.
- (ii) Switching functions. $S_j(t) := a_j(x^*(t)) + \psi_j(t)$ evolves as $\dot{S}_j = \iota_{u^*} d\alpha_j$.
- (iii) Optimality. $(\tau_j^*)'(t) \in \arg \max_{\|u_j\| \leq M_j} \langle S_j(t), u_j \rangle - C_j(\|u_j\|)$, yielding $C'_j(\|(\tau_j^*)'(t)\|) = \|S_j(t)\|$ at interior optima.
- (iv) Transversality. $\psi_j(1) = \lambda_j \nabla \phi(x^*(1))$, with $\lambda_j \phi(x^*(1)) = 0$.

The substantive feature of Proposition 6.14(i) is that each donor’s costate is driven by the aggregate velocity u^* , not by donor j ’s own contribution $(\tau_j^*)'$. Donor k ’s actions shift donor j ’s switching function S_j through the state x^* , and through that channel alter j ’s optimal direction and pace. Strategic complementarities and substitutabilities propagate across donors through the shared state, exactly as in any non-cooperative game with a public-good payoff structure. The mathematics of the propagation is the manifold-PMP machinery of Section 6.4 run n times in parallel, with each donor’s costate evolving along the common aggregate trajectory.

The free-rider problem. The substantive consequence of the public-good structure shows up sharply when donors share the same preference. Take *common preferences*: $\alpha_j = \alpha$ and $C_j = C$ for all j , with equal speed bounds $M_j = M/n$ so the joint speed limit is M . The aggregate trajectory is then a public good in the strict sense—all donors enjoy the same path—while each donor pays its private efficiency cost.

Proposition 6.15 (Free-rider under common preferences). *Under common preferences, at the symmetric Nash equilibrium each donor contributes $u_j^*(t) = u^*(t)/n$, and the aggregate optimality condition is*

$$C'(\|u^*(t)\|/n) = \|S^*(t)\|.$$

The cooperative optimum, by contrast, satisfies

$$C'(\|u^{\text{co}}(t)\|/n) = n \|S^{\text{co}}(t)\|.$$

For $n > 1$ and any switching-function value $\|S\| > C'(0)$, the cooperative aggregate speed strictly exceeds the Nash aggregate speed.

Proof. At the symmetric Nash equilibrium, all donors share the same costate (the costate ODE is the same for all j when $\alpha_j = \alpha$, with the same driving aggregate velocity), so $S_j(t) = S^*(t)$ for all j . Optimality (iii) of Proposition 6.14 gives $C'(\|u_j^*\|) = \|S^*\|$ for each j ; symmetry gives $u_j^* = u^*/n$, so $C'(\|u^*\|/n) = \|S^*\|$.

For the cooperative problem, the social planner maximizes $\sum_j V_j = n \int_\gamma \alpha - n \int_0^1 C(\|u^{\text{co}}/n\|) dt$ (by symmetry, equal contributions are optimal) over $\|u^{\text{co}}\| \leq M$. The Pontryagin condition for the aggregate gives $C'(\|u^{\text{co}}\|/n) = n \|S^{\text{co}}\|$. Strict monotonicity of $(C')^{-1}$ and $n > 1$ then give $\|u^{\text{co}}\|/n > \|u^*\|/n$ at the same switching-function value, hence $\|u^{\text{co}}\| > \|u^*\|$. $\square$

The structural content of Proposition 6.15 is the n in the cooperative optimality condition. Each donor in the Nash equilibrium equates its private marginal cost $C'(\|u_j^*\|)$ to the donor's own marginal benefit $\|S^*\|$. The cooperative planner equates the aggregate marginal cost $C'(\|u^{\text{co}}/n\|)$ to the *aggregate* marginal benefit $n\|S^{\text{co}}\|$, recognizing that each unit of aggregate progress on the trajectory contributes to all n donors' welfare. The factor n is the multiplier any single donor's contribution provides through the public-good channel; the Nash equilibrium suppresses this multiplier by construction.

Time-to-peace: n times slower. The free-rider proposition becomes a sharp time-scaling result in the leading specification.

Corollary 6.16 (Time-to-peace). *With quadratic efficiency cost $C(s) = \kappa s^2$ and common preferences, the cooperative aggregate speed is exactly n times the Nash aggregate speed at every t :*

$$\|u^{\text{co}}(t)\| = n\|u^*(t)\|.$$

Cooperative peacemaking reaches any target on $\partial\mathcal{W}$ in $1/n$ the time of the Nash equilibrium. The Nash equilibrium may fail to reach peace within the horizon $[0, 1]$ even when cooperation would.

Proof. For $C(s) = \kappa s^2$, $C'(s) = 2\kappa s$ and $(C')^{-1}(x) = x/(2\kappa)$. Proposition 6.15 gives Nash speed $\|u^*\|/n = \|S\|/(2\kappa)$, hence $\|u^*\| = n\|S\|/(2\kappa)$. Cooperative speed: $\|u^{\text{co}}\|/n = n\|S\|/(2\kappa)$, hence $\|u^{\text{co}}\| = n^2\|S\|/(2\kappa) = n\|u^*\|$. Time-to-target along the same direction is the inverse of speed, so $T_{\text{co}} = T_{\text{Nash}}/n$. $\square$

The corollary is the chapter's institutional headline. Time-to-peace scales linearly with the number of independent donors. Each additional donor acting non-cooperatively slows the collective response by a factor of n relative to a unified actor with the same aggregate budget. The mechanism is not budget inadequacy—the total resources are identical—but the failure of each donor to internalize the benefit that its own contribution confers on the other $n - 1$.

The empirical content is severe. A peacemaking effort split among five non-cooperating donors delivers peace at one-fifth the cooperative speed; if the donor horizon is shorter than the Nash time-to-peace, the conflict runs to its endogenous conclusion despite the aggregate budget being adequate for cooperative termination. The conditions for the corollary's literal n -scaling are restrictive (common preferences, quadratic cost), but the underlying logic is robust. The cooperative speed exceeds Nash speed under any strictly convex C and any preference structure satisfying the public-good property; only the magnitude of the gap depends on specification.

The institutional implication. Proposition 6.15 and Corollary 6.16 give the rational-design theory of international institutions [Koremenos et al., 2001, Abbott and Snidal, 1998] a resource-neutral foundation. The slowness of fragmented bilateral aid is not a consequence of administrative friction or scale diseconomies; it is a consequence of the peace trajectory being a public good whose acceleration each donor undervalues by a factor of n , the continuous-time analogue of the Olson and Zeckhauser [1966] alliance pathology.

The rationale is structurally distinct from the existing multilateralism literature. Keohane [1984] grounds multilateralism in the reduction of transaction costs across iterated cooperation: institutions matter because they make the gains from cooperation easier to realize. Abbott and Snidal [1998] grounds multilateralism in the centralization of information and the credibility of monitoring: institutions matter because they aggregate information that states alone could not.

The chapter's rationale is neither. Even with identical transaction costs to bilateral aid, identical information environments, identical administrative apparatus, the multi-donor peacemaking effort is n times slower than the unified equivalent. The rationale for institutional centralization here is the internalization of the peace-trajectory externality at the optimization stage, not at the transaction stage or the information stage.

Concrete institutional designs that internalize this externality include joint conditionality (donors commit to a common disbursement schedule), multilateral peace funds (a pooled budget deployed by a unified disbursement authority), and unified mandates that constrain bilateral discretion within a coordinated plan. Each delivers the cooperative speed with identical aggregate resources by structurally preventing each donor from optimizing against the others' contributions. The empirical pattern of donor fragmentation as a welfare loss is documented by [Knack and Rahman \[2007\]](#) and [Acharya et al. \[2006\]](#), who attribute the loss to administrative-cost duplication and bureaucratic friction. The framework supplies a complementary structural mechanism: even with no administrative friction, fragmentation slows the peace trajectory by the public-good factor. The clearest historical example of cooperative deployment is the Marshall Plan ([De Long and Eichengreen 1993](#)): a unified disbursement authority deploying pooled resources along a coordinated trajectory at a speed that fragmented bilateral disbursement would have decelerated.

Reading the apparatus. Three pieces of the framework's apparatus do specific work in the multi-donor analysis.

Theorem 6.5's existence argument provides the single-donor baseline that the multi-donor existence theorem Theorem 6.13 extends. Without manifold-Arzelà-Ascoli (Lemma 6.4) and the manifold structure of Chapter 3, the single-donor compactness would not be available, and the Fan-Glicksberg argument for multi-donor Nash existence would have no foundation.

Theorem 6.8's switching-function machinery, run n times in parallel, gives the Nash-Pontryagin coupled system Proposition 6.14. The substantive feature—each donor's costate driven by the aggregate velocity—is exactly the manifold-PMP machinery applied to a non-cooperative game with a shared state. Without the smooth-manifold setting of Chapter 3, the costates would not be globally well-defined, and the Nash-Pontryagin analysis would not run.

Chapter 5's peaceful-efficiency theorem (Theorem 5.8) makes the cooperative-vs-Nash speed comparison welfare-meaningful. The aggregate trajectory's contribution to donor welfare is the welfare cost of conflict Δ^* along the trajectory; the public-good structure of the trajectory means that cooperation delivers Δ^* -reduction n times faster than Nash. The institutional rationale is therefore a welfare claim with Δ^* as the cardinal scalar, not just a coordination claim with abstract benefit functions.

6.6 Looking ahead

The chapter delivered the donor's problem as control on the equilibrium manifold, and developed five substantive results through the apparatus. Theorem 6.5 delivered an optimal aid schedule without a convexity assumption. Corollary 6.6 established that aid evaluated at fixed prices cannot reduce conflict whenever the conflict primitive operates through the price system. Theorem 6.8 characterized optimal schedules through a manifold version of Pontryagin's maximum principle, with Corollary 6.9 and Proposition 6.10 as the bang-bang and gradualism patterns that emerge under absolute and speed-cost objectives. Theorem 6.13 delivered a Nash equilibrium for non-cooperative multi-donor aid, and Corollary 6.16 established that fragmented donors deliver

peace exactly n times more slowly than a cooperative arrangement with identical aggregate resources. The chapter’s content is one apparatus—joint-equilibrium economics applied to donor intervention—in five positions.

Each result inherits structure from a specific upstream chapter, which is what makes the chapter substantively leaner than its standalone-paper predecessor. Chapter 3’s manifold structure (Assumption 3.6) delivers $\tilde{E} \cong \mathbb{R}^{IL}$ as the natural ambient space for aid schedules; the chart Φ reduces manifold-PMP to flat-space PMP; the properness of $\tilde{\pi}$ converts budget bounds into compactness. Chapter 4’s comparative-statics master formula reads the equilibrium response to a transfer as one application of J^{-1} . Chapter 5’s peaceful-efficiency theorem (Theorem 5.8) supplies the peace component $\tilde{E}_p \subset \tilde{E}$ as target submanifold and certifies its welfare content under zero- or negative-sum warfare. Chapter 5’s sensitivity formula (Equation (5.10)) and three-channel decomposition make Δ^* available as the donor’s natural welfare-economic objective and locate the deployment-cost-feedback channel as exactly what partial-equilibrium thinking zeros out.

Subsequent chapters consume the chapter’s apparatus in graduated ways. Chapter 7 consumes Corollary 6.16’s resource-neutral multilateralism rationale as one mode of institutional design distinct from transaction-cost (Keohane 1984) and information-aggregation (Abbott and Snidal 1998) rationales. The same chapter consumes the cross-component motion of the manifold reading—when an aid trajectory crosses the singular locus Σ —as part of its bifurcation-as-regime-change apparatus, with the donor’s problem providing one substantive case of induced regime change. Chapter 8 consumes the manifold-Pontryagin and free-rider apparatus from a different direction: a sanctions schedule is a (target, path) institution on the equilibrium manifold (Section 7.5 of Chapter 7), with the sender’s optimization following the same Pontryagin structure the donor’s problem exhibits in this chapter, and multilateral sanctions coordination running the same free-rider apparatus as multilateral aid.

The chapter admits two readings. Read as substantive donor-intervention theory, it is a structural account of peacemaking aid with three results no partialled framework can deliver—a transversality theorem on the equilibrium manifold (Corollary 6.6), an existence-without-convexity theorem for optimal schedules (Theorem 6.5), and a coordination result that gives multilateralism a resource-neutral rationale (Corollary 6.16)—together with a characterization theorem (Theorem 6.8) that covers the optimal-schedule structure under both bang-bang and gradualism patterns. The substantive content connects to a broader empirical literature on civil war duration and aid: Fearon [2004] documents the heterogeneity of conflict duration and its covariates; Fearon and Laitin [2003] the insurgency-and-resource correlates; and the Caselli et al. [2015] interstate-resource-wars analog locates the same channel mechanism at the cross-border scale. The chapter’s structural account of when aid speeds peace and when it does not is consistent with this empirical literature and supplies a structural-formal complement to it. Read as apparatus-demonstration, the chapter is a worked extended case for the framework’s signature capability—joint-equilibrium economics applied to a substantive policy question—with the manifold structure of Chapter 3, the master sensitivity formula of Chapter 5 Section 5.4, and the welfare cost Δ^* as the throughline. Both readings are views of the same apparatus running on the same five sections.

Exercises

1. *Absolute continuity of the endowment path.* The donor's transfer schedule $\tau \in \mathcal{F}$ (Definition 6.1) determines an endowment path $\omega(t) = \omega_0 + \int_0^t \tau(s) ds$ through Equation (6.1), where $\omega_0 \in \Omega_0$ is the initial endowment. Verify that $\omega(t)$ is absolutely continuous on $[0, T]$ whenever τ is integrable, and identify where in the existence argument (Theorem 6.5) absolute continuity is required. The proof is the standard fundamental-theorem-of-calculus property for Lebesgue integrals; the substantive content is the modeling claim, namely that endowments evolve smoothly under absolutely continuous transfers, an operational assumption that distinguishes the smooth donor model from one in which transfers arrive as discrete shocks.
2. *IFT-on-manifold lift.* Given an absolutely continuous path $\omega : [0, T] \rightarrow \Omega_0$ that avoids the singular locus Σ , the donor's joint-equilibrium trajectory is the lifted path $e : [0, T] \rightarrow \tilde{E}^{\text{reg}}$ satisfying $\tilde{\pi} \circ e = \omega$. Sketch the construction of this lift in three steps: (i) at each $t \in [0, T]$, apply the implicit function theorem to the local diffeomorphism $\tilde{\pi}$ delivers under Assumption 3.17 to obtain a smooth local lift on a neighborhood of $\omega(t)$; (ii) cover $[0, T]$ with finitely many such neighborhoods using compactness; (iii) patch the local lifts using the smooth-manifold structure of $\tilde{E}^{\text{reg}}$ (Proposition 3.18 of Chapter 3). Show that the resulting lift is unique when its initial value $e(0)$ is specified, and identify what goes wrong if the path crosses Σ .
3. *Closedness of the welfare 1-form.* The donor's welfare-objective 1-form is $\alpha = -d\Delta^* \circ \tilde{\pi}$, the pull-back to $\tilde{E}$ of the differential of the conflict externality $\Delta^* : \Omega_0 \rightarrow \mathbb{R}$ (Chapter 5). First, verify that α is closed: since $\Delta^* \circ \tilde{\pi}$ is a smooth real-valued function on $\tilde{E}$, its exterior derivative is automatically closed by $d^2 = 0$. Second, identify classes of donor preferences for which the appropriate α is genuinely non-closed: non-welfare donors (e.g., a donor whose preferences trade off arming against price stability with a path-dependent weight), and multi-objective donors with non-integrable trade-offs between competing goals. The closedness or non-closedness of α is the structural feature that distinguishes commitment-only donors (closed α , integrable preferences) from strategic-evaluation donors (non-closed α , history-dependent preferences).
4. *Constructing a non-closed α .* Construct an explicit non-closed donor 1-form α on a 2-good 2-state specialization of the framework. Take a donor whose preferences depend separately on the price level p^1 and the arming level $\|m\|$, with a trade-off weight $\beta(p^1, \|m\|)$ that depends on the current $(p^1, \|m\|)$ via a non-symmetric function (e.g., $\beta = p^1 \cdot \|m\|^2$ rather than a gradient of any scalar). Compute $d\alpha$ as a 2-form and verify that at least one component is non-zero. Identify which component of the donor's preference is path-dependent: the donor cares not just about the current state but about the order in which states were visited. Open-ended; many constructions work.
5. *War region of Example 3.3 with state C as donor.* Consider Example 3.3 extended with a third state C that takes no military action but operates the donor problem. The war region for the donor is $\mathcal{W} = \{\omega \in \Omega_0 : \|m^*(\omega)\| > \bar{m}\}$, where $\bar{m} \geq 0$ is the donor's peace threshold. Using the closed-form $m^*(V_0, \theta, W) = (V_0 + \theta W)/(2k(2 + \theta))$ from Chapter 3, write $\mathcal{W}$ explicitly in (V_0, θ, W) coordinates. For the natural threshold $\bar{m} = 0$ (any positive arming counts as war), the war region is the entire domain at $V_0 > 0$; for $\bar{m} > 0$, the war

region is the set of (V_0, θ, W) for which the closed form exceeds $\bar{m}$. Identify the boundary $\partial\mathcal{W}$ and verify it is smooth where $\det J \neq 0$.

6. *Loan vs. grant: lifting loops.* A grant is a one-way endowment path $\omega : [0, T] \rightarrow \Omega_0$; a loan is a round-trip path with $\omega(0) = \omega(T)$, i.e., a loop in Ω_0 . The framework's regime structure controls whether such loops can be implemented as joint-equilibrium trajectories. First, under Assumption 3.6 (global JR), verify that every loop in Ω_0 lifts to a loop in $\tilde{E}$: the global diffeomorphism $\tilde{\pi}$ (Proposition 3.15 of Chapter 3) makes the lifting trivial, and round-trips in endowments produce round-trips in (p^*, m^*) . Second, under Assumption 3.17 (regime structure with Σ of measure zero), construct a loop in Ω_0 that lifts to a *path*, not a loop, in $\tilde{E}$: the loop crosses Σ in two places, and the system exits its initial regime component between the crossings. Substantive content: loans (round-trip endowment paths) can produce regime change that grants (one-way paths) cannot, because regime crossings are recorded in the (p^*, m^*) coordinates and cannot be undone by a counterbalancing endowment shift.
7. *Compactness of the admissible-paths set.* Lemma 6.4 establishes that the admissible-paths set $\mathcal{F}$ defined in Definition 6.1 is compact in the uniform topology, the input the existence theorem Theorem 6.5 consumes. Reproduce the proof using the Arzelà-Ascoli theorem on the manifold $\tilde{E}^{\text{reg}} \subset S_N \times M \times \Omega_0$. Three ingredients: (i) equicontinuity from the speed bound M on τ in Definition 6.1; (ii) pointwise boundedness from the constraint $\omega(t) \in \Omega_0$ along the path (reachability, Assumption 6.3); (iii) closure under uniform limits from continuity of $\tilde{\pi}$ and properness of $\tilde{\pi}$ (Lemma 3.10 of Chapter 3). Identify which ingredient fails when Assumption 6.3 is dropped.
8. *Existence fails without the speed bound.* Show by counterexample that Theorem 6.5 fails if the speed bound M on τ in Definition 6.1 is dropped. Construct a minimizing sequence of admissible schedules $\{\tau^n\}$ with $\|\tau^n\|_\infty \rightarrow \infty$ along the sequence, on which the corresponding paths $\omega^n(t)$ converge pointwise to a discontinuous limit, and the limiting “schedule” fails absolute continuity (Exercise 1). Conclude that the infimum is approached but not attained. Substantive content: the speed bound is what makes the donor problem an optimal-control problem rather than an ill-posed limit; without it, the donor's hypothetical “infinitely fast” transfer is uniformly preferred but not implementable.
9. *Uniqueness vs. multiplicity of donor minimizers.* Discuss when the donor's optimal-control problem with welfare objective $\alpha = -d\Delta^* \circ \tilde{\pi}$ delivers a unique minimizer, and when there are multiple minimizers. Three sources of multiplicity to consider: (a) regime structure, where paths in different regular components of $\tilde{E}^{\text{reg}}$ can have the same integrated welfare cost; (b) symmetric configurations, where the donor problem inherits reflection symmetry from the underlying primitives (e.g., the symmetric equilibrium of Example 3.3); (c) flat regions of Δ^* , where multiple paths produce the same welfare trajectory. The substantive interpretation of multiplicity is that the framework signals genuine ambiguity about which peace-trajectory the donor prefers, rather than imposing an arbitrary selection. Open-ended; one paragraph per source.
10. *PE analogue of Theorem 6.5.* The partial-equilibrium (PE) analogue of the donor problem imposes $D_\omega p^* = 0$ throughout, treating equilibrium prices as exogenous. State the PE analogue of Theorem 6.5 explicitly: the admissible-paths set $\mathcal{F}^{\text{PE}}$ replaces $\tilde{E}$ with a fiber-by-fiber slice, and the objective V^{PE} omits the price-feedback channel. Verify that the

PE analogue holds (it is a well-posed optimal-control problem on the PE slice). Identify what substantive content the joint-equilibrium version captures that the PE version cannot: the price-feedback channel through which transfers shift equilibrium prices and thence equilibrium arming (Corollary 6.6). Substantive content: the PE formulation makes a category mistake by treating prices as exogenous when they are equilibrium objects of the same framework.

11. *2-good 1-donor donor problem.* For a 2-good 1-donor specialization of Example 3.3, set up the donor's problem explicitly. Specify the initial endowment ω_0 , the war-region threshold $\bar{m}$, the welfare 1-form $\alpha = -d\Delta^* \circ \tilde{\pi}$, and the speed bound M . Then identify each ingredient of Theorem 6.5: (a) the compact admissible-paths set $\mathcal{F}$ (Exercise 7); (b) the continuous objective V defined by integrating α along the path; (c) the extreme-value theorem application that delivers a minimizer $\tau^* \in \mathcal{F}$. Verify that the symmetric structure of Example 3.3 delivers a unique symmetric minimizer.
12. *Conflict-through-prices on the SPT primitives.* The PE-zero corollary Corollary 6.6 requires that equilibrium deployment depend on ω only through equilibrium prices: $m^*(\omega) = \hat{m}(p^*(\omega))$ for some smooth $\hat{m}$. Verify that this conflict-through-prices hypothesis holds on the SPT primitives of Chapter 2 for a wide class of contest specifications. Specifically: identify the structural condition on (s_i, c_i) under which $\partial s_i^j / \partial m_i$ and $\partial c_i / \partial m_i$ depend on ω only through p^* at the SPT solution. The price-dependence-of-cost architecture of Chapter 2 delivers this for ratio-form Tullock contests, difference-form contests (Hirshleifer 1989), and all-pay specifications, by construction. Identify the substantive feature shared across these specifications: spoils and costs are valued at market prices, with no direct ω -dependence in either primitive.
13. *Wealth-dependent contest defeats conflict-through-prices.* Construct a wealth-dependent contest in which the prize directly scales with the belligerents' good-1 holdings, as in Example 3.3 with $\theta > 0$. Show by direct computation that the conflict-through-prices hypothesis of Corollary 6.6 fails: the equilibrium arming $m^*(\omega)$ depends on ω through both $p^*(\omega)$ (the indirect channel) and the direct θ -channel introduced by the wealth-dependent prize. What replaces the PE-zero result? Show that under partial dependence (the prize depends on a weighted average of holdings rather than on individual holdings), a sharper PE-zero analogue can be recovered by restricting attention to symmetric subsets of Ω_0 . Open-ended; the substantive question is which weakenings of conflict-through-prices preserve which weakened versions of PE-zero.
14. *Asymmetric Cobb-Douglas: closed-form p^1 .* In the two-good Tullock setup of Example 6.7 with asymmetric Cobb-Douglas weights $\alpha_B^1 \neq \alpha_C^1$, state B values good 1 differently from the donor C . Derive p^1 in closed form by aggregating demands across B, C and imposing market clearing. The result is a function of $(\alpha_B^1, \alpha_C^1, \omega_B, \omega_C, \tau)$ where τ is the donor's transfer to B . Identify the sign of $\partial p^1 / \partial \tau$ in terms of the difference $(\alpha_C^1 - \alpha_B^1)$: the transfer raises p^1 when the recipient B values good 1 less than the donor C does, and lowers it when B values good 1 more.
15. *Elasticity of m^* with respect to p^1 .* The deployment FOC at the symmetric Tullock equilibrium with quadratic cost $C(m) = \theta m^2$ gives a closed-form elasticity $d \log m^* / d \log p^1 = 1/2$. Compute this elasticity directly from the FOC. Then show that the value $1/2$ is specific

to the quadratic-cost specification: under general convex cost $c(m)$, the elasticity equals $(1/2) \cdot c'(m)/(mc''(m))$, which reduces to $1/2$ for $c(m) = \theta m^2$ but takes other values for other curvatures. Substantively: the response of arming to a price shift is governed by the cost-curvature ratio, and the empirical quasi-elasticity is informative about the cost structure of contest production rather than the contest itself.

16. *Three-channel decomposition on a transfer.* For the Tullock setup of Example 6.7, run the three-channel decomposition of Equation (5.10) from Chapter 5 explicitly on a small donor transfer $\delta\tau$ that moves p^1 by δp . The three channels are: (i) direct-resource (the transfer adds resources to B), (ii) cost-feedback (the price shift changes the cost of arming through the price-dependence of c_i), and (iii) reallocation (the welfare-weight λ^* shifts under the new equilibrium). Verify that the cost-feedback channel matches the closed-form change in the deployment cost $2\theta(m^*)^2$ at the new p^1 . The three contributions should sum to the total $\delta\Delta^*$ from Equation (5.10).
17. *PE collapse of the master sensitivity.* Show that under the conflict-through-prices hypothesis of Corollary 6.6 and PE evaluation ($D_\omega p^* = 0$ imposed), the master sensitivity formula of Chapter 5 Equation (5.10) reduces to its direct-resource and reallocation channels alone: the cost-feedback channel vanishes when prices are held fixed. Substantive content: this is the cleanest formal statement of “what the partial-equilibrium evaluation gets wrong”—it captures two of three channels of the joint sensitivity and zeroes the third by construction. Direct application of the master formula and the conflict-through-prices hypothesis.
18. *Non-Tullock contests and conflict-through-prices.* Identify several non-Tullock contests of interest: difference-form CSF (Hirshleifer 1989), all-pay auction limit (Baye et al. 1996), and wealth-dependent variants (Exercise 13). For each, verify whether the conflict-through-prices hypothesis of Corollary 6.6 holds. The substantive question is which contest specifications admit clean PE-zero corollaries and which do not. The conditions typically distinguish the standard prize-over-the-table contests (which preserve conflict-through-prices) from the contests in which the contest payoff depends on the contestants’ holdings or budget directly (which break it). Open-ended.
19. *Making peace on the cheap: GE-path savings over robust peace.* The chapter’s central insight is that the donor’s task is *cheaper* when the donor approaches peace along the equilibrium manifold $\tilde{E}$ rather than insisting that every equilibrium at the terminal endowment be peaceful. This exercise makes the comparison precise. Throughout, fix a war initial condition $e_0 \in \mathcal{W}$ and a per-unit transfer cost $\eta > 0$.
 - (a) *Two notions of cost.* The *GE-path peace cost* $P^*(e_0)$ is the value of the donor problem Equation (6.4) as defined in Section 6.1: the minimum integrated transfer cost along an admissible path $e : [0, T] \rightarrow \tilde{E}$ that reaches the war-peace boundary $\partial\mathcal{W}$. The *robust peace cost* $R^*(e_0) = \eta \cdot \inf\{\|\Delta\| : \Delta \text{ is a robust peace transfer}\}$ is the minimum transfer required to reach a terminal endowment $\omega^* = \omega_0 + \Delta$ at which the entire fiber $\tilde{\pi}^{-1}(\omega^*)$ contains only peaceful equilibria. Show that $P^*(e_0) \leq R^*(e_0)$. (Hint: any robust peace transfer Δ also serves as a GE path—the donor moves endowments to a robust-peace target, picking up a peaceful equilibrium en route—at no greater cost.)
 - (b) *Source of the strict savings.* Define the savings $\mathcal{S}(e_0) := R^*(e_0) - P^*(e_0) \geq 0$. Show that $\mathcal{S}(e_0) > 0$ strictly whenever the terminal fiber of the optimal GE path— $\tilde{\pi}^{-1}(\omega^*)$

for ω^* on the boundary $\partial\mathcal{W}$ —contains both war and peace equilibria. The argument is geometric: the GE path can stop at the war-peace boundary because it tracks a specific equilibrium (a single point on $\tilde{E}$), so the donor needs only that this particular point be peaceful, not that every equilibrium at the same ω^* be peaceful. Robust peace, by contrast, requires reaching a fiber strictly inside the peace region. The strict savings is the cost of pushing ω^* further into the peace region than the GE path needs to. Substantive content: the GE path exploits the donor’s ability to ride the equilibrium manifold without fear of equilibrium collapse onto a war fiber—a freedom the robust-peace formulation explicitly forfeits.

- (c) *Genericity of mixed fibers under JR vs. JR’ (open)*. The condition for strict savings is that the terminal fiber $\tilde{\pi}^{-1}(\omega^*)$ at $\omega^* \in \partial\mathcal{W}$ contains both war and peace equilibria. The strength of this condition turns out to depend sharply on which of the chapter’s regularity hypotheses is in force.

Under Assumption 3.6 of Chapter 3: $\tilde{\pi}$ is a global diffeomorphism (Proposition 3.15), so each ω has exactly one equilibrium. Mixed fibers cannot exist, and the weak inequality $P^* \leq R^*$ collapses to equality $P^* = R^*$. Robust peace and the GE path terminate at the same single equilibrium, and the strict-savings condition is vacuous.

Under Assumption 3.17 of Chapter 3: $\tilde{E}^{\text{reg}}$ has the regime-component structure of Proposition 3.18, and Corollary 3.19 forces finite-and-odd cardinality at regular values, with fold-catastrophe pairing (Theorem 4.5 of Chapter 4, Remark 3.20) producing mixed fibers in the boundary region $\partial\mathcal{W}$ where war and peace components meet. In this setting, mixed fibers are generic *near* $\partial\mathcal{W}$ —which is exactly the region the GE path terminates in.

The proposition’s strict savings therefore live in the JR-gen regime, not the JR regime. Under JR, the proposition delivers only the weak inequality $P^* \leq R^*$ with equality. Under JR-gen, it delivers strict savings in the boundary region of $\partial\mathcal{W}$. Discuss the implication for how the chapter’s body framing should be read: the “generic mixed fibers” claim is generic in the regime-boundary sense under JR-gen, and it is vacuous under full JR. Identify a specific JR-gen primitive specification in which mixed fibers occur and verify the strict savings on it; identify, conversely, why the analogous JR specification (e.g., Example 6.7 read under full Assumption 3.6) gives only the weak inequality. Open-ended.

20. *Pontryagin on a 1-good 2-state specialization*. Verify Theorem 6.8 (the Pontryagin characterization of optimal aid schedules) directly on a 1-good 2-state specialization of the donor problem. The reduced setting has scalar $\tau \in \mathbb{R}$ as the control, scalar $\omega \in \mathbb{R}_{++}$ as the state, and V as a real-valued objective. Identify in closed form: (a) the Hamiltonian $\mathcal{H}$ from Equation (6.7); (b) the costate variable $\psi(t)$ and its evolution from the costate equation $\dot{\psi} = -\partial\mathcal{H}/\partial\omega$; (c) the switching function $\mathcal{S}(t)$ from Equation (6.8); and (d) the optimal control $\tau^*(t) = M \cdot \text{sign}(\mathcal{S}(t))$ from the maximum principle. Confirm that the qualitative features of the optimal aid schedule (bang-bang, gradualism under cost) emerge from this reduced setting.
21. *Constant switching function under closed α* . For a closed donor 1-form $\alpha = dV$ (where V is a smooth real-valued function on $\tilde{E}$), show that the switching function $\mathcal{S}(t)$ in Theorem 6.8 is constant along the optimal trajectory. Use the switching-evolution equation Equation (6.9): when α is closed, $d\alpha = 0$, and the right-hand side of the switching evolution identically

- vanishes. Substantive content: under closed (path-independent) donor preferences, the optimal commodity mix in the transfer does not change over time. The donor commits to a fixed transfer composition determined by the gradient of V at the initial point and varies only the speed at which the composition is delivered.
22. *Switching function evolution under non-closed α .* For a non-closed donor 1-form α from Exercise 4, compute $d\alpha$ as a 2-form and identify a non-trivial component. Substitute into the switching-evolution equation Equation (6.9) and verify that $\mathcal{S}(t)$ evolves with the donor's direction of travel: the rate at which the switching function changes equals the contraction of $d\alpha$ against the velocity of the trajectory. Substantive content: non-closed (path-dependent) donor preferences make the optimal commodity mix path-dependent, because the donor's marginal rate of substitution between transfer components varies with the equilibrium states the trajectory has already visited.
 23. *Bang-bang on the 2-good Tullock case.* Show that on the 2-good Tullock case of Example 6.7, the bang-bang corollary Corollary 6.9 applies: under any closed donor preference $\alpha = dV$, the optimal aid schedule moves at speed M almost everywhere on $[0, T]$. The argument requires verifying the hypotheses of Corollary 6.9 on the example: smoothness of the trajectory (delivered by Assumption 3.17), constant $\mathcal{S}(t)$ along the optimum (Exercise 21 under closed α), and non-vanishing of $\mathcal{S}(0)$ (which holds generically under non-trivial donor preferences). Conclude that the donor moves at maximum speed along a single direction determined by V .
 24. *Three regimes under quadratic cost.* For Proposition 6.10 with quadratic transfer cost $C(s) = \eta_0 s + \kappa s^2$ (linear plus convex), the optimal aid schedule exhibits three regimes: pause (zero speed), interior (intermediate speed), and max (speed bound M). Derive each regime explicitly from the FOC for the modified Hamiltonian $\mathcal{H} - C(s)$. Identify the urgency thresholds (η_0, κ, M) at which the optimal speed transitions: (i) from pause to interior, when the welfare urgency $|\mathcal{S}(t)|$ exceeds the marginal cost η_0 ; (ii) from interior to max, when it exceeds $\eta_0 + 2\kappa M$; (iii) skipping the interior regime, when $\kappa \rightarrow 0$ collapses the interior region. Each regime has a substantive interpretation in terms of the donor's marginal welfare gain versus marginal transfer cost.
 25. *Gradualism reduces to bang-bang as $\kappa \rightarrow 0$.* Verify that Proposition 6.10's gradualism characterization reduces to Corollary 6.9's bang-bang corollary in the limit $\kappa \rightarrow 0$. Concretely: take the closed-form characterization of the three regimes from Exercise 24, send $\kappa \rightarrow 0$, and observe that the interior regime collapses (the threshold $\eta_0 + 2\kappa M$ approaches η_0 , leaving no positive-measure interior). The optimal trajectory becomes pause-or-max with switching governed by the sign of $\mathcal{S}(t) - \eta_0$. Substantive content: the gradualism in Proposition 6.10 is driven entirely by the convex part of the cost; in the purely-linear limit, the donor's optimum reverts to bang-bang.
 26. *PMP across the singular locus.* Discuss how the PMP analysis of Theorem 6.8 differs when the optimal trajectory crosses the singular locus Σ (under Assumption 3.17, but not Assumption 3.6). The trajectory is non-smooth at the crossing because $\tilde{\pi}$ ceases to be locally invertible there, and the costate equation must be re-stated. Three formulations to consider: (a) treat the crossing as an instantaneous restart, with new boundary conditions on the costate at the crossing time; (b) use generalized derivatives (Clarke subdifferentials)

at the crossing, which extend the maximum-principle machinery to non-smooth points; (c) restrict the admissible-paths set to single-component trajectories, ruling out crossings entirely. Each formulation has costs: (a) introduces a discrete jump that the framework's smooth ontology does not naturally encode; (b) expands the apparatus to non-smooth optimization; (c) restricts the donor's options. Open-ended; one paragraph per formulation.

27. *Nash existence on the 2-good Tullock case.* Verify Theorem 6.13 (existence of a Nash equilibrium among n donors) for $n = 2$ donors with quadratic costs and the 2-good Tullock case of Example 6.7. The verification reduces to checking the quasi-concavity assumption Assumption 6.12 on the example, then applying Kakutani's fixed-point theorem to the best-response correspondence. By symmetry of the example, the Nash equilibrium is also symmetric: each donor contributes half the cooperative aid, $\tau_j^{\text{Nash}} = \tau^{\text{coop}}/2$. Compute the symmetric Nash τ^{Nash} explicitly and identify the parameters of the example (V_0, θ, W, k) in which it lies in the admissible region.
28. *Gross substitutes for quasi-concavity.* Verify the gross-substitutes sufficient condition for the quasi-concavity assumption Assumption 6.12 on a 2-good Cobb-Douglas specification. Recall that gross substitutes is the condition $\partial \bar{d}^\ell / \partial p^k > 0$ for $\ell \neq k$, which Cobb-Douglas demand systems trivially satisfy. Show that under gross substitutes, each donor's payoff $U_j(\tau_j, \tau_{-j})$ is quasi-concave in τ_j holding the other donors' contributions τ_{-j} fixed. The proof uses the fact that gross substitutes implies a unique equilibrium (p^*, m^*) at every ω , which makes the donor's induced objective a smooth single-valued function of τ_j , and Cobb-Douglas concavity preserves under such compositions.
29. *Free-rider scaling for $n = 2$.* For $n = 2$ donors with common preferences, derive Proposition 6.15 (the free-rider proposition) directly from the Nash-Pontryagin conditions Proposition 6.14. Verify the $n = 2$ case of the n -times-slower scaling: each donor's first-order condition has a marginal-welfare-gain term that is half what it would be under cooperation, because each donor takes the other's contribution as fixed. The Nash equilibrium contributes only half the cooperative effort, which corresponds to a time-to-peace twice as long. Substantively: the free-rider problem in donor coordination is exactly the standard public-good free-rider problem, with the conflict externality Δ^* playing the role of the public good.
30. *Time-to-peace scaling for general convex cost.* Corollary 6.16 establishes that the Nash time-to-peace is n times the cooperative time-to-peace under quadratic cost $C(s) = \kappa s^2$. Verify exactness for the quadratic case directly. For general convex cost $C(s)$, identify the scaling factor: the Nash-to-cooperative time-to-peace ratio depends on the curvature of C through $C''(s)/C'(s)$ at the equilibrium contribution level. Specifically, $T^{\text{Nash}}(n)/T^{\text{coop}}$ approaches n when C is quadratic, and approaches 1 in the high-curvature limit (where each donor's marginal cost rises so steeply that free-riding is self-defeating). Substantive content: the free-rider penalty depends on the cost technology of aid production, not just on the number of donors.
31. *Asymmetric donors.* For asymmetric donors with different welfare weights α_j or different speed bounds M_j , the symmetric-equilibrium analysis of Proposition 6.15 no longer applies directly. Derive the Nash-Pontryagin conditions for the asymmetric case from Proposition 6.14, identifying how each donor's first-order condition depends on their own (α_j, M_j)

and on the others'. Identify when the symmetric-equilibrium analysis remains an approximation: a sufficient condition is that asymmetries are small enough that the symmetric solution remains a Nash equilibrium under perturbation (which requires the symmetric Nash to be locally stable, a condition checkable from the second-order characterization). Open-ended; the case of large asymmetries is the subject of Exercise 32.

32. *Olson–Zeckhauser “exploitation of the great by the small.”* Discuss the Olson-Zeckhauser pattern: when one donor is dominant ($M_1 \gg M_j$ for $j > 1$), the dominant donor bears a disproportionate share of the aid burden, and the smaller donors free-ride more aggressively. Locate this pattern in the asymmetric-donor extension of Exercise 31. Identify the parameter region where it emerges: small donors with low speed bounds contribute close to zero, while the dominant donor contributes essentially the full cooperative aid. Substantive content: the framework’s free-rider proposition reproduces the classic public-goods asymmetry, with the conflict externality playing the role of the public good and the speed bound playing the role of donor capacity. Open-ended; one paragraph.
33. *Welfare cost of donor fragmentation.* Compute the welfare cost of donor fragmentation for the quadratic-cost two-good Tullock case. Defined as the integrated welfare cost along the Nash trajectory minus the integrated cost along the cooperative trajectory:

$$\mathcal{C}^{\text{frag}} = \int_0^{T^{\text{Nash}}} \Delta^*(\omega^{\text{Nash}}(t)) dt - \int_0^{T^{\text{coop}}} \Delta^*(\omega^{\text{coop}}(t)) dt.$$

Compute $\mathcal{C}^{\text{frag}}$ in closed form for the 2-good Tullock case and show it is non-negative. The cost quantifies the welfare value of donor coordination: when donors coordinate, peace is reached faster and total welfare loss along the path is smaller. Identify the parameter region in which $\mathcal{C}^{\text{frag}}$ is large (high contest stakes, slow donors) versus small (low stakes, fast donors).

Chapter 7

Institutions

What does an institution do in a framework where prices and deployments are jointly determined? Two traditions in international relations have answered the question on partialled recipients. The rational-design tradition (Koremenos et al. 2001, Abbott and Snidal 1998) treats institutions as solutions to coordination problems in a strategic environment that is otherwise given: institutions matter when they shift incentives or supply information that anarchic interaction cannot provide. The structural-realist tradition (Krasner 1982, Mearsheimer 1994a) treats them as reflections of underlying power: institutions record the distribution of capabilities without independently shaping outcomes. The two positions disagree on whether institutions have independent causal force; both compute their answer on a setting where prices and deployments have already been disentangled. This chapter takes a third picture.

Institutions are constructions on the joint equilibrium manifold $\tilde{E}$ of Chapter 3, and they show up in four readings the apparatus supports. The first reads an institution as an *aggregation rule*: a continuous, anonymous, Pareto-respecting map that combines agent preferences over $\tilde{E}$ into a social ordering. The topology of $\tilde{E}$ delivered by Chapter 3 is exactly the input the Chichilnisky–Heal theorem requires; under joint regularity, the aggregation problem that Arrow [1951] foreclosed in finite-alternative settings becomes admissible on the manifold. The second reads an institution as a *gradient field*: a smooth vector field on $\tilde{E}^{\text{reg}}$ that institutional intervention implements at every regular equilibrium. The welfare-improvement direction is fixed by $-D_\omega \Delta^*$ from Chapter 5 Section 5.4; institutional design becomes a question of vector-field alignment with this gradient. The third reads an institution as a *driver of regime change*: a perturbation that moves the equilibrium across the singular locus Σ identified in Chapter 4, transitioning the system from one regime component of $\tilde{E}^{\text{reg}}$ to another. The verbal-historical IR literature on institutional reorganization recovers in this reading as the geometric content of crossings of the joint singular locus, with the index theorem of Chapter 4 as the structural constraint on how regimes can change. The fourth reads an institution as a *(target, path) pair* on the equilibrium manifold: a target equilibrium $e_1 \in \tilde{E}$ together with a path $\gamma: [0, 1] \rightarrow \tilde{E}$ from a current equilibrium e_0 to e_1 . The first three readings each capture one face of this unifying object. Preferences over (target, path) institutions are encoded by 1-forms on $\tilde{E}$ via line integrals; the fundamental group π_1 of the chamber containing e_0 and e_1 governs whether institutions adapt robustly under route changes (the JR face, simply connected chamber) or become path-pinned by their commitments to homotopy classes (the JR' face, nontrivial π_1 , with Arrow's classical impossibility theorem applying on the discrete set of homotopy classes). The political-science literature on path dependence [Pierson, 2000, Page, 2006] resolves under

this reading as mathematical content—increasing returns, critical junctures, and irreversibility become statements about the topology of chambers in $\tilde{E}^{\text{reg}}$.

What the chapter delivers is hard to find elsewhere. First, an institutional aggregation theorem on the equilibrium manifold: under joint regularity the contractibility of $\tilde{E}$ makes Chichilnisky–Heal aggregation admissible globally; under generic joint regularity the same theorem operates componentwise on each regime, with cross-regime aggregation impossibility reframing Krasner’s regime concept as a topological boundary condition. Second, a substantive direction for “institutional intervention”: $X^{\Delta^*} := -D_{\omega}\Delta^*$ is the welfare-improvement gradient, and the master sensitivity formula Equation (5.10) of Chapter 5 reads the direction in closed form. Third, a precise mathematical apparatus for regime change: bifurcations of the joint Jacobian along Σ are the formal correlate of qualitative shifts in international order, with the index theorem constraining how equilibria pair across transitions. Fourth, the (target, path) institution reading and its homotopy-theoretic structure: institutions are paths on the equilibrium manifold, preferences are 1-forms, and the fundamental group of the chamber containing the institution’s endpoints determines whether institutional adaptation is robust (the JR face, simply connected chamber, on-the-fly route changes welfare-neutral) or path-pinned (the JR’ face, nontrivial π_1 , Arrow’s classical impossibility applying to ordinal preferences over the discrete set of homotopy classes). Fifth, a resource-neutral rationale for multilateral institutions inherited from Chapter 6’s Corollary 6.16: cooperation implements the welfare gradient at full magnitude while n -fold non-cooperative aid implements the same direction at $1/n$ magnitude—the multilateralism question read geometrically as gradient-field amplification.

What the framework’s apparatus does not deliver is the calibrated empirical content of specific institutions. Which institutional design feature implements which gradient field, which membership structure corresponds to which aggregation rule, which historical event is a fold transition versus a cusp catastrophe—these are calibration questions the apparatus identifies but does not settle. Section 7.6 addresses two principal critiques the chapter’s structural reading attracts: the Lauwers [2009] critique of topological aggregation and the Mearsheimer [1994a] false-promise critique of institutionalist explanation. The framework’s response to each is methodological: the apparatus makes the critiques empirically tractable in a way the verbal-conceptual debate has not been.

The chapter is organized around the four readings, the substantive payload of each, and the engagement with critique. Section 7.1 sets up the institutional problem on the equilibrium manifold, develops the first three readings, and records what each inherits from upstream apparatus. Section 7.2 runs Chichilnisky–Heal aggregation on $\tilde{E}$ under joint regularity, with the componentwise version under generic joint regularity reframing Krasner’s regime concept. Section 7.3 reads X^{Δ^*} as the gradient field of welfare-improvement directions, develops gradient-field intervention as institutional practice, and sketches readings of Bretton Woods, the UN Security Council, and the European Coal and Steel Community. Section 7.4 reads regime change as bifurcation across Σ , recovering the verbal-historical literature (Krasner, Ikenberry, Tilly, Acemoglu and Robinson [2019]) as descriptions of motion across the singular locus. Section 7.5 introduces the unifying (target, path) reading, recovers the first three as special cases (closed-form 1-form for the gradient field, cross- Σ path for the bifurcation driver, Chichilnisky-on-1-forms for the aggregation rule), and develops the cohomological apparatus that makes the political-science path-dependence vocabulary (Pierson 2000, Page 2006, Fioretos 2011) mathematical. Section 7.6 addresses the Lauwers-style critiques of topological aggregation and the realist-dissent critique of institutionalism, identifying what the apparatus’s structural reading defends against and what

qualifies as a substantive limit. Section 7.7 hands the chapter's apparatus to the third Part 2 chapter and the conclusion that follow.

7.1 The institutional problem on the equilibrium manifold

The institutional problem in international relations has had many formulations. Krasner [1982] defined regimes as “principles, norms, rules, and decision-making procedures around which actor expectations converge” and asked when such convergence reflects underlying power versus shapes outcomes independently. Keohane [1984] reformulated the question as one of transaction costs: institutions matter when they make iterated cooperation feasible by reducing the costs of agreement. Koremenos et al. [2001] systematized the rational-design tradition: institutions matter when their design features—membership, scope, centralization, control—solve specific coordination problems states cannot solve unilaterally. Ikenberry [2001] traced the empirical record of post-war institutional settlements. Each formulation works on a setting in which the strategic environment is partialled—prices and deployments have separate existences, and the institution's job is to coordinate or constrain interaction in the half the formulation has chosen to hold fixed.

The framework's apparatus does not partial. The strategic environment is the joint equilibrium manifold $\tilde{E}$, with prices and deployments emerging together at every point. The institutional problem in framework terms is therefore not “how to coordinate the parties” or “how to reduce transaction costs at fixed prices” but rather: how does an institution intervene on a system whose state is a joint equilibrium and whose dynamics are paths on $\tilde{E}$? The chapter's task is to read the existing institutional traditions through this apparatus, identify what each tradition captures and what each misses, and develop the substantive content the joint formulation makes available.

Three readings of an institution. The framework supports three formal readings of what an institution is. The readings are not exclusive: a particular institution may admit two or all three at once. Their separation is analytic, not classificatory.

The first reading is *aggregation*. An institution is a rule $\mathcal{A} : \prod_{i \in I} \succsim_i \rightarrow \succsim_S$ that combines individual preferences over $\tilde{E}$ into a social preference ordering on $\tilde{E}$. This is the Arrow–Chichilnisky–Heal formulation (Arrow 1951, Chichilnisky 1982), specialized to preferences defined on the equilibrium manifold rather than on abstract alternative sets. The substantive content of the aggregation reading depends on the topology of $\tilde{E}$. Under Assumption 3.6 (Chapter 3), $\tilde{E}$ is contractible and the Chichilnisky–Heal continuous-aggregation theorem applies globally: a continuous, anonymous, unanimity-respecting aggregation rule exists on contractible spaces, and contractibility of $\tilde{E}$ is exactly the input the theorem takes. Under Assumption 3.17, contractibility holds componentwise on each regime, and aggregation operates within regimes; the chapter's geometric reading of regime change as cross-component motion (the third reading below) is what delineates the components.

The second reading is *gradient field*. An institution is a smooth vector field $X : \tilde{E}^{\text{reg}} \rightarrow T\tilde{E}^{\text{reg}}$ that institutional intervention implements at each equilibrium. The substantive content of the gradient reading is what direction the field points. Chapter 5 Section 5.4 identified $D_\omega \Delta^*$ as the welfare-direction-of-improvement gradient: the direction in $T\tilde{E}^{\text{reg}}$ along which the welfare cost of conflict Δ^* decreases fastest. An institution that drives the equilibrium along $-D_\omega \Delta^*$ implements welfare-improving institutional intervention by construction. The reading reorients

the rational-design tradition: institutions are not just rules of coordination but vector fields the rules implement, and the substantive design question is which gradient field the institutional architecture corresponds to.

The third reading is *driver of regime change*. An institution is a perturbation that moves the equilibrium across the singular locus $\Sigma \subset \Omega_0$ identified in Chapter 4, transitioning the system from one regime component of $\tilde{E}^{\text{reg}}$ to another. The substantive content of the regime-change reading is the formal apparatus for what regime change *is* as a mathematical phenomenon. Most institutional change in IR scholarship is described in verbal-historical terms (Ikenberry 2001, Tilly 1990, Acemoglu and Robinson 2019); the framework makes regime change geometrically precise as a bifurcation of the joint Jacobian, with the index theorem (Chapter 4) constraining how the regime structure changes across Σ . The reading reorients the structural-realist tradition: institutional changes are not free-form recompositions but structurally constrained transitions whose form the apparatus predicts.

What each reading inherits from upstream chapters. The three readings are not invented in this chapter; each comes in fully formed from the apparatus developed in earlier ones.

The aggregation reading inherits the topology of $\tilde{E}$ from Chapter 3. Under Assumption 3.6, $\tilde{E}$ is diffeomorphic to $\mathbb{R}^{IL}$ and hence contractible; this contractibility is what Chichilnisky–Heal aggregation requires as input. Under Assumption 3.17, the regime components are individually contractible on $\tilde{E}^{\text{reg}}$, and aggregation operates componentwise. The chapter’s aggregation theorem (Section 7.2) is one application of these topological facts.

The gradient-field reading inherits the welfare cost Δ^* and the sensitivity formula from Chapter 5. The cardinal welfare scalar of Definition 5.1 provides the function whose gradient the field is, and the master formula Equation (5.10) delivers the gradient as one application of J^{-1} from Chapter 4. The chapter’s gradient-field section (Section 7.3) develops what the field looks like in substantive cases.

The regime-change reading inherits the singular locus Σ and the index theorem from Chapter 4, together with the peace component $\tilde{E}_p$ from Chapter 5 Section 5.3. The bifurcation set is where the joint Jacobian becomes singular; the index theorem constrains how equilibria pair across folds; the peace component is one regime distinguished by zero-deployment geometry. The chapter’s bifurcation section (Section 7.4) reads cross-component motion as institutional regime change, with the apparatus identifying which transitions are admissible.

The fourth content piece, the multilateralism rationale, comes in from Chapter 6’s Corollary 6.16: cooperation among n donors delivers peace exactly n times faster than the non-cooperative Nash equilibrium with the same aggregate budget. The chapter develops the institutional implication (Section 7.3, in the gradient-field section, because cooperative peace trajectories implement a particular gradient field on $\tilde{E}$ that fragmented bilateral aid does not).

The unifying (target, path) reading developed in Section 7.5 inherits the path-on-equilibrium-manifold apparatus of Loi and Matta [2008, 2009, 2010, 2011], Balasko [2012] and the public-finance precedent of Garratt and Goenka [1995], extending the apparatus from the pure-exchange manifold to the joint manifold of Chapter 3. The cohomological structure $H^1(\tilde{E}^{\text{reg}}; \mathbb{R})$ inherits directly from the topology of $\tilde{E}^{\text{reg}}$ established in Chapter 4: the codimension-one stratification of Σ determines which generators H^1 has, and hence which path-aggregation obstructions appear under generic joint regularity.

The chapter’s relation to existing IR theory. The three readings each engage an existing literature.

The aggregation reading is in dialogue with the Arrow social-choice tradition (Arrow 1951) and its IR-applied descendants. The chapter’s contribution is the topology: aggregation problems on contractible spaces have well-defined answers, and the framework’s apparatus delivers contractibility (under JR) or componentwise contractibility (under JR’) as the substantive content $\tilde{E}$ adds to the abstract aggregation question.

The gradient-field reading is in dialogue with the rational-design tradition (Koremenos et al. 2001, Abbott and Snidal 1998). The chapter’s contribution is direction: the rational-design literature analyzes when institutions reduce transaction costs or aggregate information, but does not generally identify the direction in equilibrium-state-space along which design choices should push the system. $D_\omega \Delta^*$ is that direction. Welfare-improving institutional intervention is gradient descent on the conflict-externality scalar.

The regime-change reading is in dialogue with the structural-realist tradition (Krasner 1982, Mearsheimer 1994a) and the institutional-history literature (Ikenberry 2001, Tilly 1990). The chapter’s contribution is precision: where structural realism treats institutional change as recomposition of underlying power distributions and institutional history traces the verbal record, the framework provides the mathematical apparatus for what regime change *is* as a transition. The bifurcation locus Σ is the geometric place where institutional changes happen; the index theorem constrains how they happen.

What this section establishes. The institutional problem on the equilibrium manifold has three formal readings. Each reading inherits its substantive content from a specific piece of upstream apparatus: aggregation from Chapter 3’s topology, gradient field from Chapter 5’s welfare scalar, regime change from Chapter 4’s singular locus. Each reading engages an existing IR-theory tradition: the aggregation reading engages Arrow social choice, the gradient reading engages rational-design, the regime-change reading engages structural realism and institutional history. The remaining sections of the chapter develop each reading substantively, with the multilateralism rationale of Chapter 6 entering the gradient-field section as a concrete instance.

7.2 Aggregation under joint regularity

The aggregation reading of an institution asks whether agent preferences over the equilibrium manifold can be combined into a social ordering by a continuous, anonymous, unanimity-respecting rule. The classical answer is mixed. Arrow [1951] established that no such rule exists on finite-alternative settings under his axioms; the result was read in international relations as foreclosing social-choice approaches to institutional aggregation. Chichilnisky [1980] reformulated the question on topological spaces of preferences and obtained a positive result: aggregation works exactly when the space is contractible. The framework’s apparatus delivers contractibility of $\tilde{E}$ under Assumption 3.6 and componentwise contractibility under Assumption 3.17 (Chapter 3), supplying exactly the input the topological theorem requires. This section runs the theorem on the equilibrium manifold and reads the substantive content for institutional analysis.

Arrow’s impossibility, briefly. Arrow’s theorem operates on a finite set of alternatives X and profiles of preference orderings $(\succsim_i)_{i \in I}$. An aggregation rule maps every profile to a social ordering $\succsim_S$ on X . Arrow imposed three axioms: *universal domain* (the rule accepts every profile

of orderings), *independence of irrelevant alternatives* (the social ranking of x versus y depends only on each individual's ranking of x versus y), and *Pareto unanimity* (when all individuals strictly prefer x to y , so does the social ordering). The theorem states that no rule satisfying the three axioms treats individuals symmetrically without making one a dictator. The immediate IR consequence—read repeatedly through the 1970s and 1980s—was that institutions cannot consistently aggregate state preferences on questions of international order: any aggregation rule satisfying Arrow's axioms admits at least one state whose preferences override the others'.

The result fits poorly with the framework's setting in two ways. First, the alternative set in international institutional analysis is not finite. The space of feasible international configurations is Ω_0 , a continuum, and for each ω the set of joint equilibria is a fiber of $\tilde{\pi} : \tilde{E} \rightarrow \Omega_0$, also in general a continuum. Arrow's combinatorial setup does not apply directly. Second, the relevant preferences are over equilibrium states—points on $\tilde{E}$ —and these admit a smooth structure. Continuous preferences on a smooth manifold are a richer object than orderings on a finite set, and the aggregation question for continuous preferences has a different mathematical character.

The topological reformulation. Chichilnisky [1980] reformulated the aggregation question for continuous preferences on topological spaces. Take a topological space X as the alternative set, and let $P(X)$ denote a topological space of admissible preference orderings on X (e.g., the space of continuous unit-normalized gradient vector fields, when preferences are smooth and strictly monotone in some local coordinate). A *Chichilnisky aggregation rule* is a continuous map $\mathcal{A} : P(X)^I \rightarrow P(X)$ satisfying:

- *Anonymity.* $\mathcal{A}(p_1, \dots, p_I) = \mathcal{A}(p_{\sigma(1)}, \dots, p_{\sigma(I)})$ for every permutation σ of I .
- *Pareto unanimity.* $\mathcal{A}(p, p, \dots, p) = p$ for every $p \in P(X)$.

The substantive content of these axioms is the same as Arrow's, restated for the topological setting: the rule treats agents symmetrically and respects unanimity. Arrow's IIA is replaced by continuity, which captures the same intuition—small changes in preferences should produce small changes in the social ordering—in a form natural to the topological setting.

Chichilnisky and Heal [1983] proved the central result.

Theorem 7.1 (Chichilnisky–Heal). *A continuous, anonymous, Pareto-respecting aggregation rule $\mathcal{A} : P(X)^I \rightarrow P(X)$ exists if and only if $P(X)$ is contractible (homotopy-equivalent to a point).*

The theorem is a clean dichotomy. On contractible preference spaces the aggregation problem is solvable; on non-contractible spaces it is not. The role of topology is concentrated entirely in the contractibility condition: the answer to the aggregation question depends on the topology of $P(X)$ and on nothing else.

Aggregation on the equilibrium manifold under JR. The theorem applies to the equilibrium manifold once the preference space $P(\tilde{E})$ is constructed and its topology is shown to inherit from the topology of $\tilde{E}$. The construction is standard: take preferences as continuous unit-normalized gradient vector fields on $\tilde{E}^{\text{reg}}$, representing each agent's smooth, strictly monotone preference relation by the direction of preference improvement at each equilibrium. The space $P(\tilde{E})$ inherits the topology of continuous sections of the unit tangent bundle on $\tilde{E}^{\text{reg}}$, which under the manifold structure of Chapter 3 is well-behaved.

Under Assumption 3.6, $\tilde{E}$ is diffeomorphic to $\mathbb{R}^{IL}$. Contractibility transfers: $P(\tilde{E})$ is homotopy-equivalent to $P(\mathbb{R}^{IL})$, which is contractible because $\mathbb{R}^{IL}$ is contractible. Theorem 7.1 therefore applies globally, and a continuous aggregation rule exists on the full preference space.

Corollary 7.2 (Aggregation under JR). *Under Assumption 3.6, there exists a continuous, anonymous, Pareto-respecting aggregation rule $\mathcal{A} : P(\tilde{E})^I \rightarrow P(\tilde{E})$ on the equilibrium manifold.*

Proof. $\tilde{E} \cong \mathbb{R}^{IL}$ globally under Assumption 3.6 (Chapter 3), so $\tilde{E}$ is contractible. The space of continuous preferences on a contractible manifold inherits contractibility; Theorem 7.1 delivers the rule. $\square$

The substantive content is institutional: under joint regularity, international institutions can aggregate agent preferences over the full space of equilibria into a coherent social ordering without falling into Arrow’s impossibility trap. The escape route is exactly what the framework’s apparatus provides—the topology of the equilibrium manifold.

Aggregation under generic joint regularity. The clean global result of Corollary 7.2 relies on $\tilde{E} \cong \mathbb{R}^{IL}$ globally, which is the strong-form assumption. Under Assumption 3.17, $\tilde{E}^{\text{reg}}$ is a covering of $\Omega_0 \setminus \Sigma$, partitioned into regime components. Each component is contractible to its parameter coverage, contractibility of $\tilde{E}^{\text{reg}}$ as a whole fails, and the global aggregation rule of Corollary 7.2 no longer exists.

The right adaptation is componentwise. For each regime component $\tilde{E}_k \subset \tilde{E}^{\text{reg}}$, the preference space $P(\tilde{E}_k)$ is contractible, and a continuous aggregation rule $\mathcal{A}_k : P(\tilde{E}_k)^I \rightarrow P(\tilde{E}_k)$ exists by the Chichilnisky–Heal theorem applied to $\tilde{E}_k$. Aggregation operates within a regime: institutional design that takes the regime as given can aggregate preferences over the equilibria in that regime.

Corollary 7.3 (Componentwise aggregation under JR’). *Under Assumption 3.17, on each regime component $\tilde{E}_k \subset \tilde{E}^{\text{reg}}$ there exists a continuous, anonymous, Pareto-respecting aggregation rule $\mathcal{A}_k : P(\tilde{E}_k)^I \rightarrow P(\tilde{E}_k)$. A global aggregation rule on $\tilde{E}^{\text{reg}}$ does not exist whenever the partition has more than one component.*

Proof. Each $\tilde{E}_k$ is contractible to its parameter coverage (Chapter 3), so $P(\tilde{E}_k)$ is contractible and Theorem 7.1 applies. The non-existence of a global rule follows from the failure of contractibility on $\tilde{E}^{\text{reg}}$ as a whole when there are multiple components. $\square$

The componentwise result is substantively informative. *Within* a regime, aggregation is well-behaved. *Across* regime boundaries, aggregation breaks down: there is no continuous, anonymous, Pareto-respecting rule that combines preferences across components of $\tilde{E}^{\text{reg}}$. What lies across regime boundaries is not a coordination problem the institution can solve but a regime change—a mathematical phenomenon Section 7.4 below takes up through the bifurcation reading. The aggregation reading, on its own, has nothing to say about cross-regime motion; the impossibility on the multi-component manifold is a structural feature of what aggregation *is*.

Reading the substantive content. Corollaries 7.2 and 7.3 reframe two longstanding debates in international institutional theory.

The Arrow-impossibility debate read social-choice arguments as foreclosed for international institutions by the abstract result: no aggregation rule combines state preferences without dictatorship. Corollary 7.2 reopens the question. On a contractible equilibrium manifold the aggregation problem is not Arrow-impossible; the apparatus that delivers contractibility

(Chapter 3, joint regularity) is exactly what the rescuing theorem requires as input. Continuous, anonymous, Pareto-respecting aggregation across member states' preferences over international configurations is admissible whenever the strong regularity hypothesis holds.

The Krasner-regime debate (Krasner 1982) treated regimes as zones within which actor expectations converge. Corollary 7.3 reads the framework's regime components $\tilde{E}_k$ as the formal correlate of Krasner's regime concept: a regime is a contractible component of the equilibrium manifold within which preference aggregation is well-behaved, and the boundary between regimes is where aggregation breaks down. Cross-regime motion is not an aggregation problem the institution can resolve; it is the regime change that Section 7.4 treats geometrically. The verbal regime-realist tradition is recovered as a theorem about where the topological aggregation theorem applies and where it fails.

What no partial can compute. The aggregation result is not available from either of the two partialled traditions the chapter triangulates against. A partial-equilibrium analysis treats the alternative set as a static endowment-redistribution problem: states have preferences over endowment allocations, and the institution aggregates these. Endowment space Ω_0 is contractible (it is a positive orthant), so Chichilnisky–Heal applies trivially—but the result is uninformative because it does not capture the joint determination of equilibria the framework's apparatus is meant to handle. PE aggregation aggregates over Ω_0 , not over $\tilde{E}$; the aggregation problem on $\tilde{E}$ does not reduce to the aggregation problem on Ω_0 because the natural projection $\tilde{\pi}$ is not generally a bijection (under Assumption 3.17).

A conflict-side partial treats the alternative set as the militarization profile space M . Aggregation on M is Chichilnisky-admissible if M is contractible, but the aggregation question is again uninformative: it asks how to aggregate preferences over deployment profiles ignoring the equilibrium prices those deployments would induce.

Corollaries 7.2 and 7.3 aggregate over $\tilde{E}$, the joint equilibrium manifold whose existence the framework's apparatus secures. This is the substantive content: aggregation on the right space.

The Lauwers critique flagged. The topological-aggregation tradition has its own critics. Lauwers [2009] surveys the principal critiques of Chichilnisky's framework: the choice of preference-space topology (continuous unit-normalized vector fields versus other formalizations) is more substantive than typically acknowledged; the anonymity axiom in particular has been challenged for admitting limit-of-dictatorship behavior in the multi-agent limit; and the Chichilnisky–Heal theorem's positive content on contractible spaces depends on the preference-space parameterization in ways that warrant scrutiny. Section 7.6 addresses each critique in the framework's setting and identifies what survives the substantive challenges and what does not.

7.3 Gradient fields and welfare-improvement direction

The gradient-field reading of an institution asks which smooth vector field on $\tilde{E}^{\text{reg}}$ the institutional architecture implements. The welfare-improvement direction is fixed by the apparatus. $D_\omega \Delta^*$ from Chapter 5 Section 5.4 delivers the gradient of the conflict externality at every regular equilibrium; the negative of this gradient, $-D_\omega \Delta^*$, is the direction in $T\tilde{E}^{\text{reg}}$ along which the welfare cost of conflict decreases fastest. An institution that drives the equilibrium along $-D_\omega \Delta^*$ implements welfare-improving intervention by construction. This section develops the reading: defines the welfare gradient as a vector field, traces the endpoints of its integral curves, reads

cooperative aid (Chapter 6) as one institutional implementation, sketches how existing institutions correspond to gradient fields, and identifies where the reading breaks down.

The welfare gradient as a vector field. The cardinal welfare cost $\Delta^* : \Omega_0 \rightarrow \mathbb{R}$ of Chapter 5 is a smooth scalar field on the regular set; its gradient $D_\omega \Delta^*$ is a smooth covector field on $\Omega_0 \setminus \Sigma$. Pulling back through the natural projection $\tilde{\pi}$ and raising the index by an arbitrary smooth Riemannian metric on $\tilde{E}^{\text{reg}}$ produces a vector field

$$X^{\Delta^*} : \tilde{E}^{\text{reg}} \rightarrow T\tilde{E}^{\text{reg}}, \quad X^{\Delta^*}(e) = -\text{grad } \Delta^*(e), \quad (7.1)$$

the welfare-improvement gradient field on $\tilde{E}^{\text{reg}}$. The choice of Riemannian metric affects the parameterization but not the qualitative structure of the integral curves; under the chart $\Phi : \tilde{E} \rightarrow \mathbb{R}^L$ of Chapter 3 a natural metric is the Euclidean one in Φ -coordinates, and integral curves of X^{Δ^*} are the gradient flow on $\tilde{E}^{\text{reg}}$ in those coordinates.

The flow has two structural properties. First, Δ^* is monotonically non-increasing along integral curves: $\frac{d}{dt} \Delta^*(\gamma(t)) = -\|D_\omega \Delta^*(\gamma(t))\|^2 \leq 0$, with strict inequality at points where the gradient is non-zero. Second, integral curves remain in $\tilde{E}^{\text{reg}}$ until they approach the singular locus Σ or the boundary of the regular set. Both properties are standard for gradient flows on smooth manifolds.

Endpoints of the welfare flow. Where do integral curves of X^{Δ^*} go in the long run? Three answers, sorted by the geometry of the regime component the initial equilibrium sits on.

First: peace-reachable components. On a regime component containing peaceful equilibria $\tilde{E}_p \cap \tilde{E}_k$ (under zero- or negative-sum warfare, Theorem 5.8 of Chapter 5), Δ^* vanishes on $\tilde{E}_p \cap \tilde{E}_k$ and the gradient $D_\omega \Delta^*$ vanishes there too. Integral curves of X^{Δ^*} that begin in the basin of attraction of $\tilde{E}_p \cap \tilde{E}_k$ converge to the peace component asymptotically; their welfare cost converges to zero. Chapter 5 Section 5.3 certifies that endpoints in $\tilde{E}_p$ are globally Pareto optimal, so the gradient flow is welfare-improving in the strongest sense.

Second: peace-unreachable components. On a regime component that does not contain peaceful equilibria (Example 3.3 with positive prize V_0 is this case), integral curves converge to local minima of Δ^* on the component, with Δ^* strictly positive at the limit. The welfare flow improves welfare relative to the initial state but cannot drive Δ^* to zero within the component. The substantive reading of this case is the commitment-obstruction content of Chapter 5's closing paragraph on the two-state contest: the contest's marginal-benefit-at-zero structure rules out peaceful equilibria on this component, and the gradient flow identifies the cheapest non-peaceful equilibrium the institution can navigate to.

Third: cross-component endpoints. Integral curves can approach the singular locus Σ in finite or infinite time, depending on the conflict-intensity gradient near the boundary. Approach to Σ takes the welfare flow outside the regular set $\tilde{E}^{\text{reg}}$; what happens at Σ is regime change rather than continuous gradient flow. Section 7.4 below takes up this case through the bifurcation reading; for the gradient-field reading, the curve simply terminates at Σ .

Institutions as gradient-field implementations. An institution implements a gradient field on $\tilde{E}^{\text{reg}}$ by linking institutional choices to the parameter response. The substantive question is which field the institutional architecture corresponds to. A welfare-maximizing institution implements X^{Δ^*} exactly: every institutional choice drives the equilibrium in the

welfare-improvement direction at the maximum admissible rate. A constrained or fragmented institution implements a degraded field: a vector field X^{inst} that is non-decreasing in Δ^* when projected on X^{Δ^*} , but not aligned with it pointwise.

The substantive content of this reading is twofold. First, it identifies what institutional design is for: aligning X^{inst} with X^{Δ^*} . A well-designed institution has a vector-field correspondence that matches the welfare gradient closely; a poorly designed one has X^{inst} pointing elsewhere. Second, it makes institutional comparison quantitative: two institutions can be compared by the inner product $\langle X^{\text{inst}_1}, X^{\Delta^*} \rangle$ vs. $\langle X^{\text{inst}_2}, X^{\Delta^*} \rangle$ at each equilibrium, identifying which institution moves the system in the welfare-improving direction more aggressively.

The reading reorients the rational-design tradition. Koremenos et al. [2001] ask which design features solve which coordination problems; the reading asks which design features implement which gradient field. The two questions are related but distinct. A design feature can solve a coordination problem (in KLS terms) without implementing a welfare-aligned gradient field; conversely, a design feature can implement a welfare-aligned field without solving any specific coordination problem the KLS framework names. The reading complements rather than replaces the rational-design literature: the gradient-field correspondence is the welfare-direction-of-improvement check on what the institutional design is doing.

Cooperative aid as a gradient implementation. Chapter 6's Corollary 6.16 delivers a sharp instance of the reading. Under common preferences and quadratic cost, a single cooperative donor implements the gradient field X^{Δ^*} at maximum admissible speed. n non-cooperating donors implement a degraded version: their aggregate vector field at the symmetric Nash equilibrium points in the same direction as X^{Δ^*} (each donor's preference field α_j is welfare-aligned by assumption) but at $1/n$ the magnitude.

Remark 7.4 (Multilateralism as gradient-field alignment). The cooperative-vs-Nash speed comparison of Corollary 6.16 reads in the gradient-field language as: cooperation implements $X^{\text{coop}} = X^{\Delta^*}$; n -fold non-cooperative aid implements $X^{\text{Nash}} = (1/n)X^{\Delta^*}$. The vector fields point in the same direction; cooperation provides the larger magnitude. The institutional implication is that multilateral coordination is gradient-field amplification: pooled disbursement structures implement the welfare-improvement gradient at full magnitude, while fragmented bilateral aid implements the same direction at $1/n$ magnitude. The substance is direction-and-magnitude alignment, not coordination-of-incentives or transmission of information.

The remark is the chapter's substantive headline on the multilateralism question. Chapter 6's rationale was that cooperation internalizes the peace-trajectory externality at the optimization stage; Remark 7.4 reads the same content geometrically as gradient-field amplification. Both readings are views of the same dual-mandate object: the welfare gradient $-D_\omega \Delta^*$ that Chapter 5 Section 5.4 delivered.

Example 7.5 (Welfare gradient on the two-state contest). The two-state Tullock contest of Example 3.3 admits an explicit welfare-gradient calculation. At the symmetric equilibrium, the total resource burn in good 1 is $2km^* = (V_0 + \theta W)/(2 + \theta)$, with m^* the symmetric deployment. The welfare cost Δ^* scales with the burn: more deployment, more good-1 consumption foregone, larger Δ^* . Differentiating the burn with respect to the four parameters,

$$\frac{\partial(2km^*)}{\partial V_0} = \frac{1}{2 + \theta} > 0, \quad \frac{\partial(2km^*)}{\partial \theta} = \frac{2W - V_0}{(2 + \theta)^2} > 0,$$

$$\frac{\partial(2km^*)}{\partial W} = \frac{\theta}{2 + \theta} \geq 0, \quad \frac{\partial(2km^*)}{\partial k} = 0.$$

Three institutional readings follow. First, an institution that lowers the prize V_0 —a disarmament-of-stakes reduction in what the contest is over—moves the system in the welfare-improvement direction: $\partial\Delta^*/\partial V_0 > 0$, so a downward perturbation of V_0 reduces Δ^* . Second, an institution that decouples the contest from aggregate consumption by reducing θ (the share of good-1 trade the contest captures) is similarly welfare-improving. Third—and this is the structural surprise—an institution that lowers the unit cost of force production k *has zero welfare gradient*. The reason is that lower k rescales deployments inversely ($m^* \propto 1/k$) but holds the resource burn $2km^*$ fixed: cheaper force means more force, and the welfare cost—driven by the good-1 foregone—does not change. The reading distinguishes institutions that target the prize or the contest structure (welfare-improving) from institutions that target the cost of force (welfare-neutral at this approximation). The dual-mandate criterion sharpens accordingly: an institutional intervention’s welfare-direction performance depends on which parameter it perturbs, not on whether the perturbation reduces some salient input quantity.

Reading existing institutions. The gradient-field reading invites a substantive question. What gradient field does a given international institution correspond to? The chapter sketches the question without committing to specific empirical answers; the substantive case-work belongs to a calibrated chapter that this volume does not include. But three illustrative readings indicate the apparatus’s reach.

The Bretton Woods institutions (IMF, World Bank, GATT/WTO) are conventionally read as transaction-cost-reducing multilateral coordinators (Keohane 1984, Ikenberry 2001). Through the gradient-field reading, they implement an aggregated welfare-direction field on the post-WWII international economic manifold: each institution corresponds to a sectoral gradient contribution (currency, finance, trade), and the institutional architecture’s overall vector field is the sum. The architecture’s robustness comes from gradient-field amplification (the multilateralism rationale): fragmented sectoral arrangements would implement the same direction at strictly smaller magnitude.

The UN Security Council reads less cleanly. The veto-equipped P5 implement an aggregation rule that fails Theorem 7.1’s anonymity axiom (the P5 are not treated symmetrically with the rest of the membership). The gradient field the Council implements is a constrained variant of X^{Δ^*} where the welfare-improving direction is suppressed in any direction one P5 member opposes. The reading is consistent with the structural-realist literature (Mearsheimer 1994a): the Council’s welfare-improvement performance is bounded by the alignment between the welfare gradient and the P5’s individual preference fields.

The European Coal and Steel Community (and its EU successor) reads as a sustained gradient-field intervention with explicit institutional commitment to aligning the implemented field with X^{Δ^*} on a specific sectoral and geographic component. The Schuman declaration’s substantive claim was that economic integration on coal and steel made cross-border conflict materially more costly; in framework terms, the institution modifies the conflict primitive’s parameters in a direction that moves the implemented field toward X^{Δ^*} .

These readings are illustrative. They identify what the gradient-field language can do for substantive institutional analysis without claiming to settle specific empirical questions about specific institutions.

Limits of the gradient-field reading. The reading has three structural limits.

Critical points. At any critical point of Δ^* (where $D_\omega \Delta^*(e) = 0$), the gradient field X^{Δ^*} vanishes. Integral curves either start or stop at critical points. Whether a critical point is a global minimum, local minimum, saddle, or maximum determines what the flow does in its neighborhood; the welfare-improvement reading is informative only on the basins of attraction of the global minima.

The singular locus Σ . At any equilibrium on Σ , the joint Jacobian is singular, J^{-1} does not exist, and the master sensitivity formula Equation (5.10) of Chapter 5 that defines X^{Δ^*} in closed form is not available. The gradient field X^{Δ^*} extends continuously to Σ only in special cases; in general, motion across Σ is regime change, taken up in Section 7.4.

The implementation gap. The reading defines X^{Δ^*} as the welfare-improvement direction, but does not say what institutional choices implement which directions. The implementation question—which institutional design feature moves the equilibrium in which direction—is empirical, requires specification of the conflict primitive and the cost parameters, and lives outside the chapter’s abstract apparatus. A calibrated chapter that develops the implementation in specific cases is what the present chapter hands off to.

What no partial can compute. The welfare gradient X^{Δ^*} is not available from a partialled framework. A partial-equilibrium gradient holds prices fixed and computes the response of Δ^* to endowment shifts at fixed prices. Under the conflict-through-prices hypothesis of Chapter 6 Corollary 6.6, this gradient is the direct-resource channel of Equation (5.10) alone, which is zero for redistributions between recipients. PE-thinking sees no gradient at all on the welfare-cost-of-conflict along directions that move recipients without moving total endowment; the substantive welfare direction is invisible.

A conflict-side partial holds deployments fixed at the Nash equilibrium and computes welfare-cost responses to parameter shifts at fixed m . The gradient is the direct-resource and reallocation channels combined, but not the deployment-cost feedback channel. The conflict-side gradient is wrong at every equilibrium where the conflict primitive responds to parameter shifts—which includes every interesting case for the institutional reading.

The framework’s apparatus reads X^{Δ^*} as one application of J^{-1} to the welfare-direction column of $D_\omega e$, with the master sensitivity formula Equation (5.10) of Chapter 5 delivering the gradient in closed form on $\tilde{E}^{\text{reg}}$. The institutional reading then identifies what the gradient field *is for*: it is the welfare-direction-of-improvement that institutional architecture should implement. The dual-mandate punchline is that institutional design has a welfare direction, the framework’s apparatus delivers the direction, and partialled treatments cannot.

7.4 Regime change as bifurcation

The third reading of an institution is as a driver of regime change: a perturbation that moves the equilibrium across the singular locus $\Sigma \subset \Omega_0$ identified in Chapter 4, transitioning the system from one regime component of $\tilde{E}^{\text{reg}}$ to another. The reading provides the mathematical apparatus the verbal-historical IR literature on regime change has been operating without. Krasner [1982] defines regimes as zones of converged expectations; Ikenberry [2001] traces post-war institutional settlements; Tilly [1990] narrates the path-dependent state-formation trajectories; Acemoglu and Robinson [2019] diagrams the narrow corridor between strong-state and weak-state regimes. Each tradition describes regime change as a substantive phenomenon without committing to a

specific mathematical form. The framework's apparatus supplies the form: regime change is a bifurcation across Σ , with the index theorem of Chapter 4 constraining how regimes can change.

The singular locus and what happens there. Chapter 4 identified the singular locus $\Sigma \subset \Omega_0$ as the set of parameter profiles at which the joint Jacobian J becomes singular. At a regular endowment $\omega \in \Omega_0 \setminus \Sigma$, J is invertible at every equilibrium above ω , the implicit function theorem applies, and the equilibrium count above ω is finite and locally constant. At a parameter profile $\omega \in \Sigma$, J is singular at some equilibrium above ω , the implicit function theorem fails, and the equilibrium count can change as ω crosses.

The local structure at Σ admits a stratification. At a generic point of Σ the singularity is a *fold*: two equilibria above adjacent endowments merge and annihilate as ω approaches Σ from one side, with the fiber count dropping by two as the boundary is crossed. At codimension-two sub-loci the singularity is a *cusp*: two fold curves meet and the equilibrium count can change by two in either direction. At higher-codimension loci, swallowtail and butterfly catastrophes appear, with progressively richer transition geometries. The classification is the Thom catastrophe-theory hierarchy, applied to the joint equilibrium function $e : S_N \times M \times \Omega_0 \rightarrow \mathbb{R}^{L-1+I}$ of Chapter 3.

Regime components and cross-component motion. Under Assumption 3.17, $\tilde{E}^{\text{reg}}$ partitions into regime components $\tilde{E}_1, \tilde{E}_2, \dots$, each diffeomorphic to its parameter coverage in $\Omega_0 \setminus \Sigma$. A point $e \in \tilde{E}^{\text{reg}}$ belongs to a unique component; the component a particular endowment profile sits in is what the framework calls the regime at that profile.

A *regime change* in framework terms is motion of the equilibrium from one component to another. This motion cannot be smooth on $\tilde{E}^{\text{reg}}$: $\tilde{E}_1$ and $\tilde{E}_2$ are disjoint open subsets of the regular manifold, and any continuous path between them must leave $\tilde{E}^{\text{reg}}$ at some intermediate time. The point at which the path leaves the regular manifold is on Σ , the singular locus. Cross-component motion is therefore mathematically a path that crosses Σ , and substantively an event during which the system passes through a configuration where the joint Jacobian is singular and the smooth-equilibrium picture breaks down locally. This is the geometric content of regime change.

Remark 7.6 (Regimes and Krasner's definition). Krasner [1982] defined a regime as “principles, norms, rules, and decision-making procedures around which actor expectations converge.” The framework's regime components $\tilde{E}_k$ formalize the convergence-of-expectations element: within $\tilde{E}_k$, the joint equilibrium varies smoothly with parameters, and small parameter shifts produce small expectation shifts—expectations *converge* in the sense that all participants face the same local equilibrium structure. The boundary between $\tilde{E}_k$ and $\tilde{E}_{k'}$ is where convergence breaks down: small parameter shifts can produce qualitatively different equilibria depending on which side of Σ the profile is on. The verbal-historical regime concept and the formal regime component coincide in their substantive content.

The index theorem constrains how regimes change. Chapter 4's index theorem says that the signed sum of equilibrium indices over any regular fiber equals one: $\sum_{e \in \tilde{\pi}^{-1}(\omega) \cap \tilde{E}^{\text{reg}}} \text{ind}(e) = 1$ for every $\omega \in \Omega_0 \setminus \Sigma$. The signed-sum identity holds across all of $\Omega_0 \setminus \Sigma$. What it means for transitions across Σ is the substantive constraint.

When ω crosses a fold from one side to the other, the fiber count changes by two: two equilibria of opposite index either appear (on one side) or annihilate (on the other). The signed sum is preserved across the fold because the appearing or disappearing equilibria contribute $+1$

and -1 respectively, which cancel. This is what the index theorem says about fold transitions: not that the equilibrium count is fixed, but that whatever equilibria appear or disappear must come in cancelling pairs.

The substantive content for institutional analysis is sharp. A regime change cannot involve appearing or disappearing equilibria of identical index. If the framework predicts two equilibria of index $+1$ in regime $\tilde{E}_1$ and one of index -1 , a transition to regime $\tilde{E}_2$ that adds two more $+1$ equilibria and no -1 equilibria is geometrically impossible on a fold. Higher-codimension singularities admit richer transitions (cusps allow same-index pairs to merge if they pass through a crossing structure first), but each transition type has a definite signed-sum behavior. The framework reads the index theorem as the structural constraint on what regime change can do.

Reading existing IR theory. The bifurcation reading recovers four prominent literatures as descriptions of motion across Σ .

Krasner regimes are formalized as the regime components $\tilde{E}_k$ of $\tilde{E}^{\text{reg}}$, with the boundary between regimes as the crossing of Σ (Remark 7.6). The regime-change literature's central observation—that regime change is qualitatively different from incremental adjustment—is the structural feature that motion within $\tilde{E}_k$ is smooth while motion across Σ is not.

Ikenberry's post-war settlements (Ikenberry 2001) describe specific historical instances of regime selection: the post-1815 Concert, Versailles, post-1945 Bretton Woods. Through the bifurcation reading, each settlement is a moment at which the international system reorganized across Σ into a new regime component of the equilibrium manifold. The historical question of *which* settlement materialized is the question of which side of Σ the system landed on after the perturbation.

Tilly's coercion-capital paths (Tilly 1990) describe state-formation as a path-dependent trajectory through a parameter space of relative coercion-capital balances. The bifurcation reading reads each Tilly path as a trajectory through Ω_0 that crosses Σ at the moments the historical narrative identifies as transitions (centralization, fiscal-military reorganization, capital-coercion recombination). The path-dependence Tilly emphasizes is the regime-component selection at each crossing.

AR2019's narrow corridor (Acemoglu and Robinson 2019) diagrams a two-axis state-society plane partitioned into despotic, absent, and shackled-Leviathan regions. Through the bifurcation reading, the partition is a regime decomposition of a parameter manifold; the boundaries between regions are pieces of Σ ; the path-dependent Red-Queen dynamic Acemoglu and Robinson identify is motion within and between regime components, with cross-component transitions corresponding to historical moments of qualitative state-society shift. The narrow-corridor diagram is a low-dimensional projection of the bifurcation reading the framework's apparatus delivers in IL -dimensional state space.

These readings are interpretive: the apparatus identifies the mathematical structure each verbal tradition has been describing, without committing to specific historical mappings. Substantive empirical work—which conflicts ended in fold transitions, which institutional reorganizations correspond to cusp catastrophes, which path-dependencies follow swallowtail geometries—belongs to a calibrated chapter that this volume does not develop.

Example 7.7 (A regime boundary in the two-state contest). The two-state Tullock contest of Example 3.3 exhibits a regime boundary at the prize parameter $V_0 = 0$. For $V_0 > 0$, the unique symmetric equilibrium has interior deployments $m^* = (V_0 + \theta W)/(2k(2 + \theta)) > 0$: both states arm and the contest is contested, with welfare cost $\Delta^* > 0$. For $V_0 < 0$ the prize is converted

to a cost of participation, the marginal benefit of unilateral deployment at $m = 0$ becomes non-positive, and $m^* = 0$ is a Nash equilibrium with no contested counterpart on the symmetric ray. The regime to either side of the parameter slice $V_0 = 0$ is qualitatively different: contested ($V_0 > 0$, $m^* > 0$, $\Delta^* > 0$) on one side, peaceful ($V_0 < 0$, $m^* = 0$, $\Delta^* = 0$) on the other.

The transition at $V_0 = 0$ is a fold-like event in the chapter's classification. At $V_0 = 0$, the symmetric equilibrium has $m^* = \theta W / (2k(2 + \theta))$, strictly positive when $\theta > 0$ and zero when $\theta = 0$. The $\theta = 0$ subcase reduces to the Beviá and Corchón [2010] setup with $V_0 = 0$, where the contest collapses to no contest at all and the unique equilibrium is peace. The $\theta > 0$ subcase has $m^* > 0$ at $V_0 = 0$ from the endogenous-prize term θW alone, which means the contested-equilibrium continuation extends to negative V_0 until $V_0 + \theta W = 0$, i.e. until the endogenous prize term covers the negative external prize exactly. The actual fold occurs at $V_0^{\text{fold}} = -\theta W$ rather than at $V_0 = 0$: at this value the symmetric deployment m^* vanishes and the equilibrium count drops by one as V_0 decreases through V_0^{fold} .

The institutional reading is sharp. An institution that lowers the prize V_0 from a positive starting value drives the system along a path in Ω_0 toward Σ , with the regime change actually occurring at $V_0 = -\theta W$. At $\theta = 0$ this is $V_0 = 0$ (the contest disappears as soon as the external prize is removed); at $\theta > 0$ a negative prize is required to overcome the endogenous-prize term. The framework's apparatus distinguishes the two cases the verbal literature would lump together as “the contest is no longer worth fighting”: only at the fold V_0^{fold} is the regime change actually completed, and the location of V_0^{fold} depends on the share parameter θ in a way the joint formulation makes explicit.

Limits of the reading. The bifurcation reading has structural limits.

The apparatus identifies geometry, not content. Whether a particular historical event is a fold transition or a cusp crossing depends on the conflict primitive's specification and on the empirical specification of the parameter shift. The reading identifies that regime change has a definite geometric form; specific empirical claims require calibration.

Catastrophe-theoretic stratification is a generic classification. At non-generic singularities, transitions can be more complex than the fold-cusp-swallowtail hierarchy describes. The framework treats non-generic cases as boundary conditions that require specific analysis; the chapter does not develop the non-generic theory.

Cross-regime motion in finite time requires a specific driving force. The bifurcation reading says *what* motion across Σ is, not *how* it happens. Identifying which institutional or external perturbation drives the system to Σ at a particular moment is empirical content the apparatus does not supply. Section 7.3's gradient field reading identifies one mechanism (welfare-improving institutional intervention can drive the system toward Σ at the boundary of a peace-unreachable component); other mechanisms (external shocks, internal collapse, deliberate disruption) are treated by their respective substantive literatures.

What no partial can compute. The bifurcation reading is not available from a partialled framework. A partial-equilibrium analysis treats the equilibrium correspondence as a function from parameters to equilibria with no internal structure: regime change is exogenous. PE-thinking sees parameter shifts and reports new equilibria, but does not identify the singular locus where qualitative transitions happen, does not constrain how equilibria pair across transitions, and does not predict what catastrophe-theoretic type of singularity governs a given transition. The substantive content of *regime* as opposed to *state* is invisible.

A conflict-side partial works on the militarization profile space at fixed prices. The deployment-equilibrium can have its own bifurcations as parameters shift, but the bifurcation set in deployment space is not the singular locus of the joint Jacobian; the conflict-side partial misidentifies which parameter changes cause regime change because it is not seeing the cross-channel coupling Chapter 4 establishes. A conflict-only bifurcation analysis classifies the wrong transitions.

Chapter 4's singular locus Σ is the joint bifurcation set, identified by $\det J = 0$ on the joint Jacobian. The bifurcation reading consumes this directly: regime change is crossing Σ , the index theorem constrains the crossing, catastrophe-theoretic stratification classifies the crossing type. None of these is available from a partialled apparatus. The framework's joint formulation is what makes regime change a mathematical phenomenon rather than a verbal-historical description.

7.5 Institutions as paths on the equilibrium manifold

The three readings developed so far—aggregation rule (Section 7.2), gradient field (Section 7.3), driver of regime change (Section 7.4)—each capture one face of a single object the chapter has not yet named. The aggregation rule fixes a target equilibrium but is silent on how the target is reached. The gradient field induces a path but constrains it to be the gradient flow of welfare; alternative routes to the same target are not in the picture. The bifurcation reading names the crossing of Σ as the regime-change event but loses the detail of *which* crossing, *where*, and via *which* homotopy class. This section names the unifying object and shows what its analysis adds.

Institution as (target, path). Let $e_0 \in \tilde{E}^{\text{reg}}$ be the current equilibrium. An *institution* is a pair (e_1, γ) where $e_1 \in \tilde{E}$ is the target equilibrium and $\gamma: [0, 1] \rightarrow \tilde{E}$ is a continuous (in working contexts, smooth or absolutely continuous) path with $\gamma(0) = e_0$ and $\gamma(1) = e_1$. The current readings make either the target or the path the focus; the unifying reading makes both ingredients of one object. This is the manifold-paths formulation that [Loi and Matta \[2008, 2009, 2011\]](#), building on [Balasko \[1978\]](#), develop for the pure-exchange equilibrium manifold; the chapter extends the formulation to the joint equilibrium manifold $\tilde{E}$ and reads it institutionally.

Preferences over institutions. Preferences over (target, path) institutions are encoded by a 1-form $\omega \in \Omega^1(\tilde{E})$ via the line integral

$$V_\omega(\gamma) := \int_\gamma \omega = \int_0^1 \omega_{\gamma(t)}(\gamma'(t)) dt. \quad (7.2)$$

Two 1-forms ω_1, ω_2 deliver identical line integrals between every pair of fixed endpoints if and only if their difference is exact, $\omega_1 - \omega_2 = dV$ for some smooth $V: \tilde{E} \rightarrow \mathbb{R}$. The institution-preference equivalence classes modulo target-only content are therefore the de Rham cohomology classes

$$[\omega] \in H^1(\tilde{E}; \mathbb{R}) = \Omega_{\text{closed}}^1(\tilde{E}) / \Omega_{\text{exact}}^1(\tilde{E}). \quad (7.3)$$

$H^1(\tilde{E}; \mathbb{R}) = 0$ means every closed preference is target-only: line integrals telescope to differences of values at endpoints, and the institution-preference structure depends only on (e_0, e_1) . $H^1(\tilde{E}; \mathbb{R}) \neq 0$ means there are preferences genuinely over paths, not over targets alone.

Three natural classes of preference 1-form. The chapter’s existing material exhibits three distinct preference 1-forms, each instantiating a different content.

Welfare-gradient preferences. $\omega = d\Delta^*$, the differential of the welfare scalar Δ^* from Chapter 5. Always exact; the line integral telescopes to $\Delta^*(\omega_1) - \Delta^*(\omega_0)$, where $\omega_i = \pi(e_i)$ is the parameter at endpoint i . The institution-preference reduces to a target-only preference on welfare differences.

Riemannian transition-cost preferences. For a Riemannian metric g on $\tilde{E}$, the length functional $\omega(v) = (g_e(v, v))^{1/2}$ delivers transition cost as the path’s geometric length. Generically not closed; line integrals depend on the path. [Loi and Matta \[2008, 2011\]](#) construct natural Riemannian metrics on the equilibrium manifold for the pure-exchange case.

Catastrophe-counting preferences. The integer-valued preference $\omega(\gamma) = \#(\gamma \cap \Sigma)$ counts the regime crossings the path induces. Formally a current rather than a 1-form, but plays the same role: institutions that cross Σ more often are dispreferred to those that cross less. [Loi and Matta \[2009\]](#) prove existence of a path-length-minimizing γ between any two regular equilibria; [Balasko \[2012\]](#) sharpens the result with the upper bound $d(x, y) \leq 2$ for all pairs and $d = 0$ for index-1 stable pairs.

The three readings as special cases. Each of the chapter’s existing readings recovers from the unifying object under a specific restriction.

Reading 2 (gradient field) is the exact case. When $\omega = dV$ is exact, the integral telescopes and the institution-preference reduces to a preference on $V(e_1) - V(e_0)$. The gradient flow of V then determines the path uniquely from (e_0, e_1) ; (target, path) collapses to (target, induced gradient-flow). Section 7.3’s gradient-field reading is the unifying object under the restriction that the institution’s preference 1-form is exact.

Reading 3 (regime-change driver) is the cross- Σ case. When γ crosses Σ at one or more points, the institution induces regime change. The bound $d(x, y) \leq 2$ from [Balasko \[2012\]](#) caps the number of regime crossings any institution can induce. Section 7.4’s bifurcation reading is the unifying object restricted to paths with at least one Σ -crossing; the catastrophe-counting preference 1-form is the natural ranking on such institutions.

Reading 1 (aggregation rule) becomes Chichilnisky on 1-forms. The aggregation question shifts from preference profiles $P(\tilde{E})^I$ on the manifold to profiles $\Omega^1(\tilde{E})^I / \Omega_{\text{exact}}^1$ on 1-forms-modulo-exact. The Chichilnisky–Heal apparatus (Section 7.2) extends to ask whether a continuous, anonymous, unanimous aggregation rule exists on this enlarged preference space. The next paragraph establishes when it does.

Two faces of one theorem. The cohomological structure governs preference aggregation in two regimes that are most usefully read as faces of a single result rather than as a clean-vs-messy contrast.

The JR face: institutional robustness as $H^1 = 0$. Under Assumption 3.6, the equilibrium manifold $\tilde{E}$ is contractible (Corollary 3.16 of Chapter 3), hence simply connected, hence $H^1(\tilde{E}; \mathbb{R}) = 0$. By the Poincaré lemma, every closed 1-form is exact: every “consistent” institution-preference reduces to a welfare function on the manifold. Two paths between the same endpoints are homotopic relative to endpoints, and the line integral of any closed form is invariant under such homotopy by Stokes’ theorem. The substantive content is that institutions can adapt mid-stream—deform the path continuously while keeping the target fixed—without welfare consequence.

Theorem 7.8 (Aggregation under JR, path version). *Under Assumption 3.6, the institution-preference structure Equation (7.3) reduces to target-only preferences, and the aggregation rule of Corollary 7.2 extends to the (target, path) formulation without modification.*

Proof sketch. $H^1(\tilde{E}; \mathbb{R}) = 0$ by contractibility, so every preference 1-form is cohomologous to its exact part; the line integral depends only on endpoints; the aggregation problem on $\Omega^1(\tilde{E})/\Omega^1_{\text{exact}}$ reduces to the aggregation problem on smooth functions $\tilde{E} \rightarrow \mathbb{R}$, which the existing apparatus already covers. $\square$

The JR' face: institutional brittleness as nontrivial homotopy. Under Assumption 3.17, the regular set $\tilde{E}^{\text{reg}}$ is $\tilde{E} \setminus \Sigma$. Codimension-one components of Σ disconnect $\tilde{E}^{\text{reg}}$ into chambers, with the peaceful chamber containing the no-deployment equilibrium distinguished as $\tilde{E}_0$ (Definition 3.21 of Chapter 3). For any pair of regular equilibria e_0, e_1 in the same chamber, let $\Pi(e_0, e_1) := \pi_0(\Omega^{e_0, e_1}(\tilde{E}^{\text{reg}}))$ denote the set of homotopy classes of smooth paths from e_0 to e_1 in the chamber. When the chamber is simply connected, $\Pi(e_0, e_1)$ is a single class and the path-aggregation problem reduces to target aggregation. When the chamber is not simply connected—equivalently, when the chamber's first cohomology $H^1(\text{chamber}; \mathbb{R})$ is nontrivial— $\Pi(e_0, e_1)$ has multiple elements, and on-the-fly route changes that swap the homotopy class carry genuinely different line integrals of any closed-but-not-exact preference 1-form. The substantive question is what aggregation looks like over this enlarged alternative set.

The Chichilnisky-Heal apparatus does not deliver an obstruction on its own. The space of 1-forms-modulo-exact, $\Omega^1(\tilde{E}^{\text{reg}})/d\Omega^0(\tilde{E}^{\text{reg}})$, has cohomology $H^1(\tilde{E}^{\text{reg}}; \mathbb{R})$ as its quotient on the closed-form subspace; the quotient is a real vector space, hence contractible, hence Chichilnisky-Heal-aggregable as a target. The obstruction lives at a different level: aggregating *ordinal preferences* of agents over the discrete alternative set $\Pi(e_0, e_1)$, where Arrow's classical impossibility theorem applies whenever $\#\Pi(e_0, e_1) \geq 3$.

Proposition 7.9 (Arrow obstruction on path-classes under JR'). *Suppose Assumption 3.17 holds and let e_0, e_1 lie in a chamber of $\tilde{E}^{\text{reg}}$ with $\#\Pi(e_0, e_1) \geq 3$. Let each of $I \geq 2$ agents supply an ordinal ranking $\succeq_i$ on $\Pi(e_0, e_1)$. There is no aggregation rule $\Phi: (\succeq_1, \dots, \succeq_I) \mapsto \succeq_*$ that is simultaneously transitive, unanimous, non-dictatorial, and independent of irrelevant alternatives.*

Proof. Restrict each ranking $\succeq_i$ to any three classes $\gamma_a, \gamma_b, \gamma_c \in \Pi(e_0, e_1)$; Assumption 3.17 guarantees three exist by hypothesis. Arrow [1951]'s impossibility theorem applies to any finite alternative set of size ≥ 3 with finitely many agents $I \geq 2$, delivering the conclusion. $\square$

The proposition is honest about what it claims: the obstruction to path-aggregation under JR' is Arrovian on the discrete set of homotopy classes, not Chichilnisky-Heal on the continuous cohomology space. The substantive content is the same—under JR' with $\#\Pi(e_0, e_1) \geq 3$, the institutional aggregation problem strictly extends the target aggregation problem of Corollary 7.3 and inherits Arrow's classical impossibility on the path-class slice—but the mathematics are clean rather than confused.

When does the hypothesis $\#\Pi(e_0, e_1) \geq 3$ hold? The set $\Pi(e_0, e_1)$ is a torsor over the fundamental group $\pi_1(\text{chamber}, e_0)$ of the chamber, so $\#\Pi(e_0, e_1) = \#\pi_1(\text{chamber}, e_0)$. The hypothesis fails on simply connected chambers; the JR contractibility result of Corollary 3.16 delivers a trivial π_1 globally, and within-chamber simple-connectedness is the typical case under JR' when codimension-one components of Σ are unbounded hyperplane-like strata that fully

separate the manifold. The hypothesis succeeds when chambers contain *closed* codimension-one features in their interior—a Σ -stratum that loops back on itself, encircling a region of $\tilde{E}^{\text{reg}}$. The next paragraph exhibits a concrete case.

Example 7.10 (A 2-good 2-state economy with closed Σ -stratum). Consider an exchange-and-conflict economy with two states and two goods, parameterized by the endowment vector $\omega \in \mathbb{R}^2$ (interpreting the two coordinates as the relative endowment of state A in good 1 and good 2). Suppose $\det J(p, m, \omega) = q(\|\omega\|^2)$ for some smooth function q with a single isolated zero at $\|\omega\|^2 = r^2$. Then the singular locus $\Sigma \subset \Omega_0$ contains the circle $\{\omega : \|\omega\|^2 = r^2\}$, a closed codimension-one stratum. The complement $\Omega_0 \setminus \Sigma$ has two chambers: the disk-interior chamber $\{\|\omega\| < r\}$ (simply connected) and the annular exterior chamber $\{\|\omega\| > r\}$. The exterior chamber has $\pi_1 = \mathbb{Z}$, generated by a loop that encircles the singular circle once, and $H^1 = \mathbb{R}$. For any e_0, e_1 in the exterior chamber, the homotopy class set $\Pi(e_0, e_1)$ is the set of integer winding numbers around the singular circle, infinite in cardinality. Proposition 7.9’s hypothesis $\#\Pi(e_0, e_1) \geq 3$ is satisfied, and Arrow-on-winding-numbers is the substantive form the path-aggregation impossibility takes.

Two voters in the same chamber. The substantive content is sharper than “aggregation can fail.” Consider two voters in the exterior chamber of Example 7.10 with the same target e_1 but different preferred winding numbers around the singular circle—voter 1 prefers a path that loops once counterclockwise before reaching e_1 , voter 2 a path that loops once clockwise. Their disagreement is genuine institutional content, not noise: which homotopy class of intervention path the institution commits to determines, via the line-integral preferences of Equation (7.2), the cumulative welfare consequence of the institution. Proposition 7.9 applied to three or more such winding numbers delivers Arrow’s impossibility: no aggregation rule satisfying transitivity, unanimity, non-dictatorship, and IIA can resolve the disagreement. The chamber structure that Chapter 4 identifies as a regime-multiplicity feature reappears here as an aggregation-obstruction at the path level.

Two faces, one topological fact. Theorem 7.8 and Proposition 7.9 are not independent results. Both follow from the topology of the chamber containing e_0 and e_1 . The JR face says: when the chamber is simply connected, $\#\Pi(e_0, e_1) = 1$, all paths are homotopic, line integrals of closed forms agree, and institutions adapt robustly to mid-stream route changes. The JR’ face says: when the chamber has nontrivial π_1 , $\#\Pi(e_0, e_1) \geq 2$, line integrals of non-exact closed forms depend on which homotopy class is taken, and institutional adaptation is path-pinned by the cohomology class committed to. The same invariant—the fundamental group of the chamber—governs both faces, with the rank of $H^1(\text{chamber}; \mathbb{R})$ as its abelianized linear-algebraic shadow. The (p, m) framework’s structural-empirical contribution is to identify which institutions live in which face by asking where, relative to Σ , the institution operates.

The path-on-equilibrium-manifold cluster. The mathematical apparatus behind Theorem 7.8 and Proposition 7.9 sits at the intersection of two literatures. [Loi and Matta \[2008, 2009, 2010, 2011, 2018, 2021\]](#) develop path-on-equilibrium-manifold machinery for pure-exchange economies—Riemannian metrics making critical strata totally geodesic, the catastrophe-counting pseudodistance, arc-lifting under regular policy, shape-of-manifold sufficient conditions for global uniqueness—and [Balasko \[2012\]](#) sharpens the regime-quotient with the upper bound $d \leq 2$. The chapter inherits the apparatus and extends it to the joint manifold; [Loi and Matta \[2008\]](#)’s

open invitation that “the connection between metric and economics still has to be investigated” is the explicit predecessor to the institutional reading of the 1-form structure delivered here. [Garratt and Goenka \[1995\]](#) is the closest substantive economics predecessor: their finite-stage lump-sum redistribution policy staying within the regular-economy connected component is the public-finance instance of the (target, path) institution under the JR face. Their five conditions (linear taxes, finite stages, continuity, absence of Leontief reversal, stability) translate into preference-1-form properties: continuity is well-definedness of the line integral, the Leontief condition is monotone direction of the 1-form, stability is arc-liftability of equilibrium prices along the policy path [[Loi and Matta, 2010](#)].

Path dependence in political science. The political-science literature on path dependence supplies the substantive vocabulary the (target, path) formulation makes mathematical. [Pierson \[2000\]](#) grounds path-dependence in increasing-returns dynamics and identifies five substantive features—timing and sequence matter; multiple outcomes possible from similar conditions; large consequences from small events; irreversibility; critical junctures. Each maps onto a property of (target, path) institutions on the joint manifold: timing-and-sequence is path-dependence of the line integral, multiple outcomes from similar conditions is the chamber structure of $\tilde{E}^{\text{reg}}$, large consequences from small events is the codimension-one Σ -stratum geometry, irreversibility is cohomological pinning, critical junctures are Σ -crossings. [Page \[2006\]](#) adds a formal-decomposition vocabulary, distinguishing state-dependence (paths can be partitioned into finitely many states which contain all relevant information) from path-dependence proper (the full path matters); the (target, path) framework reads state-dependence as the JR-clean case the arc-lifting property of [Loi and Matta \[2010\]](#) delivers, and path-dependence proper as the JR' case where $H^1(E^{\text{reg}}; \mathbb{R}) \neq 0$. [Fioretos \[2011\]](#) positions historical institutionalism as a distinct tradition in international relations theory alongside rational-choice and sociological institutionalism; the (p, m) framework's institutional reading sits as a fourth tradition, *manifold institutionalism*, formal-mathematical at the level of joint-equilibrium-on-the-manifold.

An empirical case for the JR face. [Barton et al. \[2006\]](#) provide the canonical book-length historical-institutionalist account of GATT/WTO regime evolution. Their narrative reads, through the lens of the (target, path) formulation, as a within-chamber adaptation story: the trade regime has evolved through Uruguay Round expansion, Doha stalemate, dispute-settlement reform, and progressive incorporation of nonstate actors and developing-country interests, with each transition a re-routing of the path rather than a regime rupture. The substantive question Theorem 7.8 poses for the GATT/WTO case is whether the trade-regime equilibrium manifold restricts to a single chamber under typical parameter perturbations—in which case the institution's adaptive flexibility is enabled by topology rather than only by political-economic smoothness. [Barton et al. \[2006\]](#)'s “authoritative gap” diagnosis—the regime's inability to recast its rules in line with changing interests of powerful domestic constituencies—admits the manifold-institutionalism reading: the regime's existing chamber constrains how far the path can re-route before crossing Σ into a different chamber where the rules-and-norms package would have to be substantially different. Whether GATT/WTO has actually approached such a Σ -crossing at any point in its history—the 1971 dollar-gold suspension, the 2001 China accession, the post-2018 tariff-and-sanctions turn—is the empirical question the framework makes precise.

Looking ahead from Section 7.5. The (target, path) formulation transmits forward to subsequent chapters as the natural object the framework’s apparatus consumes. Chapter 6’s donor-aid schedules are (target, path) institutions in the special case where the path-cost 1-form is the strict-convex transition cost underlying the gradualism characterization of Proposition 6.10, and the manifold-Pontryagin apparatus of Chapter 6 is the optimization version of the (target, path) institution problem. Chapter 8 consumes the formulation as the natural object sanctions perturb: tariff schedules, sanction ramps, secondary-sanctions sequences, and deployment-cost trajectories are all (target, path) institutions, and the cohomological obstructions of Proposition 7.9 are the structural-formal reasons multilateral coordination on sanctions is harder than coordinating on targets alone. The cluster of open questions the formulation raises—tighter characterizations of $H^1(\tilde{E}^{\text{reg}}; \mathbb{R})$ from primitive data, stochastic versions of (target, path) institutions in the probability-on-manifold sense of [Loi and Matta \[2021\]](#), and computational implementations of the path-aggregation problem—is reserved for follow-up work.

7.6 Critique and defense of the topological approach

The four readings developed in Sections 7.2 to 7.5 draw on two literatures the apparatus is in dialogue with: the Chichilnisky topological social-choice tradition and the structural-realist institutionalist literature. Each carries substantive critiques the chapter must address. This section names the principal critiques and identifies the framework’s defense on each, marking what survives and what qualifies as a structural limit. Critiques of the topological-aggregation tradition transfer naturally to the (target, path) reading because that reading’s path-aggregation analysis runs an extended Chichilnisky-on-1-forms argument; the discussion below is therefore relevant to all four readings, with explicit notes where the unifying reading introduces or sharpens a concern.

The Lauwers critique of topological aggregation. [Lauwers \[2009\]](#) surveys the principal critiques of Chichilnisky’s topological aggregation framework. Three lines deserve direct response.

Preference-space topology is choice-laden. The Chichilnisky preference space $P(X)$ on a manifold X admits multiple natural topologies—continuous unit-normalized gradient vector fields under one parameterization, smooth utility functions modulo monotone transformations under another, ordinal preference correspondences under a third—and the contractibility theorem applies to one parameterization that may not transfer to another. The substantive content of the Chichilnisky–Heal theorem therefore depends on the topology choice, and the choice is not forced by the data.

The framework’s defense is that the preference-space topology is not separately ours to choose. The equilibrium manifold $\tilde{E}$ comes with the smooth structure of Chapter 3, and continuous preferences on $\tilde{E}$ inherit a natural topology from this smooth structure: the C^1 topology on smooth utility functions, or equivalently the topology of continuous sections of the unit tangent bundle. Other parameterizations of preference space are derivable but the natural one is determined by the framework’s apparatus. The Lauwers concern about choice-laden parameterization becomes a question about whether the framework’s smooth-structure choice is the right one—a question the chapter Section 7.1 engages by reading $\tilde{E}$ ’s smooth manifold structure as the substantive content the apparatus delivers.

Anonymity admits limit-of-dictatorship. [Lauwers \[2009\]](#) reports the result that anonymous Chichilnisky aggregation rules can converge to dictatorial limits as the agent count grows: the

limit rule, while continuous and Pareto-respecting on every finite-agent profile, can have an asymptotic structure that effectively privileges one agent's preferences.

The framework's defense is that the result is about the finite-to-infinite limit of the aggregation construction, not about the finite-agent rule. Corollary 7.2 delivers a finite-agent rule on $P(\tilde{E})^I$ for a fixed I , and the substantive institutional question is what this finite-agent rule looks like at the parameter values relevant to a specific institution. The asymptotic-dictatorship pathology is about the universal-rule construction across all I ; for any specific institution with finite membership, the rule does what the theorem says it does. The chapter does not need the limit construction.

Chichilnisky positive content depends on parameterization. The contractibility-implies-aggregation direction of Theorem 7.1 is general; the converse—that non-contractibility implies impossibility—depends on which preference-space topology is in force. Different topologies can yield different obstructions.

The framework's defense is parallel to the first. The framework's apparatus delivers a specific preference-space topology, and the obstruction analysis is run against that topology. Corollary 7.3's componentwise result depends on the specific way $\tilde{E}^{\text{reg}}$'s topology inherits multi-component structure from the bifurcation set Σ ; under different preference-space parameterizations the componentwise structure could read differently. The chapter takes the framework-natural topology as the substantive choice and runs the analysis against it.

The Mearsheimer false-promise critique. Mearsheimer [1994a] argues that institutions are reflections of underlying power, not independent causal forces, and that institutionalist explanations of state behavior overstate the extent to which institutions shape outcomes. On the false-promise reading, the apparent causal content of institutions is an artifact of correlation: powerful states design institutions that reflect their preferences, and institutions appear to influence behavior only because behavior tracks underlying power. The institutionalist literature, in this view, has been mistaking reflection for causation.

The framework's response is that the reflection-versus-causation distinction is a substantive empirical question the apparatus clarifies rather than answers. Section 7.3's gradient-field reading writes an institution as a vector field X^{inst} on $\tilde{E}^{\text{reg}}$ and asks how aligned it is with the welfare-improvement gradient X^{Δ^*} . The realist position translates to a specific claim about X^{inst} : it is determined by the underlying parameter profile ω rather than chosen institutionally. The institutionalist position translates to the claim that X^{inst} has degrees of freedom independent of ω that institutional design exploits. The framework's apparatus does not commit to either: it provides the geometric language in which the question can be asked, and the apparatus's tools (the inner product $\langle X^{\text{inst}}, X^{\Delta^*} \rangle$, the catastrophe-theoretic stratification of Σ) make the realist and institutionalist positions empirically distinguishable.

The substantive content of this response is methodological. The realist-institutionalist debate has often been conducted at the level of intuition—"institutions reflect power" versus "institutions have causal force"—without the apparatus to make the disagreement empirical. The framework's reading reframes the debate as a question about X^{inst} 's independence from ω , which is a parameterization that admits empirical resolution. The chapter does not resolve the debate; it identifies the form the resolution must take.

The implementation gap, again. Both critiques converge on a common observation. The framework's apparatus identifies what institutions are doing in geometric terms (aggregation

rules, gradient fields, bifurcation drivers, (target, path) pairs) but does not say which substantive institutions correspond to which geometric objects. A theory of which design features implement which gradient fields, which membership structures correspond to which aggregation rules, which institutional perturbations drive the system across which kind of Σ -stratum, which historical episodes traced by Barton et al. [2006] or Ikenberry [2001] read as within-chamber adaptations versus genuine cohomology-changing transitions, requires the calibrated apparatus a substantive applications chapter is meant to develop. The chapter does not include this calibration. The reading the chapter delivers is therefore best understood as a structural reading: the apparatus identifies what is available to be calibrated, and the calibration belongs elsewhere. The institutional content—substantively, which institutions matter and how—is empirical and downstream.

This is a real limit, not a defect to be fixed within the chapter. The framework’s apparatus is meant to deliver geometric structure under which the substantive empirical questions become precise; it is not meant to settle those questions itself. Chapter 8 is where the chapter’s apparatus meets a specific applied case; the structural-empirical bridge proper belongs to the follow-up paper that calibrates the framework to the global sanctions database.

7.7 Looking ahead

The institutional theory delivered by this chapter is, in summary form, four readings of one apparatus. An institution is an aggregation rule on the equilibrium manifold (Corollaries 7.2 and 7.3), a smooth vector field implementing some welfare-direction (Section 7.3), a driver of cross-component motion across the singular locus (Section 7.4), and a (target, path) pair on the equilibrium manifold whose preference structure is encoded by a 1-form (Section 7.5). The first three are not exclusive; a particular institution may admit two or all three at once. The fourth is unifying: each of the first three recovers from the (target, path) reading under a specific restriction (exact 1-form for the gradient field, cross- Σ path for the bifurcation driver, Chichilnisky-on-1-forms for the aggregation rule), and the cohomological structure $H^1(\tilde{E}^{\text{reg}}; \mathbb{R})$ delivers the rigorous form of what political-science usage names “path dependence.” Their separation is analytic, and each reading inherits its substantive content from a specific upstream chapter: the topology of $\tilde{E}$ from Chapter 3, the welfare gradient $D_\omega \Delta^*$ from Chapter 5 Section 5.4, the singular locus and the index theorem from Chapter 4, the multilateralism amplification reading from Chapter 6 Corollary 6.16, and the path-on-equilibrium-manifold apparatus from the Loi and Matta [2008, 2009, 2010, 2011], Balasko [2012], Garratt and Goenka [1995] cluster the chapter inherits. The chapter’s content is the apparatus’s institutional face.

Each reading reframes a debate in international relations. The aggregation reading reopens the Arrow-impossibility debate by delivering exactly the topological condition (contractibility) that the Chichilnisky–Heal theorem requires; under joint regularity, social-choice approaches to international institutions are not foreclosed. The gradient-field reading reorients the rational-design tradition by identifying the welfare-direction-of-improvement that institutional architecture should implement; institutional comparison becomes a question of vector-field alignment with X^{Δ^*} . The bifurcation reading translates the verbal-historical IR literature on regime change into a precise mathematical apparatus (Krasner regimes as components $\tilde{E}_k$, transitions as crossings of Σ , the index theorem as the structural constraint on how regimes change). The (target, path) reading translates the historical-institutionalist literature on path dependence [Pierson, 2000, Page, 2006, Fioretos, 2011, Barton et al., 2006] into a precise topological apparatus where Pierson’s

increasing-returns and critical-junctures become statements about the rank of $H^1(\tilde{E}^{\text{reg}}; \mathbb{R})$ and where the political-science distinction between robust within-chamber adaptation and brittle critical-juncture transitions becomes the JR/JR' distinction at the cohomology level.

Subsequent chapters consume the apparatus. Chapter 8 reads sanctions as (target, path) institutions on the equilibrium manifold: the bargaining-camp slice fixes both m and p , the GE-camp slice fixes m but lets p respond, and the joint formulation lets both adjust along the path. The cohomological obstructions of Proposition 7.9 become the structural-formal reasons multilateral sanctions coordination is harder than coordinating on targets alone, and the gradient-field reading of Section 7.3 identifies which sanction trajectories implement the welfare gradient and which do not. Chapter 9 returns to the program-level question of how the framework's institutional content fits into the larger research agenda; the multilateralism reading of Remark 7.4 is the chapter's sharpest concrete claim.

The chapter admits two readings. Read as substantive institutional theory, it is a structural account of what international institutions do in a joint-determination setting. Four results no partialled framework can deliver—aggregation under contractibility, $X^{\Delta*}$ as the welfare-direction gradient, regime change as bifurcation across the joint singular locus, and the cohomology-of-paths apparatus that resolves the political-science path-dependence vocabulary into mathematical content—together with a fifth result inherited from Chapter 6, the resource-neutral multilateralism rationale read as gradient-field amplification. Read as apparatus-demonstration, the chapter is a worked extended case for the framework's institutional reach: the topology of Chapter 3, the welfare scalar of Chapter 5, the singular locus of Chapter 4, the donor-coordination apparatus of Chapter 6, and the [Loi and Matta \[2008, 2009, 2010, 2011\]](#), [Balasko \[2012\]](#), [Garratt and Goenka \[1995\]](#) path-on-equilibrium-manifold cluster all combine to deliver institutional content that none of the upstream apparatus would produce in isolation. Both readings are views of the same apparatus running on the same six sections.

Exercises

1. *Institutions as sections of $\tilde{\pi}$.* A natural mathematical reading of an institution is as a section $\sigma : \Omega_0 \rightarrow \tilde{E}$ of the natural projection: an assignment of a specific equilibrium $\sigma(\omega) \in \tilde{\pi}^{-1}(\omega)$ to each parameter profile. Under Assumption 3.6 (Chapter 3), $\tilde{\pi}$ is a global diffeomorphism, and the unique inverse $\tilde{\pi}^{-1}$ is itself the unique global section. Under Assumption 3.17, fibers can have multiple equilibria, and a section must select one equilibrium per ω continuously. Identify the obstruction to global existence of a section under JR-gen: when does $\tilde{\pi}$ admit a continuous global section, and when does it admit only componentwise sections? Hint: the obstruction lives in the topology of $\tilde{\pi}$ as a covering map and relates to the connectedness and orientability of $\tilde{E}^{\text{reg}}$. Open-ended.
2. *Two sections on Example 3.3.* Suppose Example 3.3 is read in the regime where two equilibria coexist above some ω (the JR-gen regime constructed in Exercise 12 of Chapter 3). Exhibit two distinct continuous sections σ_1, σ_2 of $\tilde{\pi}$ defined on a neighborhood of that ω , and identify which institutional structures select each. The sections correspond to “war regime” and “peace regime” selections, and the institutional content is which state of affairs the framework reads as the operative equilibrium. Substantive interpretation: institutional design is the choice of section.
3. *Vector field reaching the peace component.* The chapter's gradient-field reading of institutions

makes institutional design a vector field on $\tilde{E}^{\text{reg}}$. Construct a smooth vector field X on $\tilde{E}^{\text{reg}}$ such that integral curves of X reach the peace component $\tilde{E}_0$ (Definition 3.21 of Chapter 3) in finite time from any starting point in the war region. When does the welfare gradient $X = -\nabla\Delta^*$ achieve this? Identify the additional conditions required: convexity of the welfare landscape, absence of saddle points outside $\tilde{E}_0$, and finite distance from $\tilde{E}_0$ in the relevant Riemannian metric. Open-ended; the substantive content is when the welfare gradient is by itself a successful peacemaking institution.

4. *The three readings on a single case.* The chapter offers three readings of institutions: aggregation (Section 7.2), gradient field (Section 7.3), and regime change (Section 7.4). Compare the three readings on a single substantive case: the post-1945 Bretton Woods settlement, the Concert of Europe 1815, or the post-Cold War liberal order. Identify which reading is informative and which is degenerate on the chosen case. The aggregation reading is informative when the regime is a single contractible component (no fold structure); the gradient-field reading is informative when there is a clear welfare landscape; the regime-change reading is informative when the case involves a transition across Σ . Open-ended; one paragraph per reading.
5. *Verify aggregation under JR by explicit construction.* Corollary 7.2 establishes that under Assumption 3.6, a continuous, anonymous, Pareto-respecting aggregation rule exists on the equilibrium manifold's preference space. Verify the corollary directly by constructing such a rule on a contractible 2-state preference space. The standard construction uses a barycentric average of the two states' preferences after parameterization on the contractible $\tilde{E} \cong \mathbb{R}^{IL}$. Show the rule is continuous (as a map of preference profiles), anonymous (invariant under permutation of states), and Pareto-respecting (a profile preferring e_1 to e_2 unanimously aggregates to a rule preferring e_1 to e_2). The construction works precisely because the preference space is contractible.
6. *Chichilnisky–Heal necessity direction.* Theorem 7.1 delivers an “if-and-only-if” result: a continuous, anonymous, Pareto-respecting aggregation rule on a preference space $P(X)$ exists if and only if $P(X)$ is contractible. Sketch the necessity direction: if $P(X)$ is not contractible, no such rule exists. The argument is homotopy-theoretic. The aggregation rule, viewed as a continuous map from profiles to outcomes, must be homotopic to a projection (anonymity), and the relevant homotopy group $\pi_n(P(X))$ must vanish for the projection to factor through. When $P(X)$ is not contractible, some π_n does not vanish, and the obstruction is realized as a profile that no continuous rule can resolve. [See Chichilnisky, 1980, for the original argument].
7. *Manifold with nontrivial $\pi_1(P(M))$.* Construct a smooth manifold M whose preference space $P(M)$ has nontrivial fundamental group. The natural example is $M = S^1$ (the circle): $P(S^1) \simeq S^1$ has $\pi_1(P(S^1)) = \mathbb{Z}$. Identify a specific aggregation profile on $P(S^1)$ that no continuous rule can resolve: two preferences whose convex combination wraps around S^1 in a way that cannot be continuously selected. The obstruction is visible as the impossibility of choosing a continuous winding number across the profile space. The exercise illustrates the geometric content of the Chichilnisky–Heal obstruction in a minimal example.
8. *Two distinct rules across regimes.* For the JR-gen multi-component case (Corollary 7.3), the contractibility hypothesis holds componentwise but not globally. Exhibit two distinct

continuous, anonymous, Pareto-respecting aggregation rules on the same $\tilde{E}^{\text{reg}}$ that agree within each component but disagree across regimes. Identify what makes them differ: the choice of section σ (Exercise 1) across the components determines how the rules connect at regime boundaries. Substantively: under JR-gen, the aggregation rule is not unique, and the choice of which equilibrium to select on each component is itself a constitutional question the framework cannot resolve internally.

9. *Krasner's regimes through componentwise aggregation.* Krasner's classic definition of an international regime is "principles, norms, rules, and decision-making procedures around which actor expectations converge" (Krasner 1983). Read this through Corollary 7.3: a regime in Krasner's sense corresponds to a contractible component of $\tilde{E}^{\text{reg}}$ on which a continuous aggregation rule exists. Identify which regimes in the IR canon are contractible-aggregation-admissible (Bretton Woods, the European Coal and Steel Community, the post-WTO trade regime) and which involve cross-regime cases (the great-power transitions of 1815, 1919, 1945, 1989). Open-ended; one paragraph per regime, identifying whether the case sits within a component or involves cross-component motion.
10. *Welfare gradient X^{Δ^*} on Example 3.3.* The welfare gradient $X^{\Delta^*} := -\nabla\Delta^*$ is the chapter's main operational object: the vector field on $\tilde{E}^{\text{reg}}$ along which welfare improves fastest (Equation (7.1)). For the two-state, two-good Tullock case of Example 3.3 extended with a donor state C , compute X^{Δ^*} explicitly in (V_0, θ, W) coordinates. Use the closed-form Δ^* from Exercise 1 of Chapter 5 and the manifold structure of $\tilde{E}$ from Proposition 3.15. Identify the magnitude and direction of the gradient at the symmetric equilibrium. The exercise is the worked instance the chapter's example Example 7.5 promises.
11. *Aligning institution as a vector field.* An institution can be modeled as a smooth vector field X^{inst} on $\tilde{E}^{\text{reg}}$, with the institution's welfare effect determined by the inner product $\langle X^{\text{inst}}, X^{\Delta^*} \rangle$ (Equation (7.1)). Construct X^{inst} that aligns with X^{Δ^*} on a specified subset of $\tilde{E}^{\text{reg}}$ (e.g., the boundary of the war region) and is degraded elsewhere (e.g., zero on the peace component, where no further welfare improvement is possible). Compute the welfare-improvement rate $\langle X^{\text{inst}}, X^{\Delta^*} \rangle$ as a function of position. The substantive content is the design problem of choosing X^{inst} to maximize integrated alignment along the institution's intended path of action.
12. *Multilateralism as gradient-field amplification.* Remark 7.4 interprets multilateralism as alignment of multiple institutional vector fields with X^{Δ^*} , where the magnitude of the combined response scales with the number of aligned actors. Verify the remark by deriving the n -fold magnitude reduction directly from the Nash–Pontryagin conditions of Proposition 6.14 of Chapter 6: with n donors aligned on the welfare gradient, each contributes $\|X^{\text{inst}}\|/n$ at the cooperative speed, and the time-to-peace scales as n times the cooperative case (Corollary 6.16). The exercise connects the chapter's gradient-field formulation to the optimal-control formulation of Chapter 6.
13. *Critical points of X^{Δ^*} .* A regime component $\tilde{E}_k$ on which X^{Δ^*} has multiple critical points has a non-trivial welfare topology: multiple local minima, saddle points connecting them, and basins of attraction of various sizes. Identify a regime component on which X^{Δ^*} has multiple critical points, classify each (global minimum, local minimum, saddle, local maximum), and trace basins of attraction. Open-ended; the substantive content is the

topology of the welfare landscape within a regime, which controls how institutional design's gradient flow approaches its target.

14. *Integral curves of X^{Δ^*} and the singular locus.* For a 2-good Tullock case, identify the parameter region in which integral curves of X^{Δ^*} approach the singular locus Σ in finite time, and the parameter region in which they approach in infinite time. Characterize the boundary behavior: at finite-time approach, the integral curve crosses Σ at a determinate rate (determined by the geometry of Σ near the crossing point); at infinite-time approach, the curve asymptotes to Σ without crossing. Both behaviors are substantively different institutional outcomes: the first is a deliberate regime crossing, the second is trapped welfare improvement within a regime.
15. *Welfare-alignment as inner product.* The inner product $\langle X^{\text{inst}}, X^{\Delta^*} \rangle$ measures the welfare-alignment of an institution: positive inner product means the institution's vector field is welfare-improving; zero means it is welfare-neutral; negative means it works against welfare. Compute the inner product for two stylized institutions: (a) a coordination-only institution that aligns the donors' contributions but does nothing about welfare directly; (b) a welfare-targeting institution that explicitly minimizes Δ^* . Identify the design parameter that maximizes alignment between the two institutions' field on a worked specification.
16. *Singular locus Σ for 2-state Tullock.* For a 2-state Tullock economy with parameter family (V_0, θ, W) , identify the singular locus $\Sigma \subset \Omega_0$ explicitly. Use the closed-form $\det J$ from Exercise 6 of Chapter 4 to identify the parameter values at which $\det J = 0$. Verify that fold transitions are codimension-1 by direct computation: the critical set has codimension 1 in parameter space, consistent with the Sard-Thom analysis of Remark 3.20 of Chapter 3.
17. *One-parameter family crossing a fold.* Theorem 4.5 of Chapter 4 establishes that across a fold catastrophe, two regular equilibria of opposite index merge and disappear, changing the count by -2 and preserving the signed sum at $+1$. Construct a one-parameter family $\omega(t) \in \Omega_0$ that crosses a fold of the joint equilibrium function e . Verify directly that the equilibrium count drops by two as $\omega(t)$ passes through the fold, and that the indices of the two disappearing equilibria are $+1$ and -1 , so the signed sum is preserved. The exercise instantiates fold-catastrophe pairing on a worked one-parameter family.
18. *Cusp catastrophe at Example 3.3 with V_0 varying.* For Example 3.3 with V_0 treated as a varying parameter (and other parameters held fixed), identify the cusp at which two folds meet. Characterize the parameter region of three equilibria: the cusp's interior region in the parameter plane, separated from the single-equilibrium region by two fold curves meeting at the cusp point. Body's Example 7.7 provides the worked instance. The exercise instantiates Thom's cusp catastrophe (the codimension-2 stratum of Σ) on the chapter's running example.
19. *AR2019 narrow corridor as projection of $\tilde{E}^{\text{reg}}$.* Acemoglu and Robinson [2019]'s narrow corridor diagram plots state capacity vs. society's strength, with regimes (despotism, absent state, narrow corridor) as regions in the two-dimensional plane. Read this diagram as a projection of $\tilde{E}^{\text{reg}}$'s regime decomposition onto a two-dimensional parameter slice $(\omega^{\text{state}}, \omega^{\text{society}})$. Identify what information the projection loses: the third (and higher) dimensions of Ω_0 , the within-regime gradient structure of X^{Δ^*} , and the cohomology of paths across the singular locus. Substantively: the diagram is a useful low-dimensional

summary, but the full apparatus of the framework records strictly more information about institutional trajectories.

20. *Constitutional moments and deliberate Σ -crossings.* Discuss when institutional intervention can drive the system across Σ deliberately. The rational-design literature (e.g., Koremenos–Lipson–Snidal 2001) treats institutional design as a deliberate choice subject to political and distributive constraints. “Constitutional moments” (Ackerman 1991) are deliberate cross-regime transitions of this kind. Compare the framework’s apparatus to the rational-design account: Σ -crossings as deliberate institutional design correspond to crossing-the-fold trajectories, while incremental design within a component corresponds to gradient-flow institutions. Open-ended.
21. *Exact 1-form: line integral invariance.* A 1-form ω on $\tilde{E}^{\text{reg}}$ is exact if $\omega = dV$ for some smooth $V : \tilde{E}^{\text{reg}} \rightarrow \mathbb{R}$. For an exact 1-form, verify that the line integral $\int_{\gamma} dV$ along any path γ from e_0 to e_1 equals $V(e_1) - V(e_0)$, and is invariant under reparameterization of γ . The integral depends only on the endpoints, not the path. This is the fundamental theorem of calculus on a smooth manifold, and the exercise establishes that exact 1-forms produce path-independent line integrals—the key property the chapter’s preference cohomology (Equation (7.3)) exploits.
22. *Riemannian length: a non-exact path-dependent functional.* The Riemannian length functional $\omega(v) = \sqrt{g(v, v)}$ is path-dependent on any positive-dimensional manifold, even when the manifold is contractible. Show this directly: compare the Riemannian length of a straight-line path from e_0 to e_1 in $\mathbb{R}^2$ with the length of a detour through a third point. The straight-line path is shorter, so the integrated length depends on the path. Conclude that ω is not exact: the length is not a function of the endpoints alone. The exercise illustrates that path-dependent integrals are the rule, not the exception, on smooth manifolds.
23. *Path-dependent integral on Example 3.3.* For Example 3.3 with parameter family (V_0, θ, W) , parameterize the equilibrium manifold $\tilde{E}(r)$ explicitly using the closed forms from Chapter 3. Construct a non-exact 1-form ω on $\tilde{E}(r)$, and compute $\int_{\gamma} \omega$ for two distinct paths γ_1, γ_2 between the same endpoints $e_0, e_1 \in \tilde{E}(r)$. Verify the integrals differ. The exercise gives a worked instance of path-dependence on the chapter’s running example, illustrating the cohomological content of Equation (7.3).
24. *Gradient flow as $\omega = dV$.* Show that the gradient flow of welfare V on $\tilde{E}^{\text{reg}}$ corresponds exactly to the closed-form preference $\omega = dV$ in the (target, path) formulation of Section 7.5. Specifically: when the preference 1-form is exact, $\omega = dV$, the path-cost functional $\int_{\gamma} \omega = V(e_1) - V(e_0)$ depends only on the endpoints, and the optimal path is determined by its target alone (the path-as-induced-by-target collapse). The exercise certifies that the gradient-field reading of Section 7.3 is the special case of the (target, path) formulation under exact preferences, and that the more general formulation accommodates path-dependent preferences without losing the gradient-flow case.
25. *1-form counting intersections with Σ .* For a stratified manifold $\tilde{E} = \tilde{E}^{\text{reg}} \cup \Sigma$ with codim-1 stratum Σ , construct a 1-form ω on $\tilde{E}^{\text{reg}}$ whose line integral along γ counts intersections of γ with Σ (a catastrophe-counting cost). Verify that ω is generically not closed: the integral depends on the path, not just the endpoints, because different paths can cross Σ different

numbers of times. The exercise constructs a substantive non-exact preference 1-form that captures the cost of regime change as part of institutional design.

26. *Catastrophe-counting bound.* The Loi–Matta (2009) and Balasko (2012) catastrophe-counting bound establishes that the minimal number of Σ -crossings between two points x, y in a regular-economy setting is at most 2. Verify this bound in a worked 2-good example by explicit construction of a minimal path between two points in different regime components: a path that crosses Σ once goes between adjacent components, and a path that crosses twice goes between non-adjacent components by transiting through an intermediate one. The bound depends on Σ 's codimension and the topology of the regime decomposition.
27. *Cohomology $H^1(\tilde{E}^{\text{reg}}; \mathbb{R})$ on a worked example.* Compute $H^1(\tilde{E}^{\text{reg}}; \mathbb{R})$ for a worked example with $\tilde{E} \cong \mathbb{R}^n$ and Σ a finite union of disjoint hyperplanes. The cohomology has one generator per locally-separating component of Σ , by the Mayer-Vietoris sequence applied to the decomposition. Verify this by direct computation in $n = 2$ with two disjoint hyperplanes (lines): $\tilde{E}^{\text{reg}}$ has four connected components, and H^1 has rank 2. The exercise grounds the chapter's cohomological framework in a concrete computation.
28. *Path-aggregation obstruction proof sketch.* Proposition 7.9 generalizes Chichilnisky's obstruction from preferences-on-a-manifold to preferences-on-cohomology-classes. Sketch the proof: extend the continuity-anonymity-unanimity argument from the original Chichilnisky setup (preferences in $P(M)$) to the path-cohomology setting (preferences in $H^1(\tilde{E}^{\text{reg}}; \mathbb{R})$). The key step is that the cohomology classes form a vector space (not necessarily contractible as a quotient of paths modulo exactness), and aggregation rules must be continuous on this quotient. When the cohomology has nontrivial dimension, the obstruction is realized as a profile of path-classes that no continuous, anonymous, Pareto-respecting rule can resolve.
29. *Path-cohomology disagreement.* Construct an explicit pair of voter preferences in the same regime component $\tilde{E}_k$ of $\tilde{E}^{\text{reg}}$ whose path-cohomology classes $[\omega_1] \neq [\omega_2]$ in $H^1(\tilde{E}_k; \mathbb{R})$, and verify that no continuous, anonymous, Pareto-respecting aggregation rule resolves the disagreement. The construction uses Exercise 27's cohomology computation: pick two non-cohomologous 1-forms whose integrals along the same path differ, representing two voters whose preferences disagree on path-dependent welfare structure even though they agree on endpoint values. The exercise instantiates the path-aggregation obstruction.
30. *Garratt–Goenka conditions as 1-form properties.* Garratt and Goenka [1995]'s analysis of redistribution policies imposes five conditions: linearity, finite stages, continuity, no Leontief preferences, and stability. Translate these into properties of the corresponding 1-form / preference structure in the (target, path) formulation. Linearity becomes a linear-on- $T\tilde{E}$ condition on the preference 1-form; finite stages becomes a piecewise-smooth condition on paths; continuity becomes a continuity-of-1-form condition; no Leontief becomes a generic-strict-quasiconcavity condition (cf. Assumption 2.1 of Chapter 2); stability becomes a fixed-point property of the gradient flow. The exercise locates a classical political-economy specification in the chapter's vocabulary.
31. *Loi–Matta arc-lifting as JR-clean specialization.* Loi and Matta [2010]'s arc-lifting theorem shows that under a regular policy, equilibrium prices lift uniquely along the policy path. Sketch this argument as the JR-clean special case of the (target, path) institution formulation. Under Assumption 3.6, $\tilde{\pi}$ is a global diffeomorphism, so any policy path in Ω_0

has a unique lift to $\tilde{E}$ via $\tilde{\pi}^{-1}$; arc-lifting reduces to applying the inverse function. Under Assumption 3.17, the lifting is only componentwise; the same argument works on each regime component, but cross-component lifts require additional selection. The exercise certifies that Loi-Matta's regular case is a special case of the chapter's machinery.

32. *GATT/WTO transitions through the JR face.* Discuss the GATT/WTO evolution (Barton et al. 2006) through the lens of the JR face of the framework. Identify whether the regime's transitions (Uruguay Round 1986-94, Doha Round 2001-?, dispute settlement reform 1995) read as within-chamber adaptation or candidate Σ -crossings. The within-chamber reading: most GATT/WTO transitions are incremental, smooth path adjustments within the trade regime (deepening rather than transforming). The Σ -crossing reading: some transitions (Uruguay's services-and-IP extension, the 1995 WTO institutional restructuring) may be structural enough to count as regime change. Defend a reading. Open-ended.
33. *Cheap §6.4 as (target, path) specialization.* Sketch how Chapter 6's manifold-Pontryagin formulation specializes the (target, path) institution structure to the donor-aid case. Identify the path-cost 1-form as α from Chapter 6 (the donor's preference 1-form), and the welfare-target as the peace component $\tilde{E}_0$. The donor's optimization $\min \int \alpha$ over admissible paths is exactly the (target, path) formulation restricted to the donor's specific preference structure. The exercise certifies that Chapter 6's machinery is a worked specialization of the chapter's general formulation.
34. *Sanctions as (target, path).* Anticipating Chapter 8, pose a sanctions-as-(target, path) example. The target is the compliance equilibrium (a specific point in $\tilde{E}^{\text{reg}}$ where the sanctioned state's behavior aligns with the sender's preference); the path is the sanctions schedule (ramped vs. snap, multilateral vs. unilateral); the cohomological obstructions are the multilateral-coordination difficulties (different senders' preferences may correspond to non-cohomologous 1-forms, and no continuous-anonymous-Pareto rule can aggregate them without loss). Sketch the formal correspondence and identify which features of the sanctions case the (target, path) formulation captures versus the welfare-gradient formulation of Section 7.3. Open-ended; the formalization itself is the substance, and Chapter 8 develops it in full.
35. *Limit-of-dictatorship pathology.* Lauwers [2009] establishes a pathology of Chichilnisky aggregation: in the infinite-agent limit, the aggregation rule can converge to one that assigns disproportionate weight to a single agent, even though the finite-agent rules are anonymous. State and verify the pathology by constructing a sequence of finite-agent rules whose limit assigns disproportionate weight. Identify why the pathology does not affect Corollary 7.2 or Corollary 7.3 for any fixed finite I : the chapter's results work at finite I and do not require an infinite-agent limit. The exercise locates the pathology and exempts the framework's finite-state setting from it.
36. *Alternative parameterization of $P(\tilde{E})$.* The chapter parameterizes the preference space $P(\tilde{E})$ via specific structural conventions; alternative parameterizations exist. Construct an alternative parameterization (e.g., via smooth utility functions modulo monotone transformations) and verify whether contractibility transfers from the chapter's parameterization to the alternative. The transfer holds when the two parameterizations are homotopy-equivalent; it fails when the alternative captures different topological structure. Identify

when each happens. The exercise illustrates that the chapter's contractibility result depends on the parameterization chosen, and motivates the chapter's specific choice.

37. *Mearsheimer's "institutions reflect power."* [Mearsheimer \[1994b\]](#) argues that institutions reflect underlying power distributions rather than constraining them independently. Operationalize this claim as a constraint on X^{inst} 's dependence on ω : the institutional vector field $X^{\text{inst}}(\omega)$ depends on ω in exactly the way the dominant state's preferences depend on ω (a power-reflection condition). Identify the empirical test that distinguishes the realist position from the institutionalist one in framework terms: does $\langle X^{\text{inst}}, X^{\Delta^*} \rangle$ exceed $\langle X^{\text{dom-state}}, X^{\Delta^*} \rangle$, with the gap reflecting institutional autonomy? Open-ended; the substantive content is locating the realism-institutionalism debate in the chapter's vocabulary.
38. *Power-reflecting institution and cooperative gradient.* Compute $\langle X^{\text{inst}}, X^{\Delta^*} \rangle$ for a stylized institution that perfectly reflects the dominant state's preferences ($X^{\text{inst}} = X^{\text{dom-state}}$). Identify when this rule's welfare-alignment matches the cooperative gradient X^{coop} (the field aligned with welfare for all states): this happens when the dominant state's preferences are identical to the welfare gradient, which is the special case where dominance and welfare alignment coincide. For generic primitives, the two diverge, and the gap $\langle X^{\text{coop}} - X^{\text{inst}}, X^{\Delta^*} \rangle$ measures the welfare cost of power-reflecting institutions relative to cooperative ones. Open-ended.

Chapter 8

Sanctions

What does sanctioning state j achieve, and how would we know? The question has been central to the practice of foreign policy since the League of Nations [Mulder, 2022] and central to the academic study of economic statecraft for at least four decades [Hufbauer et al., 2007a, Pape, 1997, Drezner, 1999]. Yet the formal-theoretical literature has answered the question on partialled recipients. The bargaining-and-signaling tradition (Eaton and Engers 1992, Drezner 1999, Lacy and Niou 2004, Krustev 2010, Bapat and Morgan 2009) treats sanctions as moves in an information or commitment game: prices and military deployments are taken as fixed background, and what varies is the sanction’s effectiveness as a coercive instrument. The economic-and-general-equilibrium tradition (Becko 2024, Sturm 2023, Becko and O’Connor 2025, Thoenig 2024, Mohr and Trebesch 2025) treats sanctions as price-and-quantity shocks in a smooth-GE economy: arming is exogenous or absent, and what varies is the welfare cost of trade disruption. The two traditions disagree on what sanctions do; both compute their answer on a setting in which prices and deployments have already been disentangled. This chapter takes a third picture.

A sanction in the framework’s setting is a parameter perturbation θ on the joint equilibrium manifold $\tilde{E}$ of Chapter 3; the system’s response is the joint comparative-static $D_\theta(p^*, m^*) = -J^{-1}D_\theta e$ of Chapter 4. The bargaining-camp’s analysis is the slice in which m is held fixed (and, since prices do not appear, p is fixed too); the GE-camp’s analysis is the slice in which m is held fixed but p responds to the perturbation. The joint formulation lifts both restrictions: prices and deployments respond simultaneously to the sanction, with the joint Jacobian’s cross-channel coupling delivering the response neither slice can recover. This chapter develops the joint apparatus, recovers the two camps as restricted slices, identifies what the joint formulation delivers that the slices do not, and exhibits the substantive payoff on a worked example from Chapter 3.

What the chapter delivers is hard to find elsewhere. First, a slice-recovery argument (Proposition 8.1): each tradition’s analytical framework is the joint comparative-static under a specific restriction. The bargaining-camp slice fixes both prices and deployments across the perturbation; the GE-camp slice fixes deployments and lets prices respond from the price-clearing block. The joint formulation lifts both restrictions. Second, a cross-effect proposition (Proposition 8.3): the joint response decomposes as the slice response plus a Schur-complement correction induced by the joint Jacobian’s off-diagonal blocks. The substantive surprise is the magnitude of the correction. On the canonical worked example (Example 3.3 of Chapter 3, with the prize-share parameter as the sanction), each slice gives exactly zero on its respective response variable while the joint formulation gives non-zero on both—the cross-effect is the *entire* joint response, not a

small correction to the slice analyses. Third, a multilateral-coordination result (Proposition 8.8): a symmetric coalition of n senders implements the public-good component of the sanction's effect at $1/n$ the cooperative magnitude in the non-cooperative Nash equilibrium, delivering the structural-formal account of when multilateral sanctions amplify and when they do not, and why Bapat and Morgan [2009]'s empirical observation that multilateral sanctions are not always more effective than unilateral fits the joint-equilibrium structure. Fourth, a sender-welfare structural account (Proposition 8.10): the welfare cost of imposing a sanction on the sender is itself a joint comparative-static of the welfare scalar Δ^* of Chapter 5; Becko [2024]'s optimal-sanctions formula $t_k = (1 - \lambda_F) t_k^{\text{ToT}}$ is the GE-camp's m -fixed slice optimum, and the joint formulation adds the deployment-feedback and reflexive-cost components the sanctioner-self-harm literature has been articulating without an apparatus to ground. Fifth, a candidate Σ -crossing apparatus: which sanctions push the system across the singular locus identified in Chapter 4, with dedollarization, parallel financial rails, and weaponized financial-network coercion as candidate empirical instances of the bifurcation. The financial-channel material is genuinely outside the chapter's formal apparatus (Section 8.1 bounds the chapter to trade-and-embargo sanctions; Farrell and Newman 2019 is the work that motivates the financial-channel extension this chapter does not undertake), but the framework's bifurcation language gives the structural form the financial-channel analysis would take.

What the chapter does not deliver is empirical resolution of the sanctions-effectiveness debate. The chapter cites the Hufbauer et al. [2007a] compendium and the Pape [1997, 1998]–Elliott [1998]–Baldwin exchange but does not adjudicate among them. The framework's contribution is structural-counterfactual: it gives the apparatus under which the question *do sanctions work?* becomes a question about the joint comparative-static, not about coding-rule choices. The empirical adjudication belongs to a structural-empirical follow-up that calibrates the framework to the HSE / Felbermayr global-sanctions-database episodes; this chapter is structural rather than calibrated.

The chapter is organized around the joint apparatus, the slice-recovery argument, the cross-effect proposition, and the substantive applications. Section 8.1 sets up the sanctions problem on the equilibrium manifold, anchors the empirical material in post-1914 sanctions episodes, and diagnoses the bargaining-camp / GE-camp two-tradition split. Section 8.2 parametrizes the sanction menu—tariff sanctions, embargoes, dual-use export controls, secondary sanctions—as perturbations of (ω, κ) on $\tilde{E}$, with the joint comparative-statics formula as the unifying apparatus. Section 8.3 states and proves the slice-recovery and cross-effect propositions, and works the substantive example from Chapter 3. Section 8.4 extends to multilateral and third-party-realignment material, with Sturm [2023] as the GE-camp anchor and Martin [1992b] as the bargaining-camp historical case. Section 8.5 develops the sender-welfare structural account, with Becko [2024]'s closed-form optimal-sanctions formula as the GE-camp anchor and the joint formulation's sender-side $\partial\Delta_S^*/\partial\theta$ as the framework's contribution. Section 8.6 takes up the candidate Σ -crossing material, with the dedollarization / weaponized-interdependence story as the principal empirical candidate [Farrell and Newman, 2019, Drezner et al., 2021]. Section 8.7 hands the chapter's apparatus to the conclusion and to the structural-empirical follow-up.

8.1 The sanctions problem on the equilibrium manifold

The empirical record on sanctions over the last century is substantial. Hufbauer et al. [2007a] document 174 sanctions episodes between 1914 and 2000, with declared goals ranging from

limited policy adjustments to regime change and military reversal. [Mulder \[2022\]](#) traces the rise of economic sanctions as a recognized instrument of statecraft from the Allied blockades of the First World War, through the League of Nations' interwar experiments with collective security, to the post-1945 institutionalization in the United Nations Charter. [Felbermayr et al. \[2020\]](#) compile the modern Global Sanctions Database, with 1,101 sanctions episodes from 1950 onward. The chapter's substantive empirical anchoring runs through five recurrent cases that the contemporary literature treats as benchmark: the 1990–2003 UN sanctions on Iraq, the 1979–present US sanctions on Iran, the 2014 and 2022 multilateral sanctions on Russia, the post-2017 sanctions on North Korea, and the 2018–2024 US–China tariff and export-control trajectory. Each is a parameter perturbation of considerable scope on a specific dyad or regional bloc; each has elicited a substantial academic and policy literature on whether the sanction *worked*; each remains substantively contested.

What sanctions *do* is precisely the question the formal literature has been trying to answer. Two traditions have organized the answer.

The bargaining-and-signaling tradition. The bargaining-camp formulation, traceable through [Schelling \[1960\]](#)'s analysis of threats and promises and crystallized in [Eaton and Engers \[1992\]](#)'s repeated-game model, treats sanctions as moves in an extended-form game. A sender threatens a target with economic disruption; the target chooses whether to back down or stand firm; the sender then chooses whether to follow through. The strategic environment—prices, deployments, the structure of the contest between sender and target—is taken as exogenous data, and what varies across sanctions episodes is the players' opportunity costs, patience, and conflict expectations. [Drezner \[1999\]](#) extends the framework to a parameter ρ governing the weight on relative gains in each side's utility, with the substantive prediction that adversaries (high ρ) demand frequently but extract small concessions, while allies (low ρ) demand rarely but extract large concessions when they do. [Lacy and Niou \[2004\]](#) adds issue linkage; [Krustev \[2010\]](#) adds bargaining-with-incomplete-information; [Bapat and Morgan \[2009\]](#) adds the multilateral-sanctions extension. Across the tradition, the common analytical commitment is that sanctions outcomes depend on the structure of the bargaining game, with the strategic environment as exogenous parameters.

The economic-and-general-equilibrium tradition. The GE-camp formulation, traceable through the optimal-tariff literature [[Johnson, 1953](#), [Bagwell and Staiger, 2002](#)] and culminating in [Becko \[2024\]](#)'s explicit theory of *economic sanctions as terms-of-trade manipulation*, treats sanctions as trade-policy perturbations within a neoclassical general-equilibrium framework. The sanctioning country imposes a trade tax that distorts prices; demand and supply respond; equilibrium prices and quantities adjust; welfare in both countries shifts. [Becko and O'Connor \[2025\]](#) extends the framework to a two-period model in which the sanctioning country chooses peacetime industrial and trade policies anticipating future sanctions threats. [Sturm \[2023\]](#) extends to three countries with bystander effects. [Thoenig \[2024\]](#) surveys the related literature on the quantitative-trade-and-conflict toolkit. Across the tradition, the common analytical commitment is that sanctions outcomes depend on the price-and-welfare response of a smooth-GE economy, with arming and the strategic environment as exogenous parameters or absent.

The two-camp diagnosis. The two traditions disagree on substantive questions (does sanctions effectiveness depend on bargaining-game structure or on trade-channel elasticities?), and the

disagreement has been productive. But the formal-analytical disagreement obscures a common restriction: both traditions compute their answer on a setting in which prices and deployments are not jointly determined. The bargaining-camp takes prices and deployments as exogenous, varying only the bargaining-game structure. The GE-camp lets prices respond but keeps deployments fixed. Neither lets the deployment respond to a sanction-induced shift in prices, and neither lets the price respond to a sanction-induced shift in deployment. Both restrictions are substantive: they disable cross-channel feedbacks that, on the framework's apparatus of Chapter 3, are part of the equilibrium response.

The empirical record suggests the cross-channel feedbacks are not negligible. The 2014–2024 trajectory of Russian deployments and global energy prices is exactly the kind of price-and-deployment joint adjustment that neither slice can recover. The 1990–2003 Iraqi case combined deployment changes (military infrastructure under sanctions) with profound price effects (the oil-for-food program, the regional energy market). The post-2018 US–China tariff trajectory has reshaped deployment-relevant supply chains (semiconductor capacity, rare-earth processing) in ways that fold back into the price structure of military-relevant goods. In each case, the empirical phenomenon is the joint adjustment of prices and deployments to a sanction-induced perturbation; in each case, the formal analysis on either slice misses some substantial fraction of what's happening.

What the framework's apparatus says. A sanction is a parameter perturbation θ on the joint equilibrium manifold $\tilde{E} = e^{-1}(0)$ of Chapter 3. The system's response is the joint comparative-static $D_\theta(p^*, m^*) = -J^{-1}D_\theta e$ of Chapter 4, where J is the joint Jacobian of e at the equilibrium (p^*, m^*) , evaluated under the regularity conditions of Assumption 3.6 or Assumption 3.17. Section 8.2 states the formula in display form and identifies its entry points across the sanction menu. The bargaining-camp slice corresponds to fixing $D_\theta p^* = D_\theta m^* = 0$ and asking how the bargaining outcome changes as a function of exogenous environmental parameters. The GE-camp slice corresponds to fixing $D_\theta m^* = 0$ and computing $D_\theta p^*$ from the m -fixed Jacobian restriction. The joint formulation lifts both fixings.

The apparatus delivers three substantive results that neither slice can compute. First, the bargaining-camp's exogenous environmental parameters are themselves equilibrium quantities, varying with the joint position on $\tilde{E}$. Drezner's ρ , Eaton-Engers's P_T , and Krustev's bargaining-power parameters all become endogenous outputs of the joint comparative-static at specific parameter regions of $\tilde{E}$. Second, the GE-camp's price response misses the deployment feedback: under joint formulation, a sanction-induced price change shifts the cost of military deployment, which shifts equilibrium deployments, which feeds back into prices. At the symmetric equilibrium of the canonical Example 3.3 of Chapter 3, the m -feedback is the entire price response (Section 8.3 proves the proposition). Third, the joint formulation extends straightforwardly to multilateral and third-party-realignment material that requires both slices' channels operating simultaneously (Section 8.4).

The bound the chapter accepts is on the financial channel. Financial sanctions, secondary sanctions, and weaponized interdependence [Farrell and Newman, 2019] operate through the topological structure of global financial networks (the dollar-clearing system, SWIFT, payment-rail centrality), which the framework's manifold representation does not have. The chapter's apparatus covers trade sanctions, embargoes, and dual-use export controls; the financial-channel extension is acknowledged in Section 8.6 as a candidate Σ -crossing apparatus and in Section 8.7 as material reserved for follow-up work. The boundary is substantive: it reflects a genuine limit

of the manifold-paths formulation rather than a pedagogical simplification.

What the chapter is structurally for. The chapter is the framework's response to a substantive question the empirical literature has been answering on contradictory grounds. [Hufbauer et al. \[2007a\]](#) report a 34% sanctions success rate; [Pape \[1997\]](#) re-codes the same episodes and finds 5 of 115 true successes; [Elliott \[1998\]](#) defends the original coding on substantive-policy grounds; [Baldwin \[1998\]](#) argues that the methodological disagreement reflects a deeper definitional split. The empirical disagreement is unresolved precisely because the counterfactual structure under which the question *do sanctions work?* becomes well-defined has not been agreed. The framework's contribution is to provide the structural counterfactual: a sanction's effect is the equilibrium response of the joint formulation to the sanction-induced perturbation, computable in principle from the joint comparative-static Equation (8.1) once the framework is calibrated to the relevant economy. The chapter does not perform that calibration; it gives the apparatus under which the empirical question is formal-precise.

The bargaining-camp and GE-camp slice formulations are the existing literature's best attempts at the structural question. The chapter's argument is that they are partial, not wrong. Each captures a subset of the joint formulation's response; the joint formulation captures their union and the cross-channel coupling neither slice has. Section 8.2 parametrizes the sanction menu on (ω, κ) space. Section 8.3 states the slice-recovery and cross-effect propositions and works the canonical example. The substantive payoff is the structural-counterfactual apparatus the rest of the chapter develops.

8.2 Modeling sanctions as parameter shifts

A sanction in the framework's apparatus is a parameter perturbation on the joint equilibrium manifold $\tilde{E}$. The system's response is the joint comparative-statics formula of Chapter 4 Section 4.4, which delivers the infinitesimal change in (p^*, m^*) from any perturbation of the primitive data. What changes across the sanction menu is the parameter being perturbed and the channel through which the perturbation enters the joint excess-demand map e . This section parametrizes the principal sanction types, identifies the entry points each takes into the framework's primitive data, and states the master comparative-statics formula in the form the rest of the chapter applies.

Tariff sanctions. A bilateral tariff sanction imposes a wedge between buyer and seller prices for a specific dyad and goods subset. In the framework's notation, a tariff $\tau_{ij,k} \in [0, 1)$ on country j 's good- k exports to country i acts as a distortion on the bilateral trade flow: country i 's buyers pay $(1 + \tau_{ij,k})p^k$ for the good while country j 's sellers receive p^k , with the tariff revenue accruing to country i 's government as a lump-sum redistribution to its representative agent. The framework's primitive setting in Chapter 2 treats prices as common across all market participants; the tariff sanction breaks this by introducing a country-pair-specific price wedge. The natural representation is to redefine country i 's effective endowment to incorporate the tariff revenue and country j 's effective endowment to net out the tariff incidence; the resulting effective endowment vector $\omega^\tau \in \Omega_0$ is the parameter the joint comparative-statics formula consumes. A pure tariff sanction is therefore an ω -perturbation in the framework, with the tariff parameters $\{\tau_{ij,k}\}_{(i,j,k) \in K}$ entering through the effective endowment map.

Embargoes. A trade embargo prohibits or sharply restricts a specific bilateral trade flow. The limit case of a tariff at $\tau_{ij,k} \rightarrow 1$ is one embargo representation: as the tariff approaches unity, the effective trade volume collapses and the bilateral trade in good k vanishes between i and j . A second representation operates directly on the cost-of-force function: an embargo on military-relevant goods K^{mil} raises the marginal cost of country j 's deployment, modelable as a κ_j -shift in the cost parameter Chapter 2 takes as primitive. A target country whose military deployments depend on imported inputs from sanctioning countries faces a higher κ_j post-embargo, and the joint comparative-static delivers the equilibrium response of (p^*, m^*) to the κ_j -shift. The 1990–2003 UN sanctions on Iraq, the post-2014 Western sanctions on Russia in dual-use sectors, and the post-2017 sanctions on North Korea each fit this representation: trade restrictions on goods that feed into deployment capacity raise the target's κ_j .

Dual-use export controls. A dual-use export control is an embargo restricted to a specific technology subset—semiconductors, advanced manufacturing equipment, rare-earth processing technology, civilian-aviation components with military applications. The 2018–2024 US export controls on advanced semiconductors and manufacturing equipment to China are the canonical recent case. In framework terms, dual-use export controls are κ_j -shifts on a goods subset $K^{\text{dual}} \subset K^{\text{mil}}$, with K^{dual} chosen to maximize the marginal-cost effect on the target's deployment while minimizing the marginal-cost effect on the sanctioner's own production. The substantive content of the dual-use case beyond a generic embargo is the goods-subset selection problem: which goods, when restricted, raise κ_j most steeply for fixed sanctioner welfare cost? Becko [2024]'s optimal-sanctions formula $t_k = (1 - \lambda_F) t_k^{\text{ToT}}$ delivers the GE-camp's answer to the goods-selection question; Section 8.5 extends to the joint formulation.

Secondary sanctions. A secondary sanction extends a primary sanction to apply to third parties trading with the sanctioned country. The 2018 US sanctions on Iran applied not only to US firms but to foreign firms with no US presence, on the grounds that non-compliant firms would be excluded from the dollar-clearing network and the US financial system more broadly [Farrell and Newman, 2019]. The mechanical content of a secondary sanction is to extend the sanction's ω - or κ -perturbation to a broader set of country pairs, with the bloc structure of the international system determining which third parties are bound by the secondary sanction and which are not. The natural representation in the framework's notation is a *bloc-multiplier shift*: the sanctioning bloc $B \subseteq \{1, \dots, I\}$ applies the sanction's perturbation to all bilateral pairs (i, j) with $i \in B$ and j the sanctioned country, with the secondary-sanction extension widening B to include the financial-network-bound third parties. The bloc structure interacts with the joint comparative-statics in the multilateral-coordination material of Section 8.4.

Other sanction types and the limits of the menu. The four sanction types above cover the principal trade-and-embargo material the chapter takes up. Three sanction types are *not* covered by the menu and the chapter does not formally model them. Asset freezes and financial-system access restrictions (SWIFT exclusion, dollar-clearing access denial) operate through the topological structure of global financial networks [Farrell and Newman, 2019], which the framework's manifold representation does not have. Targeted personal sanctions on specific individuals (asset freezes, travel bans, visa restrictions) operate through domestic-political channels the framework abstracts from. Arms embargoes proper—restrictions on transfers of finished weapons systems—operate through deployment-capacity channels the framework folds

into the κ_j -shift representation above; an arms embargo is a κ_j -shift on $K^{\text{arms}} \subset K^{\text{mil}}$, structurally identical to the dual-use case. The chapter's apparatus thus covers a substantively important but bounded subset of the sanctions menu; the financial-channel extensions are reserved for follow-up work (Section 8.6 treats financial-channel material as candidate Σ -crossings, but the formal apparatus covers the trade-and-embargo material only).

The master comparative-statics formula. With the sanction menu parametrized, the joint comparative-statics formula of Chapter 4 Section 4.4 delivers the equilibrium response uniformly. For any perturbation θ of the primitive data—tariff τ , cost-of-force shift κ , bloc-multiplier change, or any combination—the joint response of (p^*, m^*) satisfies

$$\begin{pmatrix} D_\theta p^* \\ D_\theta m^* \end{pmatrix} = -J^{-1} \cdot D_\theta e, \quad (8.1)$$

where J is the joint Jacobian of the excess-demand map e at the equilibrium (p^*, m^*) , and $D_\theta e$ is the direct derivative of e with respect to the sanction parameter. The formula holds at any regular equilibrium under Assumption 3.6 or Assumption 3.17: the joint Jacobian J is invertible at regular equilibria (Chapter 4 Theorem 4.3), and the implicit function theorem delivers the smooth dependence of the equilibrium on the sanction parameter on $\tilde{E} \setminus \Sigma$. At equilibria on the singular locus Σ , J is singular, the master formula breaks down, and the response of the system to the sanction is governed by the bifurcation apparatus of Section 8.6.

The substantive content of Equation (8.1) is that *every sanction in the menu produces a joint response in both prices and deployments*. The bargaining-camp slice approximation fixes both $D_\theta p^* = 0$ and $D_\theta m^* = 0$ and analyzes the bargaining game on the resulting environmental parameters. The GE-camp slice fixes $D_\theta m^* = 0$ and computes $D_\theta p^*$ from the m -fixed restriction of J . The joint formulation lifts both fixings: prices and deployments respond simultaneously, with the cross-channel structure of J determining the relative magnitudes. Section 8.3 states the slice-recovery and cross-effect propositions and works the canonical example; Sections 8.4 to 8.6 apply the master formula to multilateral coordination, sender welfare, and candidate Σ -crossings respectively.

Regularity, regimes, and what the formula buys. The master formula's deliverables depend on which regularity condition is in force. Under Assumption 3.6, $\tilde{E}$ is connected and contractible (Corollary 3.16 of Chapter 3), so the joint equilibrium is unique above each parameter profile and the master formula tracks a single smooth equilibrium correspondence across the entire parameter space. A sanction perturbs the unique equilibrium continuously, with $D_\theta(p^*, m^*)$ defined globally. Under Assumption 3.17, $\tilde{E}^{\text{reg}}$ partitions into regime components (Definition 3.21 of Chapter 3), and the master formula tracks a single smooth equilibrium correspondence within each component. A sanction that keeps the system inside its initial regime component admits the master-formula response globally; a sanction that drives the system across Σ requires the bifurcation apparatus of Section 8.6. The substantive question *which sanctions cross Σ and which do not* is itself a regime-classification question the framework's apparatus makes precise; the empirical identification of Σ -crossings in specific sanctions episodes is reserved for the structural-empirical follow-up.

8.3 Joint vs. sliced response

The chapter’s substantive headline is that the bargaining-camp and the GE-camp analyses of sanctions are not wrong but *partial*. Each tradition’s framework recovers as a special case of the joint comparative-static of Equation (8.1), with the specialness amounting to a fixing of one or both response variables. The joint formulation lifts the fixings, and the lifted response captures structure neither slice has. This section makes the claim precise. Proposition 8.1 states the slice-recovery identification: each tradition’s analysis is the joint response under a specific restriction of the master formula. Proposition 8.3 states the cross-effect result: the joint and slice responses differ by the Schur-complement correction induced by the joint Jacobian’s off-diagonal blocks. Example 8.5 exhibits the gap explicitly on the canonical Example 3.3 of Chapter 3, where the slice responses are exactly zero on their respective response variables and the joint response is the entire response.

The block-Jacobian setup. Partition the equilibrium variables into a price block $p \in S_N$ and a deployment block $m \in M$, and partition the joint excess-demand map $e : S_N \times M \times \Omega_0 \rightarrow \mathbb{R}^{L-1+I}$ accordingly:

$$e(p, m, \omega) = \begin{pmatrix} e_p(p, m, \omega) \\ e_m(p, m, \omega) \end{pmatrix},$$

with e_p the price-clearing block (the aggregate-excess-demand component) and e_m the deployment-equilibrium block (the agent-level Nash-FOC component). The joint Jacobian at any equilibrium (p^*, m^*) likewise has a block structure:

$$J = \begin{pmatrix} J_{pp} & J_{pm} \\ J_{mp} & J_{mm} \end{pmatrix}, \quad (8.2)$$

where J_{pp} is the price-on-price block, J_{mm} the deployment-on-deployment block, and J_{pm}, J_{mp} the cross-channel blocks coupling price and deployment responses. The cross-channel blocks are the substantive content of the joint formulation: when J_{pm} and J_{mp} both vanish, the two halves of the equilibrium decouple and the slice analyses recover the joint response exactly. When either is non-zero, the joint response carries cross-channel content the slices do not.

The two slice computations, formally. The bargaining-camp tradition (Eaton and Engers 1992, Drezner 1999, Lacy and Niu 2004, Krustev 2010, Bapat and Morgan 2009) holds the strategic environment fixed and analyzes the bargaining game on the resulting parameters. In the framework’s notation, this corresponds to fixing $D_\theta p^*|_B = 0$ and $D_\theta m^*|_B = 0$: the bargaining game produces outcomes (concession size, sanction-imposition probability, settlement terms) that depend on environmental parameters such as Drezner’s ρ or Eaton-Engers’s P_T, P_S , but the parameters themselves are not adjusted as a function of θ . The bargaining-camp slice response is therefore $(D_\theta p^*|_B, D_\theta m^*|_B) = (0, 0)$.

The GE-camp tradition (Becko 2024, Sturm 2023, Becko and O’Connor 2025, Thoenig 2024) holds deployments fixed and computes the price response from the price-clearing block alone. In the framework’s notation, this corresponds to fixing $D_\theta m^*|_G = 0$ and computing $D_\theta p^*|_G$ from the m -fixed restriction of J :

$$D_\theta p^*|_G = -J_{pp}^{-1} D_\theta e_p|_{m=m_0}. \quad (8.3)$$

The GE-camp computation re-solves the price-clearing equation with m held at its baseline value $m_0 = m^*(\theta_0)$ and computes the resulting price change. This is exactly Becko’s [2024]

optimal-sanctions formula derivation: the trade tax t_k that maximizes the sanction's effect at minimum sanctioner cost is computed under the assumption that the strategic environment (deployments, foreign welfare-weight λ_F) is held fixed.

Proposition 8.1 (Slice recovery). *Let θ be a smooth perturbation of the primitive data and $(p^*(\theta), m^*(\theta))$ the joint equilibrium response under the master formula Equation (8.1). The bargaining-camp slice response and the GE-camp slice response are special cases of the joint formula:*

1. *Bargaining-camp slice: $(D_\theta p^*|_B, D_\theta m^*|_B) = (0, 0)$, the joint response under the restriction that both response variables are fixed.*
2. *GE-camp slice: $D_\theta m^*|_G = 0$ and $D_\theta p^*|_G = -J_{pp}^{-1} D_\theta e_p|_{m=m_0}$, the joint response under the restriction that the deployment response is fixed at zero.*

In the special case $J_{pm} = J_{mp} = 0$ (no cross-channel coupling), the GE-camp slice response coincides with the joint response on the price component. The bargaining-camp slice never coincides with the joint response except in the trivial case where the perturbation has no effect on the equilibrium at all.

Proof. The master formula Equation (8.1) computes the joint response by inverting the full Jacobian J against the direct derivative $D_\theta e$. The bargaining-camp slice imposes $D_\theta p^*|_B = 0$ and $D_\theta m^*|_B = 0$ by fiat: the bargaining game's analysis takes the strategic environment as exogenous data, and the comparative-static of the environment is not part of the analysis. This is the joint response under the restriction that both response coordinates are zero, equivalent to ignoring the θ -dependence of (p^*, m^*) entirely.

The GE-camp slice imposes $D_\theta m^*|_G = 0$ and re-solves the price-clearing equation with m held at m_0 . The resulting $D_\theta p^*|_G$ satisfies Equation (8.3) by direct calculation: the implicit function theorem applied to the price-clearing block alone delivers $D_\theta p^*|_G = -J_{pp}^{-1} D_\theta e_p|_{m=m_0}$. This is the joint response under the restriction that the deployment-row of $-J^{-1} D_\theta e$ is set to zero.

In the case $J_{pm} = J_{mp} = 0$, the joint Jacobian is block diagonal and J^{-1} has block-diagonal structure too: $(J^{-1})_{pp} = J_{pp}^{-1}$ and $(J^{-1})_{pm} = 0$. The joint price response then equals $-J_{pp}^{-1} D_\theta e_p$, which agrees with the GE-camp slice at the baseline deployment. When either off-diagonal block is non-zero, the block-inverse formula introduces the Schur-complement correction $-(J_{pp})^{-1} J_{pm} J_{mm}^{-1} J_{mp} (J_{pp})^{-1}$, which is exactly the cross-channel content the slice misses. Proposition 8.3 below states the result. $\square$

Remark 8.2 (The bargaining-camp slice is exact only in the trivial case). Proposition 8.1 (1) is honest about the bargaining-camp slice: setting both response variables to zero recovers the joint response only when the perturbation has no equilibrium effect, i.e., when $D_\theta e \equiv 0$ at the relevant equilibrium. A genuine sanction is, by definition, a perturbation that *does* have equilibrium effects, so the bargaining-camp slice is never exact for an actually-imposed sanction. The bargaining-camp's analytical content is what happens *in the bargaining game* given the fixed environmental parameters; the framework's joint formulation supplies the comparative-static of the environment that the bargaining game runs on. The two analyses are complementary rather than competing: the bargaining-camp answers *given the equilibrium environment, what does the bargaining game predict?*; the joint formulation answers *how does the equilibrium environment respond to the sanction?*. The chapter's claim is that the second question must be answered before the first one is well-posed.

The cross-effect proposition. With the slice recovery established, the substantive question is how much of the joint response the slices miss. The block-inverse formula gives an explicit answer.

Proposition 8.3 (Cross-effect). *At any regular equilibrium $(p^*, m^*) \in \tilde{E}^{\text{reg}}$, the joint response decomposes as*

$$D_{\theta}p^* = D_{\theta}p^*|_G + \Delta_p(\theta), \quad D_{\theta}m^* = D_{\theta}m^*|_B + \Delta_m(\theta), \quad (8.4)$$

where $D_{\theta}p^*|_G$ and $D_{\theta}m^*|_B = 0$ are the GE-camp and bargaining-camp slice responses, and (Δ_p, Δ_m) are the cross-channel corrections induced by the off-diagonal blocks of J . The corrections satisfy

$$\Delta_p(\theta) = -(J_{pp})^{-1}J_{pm}D_{\theta}m^*|_J, \quad \Delta_m(\theta) = D_{\theta}m^*|_J, \quad (8.5)$$

where $D_{\theta}m^*|_J$ is the joint deployment response. The corrections vanish if and only if either (i) $J_{pm} = 0$ for Δ_p , or (ii) $D_{\theta}m^*|_J = 0$, the joint deployment is unresponsive to the sanction.

Proof. Apply the block-inverse formula to Equation (8.2). The joint price response is

$$\begin{aligned} D_{\theta}p^* &= -(J_{pp} - J_{pm}J_{mm}^{-1}J_{mp})^{-1}(D_{\theta}e_p - J_{pm}J_{mm}^{-1}D_{\theta}e_m) \\ &= -(J_{pp})^{-1}D_{\theta}e_p + [(J_{pp})^{-1}J_{pm}J_{mm}^{-1}J_{mp}(J_{pp})^{-1}D_{\theta}e_p \\ &\quad - (J_{pp} - J_{pm}J_{mm}^{-1}J_{mp})^{-1}J_{pm}J_{mm}^{-1}D_{\theta}e_m] \\ &= D_{\theta}p^*|_G - (J_{pp})^{-1}J_{pm} \cdot D_{\theta}m^*|_J, \end{aligned}$$

where the last equality uses the joint deployment response $D_{\theta}m^*|_J = -(J_{mm} - J_{mp}J_{pp}^{-1}J_{pm})^{-1}(D_{\theta}e_m - J_{mp}J_{pp}^{-1}D_{\theta}e_p)$ from the block-inverse identity. This delivers Equation (8.5) for Δ_p directly. For Δ_m , the bargaining-camp slice gives $D_{\theta}m^*|_B = 0$ by definition, so $\Delta_m = D_{\theta}m^*|_J - 0 = D_{\theta}m^*|_J$. $\square$

Remark 8.4 (The cross-effect is what the slice misses). The substantive reading of Proposition 8.3 is that the GE-camp's price response is the joint price response *minus the m -feedback correction*. When the deployment m^* responds to the sanction at all ($D_{\theta}m^*|_J \neq 0$), and when the price-clearing block is sensitive to the deployment level ($J_{pm} \neq 0$), the joint price response includes the m -feedback term $-(J_{pp})^{-1}J_{pm} \cdot D_{\theta}m^*|_J$ that the GE-camp's m -fixed computation entirely misses. The bargaining-camp's response misses both directions: it sets deployments and prices to zero comparative-statically, and the entire joint response is the bargaining-camp's slice gap.

The size of the cross-channel correction depends on the substantive structure of the framework's primitives. J_{pm} is the response of price-clearing to deployment changes (does an extra unit of deployment shift any market by burning input goods?). J_{mp} is the response of deployment-equilibrium to price changes (does the marginal deployment cost shift when the price of a deployment input changes?). Both are non-zero whenever the conflict primitive uses goods priced in the market—which is the framework's substantive case of interest. The case $J_{pm} = J_{mp} = 0$ corresponds to a contest in which deployments use goods that do not appear on the commodity market, equivalent to setting $\kappa_i = 0$ on all commodity-priced inputs. This is the limiting partial case the (p, m) framework refuses to take.

The canonical worked example. The propositions are general; the substantive payoff is sharpest on the canonical worked example of Example 3.3 in Chapter 3, where the cross-channel correction Δ_p turns out to be the *entire* joint price response.

Example 8.5 (Joint vs. slice on the two-state Tullock contest). Take the two-state Tullock contest of Example 3.3 of Chapter 3, with the prize-share parameter θ as the sanctions perturbation. A sanction that reduces θ structurally decouples the contest's prize from aggregate good-1 trade volume; this is the substantive content of the perturbation. The closed-form symmetric joint equilibrium is

$$m^* = \frac{V_0 + \theta W}{2k(2 + \theta)}, \quad p^* = \frac{(2 + \theta)W}{2W - V_0}.$$

Differentiating directly,

$$\left. \frac{\partial m^*}{\partial \theta} \right|_J = \frac{2W - V_0}{2k(2 + \theta)^2} > 0, \quad \left. \frac{\partial p^*}{\partial \theta} \right|_J = \frac{W}{2W - V_0} > 0.$$

A reduction in θ thus reduces equilibrium deployment and reduces equilibrium prices; both responses follow from the joint formulation.

The bargaining-camp slice gives zero on both responses by definition. The GE-camp slice computation requires re-solving the price-clearing equation with m held at its baseline value $m_0 = m^*(\theta_0)$. At the symmetric equilibrium $m_1 = m_2 = m_0$, the contest transfers vanish: $s_i^1(m_0, m_0) = (V_0 + \theta T^1)(m_i - m_{-i})/(2(m_1 + m_2)) = 0$ at any value of θ . The aggregate good-1 supply for consumption is $T^1 = W - 2km_0$ (independent of θ once m_0 is fixed), and the price-clearing equation with log-Cobb-Douglas demand reduces to $p^*|_{m_0} = W/(W - 2km_0)$. The crucial observation is that the prize-share θ does not appear in this expression. With m fixed at m_0 , the GE-camp price formula is θ -invariant:

$$\left. \frac{\partial p^*}{\partial \theta} \right|_G = 0.$$

The gap is striking. The joint formulation gives a positive price response of magnitude $W/(2W - V_0)$; the GE-camp slice gives zero. Proposition 8.3 attributes the gap entirely to the cross-channel correction $\Delta_p = -(J_{pp})^{-1} J_{pm} D_\theta m^*|_J$, which captures the m -feedback into the price equation through the deployment-induced shift in the resource constraint. The substantive reading is that the price response to a prize-share sanction *is entirely the m -feedback*: at symmetric equilibrium, the prize-share does not affect prices through the contest-transfer channel (because transfers vanish at symmetry), and the only way θ shifts the price is by shifting the deployment, which shifts the resource burn $2km^*$, which shifts the available good-1 supply, which shifts the price.

The bargaining-camp's gap is symmetrical. The joint deployment response is positive of magnitude $(2W - V_0)/(2k(2 + \theta)^2)$; the bargaining-camp slice gives zero. The bargaining-camp's analysis of the prize-share sanction treats the deployment as exogenous data, missing the entire joint deployment response. The GE-camp slice also misses the joint deployment response (it fixes m by construction).

Remark 8.6 (The slice gap is not a small correction). Example 8.5 delivers a sharp empirical-pedagogical point. The two slices' price responses are zero on the worked example. The joint price response is non-zero. The gap is therefore not a small correction to the slice analyses but the *whole* joint response: 100% of the equilibrium price effect of the prize-share sanction is in the cross-channel correction the slices miss.

This is a feature of the canonical worked example, not a generic fact. On other parameter directions in Example 3.3, the gap is more modest: an external-prize sanction (V_0 -shift) has

a non-zero direct effect on the price equation through the $(2W - V_0)$ denominator, so the GE-camp slice captures part of the joint response and the cross-channel correction is a non-trivial fraction. A cost-of-force sanction (k -shift) has zero joint price response (the resource burn $2km^*$ is independent of k in this specification), so all three responses agree at zero. The structural pattern is that the gap depends on which parameter is perturbed and which equilibrium-channel structure the perturbation engages. The chapter's substantive claim is not that the gap is always the entire response; it is that the joint formulation captures gap-content the slice formulations cannot.

The Drezner adversary-vs-ally pattern, structurally. Drezner [1999]'s headline finding is that adversary dyads generate frequent sanctions with small concessions, while ally dyads generate rare sanctions with large concessions when imposed. The bargaining-camp's framework derives this pattern from heterogeneity in the conflict-expectations parameter ρ across dyads: high- ρ adversaries weight relative gains heavily, low- ρ allies weight them lightly, and the bargaining-game outcomes follow. The (p, m) framework's apparatus suggests a structural origin for the heterogeneity in ρ itself: the parameter correlates with the dyad's equilibrium position on $\tilde{E}$, with high-deployment adversaries having structural responses to sanctions that produce the Drezner pattern and low-deployment allies having structural responses that produce its converse.

The structural claim is not yet a theorem, however, and the chapter does not promise one. A clean derivation of the adversary-vs-ally pattern from the joint comparative-static would require either curvature conditions on the cost-of-force function $\kappa(m_j)$ that are stronger than generic, or asymmetric specifications that break the symmetric-equilibrium structure of the canonical example. Example 8.5's symmetric two-state contest does not deliver the pattern: the deployment response $\partial m^*/\partial \theta = (2W - V_0)/(2k(2 + \theta)^2)$ is independent of V_0 at symmetric equilibrium, so the example gives the same deployment response in nominally-adversary regions (high V_0) and nominally-ally regions (low V_0). The Drezner pattern requires asymmetry the canonical example does not contain.

What the framework's apparatus does deliver is the structural form the empirical pattern would take. If the joint comparative-static $\partial m_j^*/\partial \theta$ is small in magnitude at high- m^* adversary equilibria (because the deployment is near a saturation level set by the cost-of-force curvature) and large at low- m^* ally equilibria (because the deployment is in an elastic region of the cost-of-force function), then the joint-formulation account of Drezner's pattern follows: small $\partial m_j^*/\partial \theta$ for adversaries means sanctions don't shift the target's posture much, so the sender extracts small concessions but can afford to demand frequently; large $\partial m_j^*/\partial \theta$ for allies means sanctions shift the posture substantially, so the sender extracts large concessions but demands rarely because the strategic environment is more fragile. The structural-empirical content is whether the conjectured curvature conditions hold across the relevant dyads. That is a question for calibrated work the chapter does not undertake.

The honest framing of the chapter's contribution to the Drezner-paradox literature is therefore: the framework's apparatus identifies the structural form an explanation of the adversary-vs-ally pattern would take, and it identifies the specific quantity ($\partial m_j^*/\partial \theta$) whose curvature in m^* governs whether the pattern follows from joint-equilibrium structure rather than from exogenous ρ -heterogeneity. This is the structural-counterfactual move Section 8.1 promised: the empirical literature's disagreement on whether sanctions work, and its sub-disagreement on Drezner's adversary-vs-ally pattern, becomes a question about the joint comparative-static that the framework's apparatus makes formal-precise. The empirical adjudication remains for

follow-up work that calibrates the framework to the specific dyads the literature examines.

What this means for the empirical literature. The chapter’s contribution to the empirical literature on sanctions effectiveness is not a new empirical claim. [Hufbauer et al. \[2007a\]](#)’s 34% success rate, [Pape \[1997\]](#)’s re-coded 5 of 115, and [Elliott \[1998\]](#)’s defense of the original coding all rest on what the relevant counterfactual is. The framework’s joint comparative-static is exactly the counterfactual the empirical literature has been reaching for without an apparatus to ground it. Proposition 8.1 identifies the partial counterfactuals the empirical literature has been using (bargaining-camp’s no-environment-response, GE-camp’s no-deployment-response). Proposition 8.3 states the structural reason the partial counterfactuals miss content; the Example 8.5-worked example shows that on the canonical case the missed content is the entire response.

The substantive prediction this licenses is conditional rather than universal: *when the joint comparative-static’s cross-channel terms dominate the slice responses, the empirical literature’s effectiveness counts will be biased in predictable directions.* For sanctions whose primary channel is the deployment-cost shift (embargoes, dual-use export controls), the cross-channel correction adds price content the GE-camp’s m -fixed analysis misses; the empirical literature, scoring effectiveness against the missing counterfactual, will under-attribute effects to sanctions that work through the m -feedback. For sanctions whose primary channel is the bargaining-game structure (Drezner’s adversary-vs-ally cases), the cross-channel correction adds environment-response content the bargaining-camp’s fixed-environment analysis misses; the empirical literature will under-attribute effects to sanctions that work through the joint comparative-statics. The framework’s structural-counterfactual apparatus suggests where the empirical literature’s effectiveness counts most need re-examination; the re-examination itself is calibrated work the chapter does not undertake.

8.4 Multilateral coordination and third-party realignment

The bilateral apparatus of Section 8.3 extends naturally to the multilateral case. Two structural moves are involved. First, a sanction by sender i on target j alters the equilibrium not only for the dyad (i, j) but for every state $k \notin \{i, j\}$ whose endowment, deployment, or trade-partner structure interacts with either party. This is the third-party realignment problem: how do bystander states reposition in response to sanctions imposed elsewhere, and what does the joint-equilibrium apparatus say about the response that the slice formulations cannot? Second, when multiple senders coordinate to impose sanctions jointly, the problem becomes a Nash equilibrium of the multi-sender game played on $\tilde{E}$. This is the multilateral-coordination problem: under what structural conditions does a coalition of senders implement a sanction more effectively than any single sender alone, and how does the chapter’s apparatus extend Bapat and Morgan’s [2009] bargaining-camp analysis and [Sturm’s \[2023\]](#) GE-camp three-country analysis to the joint formulation? This section takes both up.

Third-party realignment as joint comparative-statics. The master comparative-statics formula Equation (8.1) delivers the equilibrium response of *every* state’s price and deployment vector to a parameter perturbation, not just the dyad’s. For a bilateral sanction θ by sender i on target j , the response of any third-party state k is the k -restriction of the joint response:

$$\begin{pmatrix} D_{\theta} p^* \\ D_{\theta} m_k^* \end{pmatrix} = -J^{-1} \cdot D_{\theta} e, \quad k \notin \{i, j\}, \quad (8.6)$$

read off by selecting the price block (common to all states) and the k -th deployment row of the inverse-Jacobian product. The bargaining-camp slice gives zero on both responses by the slice-recovery argument of Proposition 8.1: third parties' equilibrium quantities are not in the bargaining-game model at all. The GE-camp slice gives $D_\theta m_k^*|_G = 0$ by the m -fixed restriction, but allows the price response to propagate through the third-party state's price-clearing block. The joint formulation lifts the m -fixed restriction: third parties' deployments respond to the bilateral sanction through the cross-channel coupling between the price block and each state's deployment block.

Sturm [2023]'s three-country bystander analysis is the canonical GE-camp treatment of the third-party question. In Sturm's setup, Home imposes trade taxes on Foreign while Rest of the World (RoW) trades freely with both; the analysis delivers two effects on optimal sanctions design. First, goods that Home can substitute from RoW—i.e., that RoW supplies as a buyer-side substitute for Foreign—should face *lower* sanction taxes at the optimum, because the elasticity of Home's substitution between Foreign and RoW sources lowers the sanctioner's effective cost of imposing sanctions on those goods. Second, goods that Foreign can sell to RoW as an alternative buyer should face *higher* sanction taxes at the optimum, because RoW's role as a destination dilutes the sanction's coercive impact. Sturm's analysis runs entirely on the GE-camp slice: prices respond, deployments do not. The (p, m) framework adds the third-party deployment response.

Sturm [2023]'s headline trade-off—bystander availability on the buyer side reduces the sanctioner's optimal tax; bystander availability on the seller side raises it—continues to hold under joint formulation. The joint formulation adds a structural prediction the GE-camp slice cannot make: the bystander RoW's own deployment m_{RoW}^* responds to the sanction. A bystander whose role is buyer-side substitution (Sturm's first case) sees its market access expanded by the sanction (Home buys from RoW more, paying more for the substitution); the joint comparative-static includes the deployment response to this welfare gain. A bystander whose role is seller-side outlet (Sturm's second case) sees its supply chain expanded by the sanction (Foreign sells to RoW more, accepting RoW's terms of trade); the joint comparative-static includes the deployment response to this welfare-and-strategic-positioning shift. Whether the deployment response amplifies or dampens the sanctions effect—that is, whether RoW's m_{RoW}^* response correlates with Foreign's m_F^* response—is a structural question the joint formulation makes precise.

The Russia–China counterfactual. The 2014–2024 Russia sanctions trajectory illustrates the substantive content. Western sanctions on Russia in 2014 (Crimea) and 2022 (Ukraine) reshaped Russia's trade structure substantially: Russia redirected exports toward China and India, restructured its supply of military-relevant inputs, and retained significant deployment capacity through the period. The substantive question *how do sanctions on Russia change China's optimal deployment?* is exactly the $D_\theta m_{China}^*$ component of the joint comparative-static when θ parametrizes the sanction package on Russia. Neither slice formulation can compute it: bargaining-camp doesn't have China's deployment as a variable; GE-camp lets China's prices respond but holds China's deployment fixed. The empirical question of whether China's deployment did substantively change in response to the Russia sanctions [Mohr and Trebesch, 2025, Thoenig, 2024] is in principle answerable by the joint comparative-static once the framework is calibrated to the relevant trade-and-deployment data; the chapter does not undertake that calibration. The structural-counterfactual content the chapter delivers is the apparatus under

which the empirical question becomes formal-precise.

Remark 8.7 (Third-party realignment under JR'). Under Assumption 3.17, a third-party state's response to a bilateral sanction depends on which regime component the unsanctioned state's equilibrium sits in. A bystander whose equilibrium is in the same regime component as the sanctioned dyad sees a continuous, smooth response governed by the master formula Equation (8.6). A bystander whose equilibrium sits in a different regime component sees a response governed by the bifurcation apparatus of Section 8.6: if the sanction does not push the bystander's equilibrium across Σ , the master formula applies; if it does, the bystander's response is governed by the structure of the Σ -crossing. The substantive structural content is that bystander states whose equilibria sit *near* the singular locus are candidates for cross-component motion when bilateral sanctions elsewhere shift the broader equilibrium structure—a substantive empirical claim the framework makes precise.

Multilateral coordination as a generalized Nash equilibrium problem. When multiple senders coordinate on a sanction package, the multilateral-sanctions problem becomes a generalized Nash equilibrium problem (GNEP) on $\tilde{E}$: each sender i chooses its sanction parameter θ_i to maximize its own welfare $U_i(\theta_i, \theta_{-i})$ given the others' choices, with the equilibrium of the multi-sender game characterized by the GNEP structure. The setup is structurally identical to Chapter 6's donor-coordination problem with each donor's α_j replaced by sender i 's sanction parameter θ_i , and the welfare functional replaced by sender i 's objective on the joint manifold. The Nash existence theorem of Chapter 6 Theorem 6.13 applies under the standard regularity conditions: each sender's payoff is continuous and quasi-concave in its own sanction parameter, the strategy space is non-empty compact and convex, and the joint best-response correspondence is upper hemicontinuous with non-empty convex values. The multi-sender sanction game has a Nash equilibrium on $\tilde{E}$.

Proposition 8.8 (Multilateral underprovision). *Consider a symmetric multi-sender sanction game with $n \geq 2$ senders coordinating against a common target j , each sender's welfare gain from sanctions accruing partly through the public-good channel of weakening the target's strategic position and partly through the private-good channel of bilateral trade adjustment. Under the regularity conditions of Chapter 6 Assumption 6.12, the non-cooperative Nash equilibrium of the sanction game implements the public-good component at $1/n$ the magnitude of the cooperative joint optimum, with the private-good component unaffected.*

Proof sketch. The proof is structurally identical to Corollary 6.16 of Chapter 6. The public-good component of each sender's payoff externalizes across the bloc; with n senders and symmetric incentives, each internalizes only $1/n$ of the public-good return at the non-cooperative Nash equilibrium. The private-good component (bilateral trade adjustment specific to each sender's relationship with the target) does not externalize and is implemented at the same magnitude under cooperative and non-cooperative arrangements. The cooperative joint optimum implements the public-good component at full magnitude, delivering the n -fold gap. $\square$

Remark 8.9 (The multilateralism question, revisited). Proposition 8.8 reads in the context of the empirical multilateral-sanctions debate [Bapat and Morgan, 2009, Drezner, 1999, Drezner et al., 2021]. Bapat and Morgan find empirically that multilateral sanctions are not always more effective than unilateral; the framework's joint formulation suggests why. When the public-good component of the sanction's effect is small relative to the private-good component,

multilateral coordination buys little; the senders' bilateral relationships with the target absorb most of the sanction's effect. When the public-good component is large—typically the case for sanctions targeting the target's strategic position rather than its bilateral trade flows—multilateral coordination delivers the n -fold amplification. The structural-empirical content is the relative magnitude of the public-good and private-good components for any specific sanctions episode; this is calibration material the chapter does not undertake.

The EC–Falklands case (Martin 1992). Martin [1992b]'s analysis of the 1982 European Community sanctions on Argentina during the Falklands War is the canonical historical case for the multilateral-coordination question in the bargaining-camp tradition. The substantive puzzle: the EC's sanctions on Argentina enjoyed high participation despite asymmetric exposure across member states and despite domestic political costs in several members. Martin's bargaining-camp analysis identifies the institutional structure (EC's qualified-majority voting, the role of the United Kingdom as motivated party, the linkage to broader EC-foreign-policy questions) that allowed coordination. Mearsheimer [1994a] cites the case as a counterexample to strong-form institutionalism claims: the EC coordination was fragile, contingent, and substantively driven by underlying power rather than institutional structure.

The (p, m) framework's apparatus adds a structural dimension to the case. The EC–Falklands sanctions reshaped the joint (p^*, m^*) for Argentina and the EC member states; the case is empirically a multi-sender sanction game played on a specific equilibrium manifold. Proposition 8.8's prediction is that the EC's multilateral coordination amplifies the public-good component of the sanctions' effect—specifically, the deterrent signal Argentina's standing in subsequent disputes—while the private-good component (bilateral trade adjustments specific to each EC member's relationship with Argentina) is implemented at the same magnitude under coordination as it would be under unilateral action. The historical record's observation that the EC's sanctions had substantial deterrent value while imposing modest bilateral trade costs on each individual member fits the proposition's structural prediction.

This is not a re-litigation of the EC–Falklands case. Whether the proposition's structural account fits the historical data more precisely than the bargaining-camp's institutional account or the structural-realist's power account is a calibrated empirical question the chapter does not undertake. The framework's contribution is the structural-counterfactual apparatus under which the historical case becomes a question about the joint comparative-static of a specific equilibrium configuration; the empirical adjudication remains for follow-up work.

Where this leaves the multilateral question. The multilateral coordination on a sanction is a multi-sender GNEP on the joint equilibrium manifold; the existence of Nash equilibria follows from the existing Chapter 6 apparatus; the cooperative-vs-non-cooperative gap delivers the n -fold underprovision result. Third-party realignment is the joint comparative-static of the unsanctioned states' equilibrium positions; the response is non-zero whenever the joint Jacobian's cross-channel structure couples bystander deployments to the sanctioned-dyad parameters. Both moves extend the bilateral apparatus of Section 8.3 without requiring new mathematical machinery beyond what the chapters' upstream apparatus has already established.

The framework's substantive contributions to the multilateral literature are two. First, the third-party-realignment counterfactuals (*how does sanctions on j change k 's deployment?*) are joint-equilibrium objects neither slice formulation can compute; the apparatus delivers them in principle once the framework is calibrated to specific dyads. Second, the multilateralism-

amplification result of Proposition 8.8 gives the structural-formal account of why multilateral sanctions sometimes amplify and sometimes do not—the relative magnitude of the sanction’s public-good and private-good components determines the gap, with public-good-heavy sanctions enjoying the n -fold amplification and private-good-heavy sanctions achieving little beyond unilateral. Section 8.5 takes up the sender-welfare side of the analysis; Section 8.6 takes up the candidate Σ -crossing material.

8.5 Sender welfare and the cost of sanctioning

The bargaining-camp and the GE-camp both have a notion of *sender cost*: in Eaton and Engers [1992] the sender’s cost of imposing the sanction enters the model as a stipulated parameter P_S ; in Becko [2024] the sender’s welfare loss is computed from the optimal-tariff formula at the sanctioning country’s own indifference curve. The two notions disagree on what the sender cost is—a behavioral penalty for following through on a threat, in the bargaining-camp; a welfare cost computable from neoclassical trade theory, in the GE-camp—and the disagreement reflects the slice formulations’ usual division of labor. The framework’s joint formulation gives a unified account: the sender’s welfare cost Δ_S^* from imposing a sanction is a smooth scalar field on the equilibrium manifold, and the sanction’s marginal sender cost is the joint comparative-static $\partial\Delta_S^*/\partial\theta$. This section develops the unified account, derives the GE-camp’s Becko [2024] closed-form as the m -fixed restriction, and identifies the deployment-feedback term the joint formulation adds.

The sender’s welfare cost as a smooth scalar. The chapter’s apparatus inherits from Chapter 5 a smooth scalar field $\Delta^* : \Omega_0^{\text{reg}} \rightarrow \mathbb{R}$ that records the welfare cost of conflict relative to the global Pareto frontier, and from Chapter 5 Section 5.4 a sensitivity formula $D_\omega\Delta^*$ delivering the gradient of the welfare cost in any parameter direction. The sender’s welfare cost of imposing a sanction is the state-specific component of the conflict externality: with Δ^* as defined in Chapter 5, write the per-state contribution Δ_i^* as the loss accruing to state i from operating on the post-conflict resource profile r^* rather than the full endowment, with $\Delta^* = \sum_{i \in I} \Delta_i^*$ in the welfare-weight representation of Corollary 5.5. The sender’s component Δ_S^* is the welfare cost accruing to the sanctioning state S from the sanction’s effect on its own equilibrium position. A sanction parametrized by θ shifts the joint equilibrium; the sender’s welfare cost responds via

$$\frac{\partial\Delta_S^*}{\partial\theta} = D_\omega\Delta_S^* \cdot \frac{\partial\omega}{\partial\theta} + D_p\Delta_S^* \cdot \frac{\partial p^*}{\partial\theta} + D_m\Delta_S^* \cdot \frac{\partial m^*}{\partial\theta}, \quad (8.7)$$

where the three terms are, respectively, the direct welfare effect of the sanction’s parameter shift, the welfare effect through the price response, and the welfare effect through the deployment response. The third term is the substantive content the joint formulation adds beyond the GE-camp.

The bargaining-camp’s P_S as static welfare. Eaton and Engers [1992]’s sender cost P_S is the welfare cost the sender suffers from following through on the threat: the bilateral trade is disrupted, the sender’s economic activity takes a hit, and the welfare value of the disruption is the parameter the bargaining game’s analysis takes as input. In the framework’s notation, P_S is the static welfare-cost difference $\Delta_S^*(\theta) - \Delta_S^*(0)$ at the sanction’s parameter value θ relative to no-sanction $\theta = 0$. The bargaining-camp slice fixes both prices and deployments across

the sanction; the static welfare cost is then the direct effect of the parameter shift on the welfare scalar at the unmoved equilibrium. The bargaining-camp's analysis of P_S is therefore the $D_\omega \Delta_S^* \cdot \partial\omega/\partial\theta$ term of Equation (8.7): the direct-welfare-effect-only contribution, with both price and deployment responses set to zero.

The GE-camp's optimal-sanctions formula as m -fixed. Becko [2024]'s closed-form result for optimal sanctions is the GE-camp's articulation of the sender's cost-and-benefit calculus. For a sanction-tax t_k on good k in a neoclassical trade model with sanctioning country H and sanctioned country F , Becko derives the optimal-sanctions formula

$$t_k = (1 - \lambda_F) t_k^{\text{ToT}}, \quad (8.8)$$

where t_k^{ToT} is the optimal-tariff terms-of-trade manipulation tax (the tax H would impose if it cared only about its own welfare), and λ_F is H 's willingness to pay for F 's welfare in numeraire units. $\lambda_F = 1$ corresponds to free trade (full weight on F); $\lambda_F = 0$ to pure terms-of-trade manipulation (no weight on F); $\lambda_F < 0$ to costly sanctions (H is willing to incur economic cost to lower F 's welfare). The substantive content: for any sanctioning motive ($\lambda_F < 0$), optimal sanctions target the same goods as terms-of-trade manipulation but at higher intensities; the goods-selection problem is independent of demand and supply elasticities in H .

The formula is computed under the GE-camp's m -fixed assumption. Becko's neoclassical-trade setup has no military deployments; the sanction's effect runs entirely through the price-and-quantity response of the trade-and-consumption equilibrium, with F 's strategic position taken as exogenous data. In the framework's notation, λ_F corresponds to the welfare-cost weight H places on F 's component of Δ^* : $\lambda_F = -\partial U_H / \partial \Delta_F^*$ in suitable normalization, where U_H is H 's welfare and Δ_F^* is F 's component of the conflict-cost scalar. The optimal-sanctions formula Equation (8.8) is the GE-camp's closed-form prediction of how a sanctioning country should design its trade-tax sanction on the assumption that F 's strategic position—its arming, its deployment-cost function, its conflict-feedback structure—does not respond to the sanction.

The deployment-feedback term. The joint formulation adds the deployment-feedback term that Equation (8.8) omits. Under the joint formulation, F 's deployment m_F^* responds to the sanction through the joint comparative-static of Equation (8.1), which shifts F 's strategic position and, through the cross-channel coupling, shifts the trade equilibrium H optimizes over. The sender's first-order condition for optimal sanction design on the joint formulation includes a term capturing the welfare effect of F 's deployment response, and the optimal-sanctions formula picks up an additional component beyond the terms-of-trade-manipulation calculus.

Proposition 8.10 (Optimal sanctions with deployment feedback). *Under the joint formulation of Equation (8.1), the sender's optimal-sanction first-order condition decomposes as*

$$0 = \underbrace{D_\theta \Delta_S^*|_{\text{ToT}}}_{\text{Becko 2024 term}} + \underbrace{D_\theta \Delta_S^*|_{m\text{-feedback}}}_{\text{joint-formulation correction}}, \quad (8.9)$$

where the first term is the GE-camp's terms-of-trade-manipulation calculus delivered by Equation (8.8) at the optimum, and the second term captures the welfare effect on the sender of F 's deployment response to the sanction. The second term vanishes when either (i) F 's deployment is not responsive to the sanction ($\partial m_F^* / \partial \theta = 0$), or (ii) the sender's welfare is insensitive to F 's deployment level ($D_{m_F} \Delta_S^* = 0$). Otherwise, the optimal sanction differs from Equation (8.8)'s prediction.

Proof sketch. Apply Equation (8.7) at the sender’s welfare-maximizing θ and decompose the first-order condition into the price-and-direct-effect terms (the GE-camp’s calculus, which yields Equation (8.8) at the GE-camp’s slice optimum) and the deployment-response term. Under the joint formulation, the deployment-response term enters through the cross-channel coupling $\partial m_F^*/\partial\theta$ as derived in Proposition 8.3; the sender’s welfare picks up F ’s deployment shift through $D_{m_F}\Delta_S^*$, the partial of the sender’s welfare in the deployment direction. This yields the decomposition Equation (8.9) and the conditions under which the joint correction vanishes. $\square$

Remark 8.11 (When does the deployment-feedback term matter?). Proposition 8.10 delivers a substantive structural prediction: Becko [2024]’s optimal sanctions formula is the joint optimum exactly when either the target’s deployment is unresponsive to the sanction or the sender’s welfare is insensitive to the target’s deployment. Both conditions fail in cases of substantive interest. A sanction that targets the cost-of-force structure (an embargo on military-relevant goods, or a dual-use export control) shifts $\partial m_F^*/\partial\theta$ by construction. A sender that values security as well as economic welfare values F ’s deployment level: if F deploys more, the sender’s strategic position is worse, raising Δ_S^* . The deployment-feedback term is thus expected to be non-zero in the canonical sanctions episodes the chapter cites (post-2014/2022 Russia, post-2017 North Korea, post-2018 US–China dual-use export controls), and the GE-camp’s optimal-sanctions formula misses substantive content the joint formulation captures.

The sanction’s reflexive cost. The sanction’s effect on the sender’s own deployment is a secondary structural consequence the bargaining-camp and the GE-camp both miss. A sanction that disrupts bilateral trade in good k raises the domestic price of k in the sender’s economy. If k is a military-relevant good, the increase in domestic p_k raises the sender’s own cost of force production, shifting the sender’s deployment m_S^* via the joint comparative-static. The sender’s welfare picks up this reflexive cost through $D_{m_S}\Delta_S^*$, completing the welfare decomposition of Equation (8.7).

The 2018–2024 US export controls on advanced semiconductors to China illustrate the reflexive cost. The export controls disrupted US semiconductor supply chains domestically, raising US prices for advanced chips, with downstream effects on US-domestic deployment-relevant manufacturing capacity. The GE-camp’s analysis captures the price effect; the bargaining camp’s analysis captures neither the price nor the deployment shift. The joint formulation captures both: the joint comparative-static delivers $\partial m_S^*/\partial\theta$ alongside $\partial m_F^*/\partial\theta$, and the sender’s welfare picks up the reflexive cost through the $D_{m_S}\Delta_S^* \cdot \partial m_S^*/\partial\theta$ component of Equation (8.7). The substantive empirical content of the reflexive cost is what the policy literature on *sanctioner self-harm* and *de-risking* [Farrell and Newman, 2019, Becko and O’Connor, 2025] has been articulating without the structural-formal apparatus to ground it.

Sender welfare under multilateral coordination. The multilateral case of Section 8.4 adds a structural layer to the sender-welfare analysis. Each sender in a multilateral sanctions coalition picks up the sanction’s reflexive cost on its own deployment and prices, with the cost typically smaller per sender as the coalition spreads the disruption across multiple economies. The cooperative-vs-non-cooperative gap of Proposition 8.8 thus has two components: the public-good amplification on the target’s strategic position (the n -fold result), and the per-sender cost-distribution benefit on the reflexive-cost component (each sender’s reflexive cost is reduced when the disruption is shared across the coalition). A coalition of n symmetric senders implements the public-good component at full magnitude (relative to the non-cooperative $1/n$) and the

reflexive-cost component at $1/n$ the per-sender magnitude (relative to going alone). Multilateral coordination thus dominates unilateral on both dimensions when the public-good and the cost-distribution benefits are both substantive.

The framework’s contribution to the sender-welfare-and-multilateral literature is the structural-formal apparatus under which the two effects combine. The bargaining-camp’s Drezner [1999] adversary-vs-ally analysis identifies one structural source of variation in P_S across dyads; the GE-camp’s Becko [2024] terms-of-trade analysis identifies another (the goods-selection problem); the joint formulation adds the deployment-feedback and reflexive-cost components. The empirical adjudication of which component dominates in any specific sanctions episode remains for follow-up calibrated work; the structural-counterfactual apparatus is what the chapter delivers.

Looking ahead. The sender-welfare analysis of this section completes the chapter’s bilateral apparatus. Section 8.6 takes up the candidate Σ -crossing material—sanctions that push the system across the singular locus into a different regime component— with the dedollarization, weaponized-interdependence, and financial-network material as candidate empirical cases. The bifurcation apparatus is the chapter’s response to the substantive question of when sanctions induce structural regime change rather than within-regime adjustment, and it inherits the bifurcation-as-regime-change reading of Chapter 7 Section 7.4.

8.6 Sanctions as candidate Σ -crossings

The five preceding sections have analyzed sanctions as within-regime perturbations: the master comparative-statics formula Equation (8.1) delivers $D_\theta(p^*, m^*)$ at any regular equilibrium, the slice-recovery and cross-effect propositions of Section 8.3 characterize what the slice formulations capture and miss, the multilateral and sender-welfare extensions of Section 8.4 and Section 8.5 extend the apparatus to multi-state and sender-side analyses. The implicit assumption throughout has been that the sanction keeps the joint equilibrium inside its initial regime component of $\bar{E}^{\text{reg}}$. A sanction that drives the system across Σ —into a different regime component, with a qualitatively distinct equilibrium structure—requires a different apparatus. This section takes up that case.

The chapter’s stance on Σ -crossings is light by design. The substantive bifurcation apparatus—catastrophe-theoretic stratification of Σ , the index theorem’s constraints on how regimes can change, the structural-formal account of regime change as a mathematical phenomenon—was delivered in Chapter 4 Section 4.5 and Chapter 7 Section 7.4. The chapter’s role here is to identify which sanctions are candidates for inducing Σ -crossings and to gesture at the substantive empirical material the candidate-crossings literature has been articulating. The financial-channel material—weaponized interdependence, dedollarization, parallel payment rails—is the principal candidate site for Σ -crossings in the contemporary sanctions environment, and the literature on it is the most substantively rich. The chapter’s apparatus does not formally model the financial channel (per the boundary stated in Section 8.1), but the bifurcation framing identifies what the formal model would have to deliver.

When does a sanction cross Σ ? A bilateral sanction by sender i on target j shifts the joint equilibrium by the master-formula amount $D_\theta(p^*, m^*)$. The sanction crosses Σ when the resulting trajectory in Ω_0 leaves the initial regime component of $\bar{E}^{\text{reg}}$ and enters a different component. The substantive content of *which* sanctions cross Σ is structural: the joint Jacobian

J becomes singular ($\det J = 0$) on Σ , the equilibrium count above the parameter profile changes as the trajectory crosses, and the nature of the change is governed by the catastrophe-theoretic stratification of Σ (Section 4.6). Three structural conditions characterize candidate Σ -crossings.

First, the sanction's parameter perturbation θ has to have non-trivial effect along directions that bring the equilibrium toward Σ . A sanction whose comparative-static keeps the system far from the singular locus does not induce a crossing; one whose comparative-static moves the system toward Σ in a direction transversal to the locus is a candidate crossing.

Second, the magnitude of the perturbation has to be sufficient to traverse the distance from the initial equilibrium to the nearest Σ -stratum. For local perturbations, the within-regime master formula applies; the chapter's apparatus delivers the response. For global perturbations large enough to cross, the bifurcation apparatus replaces the master formula at the crossing point.

Third, the structure of the Σ -stratum the trajectory crosses determines what kind of regime change occurs. Fold transitions (codimension one) are the canonical case: two equilibria of opposite index merge and annihilate as the trajectory crosses, with the equilibrium count dropping by two and the index theorem of Chapter 4 constraining the pairing. Cusp catastrophes (codimension two) admit richer transitions in which the equilibrium count can change in either direction depending on which pair of fold curves the trajectory crosses. Higher-codimension singularities admit progressively more complex transition geometries.

The dedollarization candidate. The most substantive candidate Σ -crossing in the contemporary sanctions environment is dedollarization: the shift of international transactions away from the US dollar-clearing network and toward parallel financial infrastructures. [Farrell and Newman \[2019\]](#) identifies the dollar-clearing network as the dominant chokepoint sanctioning countries can weaponize; the panopticon and chokepoint effects of weaponized interdependence operate through the asymmetric position of the US in the global financial network. [Drezner et al. \[2021\]](#)'s edited volume catalogues specific applications: secondary sanctions on Iran (2018), SWIFT exclusion of Russian banks (2022), the threatened use of dollar-clearing access denial as a coercive tool in US–China economic disputes. The substantive question is whether the cumulative pressure of weaponized dollar-clearing pushes the international financial system across a Σ -stratum into a regime in which non-dollar-clearing networks (yuan-based, BRICS-coordinated, crypto-mediated) provide a substantively independent alternative.

The framework's apparatus does not formally model the financial channel, so the chapter cannot deliver the bifurcation analysis directly. What it can deliver is the structural form the analysis would take if the framework's manifold representation were extended to include financial-network position as a primitive. The analysis would identify a financial-network parameter analogous to κ_j in the cost-of-force structure, trace the joint equilibrium's response to weaponization of the dollar-clearing chokepoint, and identify the singular-locus geometry where the equilibrium structure changes qualitatively. The empirical question—whether the post-2018 trajectory has crossed Σ —would translate into a question about the financial-network parameter's location relative to the singular-locus stratification.

The chapter does not adjudicate the empirical question. What it does is identify dedollarization as the principal candidate Σ -crossing in contemporary sanctions practice and gesture at the formal-extension work the candidate analysis would require. The structural-empirical follow-up to which the chapter defers is exactly this work: extending the framework to financial-network primitives and running the bifurcation analysis on the calibrated extension.

Other candidate Σ -crossings. Three other candidate sites for sanctions-induced regime change deserve brief mention.

The post-2022 Russian economic decoupling. The cumulative effect of the 2014 and 2022 sanctions on Russia plus Russia’s military-economic restructuring may constitute a regime change in the sense the framework’s apparatus formalizes. Thoenig [2024] surveys the contemporary quantitative-trade-and-conflict toolkit from the GE-camp’s perspective; the toolkit can be applied to ask whether Russia’s post-2022 trade structure represents a discontinuous shift or a within-regime adjustment. The framework’s bifurcation apparatus delivers the structural-formal version of the same question: is the post-2022 Russian equilibrium in the same component of $\tilde{E}^{\text{reg}}$ as the pre-2014 Russian equilibrium, or has the trajectory crossed Σ ?

The post-2018 US–China technology decoupling. The 2018–2024 US export controls on advanced semiconductors and the parallel Chinese investment in domestic semiconductor capacity may represent a regime change in the technology-and-trade sub-system. The substantive question is whether China’s post-2024 chip production trajectory represents a within-regime substitution (Chinese chips replacing US chips at marginally higher cost) or a structural shift to a parallel technology-supply regime (Chinese chips on a different technology trajectory, with different downstream implications for military-relevant manufacturing). The framework’s bifurcation apparatus identifies the structural question; the empirical adjudication is calibrated work the chapter does not undertake.

The interwar collective-security crisis. The League of Nations sanctions on Italy (1935–36) over Abyssinia, the failed sanctions on Japan over Manchuria, and the broader collapse of the interwar collective-security arrangement represent a historical regime change at the international-system level. Mulder [2022]’s historical analysis identifies the specific failures of the interwar sanctions episodes that contributed to the collapse. The framework’s bifurcation apparatus reads the collapse as a Σ -crossing of the international-system equilibrium manifold: the system’s pre-1935 configuration in a regime component supporting collective-security commitments, the system’s post-1936 configuration in a different component where the commitments could not be sustained. The formal-historical analysis remains for follow-up work; the chapter’s apparatus identifies the structural form the analysis would take.

The bifurcation apparatus, briefly. The apparatus the candidate-crossings analysis consumes is the bifurcation machinery of Chapter 7 Section 7.4. A sanction-induced trajectory crosses Σ at a specific parameter profile; the local geometry of Σ at the crossing point determines the catastrophe-theoretic type of the transition (fold, cusp, swallowtail). The index theorem of Chapter 4 (Theorem 4.5) constrains how the equilibrium count changes across the transition: the signed sum of equilibrium indices is preserved across regular fibers, so equilibria appearing or disappearing across Σ must come in cancelling pairs (fold) or higher-order combinations (cusp, swallowtail). The substantive content of the apparatus for sanctions analysis is that regime change is not free-form: the catastrophe-theoretic stratification constrains which transitions are admissible, and the index theorem constrains the equilibrium-count arithmetic at each transition.

For sanctions specifically, the bifurcation apparatus delivers three structural commitments. First, candidate Σ -crossings are not generic: the sanction has to perturb in a direction transversal to a specific stratum of the singular locus. Second, the index of the disappearing or appearing equilibria across the crossing is constrained by the index theorem; not every regime-change story the verbal-historical literature imagines is geometrically admissible. Third, the post-crossing regime is not free: it has to be a component of $\tilde{E}^{\text{reg}}$ adjacent to the initial component across the

specific Σ -stratum the sanction crossed. The historical-empirical question of which sanctions episodes cross Σ and which post-crossing regime they land in becomes a structural-geometric question the framework's apparatus makes formal-precise.

The financial-channel boundary, restated. The chapter's apparatus does not formally model the financial channel. This is a substantive limit: the framework's manifold representation has (ω, κ) -space primitives but not financial-network topology, and the candidate Σ -crossings of dedollarization, weaponized interdependence, and parallel-payment-rail emergence operate through a substantive domain the framework currently does not have apparatus for. Farrell and Newman [2019] and the Drezner et al. [2021] volume make the case for the substantive importance of the financial channel; the chapter takes the case seriously and identifies financial-channel material as the principal site of contemporary candidate Σ -crossings.

The structural-empirical follow-up paper to which the chapter defers is, in part, the formal-extension work that would equip the framework with financial-network primitives. The extension would add a network-topology layer to the manifold representation, with the dollar-clearing centrality and SWIFT chokepoint as primitive parameters analogous to the κ_j cost-of-force parameters in the chapter's apparatus. The bifurcation analysis on the extended framework would deliver the candidate- Σ -crossing analysis the chapter gestures at here without formally executing.

Looking ahead. The bifurcation apparatus of this section closes the chapter's substantive arc. The bilateral, multilateral, and sender-welfare analyses of the preceding sections covered within-regime perturbations on the joint equilibrium manifold; this section identifies the candidate cross-regime cases and the structural form their analysis would take. Section 8.7 takes up the chapter's position in the broader research program: how the structural-counterfactual apparatus the chapter delivers relates to the empirical sanctions literature, what the framework's contribution is and what remains for follow-up work, and how the chapter's apparatus connects to the program-level questions Chapter 9 returns to.

8.7 Looking ahead

The sanctions theory delivered by this chapter is, in summary form, a structural-counterfactual apparatus. A sanction is a parameter perturbation θ on the joint equilibrium manifold $\tilde{E}$; the system's response is the joint comparative-static $D_\theta(p^*, m^*) = -J^{-1}D_\theta e$ of Equation (8.1). The bargaining-camp tradition's analysis is the slice in which both prices and deployments are held fixed (Proposition 8.1 (1)); the GE-camp's analysis is the slice in which deployments are held fixed (Proposition 8.1 (2)); the joint formulation lifts both fixings, with the cross-channel terms of the joint Jacobian capturing structural content the slices miss (Proposition 8.3). On the canonical worked example of Example 8.5, the cross-effect is the entire joint response. The chapter's structural payload is the apparatus under which the question *do sanctions work?* becomes a question about the joint comparative-static rather than about coding-rule choices in the empirical literature.

Each section reframes a debate. The bilateral apparatus of Section 8.3 reframes the empirical-effectiveness debate (Hufbauer et al. 2007a, Pape 1997, Elliott 1998, Baldwin 1998) as a question about which counterfactual the empirical coding rules are implicitly using; the

framework’s joint comparative-static makes the question precise. The multilateral and third-party material of Section 8.4 reframes [Sturm \[2023\]](#)’s GE-camp three-country analysis and [Bapat and Morgan \[2009\]](#)’s bargaining-camp multilateral analysis as restricted slices of the joint response, with the n -fold underprovision result of Proposition 8.8 delivering the structural-formal account of when multilateral sanctions amplify and when they do not. The sender-welfare analysis of Section 8.5 reframes [Becko \[2024\]](#)’s optimal-sanctions formula as the GE-camp’s m -fixed slice optimum, with the joint-formulation correction of Proposition 8.10 adding the deployment-feedback and reflexive-cost components the literature has been articulating without an apparatus to ground. The bifurcation material of Section 8.6 reframes the weaponized-interdependence and dedollarization literatures ([Farrell and Newman 2019](#), [Drezner et al. 2021](#), [Mulder 2022](#)) as candidate Σ -crossings, with the catastrophe-theoretic stratification and the index theorem constraining admissible regime changes.

What the framework’s apparatus does not deliver. The chapter’s structural-counterfactual apparatus does not adjudicate the empirical-effectiveness debate. The HSE-database success rate, Pape’s recoded count, and the relative effectiveness of multilateral versus unilateral sanctions all rest on calibrated empirical work the chapter does not undertake. The framework supplies the apparatus under which the empirical questions become formal-precise; it does not supply the calibration. The structural-empirical follow-up to which the chapter defers is the calibrated work that closes this gap: a quantitative calibration of the framework’s primitives to the HSE–Felbermayr-database episodes, a structural-econometric identification of the conflict-cost primitives from observational data, an empirical adjudication of which counterfactual best fits the historical record across sanctions categories. The follow-up is reserved for econ-paper-length treatment in the structural-trade-and-conflict empirical tradition the chapter cites; the chapter’s role is to identify the structural form that calibrated work would take.

The financial-channel material is similarly deferred. The framework’s (ω, κ) -space primitives have no financial-network topology, and the candidate Σ -crossings of dedollarization and weaponized interdependence operate through a substantive domain the chapter does not formally model. Section 8.6 gestures at the formal-extension work the financial-channel analysis would require; the extension itself is reserved for follow-up.

The geoeconomics field, briefly. The chapter’s apparatus sits inside the broader geoeconomics field as that field has been articulated in the contemporary literature. [Mohr and Trebesch \[2025\]](#)’s 2025 *Annual Review of Economics* survey identifies five subfields of geoeconomics (geoeconomic policy tools, geopolitics of trade, geo-international finance, geopolitical risk, economics of war) and traces a long lineage including Hicks, Hirschman, Kantorovich, Keynes, and Schelling. [Thoenig \[2024\]](#) develops the quantitative-toolkit version of the trade-and-war analysis at the center of the field’s substantive case material. The framework’s contribution to the geoeconomics field is the joint-formulation apparatus the field’s existing toolkits have been operating without: the bargaining-camp and the GE-camp formulations have both been occupying parts of the geoeconomics intellectual space, but the joint formulation brings them together on a common manifold and identifies what each has been missing. The chapter is one of two pieces of geoeconomics-flavored material in the manuscript: Chapter 6 (the peacemaking-on-the-cheap analysis) covers the donor-coordination and welfare-improvement direction; Chapter 8 (this chapter) covers the sanctions-as-coercion direction. Together they mark the framework’s geoeconomics footprint: a structural-formal apparatus the field has been reaching for.

Open theoretical extensions. Three substantive extensions remain open.

First, alliance-formation under sanctions. The chapter's bloc structure of Section 8.2 took the alliance configuration as exogenous data: the sanctioning bloc B and the target's relationships with non-aligned states were primitives of the analysis. The endogenous alliance-formation problem—which states join the sanctioning bloc, which target-aligned states defect under sanctions pressure, how the bloc structure responds to sanctions shocks—is a natural extension that the framework's apparatus admits but the chapter does not develop. The structural-formal version would extend the multilateral GNEP analysis of Proposition 8.8 to include bloc-membership choices as endogenous variables.

Second, dynamic-arming response paths. The chapter's apparatus is static: the joint comparative-static delivers the equilibrium response to a parameter perturbation at a single point in time. Sanctions in practice unfold over time, with the sanctioning country adjusting the sanction's intensity and scope in response to the target's behavior, and with the target adjusting its deployment trajectory in response to the sanction's evolution. A dynamic extension of the chapter's apparatus would extend the manifold-paths formulation of Chapter 7 Section 7.5 to sanctions-as-(target, path)-institutions, with the cohomological obstructions of Proposition 7.9 delivering structural-formal reasons multilateral coordination on sanctions is harder than coordinating on targets alone. The structural extension is straightforward in principle; the empirical calibration of the dynamic apparatus is the substantively challenging part.

Third, financial-network propagation. The chapter's (ω, κ) -space primitives do not include financial-network topology, and the candidate Σ -crossings of Section 8.6 are gestures rather than formal claims. A formal-extension paper would equip the framework with financial-network primitives—dollar-clearing centrality, SWIFT chokepoint, payment-rail dependency—and run the bifurcation analysis on the extended framework. The extension would let the framework deliver a structural-formal account of weaponized interdependence the Farrell and Newman [2019] thesis has been articulating without the manifold-paths apparatus the rest of the framework provides.

The chapter's two readings. The chapter admits two readings. Read as substantive sanctions theory, it is a structural account of what sanctions do in a joint-determination setting. Five results no partialled framework can deliver—the slice-recovery proposition, the cross-effect proposition, the multilateral-underprovision proposition, the optimal-sanctions-with-deployment-feedback proposition, the candidate- Σ -crossings analysis—combine to deliver a structural-counterfactual apparatus the empirical sanctions literature has been operating without. Read as apparatus-demonstration, the chapter is a worked extended case for the framework's reach into the sanctions-and-economic-coercion domain: the topology of $\tilde{E}$ from Chapter 3, the master comparative-statics formula from Chapter 4, the welfare scalar of Chapter 5, the donor-coordination apparatus of Chapter 6, and the bifurcation reading of Chapter 7 all combine to deliver sanctions content that none of the upstream apparatus would produce in isolation. Both readings are views of the same apparatus running on the same six sections. Chapter 9 returns to the program-level questions the chapter raises about the framework's substantive contributions and its bounded reach into the geoeconomics field.

Exercises

1. *Bargaining-camp vs. GE-camp slice on a sanctions episode.* The chapter argues that real sanctions episodes exhibit both price responses (the GE-camp's domain) and deployment

responses (the bargaining-camp's domain), and that neither sliced formulation captures the joint response in full. Identify a sanctions episode from [Hufbauer et al. \[2007b\]](#) or [Felbermayr et al. \[2020\]](#) in which both the price response and the deployment response are substantively visible—e.g., the post-2014 sanctions on Russia (visible price effects on energy markets and visible deployment effects in eastern Ukraine) or the post-1990 sanctions on Iraq (visible market disruption and visible military-deployment shifts). Argue informally which slice recovers more of the response, and where the cross-channel content lives. Open-ended.

2. *Drezner's conflict-expectations parameter ρ .* [Drezner \[1999\]](#)'s analysis of sanctions effectiveness uses a conflict-expectations parameter ρ to distinguish the adversary case (high ρ , sanctions backfire) from the ally case (low ρ , sanctions work). Translate ρ into the framework's apparatus. Identify which equilibrium quantity ρ structurally corresponds to. The natural candidate is a deployment-elasticity coefficient on the cross-channel block $J_{m,p}$ (Equation (8.2)): ρ governs how strongly the target's deployment m^* responds to a price perturbation p . Open-ended.
3. *Three parameter directions on Example 3.3.* Identify three parameter perturbations θ on Example 3.3 that exhibit qualitatively different slice behavior. (a) A perturbation the bargaining-camp slice handles cleanly (i.e., where $D_\theta p^* = 0$); the k -perturbation is the natural candidate, since p^* is k -invariant by Exercise 2 of Chapter 3. (b) A perturbation the GE-camp slice handles cleanly (i.e., where $D_\theta m^* = 0$); harder to find on the symmetric example, but a perturbation that holds the contest prize fixed while shifting endowments will work in some configurations. (c) A perturbation neither slice handles cleanly; the V_0 -perturbation is the natural candidate, since both p^* and m^* respond non-trivially. Verify each by direct closed-form computation. Open-ended.
4. *Post-2018 US–China tariffs through the framework's lens.* Discuss the post-2018 US–China tariff trajectory through the framework's lens. Identify which channels the apparatus captures: the trade channel via tariff-induced shifts in ω^τ (the effective-endowment map of Section 8.2), the deployment channel via shifts in defense and dual-use industrial capacity, and the financial channel via dedollarization pressures on cross-border payment systems. Identify which channels remain out of scope: specifically, the financial channel sits outside the framework's (ω, κ) -space and would require an extension to a financial-network primitive (Exercise 32). Open-ended.
5. *Tariff sanction τ on Example 3.3.* Section 8.2 models sanctions as parameter shifts on the equilibrium manifold via the effective-endowment map ω^τ . For Example 3.3 (Ch 3), construct a tariff sanction τ on good 1 trade between states 1 and 2 explicitly: the sanctioning state imposes a tariff that reduces the effective endowment of good 1 available for cross-state exchange by a factor $(1 - \tau)$. Compute the effective endowment map ω^τ from the original symmetric endowments. Verify $\omega^\tau \in \Omega_0$ for τ in some interval $[0, \tau_{\max})$, with $\tau_{\max}$ determined by the interior-of- Ω_0 condition. The exercise instantiates the chapter's general parameter-shift formulation on the running example.
6. *Arms embargo as κ_j -shift.* A second canonical sanction is an arms embargo, modeled as a shift in the cost-of-force parameter κ_j for the embargoed state. For Example 3.3, construct a κ_2 -shift representing an arms embargo on state 2. Compute the joint comparative-static

$D_\kappa(p^*, m^*) = -J^{-1}D_\kappa e$ explicitly from Equation (8.1). The embargo raises κ_2 , which makes deployment more expensive for state 2 and shifts the joint equilibrium toward lower m_2^* and correspondingly different p^* . The exercise illustrates the κ -shift channel of sanctions as a parameter perturbation.

7. *Tariff vs. embargo equivalence at the limit.* Show that a tariff $\tau \rightarrow 1$ on the embargoed-goods subset produces the same equilibrium response as a $\kappa_j \rightarrow \infty$ shift on the same subset, up to measure-zero technical issues at the boundary of Ω_0 . The argument runs through the effective-endowment map: as $\tau \rightarrow 1$, the cross-state trade in those goods vanishes, and the embargoed state's effective endowment converges to its autarky endowment, which is the same as the limit of the $\kappa_j \rightarrow \infty$ shift. Identify which formal representation is more useful for which substantive purpose: the tariff τ is more natural for partial-coverage sanctions; the κ -shift is more natural for total embargoes. Open-ended.
8. *Secondary-sanction bloc-multiplier shift.* For a 3-state economy with one sanctioning state and two potential bystanders, construct a secondary-sanction bloc-multiplier shift: the sanctioning state imposes secondary sanctions on bystanders that trade with the target, parameterized by a bloc-multiplier β that scales the effective sanction reach. Derive the conditions under which a bystander joins the sanction bloc voluntarily (when joining yields a higher equilibrium utility) versus involuntarily (when β is large enough that not joining costs more than joining). The exercise generalizes the secondary-sanctions literature (Felbermayr et al. 2020) into the framework's formal vocabulary.
9. *Master formula breaks at Σ .* The master formula Equation (8.1) delivers $D_\theta(p^*, m^*) = -J^{-1}D_\theta e$ at any $(p, m, \omega) \in \tilde{E}^{\text{reg}}$. Verify that the formula breaks down at a singular equilibrium $e \in \Sigma$ (where J is singular and J^{-1} does not exist). Identify the geometric structure that replaces it: at Σ , the comparative-static analysis becomes catastrophe-theoretic per Remark 3.20 of Chapter 3 and Section 4.6 of Chapter 4. The substantive content is that smooth sanctions analysis applies on $\tilde{E}^{\text{reg}}$, and Σ -crossings (Section 8.6) require the singular machinery.
10. *Slice recovery by direct calculation.* Proposition 8.1 establishes that the GE-camp slice formula Equation (8.3) arises from the master formula by holding $D_\theta m^* = 0$ and inverting only the J_{pp} block. Verify the proposition by direct calculation. Write out the joint Jacobian J as a 2×2 block matrix of Equation (8.2); apply the master formula $-J^{-1}D_\theta e$; impose $D_\theta m^* = 0$ and isolate the price response; verify the resulting expression matches Equation (8.3) exactly. The exercise instantiates the slice-recovery result on the chapter's running example.
11. *Schur-complement correction.* Proposition 8.3 derives the cross-effect as a Schur-complement correction to the GE-camp slice. Derive the correction by direct block-matrix manipulation. From the joint formula $D_\theta p^* = -[S^{-1}]D_\theta \bar{d} - [-J_{mm}^{-1}J_{mp}S^{-1}]D_\theta r$ of Equation (4.3), verify $\Delta_p = -(J_{pp})^{-1}J_{pm} \cdot D_\theta m^*|_J$ as the Schur-complement correction in Equation (8.5). The exercise certifies the chapter's cross-effect formula directly from the Chapter 4 master apparatus.
12. *Joint Jacobian on Example 3.3.* For Example 3.3 with parameters (V_0, θ, W, k) , compute the joint Jacobian J explicitly at the symmetric equilibrium. Use the closed-form deployment $m^* = (V_0 + \theta W)/(2k(2 + \theta))$ and the closed-form blocks $J_{pp}, J_{pm}, J_{mp}, J_{mm}$ from Exercise

- 4 of Chapter 4. Verify that $J_{pm} = D_m \bar{d} \neq 0$ at $\theta > 0$ (so the cross-channel correction is non-trivial). The exercise quantifies the cross-channel block on the running example, where θ controls how much of the contest prize tracks aggregate consumption.
13. *GE-camp price formula at fixed m_0 .* The GE-camp slice fixes deployment at some baseline m_0 and price-clears the resulting exchange economy. Show that for Example 3.3 with log-Cobb-Douglas demand and the symmetric-Tullock contest's transfer-cancellation property at symmetry, the GE-camp slice's price formula reduces to $p^*|_G = W/(W - 2km_0)$ by direct derivation. The substantive content: the GE-camp's price response to a tariff sanction tracks how much resource is held off-market by the deployment m_0 , and the framework's joint formulation adds a correction for how m_0 itself responds to the sanction.
 14. *Three parameter directions and the gap pattern.* Explore which parameter directions in Example 3.3 give zero gap between joint and sliced responses, versus non-zero gap. Identify three such directions: (a) one where the gap is the entire response (a θ -perturbation, where the cross-channel block dominates because deployment responds more than prices); (b) one where the gap is a non-trivial fraction of the response (a V_0 -perturbation, where both channels respond and the cross-correction is intermediate); (c) one where there is no gap (a k -perturbation, where p^* is k -invariant and the deployment-only channel suffices). Explain the structural pattern: the gap tracks how strongly $D_\theta m^*$ couples back into prices via the cross-channel block J_{pm} .
 15. *Drezner's ρ and adversary-vs-ally pattern.* Building on Exercise 2, translate Drezner's conflict-expectations parameter ρ into the framework's apparatus and identify what curvature conditions on $\kappa(m_j)$ would deliver the adversary-vs-ally pattern from joint comparative-statics. Drezner's adversary case (high ρ , sanctions backfire) corresponds to a parameter regime in which the target's deployment response amplifies the welfare cost of sanctions; the ally case (low ρ) corresponds to a regime where it dampens. The structural distinction lives in the sign of $D_\theta m^*$ from the joint formula, modulated by the sign of the cross-channel block. Open-ended.
 16. *Sanctioned vs. non-sanctioned counterfactual on the two-state contest.* Construct a worked sanctioned-vs-non-sanctioned counterfactual on Example 3.3 with a specific parameter perturbation θ_* representing a tariff sanction by state 1 on state 2. Compute three quantities: the bargaining-camp's predicted concession size (from the bargaining-of-war literature), the GE-camp's predicted welfare cost (from Equation (8.3)), and the joint formulation's predicted equilibrium response (from Equation (8.1)). Identify which differences between the three predictions are attributable to the cross-channel correction Equation (8.4). The exercise instantiates the chapter's central comparison (Example 8.5) on a worked parameter perturbation.
 17. *Asymmetric extension and Drezner pattern.* For an asymmetric extension of Example 3.3 with $\omega_1 \neq \omega_2$ by some asymmetry parameter ε , compute the joint comparative-static $\partial m_j^* / \partial \theta$ for $j = 1, 2$ separately under a sanctions perturbation θ . Identify the parameter region in which the deployment responses qualitatively differ across the two states (the analog of Drezner's adversary-vs-ally pattern from Exercise 15). Substantively: the state with smaller ω is the "smaller" party, and its deployment response under sanctions can have opposite sign from the larger party's, depending on which state's contest-side constraint is binding.

18. *Third-party comparative-static by direct computation.* Equation (8.6) expresses third-party deployment responses to bilateral sanctions through the master formula. Verify the equation by direct block-Jacobian computation. For an n -state economy, partition J into per-state blocks; show that the master formula's response on state k 's deployment row is non-zero whenever the cross-channel blocks J_{p,m_k} or $J_{m_k,p}$ are non-zero. The exercise establishes that third-party responses are generic in the joint formulation, even when the bilateral-sanction setup is between only two of the n states.
19. *Sturm 2023's three-country setup.* For Sturm [2023]'s three-country setup (Home, Foreign, RoW), specify the parameter perturbation that captures Western sanctions on Russia with India/China as bystanders. Compute $D_{\theta}p^*$ and $D_{\theta}m_k^*$ for $k \in \{\text{Russia, India, China}\}$ under the joint formulation. Compare with Sturm's GE-camp (m -fixed) results. The exercise quantifies how much of the bystander response Sturm's analysis captures versus how much the joint formulation adds through the cross-channel block.
20. *n -fold underprovision via Nash FOCs.* Proposition 8.8 establishes that a coalition of n symmetric senders underprovides sanctions effort by a factor of n relative to the cooperative benchmark. Derive the result explicitly from the Nash first-order conditions. Identify which payoff structure (linear, Cobb-Douglas, CES) admits the clean public-good/private-good decomposition the proposition assumes: linear utilities make the free-rider problem additive in a clean way; Cobb-Douglas does not. The exercise certifies the proposition's structural content and locates its dependence on the payoff form.
21. *EC–Falklands sanctions case.* For the EC–Falklands sanctions case (Martin 1992a), identify the substantive public-good and private-good components of the EC's sanction package on Argentina. Discuss whether Proposition 8.8's structural prediction (public-good amplification via cooperation, private-good unchanged) fits the historical record. The public-good component is the EC's collective deterrent posture; the private-good component is each member state's bilateral trade interest. Martin's empirical analysis suggests amplification on the deterrent side (visible in the EC's collective bargaining position) but not on the bilateral trade side. Open-ended.
22. *Bystander cross-component motion under JR' .* Remark 8.7 flags that under Assumption 3.17, a bilateral sanction can push a bystander's equilibrium across Σ into a different regime component, producing structural regime change rather than smooth adjustment. Construct an explicit example of this phenomenon: a state whose equilibrium sits near Σ such that a bilateral sanction elsewhere pushes its equilibrium across Σ . Identify the substantive interpretation: the bystander's regime configuration changes (e.g., from a coalition member to a neutral stance, or from an open economy to a partial-autarky regime), and the change is discrete rather than incremental. Open-ended.
23. *Calibration strategy for $D_{\theta}m_{\text{China}}^*$.* For the Russia–China counterfactual that motivates much of the post-2014 sanctions literature, propose a calibration strategy that would deliver $D_{\theta}m_{\text{China}}^*$ (China's deployment response to Western sanctions on Russia) empirically. Identify which framework primitives need to be calibrated: the cost-of-force parameters κ_j , the contest-success-function curvatures, the cross-state substitution elasticities J_{p,m_k} . Identify which data sources would be useful: defense-spending data for κ_j , trade data for substitution elasticities, contest data (e.g., Felbermayr et al. 2020's sanctions-effectiveness panel) for contest-success-function curvatures. Open-ended.

24. *Sender's welfare cost decomposition.* Equation (8.7) decomposes the sender's welfare cost Δ_S^* into terms running through endowments $\omega(\theta)$, prices $p^*(\theta)$, and deployments $m^*(\theta)$. Verify the decomposition by direct differentiation. Identify which terms vanish under the bargaining-camp slice ($\partial p^*/\partial\theta = \partial m^*/\partial\theta = 0$, so only the direct ω -term survives) and which under the GE-camp slice ($\partial m^*/\partial\theta = 0$, so the ω -term and the p -term survive but the m -term vanishes). The joint formulation includes all three. The exercise locates the partialled welfare-cost calculations as restrictions of the joint expression.
25. *Becko's optimal-sanctions formula.* Equation (8.8) states Becko [2024]'s optimal-sanctions formula $t_k = (1 - \lambda_F)t_k^{\text{ToT}}$ from the GE-camp slice's first-order condition. Derive the formula directly from the GE-camp's optimization over sanction intensity t_k . Verify that the formula is independent of demand and supply elasticities in the sanctioning country (the GE-camp's terms-of-trade calculus depends only on the target's welfare elasticity λ_F). The exercise instantiates Becko's headline result and identifies the elasticity assumptions it implicitly imposes.
26. *Optimal sanctions with deployment feedback.* Proposition 8.10 extends Becko's formula with a deployment-feedback correction Equation (8.9). Apply the proposition to a parameterized version of Example 3.3 with sender H and target F. Compute the GE-camp Becko term and the joint-formulation correction explicitly. Identify parameter regions where the correction is larger than the Becko term: specifically, when $D_\theta m^*$ is large (Drezner's adversary regime, Exercise 15) and the cross-channel block J_{pm} couples deployment back into welfare strongly. Open-ended.
27. *Reflexive sanctions cost.* Construct an example of a sanction with non-zero reflexive cost: the sanctioning state H imposes an export control on a military-relevant good k that H also produces domestically. Compute $\partial m_H^*/\partial\theta$ (H's own deployment response to its own sanction) and the welfare component $D_{m_H}\Delta_S^* \cdot \partial m_H^*/\partial\theta$ (the reflexive welfare cost). Substantively: a sanction on a dual-use good imposes costs on the sanctioning state's own defense industrial base, and the joint formulation captures this as a non-trivial term in the sender's welfare-cost decomposition.
28. *Post-2018 US-China semiconductor controls.* For the post-2018 US-China semiconductor export controls, identify the substantive components of the sender's welfare cost: (a) the GE-camp's terms-of-trade calculus (the value of the lost export market); (b) the joint formulation's deployment-feedback term (how the controls affect China's deployment of dual-use semiconductor capacity, which feeds back into US security cost); (c) the reflexive cost on US-domestic semiconductor capacity (Exercise 27). Discuss which components the policy debate has emphasized (mostly (a)) and which have been missed (mostly (b) and (c)). Open-ended.
29. *n^2 -fold relative advantage of cooperation.* For a multilateral coalition of n symmetric senders, verify that the public-good amplification ($\times n$ from Proposition 8.8) and the reflexive-cost distribution ($\div n$, since reflexive costs spread across the coalition) combine multiplicatively to deliver an n^2 -fold relative advantage of cooperation over going alone in the public-good-only limit. The result is the strongest argument the framework offers for multilateral cooperation in sanctions: combining public-good amplification and cost-distribution gives quadratic returns to coalition size.

30. *Sanctions trajectory crossing Σ on the two-state contest.* Construct an explicit example of a sanctions trajectory in Example 3.3 that crosses Σ . Identify the parameter direction (V_0, θ, W, k) and the magnitude of the perturbation required. Compute the index structure of the disappearing or appearing equilibria across the crossing, using Theorem 4.5 of Chapter 4: the disappearing pair must have indices +1 and -1 to preserve the signed sum. The exercise instantiates the chapter's Σ -crossing reading of sanctions on the running example.
31. *Post-2018 US–China semiconductor decoupling.* For the post-2018 US–China semiconductor decoupling, identify structural primitives that would need to be in the framework to formalize the candidate Σ -crossing. The minimal extension involves: a state-specific contest-feedback parameter indexing technological capacity, a cross-state coupling that captures supply-chain interdependence, and a regime-component structure in E^{reg} separating the “integrated supply chain” regime from the “decoupled parallel” regime. Discuss what data sources would be required for calibration: trade-flow data, technology-transfer restrictions, defense-industrial capacity. Open-ended.
32. *Dedollarization and a financial-network primitive.* Dedollarization—the partial retreat from dollar-denominated international payments after 2022 sanctions—falls outside the framework's (ω, κ) -space. Propose a financial-network primitive that extends the framework. Identify the analog of the cost-of-force parameter κ_j (a cost-of-financial-routing parameter η_j representing the cost of using non-dollar payment systems), and the analog of the singular locus Σ (a financial regime boundary separating dollar-dominant configurations from multi-currency configurations). The exercise sketches how the framework would extend to capture financial sanctions and dedollarization explicitly. Open-ended.
33. *Index theorem constraint on regime change.* Theorem 4.5 of Chapter 4 constrains regime change through the signed-sum identity: not every pre-/post-crossing equilibrium pairing is admissible, since the indices must match across the crossing. For Example 3.3 with V_0 crossing through 0 (the boundary between peace and war regimes), identify which equilibria appear/disappear at the crossing and verify the index theorem applies. At $V_0 = 0$, the peaceful equilibrium $m^* = 0$ exists; for $V_0 > 0$, the symmetric arming equilibrium emerges. The crossing is not a fold (the peaceful and arming equilibria do not merge), but a more delicate boundary transition. Discuss what the index theorem constrains here.
34. *Mulder 2022 and interwar collective security.* For Mulder [2022]'s historical analysis of the interwar collective-security collapse, formalize the international-system parameters that would represent the regime configurations pre- and post-1936 (the year of the Italian invasion of Ethiopia, when League of Nations sanctions failed). Discuss what catastrophe-theoretic transition type (Remark 3.20) best fits the historical record: a fold catastrophe (two equilibria merge and disappear), a cusp (three equilibria collapse to one), or a higher-order singularity. The collapse of collective security looks fold-like (a coalition equilibrium and a holdout equilibrium merge as enforcement becomes infeasible), but the historical record exhibits enough complexity (multiple shocks, multiple actors) that a cusp may fit better. Open-ended.

Chapter 9

Of wealth and power

The book opened with one master claim and four supporting legs. The international political-economic system is one phenomenon, not two phenomena coupled. States are economically embedded at the agent level: their optimization problems do not separate into a consumption decision at given prices and a deployment decision at given resources. The equilibrium object is one manifold at the system level: prices and military deployments are simultaneously fixed by the joint strategic choices of states on the smooth zero set $\tilde{E} := e^{-1}(0)$. The master comparative-static $-J^{-1}D_{\theta}e$ delivers counterfactuals—welfare, aid, institutions, sanctions—no partialled tradition can compute. What the verbal-historical international-relations tradition calls regime change has formal correlate Σ -crossing on $\tilde{E}$, and the index theorem on the joint Jacobian classifies the crossings.

The seven body chapters built the apparatus. Three of them deployed it on substantive applications. The work the master claim required is done. The present chapter notes what was delivered, addresses the gap-naming-predecessor cluster from the chapter introduction directly, and flags the classes of phenomena the framework holds fixed for the next generation of the program.

Embeddedness. Chapter 2 established that no state’s optimization problem separates into a consumption decision at given prices and a deployment decision at given resources. The two architectural choices identified in Section 1.4—the opportunity-cost function c_i taking prices as a primitive argument, the battle function s_i taking consumption-after-cost as a primitive argument—couple the consumption and deployment decisions through the entanglement matrix $R_i = I + D_{x_i}s_i$ inside the state’s first-order conditions. The disembedded special case $R_i = I$, in which the partialled methods recover the joint solution, is the case in which the battle function ignores consumption—a knife-edge condition no real international system satisfies. Polanyi [1944] was right about the substantive impossibility; Chapter 2 named the formal location.

The joint manifold. Chapter 3 assembled the joint manifold $\tilde{E} \subset S_N \times M \times \Omega_0$ as a single smooth object, established existence via homotopy from a Walrasian endpoint, and produced a global chart Φ exhibiting $\tilde{E}$ as diffeomorphic to $\mathbb{R}^{IL}$. Under generic joint regularity—earned from primitives via parameterized Sard rather than imposed by axiom—the singular locus Σ is closed and Lebesgue-null, and the regular set $\tilde{E}^{\text{reg}}$ partitions into regime components. Chapter 4 developed the master comparative-static

$$D_{\theta}(m^*, p^*) = -J^{-1}D_{\theta}e$$

and the index-theorem constraint on equilibrium pairings across Σ -crossings. What the chapter introduction promised at the system level—one manifold rather than two coupled, comparative statics from one inverse rather than from sequential block-diagonal inverses—was delivered.

Counterfactuals: one operator, four uses. The same operator J^{-1} did substantive work in four contexts. Chapter 5 delivered the welfare gradient $D_\omega \Delta^*$ —the cardinal welfare cost of conflict at any joint equilibrium—and relocated the bargaining-theory-of-war’s premise that war is inefficient from assumption to theorem under explicit conditions on the conflict technology. Chapter 6 delivered the donor’s gradient field, the transversality result that aid evaluated at fixed prices cannot reduce conflict because the partial-equilibrium direction has zeroed out the operative channel by construction, and the resource-neutral $1/n$ multilateralism rationale. Chapter 7 delivered the institutional vector field $X^{\Delta^*} := -D_\omega \Delta^*$, the homotopy structure of paths-on- $\tilde{E}$, and the Chichilnisky–Heal aggregation theorem operating globally under joint regularity and componentwise under generic joint regularity. Chapter 8 delivered the slice-recovery proposition that bargaining-camp and exchange-side formulations are restrictions of the joint comparative-static, with the cross-effect on the canonical worked example being the *entire* joint response rather than a small correction. Four substantive results that no combination of partialled methods produces, derived as four parameter directions of one operator.

Regime change is geometric. Chapter 4’s catastrophe-theoretic stratification of Σ and Chapter 7’s bifurcation analysis cashed out the regime-change-is-geometric claim. Krasner [1982]’s regimes are connected components of $\tilde{E}^{\text{reg}}$. Ikenberry [2001]’s post-major-war institutional settlements—1815, 1919, 1945—are Σ -crossings driven by parameter shifts of sufficient magnitude. Tilly [1990]’s three-path state-formation model is a parameter-coverage claim about which components are reachable from which historical environments. Acemoglu and Robinson [2019]’s narrow corridor is a low-dimensional projection of the framework’s component structure. Smith [1776]’s natural-vs-retrograde distinction, recovered by Hont [2005], is the eighteenth-century anticipation. The verbal-historical tradition described what the joint Jacobian’s geometry classifies. The vocabulary the tradition lacked was the formal apparatus the framework supplies.

The certification. The chapter introduction named a gap-naming-predecessor cluster whose authors had repeatedly identified the joint-pursuit thesis without supplying the apparatus. List [1841] pressed the cosmopolitical-versus-political distinction in 1841. Strange [1970] named the mutual neglect that founded the field of international political economy in 1970. Gilpin [1975] articulated the reciprocal-and-dynamic interaction of wealth and power as the field’s defining object in 1975. Keohane and Nye [1987] acknowledged in 1987 that less had been done with the liberal half of their attempted synthesis than with the realist half. Mansfield [1994] named in 1994 the lack of a unified deductive framework that incorporates features of both the distribution of power and trade. Each named the gap; none supplied the apparatus.

The book’s claim is that the apparatus has now been supplied. The framework’s joint (m, p) formulation is the unified deductive framework Mansfield identified as missing. The two-level joint determination—agent-level via the price-dependence of c_i and the consumption-dependence of s_i ; system-level via the manifold $\tilde{E}$ —is the structural-formal articulation of what Strange named verbally as mutual neglect. The four substantive uses of $-J^{-1}D_\theta e$ —welfare, aid, institutions, sanctions—are the demonstration that the apparatus is not ornamental but operative: it produces results no combination of partialled methods can produce, on questions the older tradition cared

about and the contemporary literature still does. Gilpin’s joint-and-reciprocal pursuit of wealth and power is the substantive content; the framework is the formal apparatus that content required. The book has done what the gap-naming predecessors called for.

Future work: technology change. The framework’s apparatus describes regime change driven by parameters within a fixed structural setup; the functional forms of c_i and s_i are held fixed in the present book. Technology change is the most consequential class of phenomena the held-fixed assumption brackets out, but it is also the class the framework’s joint-determination structure is most naturally suited to host. The standard reading runs from technology to military capability: the Industrial Revolution drove the cost reductions and material innovations on which the modern army depended, and the Military-Revolution literature reads fortifications-and-cannons, mass-mobilization, nuclear displacement of land-mass-based military advantage, and the contemporary precision-strike-and-cyber-and-AI domains as sequential shifts in the contest technology. Satia [2018]’s *Empire of Guns* inverts the standard reading. The eighteenth-century British arms trade—the gun-trade infrastructure on which the Birmingham metalworkers, the iron foundries, the financial system, and the colonial supply chains all depended—was not a downstream consumer of industrial capacity but an upstream driver of it. Military demand made the industrial revolution that the standard reading credits with making the military.

The substantive content the framework’s joint-determination apparatus is built to host is precisely this kind of joint-reciprocal phenomenon. Technology change parameterized as a shock to both c_i (through the productive infrastructure that mobilization rests on) and s_i (through the contest technology mobilization deploys) is not a structural-setup shift but a joint parameter direction the comparative-static $-J^{-1}D_{\theta}e$ already handles. Satia’s reading is the historiographical demonstration that the joint-reciprocal direction is the empirically correct one for the case the Military-Revolution literature has long debated. The future-work scope here is therefore less an extension of the apparatus than an application of it: parameterize the technology-state variable, run the joint comparative-static, and recover what the verbal historiography has been describing as a joint-reciprocal phenomenon all along. What the framework does need beyond its current scope is the class of technology shifts that change the *form* of c_i or s_i —a new contest medium, a new commodity dimension—and those are genuinely outside the parameter-direction reach. The substantive line between parameter shift and form shift is where the technology-change extension does its work.

Future work: financial-network topology. The framework’s commodity space and contest structure capture trade and force as continuous-variable quantities. They do not capture the network topology of financial relations through which contemporary economic coercion increasingly operates. Farrell and Newman [2019]’s weaponized-interdependence framework identifies the central nodes of the global financial network (clearinghouses, payment systems, reserve currencies) whose asymmetric-interdependence positions concentrate economic-coercive capability in ways the dyad-level apparatus does not capture. Drezner et al. [2021] extends the framework into a research program on weaponized interdependence as the contemporary form of economic coercion; Mohr and Trebesch [2025]’s geoeconomics review consolidates the 2024–25 state of the literature. On the formal-trade-theory side, Becko [2024] and Sturm [2023] develop optimal-sanctions analyses under network topologies—the structural neighborhood the framework’s network-augmented extension would engage directly. Section 8.6 flagged dedollarization and weaponized-interdependence as candidate Σ -crossings the framework can identify structurally without analyzing in detail; the

network-augmented apparatus is what would let the framework analyze them.

Future work: explicit dynamics in time. The framework is comparative-statics with bifurcations, not a dynamical system over time. It supplies the structural classification of regime change but not its temporal content: when a Σ -crossing occurs, how fast it propagates, whether the system overshoots, what oscillatory behavior follows, what kinds of stable cycles the joint manifold supports. The natural research-program neighborhood is the systemic-cycles literature. [Modelske \[1987\]](#)’s long-cycles theory of world leadership identifies a roughly century-long cycle of hegemonic rise, maturity, delegitimation, and deconcentration that the framework’s joint manifold could be the state space for. [Goldstein \[1988\]](#)’s war-and-peace-cycles synthesis surveys the broader historical-statistical literature on cycles in international war and economic activity. The framework’s joint manifold is the right state space for the dynamics the systemic-cycles tradition has identified verbally; what’s missing is the flow on the manifold whose orbits would trace the cycles. The dynamical-systems extension is the natural formal addition; the substantive content the extension would track has been mapped in advance by the systemic-cycles literature.

Future work: endogenous formation. The framework takes the number of states I as exogenous and preferences u_i as primitive. State births and deaths, consolidation and fragmentation, preference formation through cultural and discursive processes, and the constructivist program of identity-formation are held outside the framework’s apparatus by the choice to take the primitives as given. The natural research-program neighborhood is constructivism. [Wendt \[1992\]](#) argued that anarchy is what states make of it: not a structural primitive of the international system but a social construction, and one that admits multiple constructions. [Wendt \[1999\]](#) developed the constructivist program at book length, with sovereignty and identity as endogenous outcomes of state interaction rather than fixed inputs. The framework’s structural-formal reading of anarchy is a third way between the constructivist “what states make of it” and the realist “what the structure uniquely determines”: anarchy is a specific reachable equilibrium type on the joint manifold whose reachability depends on the system’s primitives. The extension that endogenizes the primitives—preference formation through cultural processes, identity formation through interaction, state formation through coercion-and-capital trajectories the way [Tilly \[1990\]](#) reads European political development—would be the constructivist-friendly version of the apparatus. Chapter 7’s engagement with the historical-institutionalism literature—[Pierson \[2000\]](#), [Page \[2006\]](#), [Fioretos \[2011\]](#)—is the framework’s first move in this direction; the constructivist extension goes further.

Future work: empirical calibration. The framework is a structural-counterfactual apparatus, not a predictive-empirical one. The decades of methodology-of-economics debate over Friedman’s instrumentalism and the multi-decade specification fight in the dyadic commercial-peace literature, both engaged in Section 1.5, are engagements with empirical-predictive questions the framework does not directly address. [Lucas \[1976\]](#) grounds the methodological status: the framework’s structural primitives (s_i , c_i , u_i) are deep in Lucas’s sense, and the comparative-static formula computes responses to parameter perturbations at these primitives rather than at policy-dependent reduced-form coefficients. The framework’s empirical content survives the Lucas critique structurally; what remains is the calibration step in which specific values of the primitives are assigned to specific historical or contemporary configurations and the framework’s predictions are checked against the empirical record. The structural-empirical follow-up is the

natural next step, and it is the work the present book defers rather than the work the present book denies.

The framework is one apparatus. The master claim is one observation. The two together constitute what the book has built and what the book has shown the apparatus can do. Five research scopes—technology change, financial-network topology, explicit dynamics, endogenous formation, empirical calibration—are what the next generation of the program inherits. Each is a natural extension of the joint-determination apparatus rather than a departure from it. The apparatus is the right shape; the extensions are the future the apparatus enables.

The substantive thesis the book defends is older than the apparatus by two centuries. List in 1841, Strange in 1970, Gilpin in 1975, Keohane and Nye in 1987, Mansfield in 1994—five voices across a century and a half, naming the same absence in the same field. What each named, the book has supplied. The joint pursuit of wealth and power, named verbally for two centuries, has now been formalized.

Appendix A

Mathematical primer

This appendix collects the mathematical apparatus the body chapters use without introducing in-text. The book’s substantive content lives at the intersection of two literatures with different mathematical baselines—formal international relations and smooth general-equilibrium theory—and the apparatus draws on tools that one literature treats as standard and the other treats as new. The primer’s job is to make the apparatus accessible to readers trained in either literature without forcing either to read a separate textbook first.

Audience. The reader is assumed to be comfortable with real analysis on $\mathbb{R}^n$ —continuity, convergence, compactness—and with constrained optimization at the level of [Mas-Colell \[1985\]](#) or [Sundaram \[1996\]](#): Lagrangians, Karush–Kuhn–Tucker conditions, the standard envelope theorem, basic linear and matrix algebra. The reader is also assumed to be comfortable with the body of elementary game theory the book uses without reviewing: Nash equilibrium, best-response correspondences, mixed strategies as a specialization of expected utility.

What the reader is *not* expected to know is the differential-topological apparatus the framework’s structural arguments turn on:

- the total derivative Df as a linear map (rather than a row of partial derivatives);
- smooth manifolds embedded in $\mathbb{R}^n$ as a setting beyond Euclidean space, with their tangent spaces and smooth submanifold structure;
- Sard’s theorem and its parameterized form, the differential-topological engine of the regular-economies tradition;
- differential 1-forms, line integrals, the exterior derivative, and de Rham cohomology;
- Pontryagin’s maximum principle on a smooth manifold as a generalization of the calculus-of-variations on $\mathbb{R}^n$;
- the Chichilnisky–Heal aggregation theorem and the topological obstruction to continuous, anonymous, Pareto-respecting preference aggregation.

These tools are introduced from scratch here, using the book’s own objects as running examples wherever possible. The treatment is cursory rather than comprehensive: the goal is to make the body chapters’ arguments parseable, not to substitute for a textbook on any of these areas. References are given for readers who want to go further.

Notation. Throughout the appendix, the symbols $X_i, M_i, \Omega_0, S_N, \tilde{E}$ retain their meanings from the body chapters: $X_i = \prod_\ell (b_i^\ell, \infty)$ is state i 's consumption set (Chapter 2), $M_i \subset \mathbb{R}_+$ its deployment space, $\Omega_0 = \mathbb{R}_{++}^L$ the open endowment space, $S_N = \{p \in \mathbb{R}_{++}^L : p^L = 1\}$ the numéraire-normalized open price simplex, and $\tilde{E} := e^{-1}(0) \subset S_N \times M \times \Omega_0$ the joint equilibrium manifold (Chapter 3). When a symbol like X appears in a generic primer definition (“let X be a smooth manifold”), the generic usage is clear from context; the body's X_i retains its specific meaning.

Organization. The primer is organized by mathematical area. Each body section opens with a brief announcement of where in the book the apparatus is used, presents the technical content in self-contained form (definitions, statements, occasional proof sketches), and closes with an “application to the book” beat that pins the apparatus to specific theorems and propositions in the body chapters. The use-map below provides a quick reference for navigating to the section that handles a particular tool. The synthesis section Section A.10 traces the master comparative-static formula $D_\theta(p^*, m^*) = -J^{-1}D_\theta e$ through the apparatus, activating every body section in a single calculation.

A.1 Use-map

The table below maps each appendix section to the body chapters that consume it and the principal results it powers. Readers seeking a specific tool should navigate directly to the relevant section; readers wanting a coherent walkthrough should read the synthesis at Section A.10 first and dip into body sections as needed.

§	Used in	Powers
§A.2 <i>Notation</i>	all	symbols $\mathbb{R}, \mathbb{R}_{++}, S_N, \Omega_0, \tilde{E}$; profile notation $(m_1, \dots, m_I), m_{-i}$; Lebesgue measure-zero.
§A.3 <i>Differential calculus</i>	2–6	total derivative Df as linear map (central conceptual jump); Jacobians R_i, J ; chain rule; IFT , powering joint best-response of Chapter 2 and master CS formula of Chapter 4.
§A.4 <i>Linear and matrix algebra</i>	3, 4, 8	operator norm, spectral radius; block matrices; Schur decomposition (Lemma 3.4, Chapter 3); block-inverse formula (Equation (4.3)), backbone of sliced-vs-joint analysis in Chapter 8.
§A.5 <i>Smooth manifolds and differential topology</i>	3, 4, 6, 7	smooth submanifolds of Euclidean space; tangent space; regular values; preimage theorem; Sard's theorem and parameterized Sard, powering Debreu–Mas-Colell genericity (Theorem 4.3 + Lemma 4.2); smooth coverings, powering Proposition 3.15.
§A.6 <i>Singularity theory and the index theorem</i>	3, 4, 7, 8	mod-2 degree of a proper smooth map, powering existence (Theorem 3.12); index theorem and signed-sum identity (Theorem 4.5), powering Corollary 3.19; Sard–Thom stratification of Σ (Remark 3.20), powering catastrophe readings in Section 4.6, Section 7.4, Section 8.6.

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(Section A.1, continued)

§	Used in	Powers
§A.7 <i>Differential forms, line integrals, cohomology</i>	6, 7	1-forms on smooth manifolds; line integrals; exterior derivative; closed-vs-exact; Poincaré lemma , powering closed-preference reading of Section 6.1; de Rham H^1 , π_1 , Mayer–Vietoris, powering path-aggregation obstruction (<i>Proposition 7.9</i>).
§A.8 <i>Manifold optimal control</i>	6	cotangent bundle, 1-form fields; Pontryagin maximum principle on a smooth manifold, powering <i>Theorem 6.8</i> ; switching function, singular arcs, bang-bang; strict convexity in CoV, powering <i>Proposition 6.10</i> .
§A.9 <i>Welfare and social-choice apparatus</i>	5, 7	Pareto frontier as smooth manifold; utility-possibility set and frontier diffeomorphism; value functions; Negishi representation , powering <i>Corollary 5.5</i> ; First Welfare Theorem (conditional, <i>Proposition 5.4</i>); Arrow’s impossibility; Chichilnisky–Heal (<i>Theorem 7.1</i>); homotopy equivalence.
§A.10 <i>Synthesis: master CS formula</i>	all	single calculation activating every body section. The formula $D_\theta(p^*, m^*) = -J^{-1}D_\theta e$ invokes IFT (§A.3), Schur and block-matrix algebra (§A.4), smooth-manifold structure of $\bar{E}$ (§A.5), Sard genericity (§A.5), and the index theorem (§A.6).

The use-map is for navigation, not assessment of importance. Section A.3 (differential calculus) and Section A.5 (smooth manifolds and differential topology) are the two longest sections, because they treat the two largest conceptual jumps for the audience. Section A.8 (manifold optimal control) is shorter because the audience that needs Pontryagin self-selects into reading Chapter 6’s asterisked sections Section 6.4, where the chapter develops the machinery in body in any case.

Conventions. Definitions, propositions, and theorems are numbered within section. The treatment of each concept is calibrated by its body use: notational conventions get a sentence, working tools get a paragraph or two, and load-bearing theorems (IFT, Sard, the Schur identity, the Poincaré lemma, the index theorem, Pontryagin’s maximum principle, Chichilnisky–Heal) get boxed statements with reference-quality citations to the literature. Proof sketches appear when the proof is short and illuminating; longer proofs are pointer-only, with references to Hirsch [1976], Guillemin and Pollack [1974], Mas-Colell [1985], Balasko [2009], Liberzon [2011a], and Hatcher [2002] as the standard further-reading sources.

A.2 Notation conventions

This section collects the symbols and conventions the body chapters use repeatedly. The conventions are not idiosyncratic; they follow the smooth-general-equilibrium tradition of Mas-Colell [1985] and Balasko [2009].

Number systems. $\mathbb{R}$ is the real line; $\mathbb{R}_+ = [0, \infty)$ the non-negative reals; $\mathbb{R}_{++} = (0, \infty)$ the positive reals. $\mathbb{R}^n$ is Euclidean n -space; $\mathbb{R}_+^n$ and $\mathbb{R}_{++}^n$ are respectively the non-negative and positive orthants. The vector inequality $x \geq y$ means $x^\ell \geq y^\ell$ for every ℓ ; $x > y$ means $x \geq y$ and $x \neq y$; $x \gg y$ means $x^\ell > y^\ell$ for every ℓ .

Endowment and price spaces. $\Omega_0 = \mathbb{R}_{++}^{IL}$ is the open endowment space: $\omega \in \Omega_0$ is a profile $(\omega_1, \dots, \omega_I)$ with $\omega_i \in \mathbb{R}_{++}^L$ for each state i . The numéraire-normalized open price simplex is $S_N = \{p \in \mathbb{R}_{++}^L : p^L = 1\}$, identified with $\mathbb{R}^{L-1}$ via the chart $p \mapsto (p^1, \dots, p^{L-1})$. The Walrasian convention—good L as numéraire—is fixed throughout.

Bounded-below consumption sets. State i 's consumption set is $X_i = \prod_{\ell} (b_i^{\ell}, \infty) \subset \mathbb{R}^L$, a product of open half-lines bounded below by the survival floors $b_i^{\ell} \in \mathbb{R}$. The product space $X = \prod_i X_i$ holds the full allocation profile $(x_1, \dots, x_I)$.

Profile notation. For any indexed object $z = (z_1, \dots, z_I)$, $z_{-i} := (z_1, \dots, z_{i-1}, z_{i+1}, \dots, z_I)$ is the profile of others. This convention applies to deployments m , allocations x , endowments ω , and any other state-indexed object. We write $(z_i; z_{-i}) = z$ when emphasizing that i is the focal state.

Lebesgue measure-zero. A subset $A \subset \mathbb{R}^n$ has *Lebesgue measure zero* if for every $\varepsilon > 0$ there is a countable cover of A by open boxes whose total volume is less than ε . Equivalently, A has measure zero iff its n -dimensional Lebesgue measure is zero. The framework's regularity hypotheses (Assumption 3.17) and the genericity arguments of Chapter 4 are formulated in this measure-zero sense.

Smooth maps. $f : U \rightarrow V$ is *smooth* (C^∞) if U, V are open subsets of Euclidean spaces (or smooth manifolds, see Section A.5) and f has continuous partial derivatives of every order. The book uses smoothness uniformly: every primitive in Chapter 2 (utility u_i , battle function s_i , cost c_i) is assumed smooth, and the joint equilibrium function $e(p, m, \omega)$ inherits smoothness from the primitives.

A.3 Differential calculus for the apparatus

This section treats the differential-calculus tools the apparatus chapters use without introducing in-text. The reader has seen partial derivatives in the standard multivariable-calculus sense, $\partial f / \partial x^\ell$. The book uses partial derivatives, but it also uses the *total derivative* Df as a linear map, the chain rule for composed smooth maps, the implicit function theorem in its multivariable form, and Jacobians of vector-valued functions of vector arguments. The conceptual move from “a partial derivative is a component” to “the total derivative is a linear map” is, per the user's flag, the central conceptual jump for the audience; this section treats it generously.

A.3.1 The total derivative as a linear map

For a smooth function $f : \mathbb{R}^n \rightarrow \mathbb{R}$, the total derivative at a point x_0 is the linear map $Df(x_0) : \mathbb{R}^n \rightarrow \mathbb{R}$ that best approximates f near x_0 :

$$f(x_0 + h) = f(x_0) + Df(x_0) \cdot h + o(\|h\|).$$

In coordinates, $Df(x_0)$ is the row vector of partial derivatives, $Df(x_0) = (\partial f / \partial x^1, \dots, \partial f / \partial x^n)$, acting on tangent vectors $h \in \mathbb{R}^n$ by the inner product. The total-derivative reading is what makes the chain rule *compositional* rather than bookkeeping.

For a smooth vector-valued function $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$, the total derivative at x_0 is a linear map $Df(x_0) : \mathbb{R}^n \rightarrow \mathbb{R}^m$. In coordinates, $Df(x_0)$ is the $m \times n$ Jacobian matrix $[\partial f^i / \partial x^j]$. A tangent vector $h \in \mathbb{R}^n$ is taken to the tangent vector $Df(x_0) \cdot h \in \mathbb{R}^m$.

Why the linear-map reading matters. The body's chain rules are easier to read in linear-map form than in component-by-component form. Consider Chapter 2's chain rule for the consumption FOC: the FOC has the form $Du_i(x_i^*) \cdot R_i = \tilde{p}_i$, where $Du_i(x_i^*) : \mathbb{R}^L \rightarrow \mathbb{R}$ is the total derivative of u_i , $R_i : \mathbb{R}^L \rightarrow \mathbb{R}^L$ is a linear map (I plus the Jacobian of s_i in x_i), and $\tilde{p}_i : \mathbb{R}^L \rightarrow \mathbb{R}$ is the effective-price linear functional. Read componentwise, the formula is a dense system of L equations. Read as composition of linear maps, it is a single relation: the total-derivative covector Du_i , pulled back through R_i , equals the effective-price covector. The structural content of the formula is the composition; the component computation is bookkeeping.

A.3.2 Jacobians of vector functions

For a smooth $f : U \rightarrow \mathbb{R}^m$ on $U \subset \mathbb{R}^n$, the *Jacobian* of f at $x \in U$ is the matrix representation of the total derivative $Df(x)$:

$$J_f(x) = [\partial f^i / \partial x^j]_{i,j} \in \mathbb{R}^{m \times n}.$$

We use $Df(x)$ for the linear-map reading and $J_f(x)$ for the matrix reading; they are the same object. When the function or the variable is clear from context, we omit the subscripts: the joint Jacobian $J = D_{(p,m)}e$ in Chapter 4 is the matrix $[\partial e^i / \partial p^j, \partial e^i / \partial m^j]$ of partial derivatives of e with respect to (p, m) .

Restricted Jacobians. When the function depends on multiple arguments, *restricted* or *partial* Jacobians take the derivative with respect to a subset of the arguments. $D_{x_i}s_i$ is the Jacobian of $s_i(m, x, \omega)$ with respect to x_i alone, holding m and x_{-i} and ω fixed. $D_p c_i$ is the Jacobian of $c_i(m_i, p, \omega_i)$ with respect to p alone. The body chapter notation $J_{pp} = D_p \bar{d}$, $J_{pm} = D_m \bar{d}$, $\dots$ in Chapter 4 is the systematic block-decomposition of the joint Jacobian into restricted Jacobians.

A.3.3 The chain rule

The chain rule for total derivatives reads as composition of linear maps. For smooth $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $g : \mathbb{R}^m \rightarrow \mathbb{R}^p$, the composite $g \circ f : \mathbb{R}^n \rightarrow \mathbb{R}^p$ has total derivative

$$D(g \circ f)(x) = Dg(f(x)) \circ Df(x),$$

which in matrix form is the product of Jacobians $J_{g \circ f}(x) = J_g(f(x)) \cdot J_f(x)$. The component-wise version, $\partial(g \circ f)^i / \partial x^j = \sum_k (\partial g^i / \partial y^k)(\partial f^k / \partial x^j)$, is the same statement read as matrix-element multiplication. The body chapters use the chain rule routinely; in linear-map form it is one symbol.

A.3.4 The implicit function theorem

The implicit function theorem (IFT) is the apparatus's most load-bearing differential-calculus tool. It produces the joint best-response of Chapter 2 from the SPT first-order conditions, and it produces the master comparative-statics formula $D_\theta(p^*, m^*) = -J^{-1} D_\theta e$ of Chapter 4 from the equilibrium identity $e(p^*, m^*, \omega) \equiv 0$.

Theorem A.1 (Implicit function theorem). *Let $F : \mathbb{R}^n \times \mathbb{R}^k \rightarrow \mathbb{R}^n$ be smooth, and let $(x_0, \theta_0) \in \mathbb{R}^n \times \mathbb{R}^k$ satisfy $F(x_0, \theta_0) = 0$. If the partial Jacobian $D_x F(x_0, \theta_0) \in \mathbb{R}^{n \times n}$ is invertible, then there exist open neighborhoods $U \ni x_0$ and $V \ni \theta_0$ and a unique smooth function $x^* : V \rightarrow U$ with $x^*(\theta_0) = x_0$ and $F(x^*(\theta), \theta) = 0$ for every $\theta \in V$. Moreover,*

$$D_\theta x^*(\theta_0) = -[D_x F(x_0, \theta_0)]^{-1} \cdot D_\theta F(x_0, \theta_0).$$

[Rudin, 1976, Theorem 9.28] is a standard reference for the multivariable IFT; Hirsch [1976, Section 3.2] treats the manifold version we use in Section A.5.

Reading the IFT. The hypothesis is invertibility of the x -block of the Jacobian. The conclusion is two statements, packaged together: existence of a smooth local solution $x^*(\theta)$, and a formula for its derivative. The formula is the master CS formula in disguise: take $F = e$ (the equilibrium function), $x = (m, p)$ (the equilibrium variables), $\theta = \omega$ (the parameters), and the hypothesis is invertibility of $J = D_{(p,m)} e$, exactly the joint regularity assumption of Assumption 3.6; the conclusion is the local equilibrium function $(p^*(\omega), m^*(\omega))$ together with $D_\omega(p^*, m^*) = -J^{-1} D_\omega e$. Section 4.1 derives the master formula by exactly this argument.

A.3.5 Application to the book

The total-derivative-as-linear-map reading is the standing convention throughout the apparatus chapters. The chain rule appears in Section 2.2 in the derivation of the joint FOC; in Section 4.1 in the derivation of the master formula; and in Section 6.1 in the differentiation of the welfare 1-form α along an admissible path. The IFT is invoked in three load-bearing arguments: the joint best-response of Chapter 2 (existence and smoothness of μ_i, f_i as functions of (p, m_{-i}, ω)), the diffeomorphism of Chapter 3 ($\tilde{\pi} : \tilde{E} \rightarrow \Omega_0$ as a global diffeomorphism under Assumption 3.6), and the master CS formula of Chapter 4. Restricted Jacobians appear pervasively, organized by the block-matrix conventions of Section A.4 below.

A.4 Linear and matrix algebra

This section treats two specific tools from linear and matrix algebra that the apparatus uses heavily: operator norms (with spectral radius as a special case), and the block-matrix algebra of the joint Jacobian's Schur decomposition. The reader is assumed comfortable with basic linear algebra (matrix multiplication, determinants, eigenvalues, the rank-nullity theorem); what we add is the specific block-decomposition machinery the joint formulation requires.

A.4.1 Operator norm and spectral radius

For a linear map $A : \mathbb{R}^n \rightarrow \mathbb{R}^n$ (equivalently, an $n \times n$ matrix), the *operator norm* is

$$\|A\|_{\text{op}} := \sup_{\|x\|=1} \|Ax\|,$$

where $\|\cdot\|$ on $\mathbb{R}^n$ is the Euclidean norm. The operator norm equals the largest singular value of A and satisfies the submultiplicative property $\|AB\|_{\text{op}} \leq \|A\|_{\text{op}} \cdot \|B\|_{\text{op}}$.

The *spectral radius* of A is the maximum modulus of its eigenvalues: $\rho(A) := \max_i |\lambda_i(A)|$. It satisfies $\rho(A) \leq \|A\|_{\text{op}}$, with equality when A is normal. For any matrix norm $\|\cdot\|$ and any $\varepsilon > 0$, $\rho(A) = \lim_{k \rightarrow \infty} \|A^k\|^{1/k}$ (Gelfand's formula).

Neumann series. If $\|A\|_{\text{op}} < 1$ (or more generally if $\rho(A) < 1$), then $I + A$ is invertible and

$$(I - A)^{-1} = \sum_{k=0}^{\infty} A^k, \quad (I + A)^{-1} = \sum_{k=0}^{\infty} (-A)^k.$$

This is the Neumann series representation of the inverse. Chapter 2's sufficient condition for the conflict-feedback bound Assumption 2.6—namely $\|D_{x_i} s_i\|_{\text{op}} < 1$ —uses the Neumann series to invert $R_i = I + D_{x_i} s_i$ explicitly when the condition holds.

A.4.2 Block matrices and the Schur complement

For an $n \times n$ matrix J partitioned into four blocks

$$J = \begin{pmatrix} A & B \\ C & D \end{pmatrix}, \quad A \in \mathbb{R}^{a \times a}, B \in \mathbb{R}^{a \times b}, C \in \mathbb{R}^{b \times a}, D \in \mathbb{R}^{b \times b}, \quad a + b = n,$$

the *Schur complement* of D in J is $S_D := A - BD^{-1}C$ (defined when D is invertible). Symmetrically, the Schur complement of A is $A_D := D - CA^{-1}B$ (defined when A is invertible).

Theorem A.2 (Schur block-decomposition). *Let $J = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$ be $n \times n$ with D invertible. Then J admits the factorization*

$$J = \begin{pmatrix} I & BD^{-1} \\ 0 & I \end{pmatrix} \begin{pmatrix} S_D & 0 \\ 0 & D \end{pmatrix} \begin{pmatrix} I & 0 \\ D^{-1}C & I \end{pmatrix}$$

where $S_D = A - BD^{-1}C$. Consequently, $\det J = \det S_D \cdot \det D$, and J is invertible if and only if S_D is invertible. When both are invertible,

$$J^{-1} = \begin{pmatrix} S_D^{-1} & -S_D^{-1}BD^{-1} \\ -D^{-1}CS_D^{-1} & D^{-1} + D^{-1}CS_D^{-1}BD^{-1} \end{pmatrix}.$$

The proof is direct multiplication of the three factors and inverting each. The Schur factorization is the operational backbone of the chapter's joint-vs-sliced comparison: the S_D^{-1} block of J^{-1} is the GE-camp's reduced-economy inverse; the D^{-1} block is the bargaining-camp's deployment inverse; the off-diagonal blocks are the cross-channel corrections that the partialled formulations zero out.

A.4.3 Application to the book

The joint Jacobian of Chapter 4 is partitioned into the 2×2 block structure $J = \begin{pmatrix} J_{pp} & J_{pm} \\ J_{mp} & J_{mm} \end{pmatrix}$ according to whether each block differentiates in the price or deployment direction. Lemma 3.4 of Chapter 3 is exactly Theorem A.2 applied to this J , with the Schur complement $S = J_{pp} - J_{pm}J_{mm}^{-1}J_{mp}$ identified as the Jacobian $D_p \hat{d}$ of the reduced excess demand $\hat{d}(p, \omega) := \bar{d}(p, m^*(p, \omega), \omega)$. The block-inverse formula Equation (4.3) of Chapter 4 is the explicit four-block expression of J^{-1} from Theorem A.2, and it is the formula that Chapter 8's slice-vs-joint comparisons consume. The Neumann series invertibility criterion is what Chapter 2's operator-norm sufficient condition ($\|D_{x_i} s_i\|_{\text{op}} < 1 \Rightarrow R_i$ invertible) instantiates.

A.5 Smooth manifolds and differential topology

The framework's central object is the joint equilibrium manifold $\tilde{E} = e^{-1}(0) \subset S_N \times M \times \Omega_0$. This is not a subset of any $\mathbb{R}^n$ in the obvious sense—it is the zero set of a smooth function on an open subset of Euclidean space—but it inherits a smooth-manifold structure that makes the apparatus chapters' arguments (existence, regularity, diffeomorphism, regime-component decomposition) precise. This section introduces the differential-topological vocabulary the body chapters use without introducing. The audience is assumed comfortable with $\mathbb{R}^n$ as a topological space (open sets, continuity, compactness) but not with smooth manifolds beyond it.

A.5.1 Smooth submanifolds of Euclidean space

Definition A.3 (Smooth submanifold). A subset $M \subset \mathbb{R}^N$ is a *smooth submanifold of dimension d* if for every $x \in M$ there is an open neighborhood $U \ni x$ in $\mathbb{R}^N$ and a smooth map $\phi : U \rightarrow \mathbb{R}^{N-d}$ such that

- (1) $M \cap U = \phi^{-1}(0)$, and
- (2) $D\phi(x) : \mathbb{R}^N \rightarrow \mathbb{R}^{N-d}$ has rank $N - d$ at every $x \in M \cap U$.

The intuition: M is locally cut out as the zero set of $N - d$ smooth equations whose Jacobian has full rank. The full-rank hypothesis is the regular-value condition that makes the implicit function theorem applicable.

Tangent space. The *tangent space* to M at $x \in M$ is the d -dimensional subspace $T_x M := \ker D\phi(x) \subset \mathbb{R}^N$. Equivalently, $T_x M$ is the set of velocity vectors $\gamma'(0)$ for smooth curves $\gamma : (-\varepsilon, \varepsilon) \rightarrow M$ with $\gamma(0) = x$. Tangent vectors are the geometric objects the apparatus's gradient fields (Section 7.3) and Pontryagin costate (Section 6.4) live on.

A.5.2 Regular values and the preimage theorem

For a smooth map $f : U \rightarrow \mathbb{R}^m$ on $U \subset \mathbb{R}^n$, a point $y \in \mathbb{R}^m$ is a *regular value* of f if for every $x \in f^{-1}(y)$, the Jacobian $Df(x) : \mathbb{R}^n \rightarrow \mathbb{R}^m$ has full rank m . A non-regular value is called a *critical value*.

Theorem A.4 (Preimage theorem). *Let $f : U \rightarrow \mathbb{R}^m$ be smooth on an open $U \subset \mathbb{R}^n$, and let $y \in \mathbb{R}^m$ be a regular value of f with $n \geq m$. Then $f^{-1}(y)$ is a smooth submanifold of U of dimension $n - m$.*

[Guillemin and Pollack, 1974, Section 1.4]. The preimage theorem and the implicit function theorem Theorem A.1 are the same theorem in two costumes: the preimage theorem is the IFT applied locally at every point of $f^{-1}(y)$ to produce a smooth chart, and patched. The body chapters use the preimage theorem to establish that $\tilde{E} = e^{-1}(0)$ is a smooth submanifold of $S_N \times M \times \Omega_0$ (Proposition 3.8 of Chapter 3).

A.5.3 Sard's theorem and the parameterized form

Sard's theorem is the key differential-topology input to the regular-economies tradition. It guarantees that the critical values of a smooth map are “rare” in a measure-theoretic sense.

Theorem A.5 (Sard’s theorem). *Let $f : U \rightarrow \mathbb{R}^m$ be smooth on an open $U \subset \mathbb{R}^n$. The set of critical values of f has Lebesgue measure zero in $\mathbb{R}^m$.*

[Hirsch, 1976, Theorem 3.1.3]; [Guillemin and Pollack, 1974, Section 1.7]. The theorem says that even when critical *points* are plentiful (a smooth map can have many critical points), the *values* they map to are negligible in measure.

The version the framework’s genericity arguments need is the parameterized one.

Theorem A.6 (Parameterized Sard). *Let X, Y, Ω be smooth manifolds and $f : X \times \Omega \rightarrow Y$ smooth. If $0 \in Y$ is a regular value of f , then 0 is a regular value of the restriction $f_\omega : X \rightarrow Y$ for almost every $\omega \in \Omega$ (i.e., outside a set of ω of Lebesgue measure zero).*

[Guillemin and Pollack, 1974, Theorem 3.1]. The parameterized version is the input Theorem 4.3 of Chapter 4 consumes: f is the joint equilibrium function e , X is $S_N \times M$, Ω is Ω_0 , and the conclusion is that for almost every ω , 0 is a regular value of e_ω , which is exactly the joint regularity condition Assumption 3.6 at ω .

A.5.4 Smooth proper maps and covering maps

A continuous map $f : X \rightarrow Y$ between topological spaces is *proper* if the preimage $f^{-1}(K)$ of every compact $K \subset Y$ is compact in X . Properness is the hypothesis that lets the framework’s existence arguments transport from one parameter to another via homotopy.

A smooth map $f : X \rightarrow Y$ between smooth manifolds of equal dimension is a *smooth covering map* if it is proper and a local diffeomorphism (i.e., $Df(x)$ is invertible at every $x \in X$). A proper local diffeomorphism is automatically a covering map, with finitely many sheets over each connected component of Y .

Theorem A.7 (Simply-connected coverings have one sheet). *If $f : X \rightarrow Y$ is a smooth covering map between connected smooth manifolds, with X and Y both simply connected, then f has exactly one sheet, i.e., f is a diffeomorphism.*

[Hatcher, 2002, Proposition 1.39]. This is the engine of Proposition 3.15 of Chapter 3: $\tilde{\pi}$ is a proper local diffeomorphism under Assumption 3.6, and both $\tilde{E}$ (via the global chart Φ) and Ω_0 (as a convex open subset of Euclidean space) are simply connected, so $\tilde{\pi}$ has one sheet—it is a global diffeomorphism.

A.5.5 Application to the book

The smooth-manifold structure of $\tilde{E}$ is invoked throughout the book. Proposition 3.8 of Chapter 3 establishes $\tilde{E}$ as a smooth submanifold of dimension IL via the preimage theorem. Proposition 3.15 of Chapter 3 establishes $\tilde{\pi} : \tilde{E} \rightarrow \Omega_0$ as a global diffeomorphism via the simply-connected covering argument. Theorem 4.3 of Chapter 4 establishes that the singular locus Σ has Lebesgue measure zero via parameterized Sard, with Lemma 4.2 supplying the non-triviality input the proof requires. Section 6.1 lifts admissible-paths in Ω_0 to smooth paths in $\tilde{E}^{\text{reg}}$ via the implicit-function-theorem-on-manifold form (local diffeomorphism + paths-stay-in-regular-set), and Chapter 7 reads regime components as connected components of $\tilde{E}^{\text{reg}}$ in the manifold sense.

A.6 Singularity theory and the index theorem

Where Section A.5 treats the smooth-manifold structure of $\tilde{E}$ at regular points, this section treats the structures at singular points and the finitely-many-equilibria results that come with them. The key tools are the mod-2 degree of a smooth proper map (which delivers existence and odd cardinality of the equilibrium count), the index of a regular zero (which refines the count to a signed sum), and the Sard–Thom catastrophe-theoretic classification of singular sets (which structures the boundaries between regimes).

A.6.1 Mod-2 degree of a smooth proper map

For a smooth proper map $f : X \rightarrow Y$ between connected smooth oriented manifolds of equal dimension, the *mod-2 degree* of f at a regular value $y \in Y$ is

$$\deg_2(f) := \#f^{-1}(y) \pmod{2}.$$

The mod-2 degree is independent of the regular value y chosen (as y varies over the regular values, the count changes only in pairs, so the parity is invariant), and is invariant under proper homotopy of f . For a proper local diffeomorphism (covering map), the mod-2 degree equals the number of sheets mod 2.

The mod-2 degree of $\tilde{\pi} : \tilde{E} \rightarrow \Omega_0$ is one, established via the homotopy-to-the-Walrasian-endpoint of Theorem 3.12 of Chapter 3. This is the source of Corollary 3.19: the count of joint equilibria at any regular ω is finite and odd.

A.6.2 Index of a regular zero

For a smooth proper map $f : X \rightarrow Y$ between oriented manifolds of equal dimension, the *index* of f at a regular zero $x \in f^{-1}(0)$ is

$$\text{ind}_x(f) := \text{sign}(\det Df(x)) \in \{-1, +1\}.$$

The *signed degree* of f at a regular value y is the signed sum

$$\deg(f; y) := \sum_{x \in f^{-1}(y)} \text{ind}_x(f).$$

Like the mod-2 degree, the signed degree is independent of the regular value chosen and is invariant under proper homotopy.

Theorem A.8 (Signed-sum identity). *Let $f : X \rightarrow Y$ be a smooth proper map between connected oriented manifolds of equal dimension. The signed degree of f is constant across regular values of f , and changes only via fold-catastrophe pairings (creation or annihilation of opposite-index pairs) at critical values.*

[Guillemin and Pollack, 1974, Section 1.5]; [Milnor, 1965, Chapter 5]. The signed-sum identity is the input to Theorem 4.5 of Chapter 4: the framework’s joint projection $\tilde{\pi}$ has signed degree one (transported from the Walrasian endpoint), and the index theorem says that for every regular $\omega \in \Omega_0 \setminus \Sigma$, the signed sum of indices over $\tilde{\pi}^{-1}(\omega) \cap \tilde{E}^{\text{reg}}$ equals +1.

A.6.3 Catastrophe-theoretic stratification of Σ

The set of critical values of a smooth proper map—the *singular locus*—generically has a stratified structure described by Thom’s classification of elementary catastrophes. For a smooth proper map between equal-dimensional manifolds, the generic singular locus has codimension-one fold strata, with higher-codimension catastrophes (cusps, swallowtails, butterflies, hyperbolic and elliptic umbilics) along intersections of folds.

Theorem A.9 (Thom transversality, applied form). *For a generic smooth proper map $f : X \rightarrow Y$ of equal-dimensional manifolds, the critical set is a stratified subset of X whose strata have codimension at least 1, with codimension-1 fold strata generically the bulk of the critical set, and the codimension- k stratum classified by Thom’s k th elementary catastrophe.*

[Poston and Stewart, 1978, Chapters 6–7]; [Arnold, 1992, Chapter 10]. The framework consumes only the codimension-1 fold structure operationally (Theorem 4.5 and the index-pairing property across folds); the higher-codimension catastrophes are classification-only references at Remark 3.20 of Chapter 3 and Section 7.4.

Fold catastrophe. The local normal form for a fold catastrophe is $f(x; \omega) = x^3 - \omega x$ in one local coordinate x on X and one local parameter $\omega \in Y$. The critical set $\partial f / \partial x = 0$ gives $x^2 = \omega/3$, a parabolic curve in (x, ω) -space; the singular locus in ω -space is $\Sigma = \{0\}$. For $\omega > 0$, two real solutions exist (an opposite-index pair); for $\omega < 0$, no real solution exists; at $\omega = 0$ the two solutions coalesce at $x = 0$, a fold catastrophe. The framework’s regime structure at codimension-1 fold strata takes this local geometry as its operational model.

A.6.4 Application to the book

The mod-2 degree powers Theorem 3.12 of Chapter 3 (existence of a joint dual-competitive equilibrium for every $\omega \in \Omega_0$) and Corollary 3.19 (odd cardinality at regular values). The index theorem (Theorem 4.5 of Chapter 4) refines the count to a signed sum equal to +1, and Corollary 4.6 extracts the unique-equilibrium result under Assumption 3.6. The catastrophe-theoretic stratification of Σ structures the regime-change reading of Section 7.4 (regime boundaries as fold strata, with cusps as codimension-2 meeting points) and the candidate Σ -crossing analyses of Section 8.6 (sanctions trajectories that cross Σ deliberately).

A.7 Differential forms, line integrals, cohomology

The deployment chapters Chapter 6 and Chapter 7 use differential 1-forms as the natural language for preferences-on-paths, and the path-aggregation obstruction of Section 7.5 turns on de Rham cohomology and the fundamental group of the equilibrium manifold. This section introduces the apparatus. The audience is assumed comfortable with smooth manifolds (Section A.5) and the chain rule (Section A.3); what we add is the differential-forms vocabulary.

A.7.1 Differential 1-forms on a smooth manifold

Definition A.10 (Cotangent space and 1-form). For a smooth manifold M and a point $x \in M$, the *cotangent space* T_x^*M is the dual space of the tangent space T_xM : the space of linear functionals $T_xM \rightarrow \mathbb{R}$. A *differential 1-form* on M is a smooth assignment $\alpha : x \mapsto \alpha_x \in T_x^*M$ to each point.

In coordinates on a chart $(x^1, \dots, x^d)$ of M , a 1-form has the local expression $\alpha = \sum_i a_i(x) dx^i$, where $a_i : U \rightarrow \mathbb{R}$ are smooth functions and dx^i are the coordinate basis 1-forms (the linear functionals dual to $\partial/\partial x^i$). The total derivative df of a smooth function $f : M \rightarrow \mathbb{R}$ is the 1-form whose coordinate expression is $df = \sum_i (\partial f / \partial x^i) dx^i$.

A.7.2 Line integrals and parameterization-independence

For a smooth path $\gamma : [a, b] \rightarrow M$ and a 1-form α on M , the *line integral* of α along γ is

$$\int_{\gamma} \alpha := \int_a^b \alpha_{\gamma(t)}(\gamma'(t)) dt,$$

where $\gamma'(t) \in T_{\gamma(t)}M$ is the velocity vector of the path. The line integral is independent of the parameterization of γ : any orientation-preserving smooth reparameterization gives the same integral. For $\alpha = df$ exact, the fundamental theorem of calculus on manifolds gives

$$\int_{\gamma} df = f(\gamma(b)) - f(\gamma(a)),$$

which depends only on the endpoints, not the path.

A.7.3 Closed and exact 1-forms; the Poincaré lemma

A 1-form α is *closed* if its exterior derivative $d\alpha$ vanishes. In local coordinates, $\alpha = \sum_i a_i dx^i$ is closed iff $\partial a_i / \partial x^j = \partial a_j / \partial x^i$ for every i, j (the symmetry-of-cross-partials condition). A 1-form α is *exact* if $\alpha = df$ for some smooth function f . Every exact form is closed (since $d^2 = 0$); the converse is not always true.

Theorem A.11 (Poincaré lemma). *On a contractible smooth manifold M , every closed 1-form is exact: if $d\alpha = 0$, then $\alpha = df$ for some smooth $f : M \rightarrow \mathbb{R}$.*

[Lee, 2013, Theorem 17.14]. Contractible manifolds include $\mathbb{R}^n$, all open star-shaped subsets of $\mathbb{R}^n$, and—most relevant here—the joint equilibrium manifold $\tilde{E}$ under Assumption 3.6 (Corollary 3.16 of Chapter 3). On non-contractible manifolds, closed 1-forms can fail to be exact, and the failure is measured by the de Rham cohomology group $H^1(M; \mathbb{R})$.

A.7.4 Fundamental group and de Rham H^1

For a smooth manifold M and a basepoint $x_0 \in M$, the *fundamental group* $\pi_1(M, x_0)$ is the group of homotopy classes of loops based at x_0 , with concatenation of loops as the group operation. M is *simply connected* if $\pi_1(M, x_0) = \{e\}$ for some (equivalently, every) basepoint x_0 . Equivalently, every loop in M contracts to a point.

Definition A.12 (de Rham H^1). The first *de Rham cohomology group* of M is the real vector space

$$H^1(M; \mathbb{R}) := \frac{\{\alpha : d\alpha = 0\}}{\{df : f \in C^\infty(M)\}}$$

of closed 1-forms modulo exact 1-forms.

The Poincaré lemma says $H^1(M; \mathbb{R}) = 0$ on contractible manifolds. On non-contractible manifolds, $\dim H^1$ counts the independent obstruction classes to the fundamental theorem of calculus: for each non-zero cohomology class $[\alpha]$, the line integral $\int_{\gamma} \alpha$ depends on the homotopy class of γ (not just the endpoints), and the dependence is captured by $\pi_1(M)$.

Mayer–Vietoris and stratified manifolds. For a manifold M with a stratified-singular subset Σ (e.g., $\tilde{E}$ minus the singular locus from Assumption 3.17), the de Rham cohomology of the regular part $M \setminus \Sigma$ can be computed by a Mayer–Vietoris long exact sequence in cohomology. For Σ a finite union of disjoint codim-1 hyperplanes in $M = \mathbb{R}^n$, the result is $H^1(M \setminus \Sigma) = \mathbb{R}^k$ where k is the number of locally separating hyperplanes. [Hatcher, 2002, Section 3.A] treats the Mayer–Vietoris machinery in the topological-cohomology version, which agrees with de Rham cohomology by the de Rham theorem.

A.7.5 Application to the book

Section 6.1 introduces the donor’s preference as a 1-form α on $\tilde{E}$, with closed-form ($d\alpha = 0$, $\alpha = dV$ for some V) corresponding to path-independent preferences and non-closed corresponding to path-dependent ones. The Poincaré lemma applies under Assumption 3.6 (the Carroll working paper’s working hypothesis): on the contractible $\tilde{E}$, closed-preference 1-forms are exact, and the donor’s optimization reduces to gradient flow of V . Under Assumption 3.17, the regime structure of $\tilde{E}^{\text{reg}}$ generally has nontrivial H^1 , and non-cohomologous closed 1-forms represent genuinely path-dependent preferences.

Section 7.5 takes the cohomology one step further: the path-aggregation obstruction of Proposition 7.9 uses $H^1(\tilde{E}^{\text{reg}}; \mathbb{R})$ as the natural alternative space on which voter preferences live, and the Mayer–Vietoris computation tells us H^1 has one generator per locally-separating component of Σ . The path-aggregation obstruction is exactly Chichilnisky’s continuous-anonymity-Pareto obstruction restated on the cohomology classes: when $\dim H^1 > 0$, no continuous anonymous Pareto-respecting rule resolves the cohomological disagreement.

A.8 Manifold optimal control

Section 6.4 develops the Pontryagin maximum principle on the joint equilibrium manifold as the operational tool for characterizing optimal aid schedules. The audience trained in standard optimal control (Lagrangian methods, the maximum principle on $\mathbb{R}^n$) needs the manifold adaptation; the audience without optimal-control background needs the apparatus from scratch. This section provides both, in the cursory form sufficient for Chapter 6’s arguments.

A.8.1 Cotangent bundle and 1-form fields

For a smooth manifold M , the *cotangent bundle* is the disjoint union $T^*M = \bigsqcup_{x \in M} T_x^*M$ with the natural smooth-manifold structure: a smooth section $\psi : M \rightarrow T^*M$ assigning a covector $\psi(x) \in T_x^*M$ to each point x is exactly a 1-form on M (Definition A.10). For a smooth path $\gamma : [0, T] \rightarrow M$, a *costate* is a smooth section $\psi : [0, T] \rightarrow T^*M$ along γ , i.e., $\psi(t) \in T_{\gamma(t)}^*M$ for each t . The pair $(\gamma(t), \psi(t)) \in T^*M$ is the state-costate object the Pontryagin maximum principle’s costate equation governs.

A.8.2 The Pontryagin maximum principle on a manifold

Consider the optimal-control problem

$$\min_{\gamma, u} \int_0^T L(\gamma(t), u(t)) dt \quad \text{s.t.} \quad \dot{\gamma}(t) = X(\gamma(t), u(t)), \quad \gamma(0) = e_0, \quad u(t) \in U,$$

where $\gamma : [0, T] \rightarrow M$ is a smooth path on a manifold M , $u(t) \in U \subset \mathbb{R}^k$ is the control, X is a smooth vector field depending on the control, and L is the running cost. The *Hamiltonian* is

$$\mathcal{H}(x, \psi, u) := \psi(X(x, u)) - L(x, u),$$

where $\psi \in T_x^*M$ contracts with the velocity $X(x, u) \in T_xM$ via the duality pairing.

Theorem A.13 (Pontryagin maximum principle, manifold form). *Let (γ^*, u^*) solve the optimal-control problem above. Then there exists a smooth costate $\psi^* : [0, T] \rightarrow T^*M$ along γ^* such that:*

- (1) Costate evolution: $\dot{\psi}^*(t) = -\partial\mathcal{H}/\partial x|_{(\gamma^*, \psi^*, u^*)}$ in coordinates;
- (2) Hamiltonian maximization: $\mathcal{H}(\gamma^*(t), \psi^*(t), u^*(t)) = \max_{u \in U} \mathcal{H}(\gamma^*(t), \psi^*(t), u)$ for almost every t ;
- (3) Transversality: boundary conditions on ψ^* at $t = T$ depending on the terminal-cost structure.

[Liberzon, 2011a, Chapter 4]. The manifold form is the standard $\mathbb{R}^n$ form transported to the smooth-manifold setting; the costate equation in coordinates becomes the standard form, and the Hamiltonian maximization is unchanged. The transversality condition depends on the specific terminal constraint (free terminal, fixed terminal, terminal on a submanifold).

A.8.3 Switching function and singular arcs

When the Hamiltonian is linear in the control u (so that $\mathcal{H} = \psi(X(x)) + S(x, \psi) \cdot u - L(x)$ for some *switching function* $S(x, \psi)$), the maximization condition delivers:

- bang-bang control on the boundary of U in the direction $S/\|S\|$ when $S \neq 0$;
- ambiguity (*singular arc*) when $S = 0$, which must be resolved by higher-order conditions.

The switching function evolves along the optimal trajectory according to $\dot{S}(t) = \iota_{u^*(t)} d\alpha$, where $d\alpha$ is the exterior derivative of the relevant 1-form and ι_v is the interior product of vectors with forms (the contraction $\iota_v \omega(w_1, \dots, w_{k-1}) := \omega(v, w_1, \dots, w_{k-1})$). For closed α ($d\alpha = 0$), the switching function is constant; for non-closed α , the switching function evolves with the direction of travel.

A.8.4 Strict convexity and the gradualism characterization

When the Lagrangian $L(x, u)$ is strictly convex in the control u (e.g., a quadratic transfer cost $L = \kappa\|u\|^2$), the Hamiltonian maximization has a unique interior optimum at each t , and the optimal control is smooth (no bang-bang, no singular arcs). This is the smooth-control case the calculus of variations classically handles; the Pontryagin form reduces to the Euler–Lagrange equations on the manifold. Proposition 6.10 of Chapter 6 is a three-regime characterization (pause, interior, max) that specializes the smooth-control case to a piecewise structure when the cost is convex but not strictly convex over the full control range.

A.8.5 Application to the book

Section 6.4 develops the Pontryagin characterization on the joint equilibrium manifold. $M = \tilde{E}$, $u(t) = \tau(t)$ is the donor's transfer schedule, and the running cost is the welfare gradient along the trajectory. Theorem 6.8 is exactly Theorem A.13 specialized to this setting. The bang-bang corollary *Corollary 6.9* specializes the linear-Hamiltonian case (constant switching under closed $\alpha = dV$). The gradualism proposition *Proposition 6.10* specializes the strict-convexity case under quadratic transfer cost. The switching-function evolution $\dot{S} = \iota_{u^*} d\alpha$ appears at *Equation (6.9)* and powers the chapter's distinction between path-independent (closed α , constant switching) and path-dependent (non-closed α , evolving switching) preferences.

A.9 Welfare and social-choice apparatus

Chapter 5 and Chapter 7 draw on apparatus from welfare economics and social choice theory. This section collects the specific results those chapters consume: the Pareto frontier as a smooth manifold, the utility-possibility set and its frontier diffeomorphism, the value function in constrained optimization, the Negishi welfare-weight representation, the First Welfare Theorem in its conditional form, Arrow's impossibility theorem, and the Chichilnisky–Heal aggregation theorem.

A.9.1 Pareto frontier and utility-possibility set

For a resource-feasible set $\Omega(r) := \{x \in X : \sum_i x_i \leq r\}$ at total resource profile $r \in \mathbb{R}_{++}^L$, the *conditional Pareto set* is $\mathcal{P}(r) := \{x \in \Omega(r) : x \text{ is Pareto optimal}\}$. Under strict concavity of every u_i (Assumption 2.1(2) of Chapter 2), $\mathcal{P}(r)$ is a smooth $(I-1)$ -manifold embedded in $\mathbb{R}^{IL}$. The *utility-possibility set* at r is the closed set $\bar{\mathcal{U}}(r) := \text{cl}\{u(x) : x \in \Omega(r)\} \subset \mathbb{R}^I$, and its boundary $\partial\bar{\mathcal{U}}(r)$ is the *utility-possibility frontier*.

Theorem A.14 (Pareto-frontier diffeomorphism). *Under Assumption 2.1 of Chapter 2, the utility map $x \mapsto (u_1(x_1), \dots, u_I(x_I))$ restricts to a smooth diffeomorphism $\mathcal{P}(r) \rightarrow \partial\bar{\mathcal{U}}(r)$.*

[Balasko, 2009, Proposition A.6.2]. The diffeomorphism is what makes the conditional First Welfare Theorem statable as “the equilibrium utility profile lies on the boundary of the utility-possibility set”—a geometric statement.

A.9.2 Value function and Negishi weights

The value function

$$v(r, u_{-I}) := \max\{u_I(x_I) : x \in \Omega(r), u_i(x_i) \geq u_{-I,i} \text{ for } i < I\}$$

is the maximum utility state I achieves when the resources are r and the others' utilities are pinned at u_{-I} . v is smooth on the open subset where the constraint set is non-empty, and strictly increasing in r [Balasko, 2009, Proposition A.5.1].

The *Negishi welfare-weight representation* expresses the Pareto frontier $\mathcal{P}(r)$ via a smooth bijection $T(r, \cdot) : \mathbb{R}_{++}^{I-1} \rightarrow \mathcal{P}(r)$ parameterized by welfare weights $\lambda \in \mathbb{R}_{++}^{I-1}$. At any joint equilibrium (p^*, m^*, x^*) at ω , the welfare weights are $\lambda_i^* \propto 1/\mu_i^*$, where μ_i^* is the consumer-side Lagrangian multiplier for state i 's budget constraint at the equilibrium [Balasko, 2009, Proposition A.6.1].

A.9.3 First Welfare Theorem

Theorem A.15 (First Welfare Theorem, classical form). *A competitive equilibrium of a smooth exchange economy with locally non-satiated preferences is Pareto optimal: no other feasible allocation Pareto-dominates the equilibrium allocation.*

[Arrow, 1951]; see Mas-Colell et al. [1995, Chapter 16] for the modern textbook treatment. The conditional First Welfare Theorem of Chapter 5 (*Proposition 5.4*) is Theorem A.15 applied at the post-conflict resource profile r^* rather than at the full endowment ω . The conditional formulation is necessary because the joint equilibrium does not, in general, attain the full-endowment Pareto frontier; the conflict externality Δ^* measures the displacement.

A.9.4 Arrow’s impossibility theorem

Theorem A.16 (Arrow’s impossibility theorem). *On a finite alternative set with at least three elements, no preference-aggregation rule satisfies all four of: universal domain, independence of irrelevant alternatives, Pareto unanimity, and non-dictatorship.*

[Arrow, 1951]. Arrow’s theorem applies to discrete alternative sets; the continuous-domain analogue is Chichilnisky–Heal below. The framework’s path-aggregation obstruction in Section 7.5 invokes Arrow on the finite-but-three-or-more set of homotopy classes of paths between fixed endpoints in a multi-component $\tilde{E}^{\text{reg}}$.

A.9.5 Chichilnisky–Heal aggregation theorem

Theorem A.17 (Chichilnisky–Heal). *Let X be a smooth manifold and $P(X)$ the space of smooth preference profiles on X , equipped with the topology of continuous sections of the unit tangent bundle. A continuous, anonymous, Pareto-respecting aggregation rule $\mathcal{A} : P(X)^I \rightarrow P(X)$ exists if and only if $P(X)$ is contractible.*

[Chichilnisky, 1980, Chichilnisky and Heal, 1983]. Contractibility is a homotopy-theoretic condition: $P(X)$ is contractible if it is homotopy-equivalent to a point. The framework’s *Corollary 7.2* applies the theorem to the contractible $\tilde{E}$ under Assumption 3.6; *Corollary 7.3* applies it componentwise to the multi-component $\tilde{E}^{\text{reg}}$ under Assumption 3.17.

A.9.6 Application to the book

Section 5.1 invokes Theorem A.14 to identify the conflict externality Δ^* as a displacement on the utility-possibility frontier; Section 5.2 invokes the value function and the Negishi welfare-weight representation to deliver *Corollary 5.5* and *Proposition 5.4* as the conditional analogue of Theorem A.15. Section 7.2 invokes Theorem A.16 as the discrete-alternative-set baseline and Theorem A.17 as the smooth-manifold extension that the framework’s contractibility-of- $\tilde{E}$ result (*Corollary 3.16* of Chapter 3) makes applicable.

A.10 Synthesis: the master comparative-static formula

This synthesis section walks through the master comparative-static formula

$$D_{\theta}(p^*, m^*) = -J^{-1}D_{\theta}e$$

of Section 4.1, identifying which piece of the appendix's apparatus each step uses. The formula is the appendix's organizing example: a single calculation that activates every body section. A reader who has worked through the earlier sections should find each step here a direct application; a reader who came to this synthesis first should treat it as a roadmap and dip back into the body sections for any step that is unclear.

A.10.1 Setup: the equilibrium identity

The joint equilibrium function $e : S_N \times M \times \Omega_0 \rightarrow \mathbb{R}^{L-1+I}$ is defined by

$$e(p, m, \omega) := (\bar{d}(p, m, \omega), r(p, m, \omega)),$$

where $\bar{d} \in \mathbb{R}^{L-1}$ is the aggregate excess-demand vector (in the $L - 1$ non-numéraire goods, after Walras' law absorbs the L th) and $r \in \mathbb{R}^I$ is the Nash residual. The equilibrium manifold $\tilde{E} := e^{-1}(0)$ is the zero set of e . Under the SPT primitives of Chapter 2 and the regularity hypotheses of Chapter 3, $\tilde{E}$ is a smooth submanifold of $S_N \times M \times \Omega_0$ of dimension IL (Proposition 3.8 of Chapter 3, via the preimage theorem Theorem A.4 of Section A.5).

What we want. A parameter perturbation θ —an endowment shift, a contest-parameter shift, a tariff sanction, a donor transfer—moves the equilibrium (p^*, m^*) . The master formula expresses the response $D_\theta(p^*, m^*)$ as a single matrix application $-J^{-1}D_\theta e$.

A.10.2 Step 1: differentiate the identity

At any equilibrium $(p^*, m^*, \omega) \in \tilde{E}$, we have the identity $e(p^*, m^*, \omega) \equiv 0$. If $J = D_{(p,m)}e(p^*, m^*, \omega)$ is invertible (the joint regularity hypothesis Assumption 3.6), then by the implicit function theorem (Theorem A.1 of Section A.3), there exist smooth local solutions $p^*(\omega)$ and $m^*(\omega)$ defined on a neighborhood of ω with $e(p^*(\omega), m^*(\omega), \omega) \equiv 0$.

Differentiating the identity in ω using the chain rule (Section A.3.3):

$$J \cdot D_\omega(p^*, m^*) + D_\omega e = 0,$$

so that $D_\omega(p^*, m^*) = -J^{-1}D_\omega e$. This is the master formula for $\theta = \omega$; parameter perturbations of other primitives (contest parameters, tariffs, etc.) follow the same logic with the appropriate $D_\theta e$.

What we used. The IFT (Theorem A.1, §A.3) delivers existence of the local solution. The chain rule for total derivatives (Section A.3.3) delivers the differentiated identity. The smooth-manifold structure of $\tilde{E}$ (Section A.5.1) ensures the local solution stays on the manifold.

A.10.3 Step 2: the block structure of J

The joint Jacobian partitions naturally into a 2×2 block matrix

$$J = \begin{pmatrix} J_{pp} & J_{pm} \\ J_{mp} & J_{mm} \end{pmatrix}, \quad J_{pp} = D_p \bar{d}, \quad J_{pm} = D_m \bar{d}, \quad J_{mp} = D_p r, \quad J_{mm} = D_m r.$$

The diagonal blocks are within-channel (price-price and deployment-deployment); the off-diagonal blocks are cross-channel. Inverting J via the Schur block-decomposition (Theorem A.2 of Section A.4) yields J^{-1} as the four-block expression

$$J^{-1} = \begin{pmatrix} S^{-1} & -S^{-1}J_{pm}J_{mm}^{-1} \\ -J_{mm}^{-1}J_{mp}S^{-1} & J_{mm}^{-1} + J_{mm}^{-1}J_{mp}S^{-1}J_{pm}J_{mm}^{-1} \end{pmatrix},$$

where $S = J_{pp} - J_{pm}J_{mm}^{-1}J_{mp}$ is the Schur complement. The diagonal blocks of J^{-1} recover the partialled inverses of the bargaining-camp (J_{mm}^{-1}) and GE-camp (S^{-1}); the off-diagonal blocks of J^{-1} are the cross-channel corrections that the partialled formulations zero out.

What we used. Block matrices and the Schur block-decomposition (§A.4). The identification $S = D_p\hat{d}$ (the Schur complement equals the reduced excess-demand Jacobian) is *Lemma 3.4* of Chapter 3.

A.10.4 Step 3: when J is invertible

The master formula requires J to be invertible at the equilibrium where we apply it. The framework offers two regularity hypotheses, in increasing order of strength.

Under Assumption 3.6: J is invertible at every $(p, m, \omega) \in \tilde{E}$. This is the strong working hypothesis used for the global-diffeomorphism result of *Proposition 3.15* (via Theorem A.7 of Section A.5).

Under Assumption 3.17: J is invertible at every $(p, m, \omega) \in \tilde{E}$ outside a measure-zero singular locus $\Sigma \subset \Omega_0$. This is the standard hypothesis of regular-economy theory. *Theorem 4.3* of Chapter 4 establishes Σ 's measure-zero character via parameterized Sard's theorem (Theorem A.6 of Section A.5), with the non-triviality input *Lemma 4.2* supplying the SPT-specific verification that the relevant rank condition holds.

What we used. Sard's theorem and parameterized Sard (§A.5). The smooth-manifold structure of $\tilde{E}$ and the regular part $\tilde{E}^{\text{reg}}$ (§A.5).

A.10.5 Step 4: counting equilibria

The framework's existence and cardinality results are consequences of the mod-2 degree of $\tilde{\pi} : \tilde{E} \rightarrow \Omega_0$: *Theorem 3.12* of Chapter 3 establishes $\deg_2(\tilde{\pi}) = 1$ via a homotopy to the Walrasian-endpoint exchange economy (where the standard mod-2 degree result applies). *Corollary 3.19* specializes: the number of joint equilibria at any regular ω is finite and odd.

The signed refinement is the index theorem *Theorem 4.5* of Chapter 4: the signed sum of indices over $\tilde{\pi}^{-1}(\omega) \cap \tilde{E}^{\text{reg}}$ is $+1$, transported from the same Walrasian-endpoint via the proper-homotopy invariance of signed degree (Theorem A.8 of Section A.6).

What we used. Mod-2 degree (§A.6). Index theorem and signed-sum identity (§A.6).

A.10.6 Step 5: the master formula in deployment chapters

The deployment chapters consume the master formula in different ways. Chapter 5 differentiates the conflict externality Δ^* along smooth paths in Ω_0 , giving the sensitivity formula

Equation (5.10); the calculation uses the master formula's response $D_\omega(p^*, m^*)$ together with the Negishi welfare-weight representation (§A.9). Chapter 6 lifts the master formula to the donor's optimal-control problem on $\tilde{E}$; the Pontryagin maximum principle (§A.8) characterizes the optimal donor schedule, with the switching function evolution governed by the closed-vs-non-closed status of the donor's preference 1-form (§A.7). Chapter 7 reads institutional design as a vector field on $\tilde{E}^{\text{reg}}$ aligned with the welfare gradient X^{Δ^*} , with the path-aggregation obstruction (§A.7, §A.9) bounding which institutions deliver continuous, anonymous, Pareto-respecting aggregation across regimes. Chapter 8 runs the master formula on sanctions parameters, with the sliced-vs-joint distinction following directly from the Schur block-decomposition (§A.4).

A.10.7 Closing

The master comparative-static formula is the apparatus's central operational tool, and every body section of this primer enters its derivation or its application: the IFT (§B), block-matrix algebra (§C), smooth-manifold structure (§D), Sard genericity (§D), the index theorem (§E), differential forms and cohomology (§F), Pontryagin (§G), and welfare-and-social-choice (§H). The structure of the apparatus is exactly the structure required to make the master formula's claim—that joint comparative statics is one matrix application, with both the partialled inverses and their cross-channel corrections legible inside J^{-1} —both well-defined and operationally usable. The body chapters compute with this formula on every parameter shift the framework admits; the appendix collects the apparatus that makes the formula precise.

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